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Your Learning Club

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Why Choose Us?

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What People Say About CP?

Jordon Smith

I am happy to have chosen Certification Planner for the PMP certification training.

Jay Walker

I took a Lean Six Sigma Green Belt (LSSGB) course with Certification Planner.

Jessica Williams

Awesome learning experience & excellent instructor having knowledge about subject.

Andrea Jones

Had a great experience with Certification Planner.

Ricky Harrington

Certification Planner has done a great job to design entire course material and classes.

Carol Johnson

Joined LSS course & very happy with training and the material they offered for the same.

Barbara Hewitt

Have attended PMP class. Was great learning experience with an outstanding instructor.

Sophia Wilson

I would definitely recommend CP to my friends and colleagues.

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Your comfort is our priority. Hence, we have designed our training programs, keeping every individual in mind.
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ACCREDITATION

Project Management Institute

IASSC
LEAN SIX SIGMA CERTIFICATION

AXELOS
GLOBAL BEST PRACTICE

PeopleCert®

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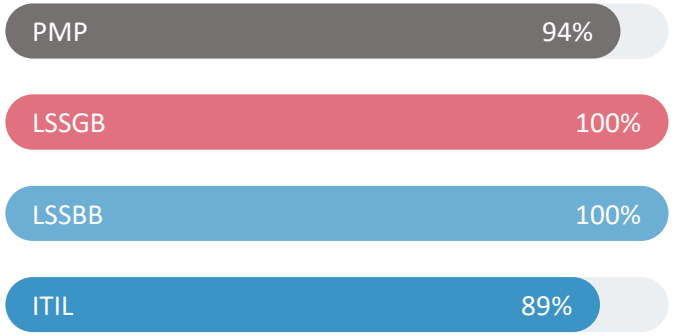
About Certification Planner

Our creative approach and progressive thinking makes us stand out from the crowd. Certification Planner is an organization that is driving progress by offering training on advanced and trending subjects. Being a responsible training provider, we always think about what further contribution we can make in assisting people to shape-up their career. Our clients are our priority, hence every new initiative at CP starts with a question: how can we deliver better to our customers? CP has a clear vision of what we want to achieve, and we guide our business towards the same direction.

At CP, we believe learning and adapting new technologies and upgrading skillset is the only way to achieve sustainable growth

CEO
Suhaan Kkhanna
LSSGB, LSSBB, ITIL Foundation

Success Rate:



Veteran Planner

"Power to Heroes"

An initiative to support and upskill transitioning veterans into the corporate world.

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Live Training

There is a real difference between hearing and listening. We hope this course material provides you with the reinforcement of the DMAIC process information gained through live training. If you are listening to the recorded session of this course, we highly recommend that you also participate in a live class. CertificationPlanner offers both in-person and live virtual (on-line) training courses. The ability to interact with a live instructor and other students exercises additional areas in your brain, to better reinforce and commit the knowledge presented. As a live student, you will have the ability to ask questions that interest you, obtain examples that relate to your area or expertise, access after class help, revisit questions on the material after class, obtain any resources provided by fellow students, as well as get tips on how to take the exam.

Contact Certification Planner(CP) today for more information.
Email: Support@certificationplanner.com Phone: +1855.322.1201 or
www.certificationplanner.com

Once again welcome to Certification Planner. Let us know if you need any assistance.

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About This Tutorial

There are a total of 6 lessons in the Dual Lean Six Sigma Green and Black Belt certification course.

1. Overview of Lean and Six Sigma
2. Define Phase
3. Measure Phase
4. Analyze Phase
5. Improve Phase
6. Control Phase
7. Practice Exams
8. Final Exam

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Lesson 00 –
Overview of Lean and Six Sigma

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Organizational Objectives

- Successful organizations set and strive to achieve objectives
  - An organizational objective reflects the organization's goals, mission, and vision
- In addition, these objectives should consider:
  - Where an organization currently sees itself in an industry and where it wants to be
  - What are its strengths, weaknesses, opportunities, and threats?

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Lean Six Sigma Yellow Belt

- These are the project-specific, full-or part-time resources that provide process and cross-functional knowledge, as well as help to sustain the gains. They have co-ownership of the project with the Six Sigma Experts and are responsible for the quality of the work and results.

- This team also plays the critical role of translating the process gains from Six Sigma to other areas of the business after the specific project has been completed. This is the true leverage of Six Sigma methodology!

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Deliverables of a Yellow Belt

- A Yellow Belt has basic knowledge of Six Sigma
- They do not lead projects on their own, as does a Green Belt or Black Belt.
- YB participates as a core team member or subject matter expert (SME) on DMAIC project or projects. Supports Green Belt or Black Belt in developing process maps, helping with data capture, facilitating simulation, and improvements.
- YBs may often be responsible for driving smaller process improvement projects using Lean tools or best practice sharing in their processes.

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Benefits of being a Yellow Belt for Organization:

- For any project, Yellow Belts are those individuals who are the Subject Matter Experts (SME's) of their respective processes and also have the basic know-how of Six Sigma. They are the spokes of a wheel and can help drive any Six Sigma process to closure by using their process expertise. Organizations can greatly benefit by choosing the right YB's for the right projects.

- For yourself and your career, YB's get exposure of channelizing their Subject knowledge to process improvement opportunities yielding tremendous benefits for self understanding.

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Lean Six Sigma Green Belt

- Six Sigma Green Belts are the engine of Six Sigma projects. Black Belt's support the efforts of the broader business teams to identify and implement change. The GB's are part-time Six Sigma Project Leaders. They are responsible for scoping the projects, leading the project team, calling for help when needed, managing interfaces with business leaders, and ensuring sustainable results.

- The goal of GB's is to translate the value of Six Sigma to the specific work environment and problems.

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Deliverables of a Green Belt

- Project Execution
- Team and Project Structuring
- Six Sigma Project Results
- Share Best Practices

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Benefits of being a Green Belt for Organization:

- GB's have authority in their respective processes and can get the work done effectively. This is a very critical aspect for the organization as it builds its process improvement structure within each process.

- For yourself and your career, GB's receive exposure to senior management directly by virtue of the projects and get the opportunity to make a difference in the organization.

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Lean Six Sigma Black Belt

- Six Sigma BB's are full-time/part time project leaders and mentors of the business, including Green Belts and other associates. They have tactical responsibility for executing specific projects and ensuring that the results are captured, the changes are owned by the Champions (Sponsors), and the changes are sustained. They will also lead Six Sigma knowledge transfer to both full- and part-time participants.

- BB's are expected to create an environment of open, honest debate of facts. They challenge the status quo where appropriate and share (and seek) ideas across boundaries.

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Deliverables of a Black Belt

- Lean and Six sigma strategy and roll-out plan for the given process/area
- Execute Projects
- Help and guide Project Resources/ Help remove project level Barriers
- Team and Project Structuring
- 6 sigma Project Results
- Mentor Green Belts
- Share Best Practices

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Benefits of being a Black Belt for Organization and for self-career:

- BB’s are responsible for taking the process improvements to the next level in the organization.

- BB’s are highly trained on improving results for the organization using statistical analysis and Six Sigma tools. Hence, they have a very lucrative career path ranging from Business Analysts to Process Improvement experts.

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Profit: The Common Objective

- Critical to every organization's operational viability and success is its profitability
- Lean Six Sigma practices share the organization's objectives to maximize profits, with its primary focus being "Savings" (Cost reduction)

- At the most basic level, profitability can be demonstrated with the following formula:

Revenue	$ 100,000
- Costs	- $ 30,000
Profit	$ 70,000

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Profits & Lean Six Sigma

Following is a high-level look at how different Lean Six Sigma practices interact with this profit model

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Course Overview

- This course will provide an extensive amount of information, insight, expectations, and recommendations

- The ultimate purpose of obtaining at Lean Six Sigma Green Belt is to continually seek valid, quality improvement efforts with educated and skilled methods

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Lesson Completed

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1.0 Define

The "Define" phase is about:

- ✓ **Selecting high-payback projects**
- ✓ **Identifying the underlying metric(s) that will measure project success**

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Define Phase

Six Sigma: The Basics

- The Many Meanings of Six Sigma
- History of Six Sigma & Continuous Improvement
- Roles & Responsibilities
- Project Team Selection

Six Sigma: The Fundamentals

- Voice of the Customer (VOC)
- Critical to Quality (CTQ)
- Affinity Diagrams & Pareto Analysis (80:20)
- Cost of Poor Quality (COPQ)
- Six Sigma Projects: DMAIC
- Basic Six Sigma Metrics

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Define Phase

Project Management Basics

- Change
- Business Case & Project Charter
- Cost Benefit Analysis
- Project Risk Management
- Failure Modes & Effects Analysis (FMEA)

Lean

- Understanding Lean
- Lean: History
- Lean vs. 7 Deadly Wastes
- Lean & Six Sigma

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1.1 Six Sigma: The Basics

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Primer: DMAIC

Six Sigma project phases are referenced as **DMAIC**

- **DEFINE** the customer, their critical to quality (CTQ) requirements, and the core business process involved. Select the appropriate responses (the “Y’s”) to be improved
- Data must be gathered to **MEASURE** the performance of the core business process involved
- **ANALYZE** the data collected to identify the root causes of defects, defectives, or significant measurement deviations whether in or out of specifications and opportunities for improvement
- **IMPROVE** the target process by designing creative solutions to fix and prevent problems, reduce variability, or eliminate the cause
- With the desired improvements in place, monitor and **CONTROL** the process to sustain the improvements

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1.1.1 The Many Meanings of Six Sigma

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What is “Six Sigma”?

Though the statistical term “sigma” refers to a measure of variance, it has evolved as a discipline to include many aspects of business performance and improvement

Following are various facets of “Six Sigma”:

- Measure of Variance
- Process Measurement
- Business Strategy & Approach
- Methodology
- Philosophy
- Benchmark
- Project
- Tools

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Measurement: Variance = Bad

Variance away from a target value increases the chances of a product, service, or etc. failing, being rejected, or not purchased by a customer

- You will come to learn that **variation is the enemy of any quality process**.
  - ➢We need to understand, manage, and minimize process-based variation

- A Six Sigma "**goal**" is for a process to **minimize variation**

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Six Sigma Performance Measurements

LT Sigma Level	ST Sigma Level	C_p	DPMO	LT Sigma Level	ST Sigma Level	C_p	DPMO	LT Sigma Level	ST Sigma Level	C_p	DPMO
4.5	**6**	**2.00**	3.4	2.8	4.3	1.43	2,555	1.1	2.6	0.87	135,666
4.4	5.9	1.97	5	2.7	4.2	1.40	3,467	1	2.5	0.83	158,655
4.3	5.8	1.93	9	2.6	4.1	1.37	4,661	0.9	2.4	0.80	184,060
4.2	5.7	1.90	13	2.5	**4**	**1.33**	6,210	0.8	2.3	0.77	211,855
4.1	5.6	1.87	21	2.4	3.9	1.30	8,198	0.7	2.2	0.73	241,964
4	5.5	1.83	32	2.3	3.8	1.27	10,724	0.6	2.1	0.70	274,253
3.9	5.4	1.80	48	2.2	3.7	1.23	13,903	0.5	**2**	**0.67**	308,538
3.8	5.3	1.77	72	2.1	3.6	1.20	17,864	0.4	1.9	0.63	344,578
3.7	5.2	1.73	108	2	3.5	1.17	22,750	0.3	1.8	0.60	382,089
3.6	5.1	1.70	159	1.9	3.4	1.13	28,717	0.2	1.7	0.57	420,740
3.5	**5**	**1.67**	233	1.8	3.3	1.10	35,930	0.1	1.6	0.53	460,172
3.4	4.9	1.63	337	1.7	3.2	1.07	44,565	0	1.5	0.50	500,000
3.3	4.8	1.60	483	1.6	3.1	1.03	54,799	-0.1	1.4	0.47	539,828
3.2	4.7	1.57	687	1.5	**3**	**1.00**	66,807	-0.2	1.3	0.43	579,260
3.1	4.6	1.53	969	1.4	2.9	0.97	80,757	-0.3	1.2	0.40	617,911
3	4.5	1.50	1,350	1.3	2.8	0.93	96,800	-0.4	1.1	0.37	655,422
2.9	4.4	1.47	1,866	1.2	2.7	0.90	115,070	-0.5	**1**	**0.33**	691,462

- This table shows each sigma level's corresponding DPMO (defects per million opportunities) and short-term Capability Index (C_p)
- The higher the sigma level, the lower the defect rate and DPMO

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Six Sigma: Philosophy

The Six Sigma philosophy is that of **"Continual Improvement" through** the use of **data** and specific **variation reduction techniques,** resulting in **breakthrough improvements**

Competitive Advantage is gained by meeting customer expectations (products and services that perform within spec limits) in the most **efficient** and **effective** way possible

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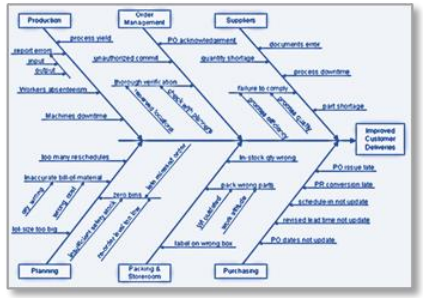
Six Sigma: Tools

Six Sigma utilizes specific Tools for business problem solving

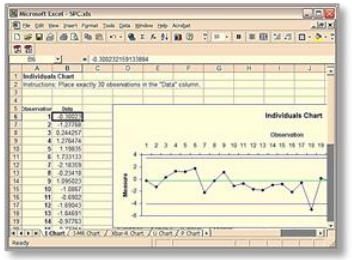
- Six Sigma tools are used to scope and select projects, modify and/or design new processes, improve current processes, decrease downtime and improve customer response time

FMEA

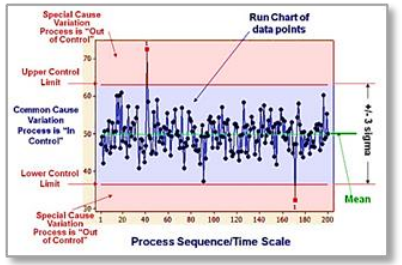
Matrices



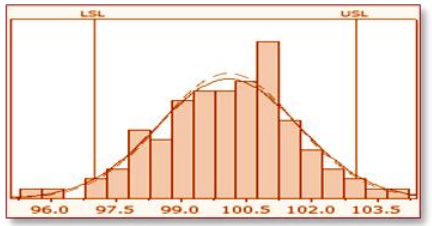
Fishbone



Data Analysis



SPC



Histogram

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Six Sigma: Business Strategy

Six Sigma as a Business Strategy is the integration of its philosophy into action, in order to achieve corporate objectives.

- It links the outputs of the processes conducted within an organization to the expectations of the customers
  - ➢ The strategy is realized as waste and variation expenses are reduced, and customer satisfaction increases thus improving operating margins by as much as 50%.

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1.1.2 Problem Solving Strategy: Y=f(x)

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Six Sigma: Approach

The Six Sigma approach is built around the Cause and Effect model

- Where **you want to change the Effect, you focus on the Cause(s)**
  - If done properly, the Effect, as a dependent product, will reflect the correct control of the independent Cause(s)

- The first portion of this approach consists in the general identification of these Causes and Effects, including:
  - Effects: Identifying the **customer requirements** for the **process outputs**
  - Causes: Identifying the various **processes** involved in delivering those specific requirements
  - Causes: Identifying the respective **inputs** that go into the processes that impact the required characteristics

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Six Sigma: Approach

The second portion of the Six Sigma approach consists of:

- Establishing the **specific** and **measureable** characteristics of the customer's requirements (CTQ's to be discussed later)
- **Prioritization**, by the customer, of their requirements
- Identify which processes **impact** the characteristics, or value, of the customer product the most
- **Prioritize** and **streamline** process efficiency
- Progressively identify the **critical inputs** by:
  - Identifying those which have a significant influence on the customer product and prioritize them for any needed improvements
  - Identifying those which have little to no significant influence on the customer product and eliminating them from the project focus

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Six Sigma Approach: Output “Y”

OUTPUT: The customer product characterized by its ability to meet requirements

- Outputs are:
  - **Dependent:** Changed directly or indirectly by either a process and/or an input
  - Also known as “**Y**”
  - Y is the measure of process performance that you are trying to improve
  - The result or effect of various causes
  - Measure or output of a process
  - Y is usually the primary metric
    - Prioritized and critical outputs are also known as **Key Process Output Variables (KPOV)**

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Six Sigma Approach: Input “X”

INPUT: The general materials and efforts that go through a series of steps (process) in order to positively influence the creation of an output that meets customer demands

Inputs are:

- **Independent:** Need to be changed directly
- Either supplier provided or an output from another process
- Known as “**X**” or a/the **Cause**
- X’s affect the Y
- Prioritized and most influential inputs are also known as **Key Process Input**
- **Variables (KPIV)** and/or **Critical X’s**

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Six Sigma Approach: Function “f(x)”

PROCESS: The process(es) act as functions that **transform input(s) “X”** into an **output(s) “Y”** that meets customer requirements

- These functions include business systems, processes, sub-processes, steps, or etc.

Processes are:

- Designated using the transfer function syntax: “**f(x)**”
  - ➢Referred to as a/the function; function of x; “f of x”
- Active and procedural, e.g. has steps, stages, or sub-processes
- Primary factor in efficient productivity
- Need to be evaluated for improvement in efficiency, quality, yield, and necessity

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Six Sigma Objective

The **objective** in a Six Sigma project is to identify the **critical input x**'s that have the greatest influence on the output "y" and **adjust them so that the output "y" improves**

- In simple terms: My process "f(x)" output "y" is dependent on various input "x's"

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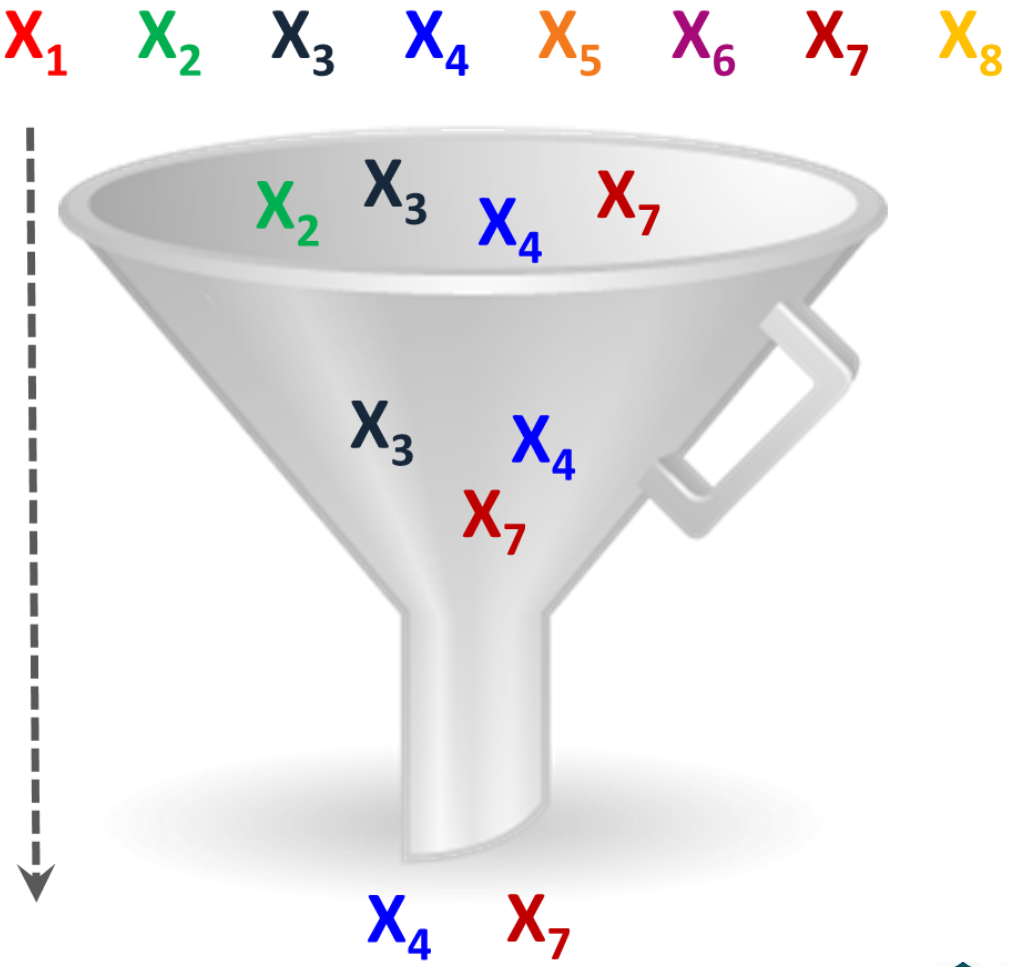
Six Sigma Approach: Vital Few x's

- With this approach, all potential x's are identified, evaluated, and respectively improved, through the Six Sigma "DMAIC" methodology (discussed soon)

- The x's should be narrowed down until the vital few x's that significantly influence "on-time pizza deliveries" are identified
  - A funnel is the most commonly utilized imagery to demonstrate the "filtering" of all possible x's down to the vital few



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1.1.3 History of Six Sigma & Continuous Improvement

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Quality Gurus

QUALITY GURU	CONTRIBUTION
Philip B. Crosby	• Author: "Quality is Free" • Do things right the first time • Four cardinal rules of quality management: 1. Performance standard of zero defects 2. Quality = Conformance to Requirements 3. Quality achieved by preventing defects 4. Measure of Quality's value = Cost of Nonconformance
George D. Edwards	• First president of the American Society for Quality • Head of inspection for the engineering department for Bell Telephone Laboratories

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Quality Gurus

QUALITY GURU	CONTRIBUTION
Walter A. Shewhart	• Control charts • Plan-Do-Check-Act (PDCA) • Use of statistics for improvement • Top engineer at Western Electric and Bell Telephone Laboratories • Author: "Economic Control of Quality of Manufactured Product"

W. Edwards Deming	• Plan-Do-Check-Act (PDCA) -> Plan-Do-Study-Act (PDSA)
	• "Seven deadly diseases of the workplace"
	  • Lack of consistency in planning; short-term profit emphasis, and etc.
	• "14 points for management"
	  • Consistency of purpose, embrace the new quality philosophy, and etc.

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Quality Gurus

QUALITY GURU	CONTRIBUTION
Armand V. Feigenbaum	• Promoter of Total Quality Management (TQM) • Nine M's of Quality • Markets, Management, Men, Money, Motivation, Materials, Machines, Modern Info, and Mounting • Quality standards tied to what the Customer is willing to pay for

Joseph M. Juran	• Quality Control from the Customer perspective
	• Management involvement in quality improvement
	• Quality Trilogy, including:
	  • Use of statistical control methods to mitigate problems
	  • Use of breakthrough sequence for resolving problems
	  • Annual quality reviews for the purpose of institutionalizing quality control efforts

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Quality Gurus

QUALITY GURU	CONTRIBUTION
Kaoru Ishikawa	• Total Quality Control (Japan) • Cause & Effect / Fishbone / Ishikawa Diagram for Root Cause Analysis • Quality Circles • Employees should be involved as much as possible in quality management • Company Wide Quality Control (CWQC) • Hoshin Kanri – a strategic management methodology
Genichi Taguchi	• Advocate of quality management (Japan) • Author: “Design of Experiments” • Signal-to-Noise measurement • Worked for Japan’s Ministry of Public Health and Welfare • Promoted use of statistics in quality management

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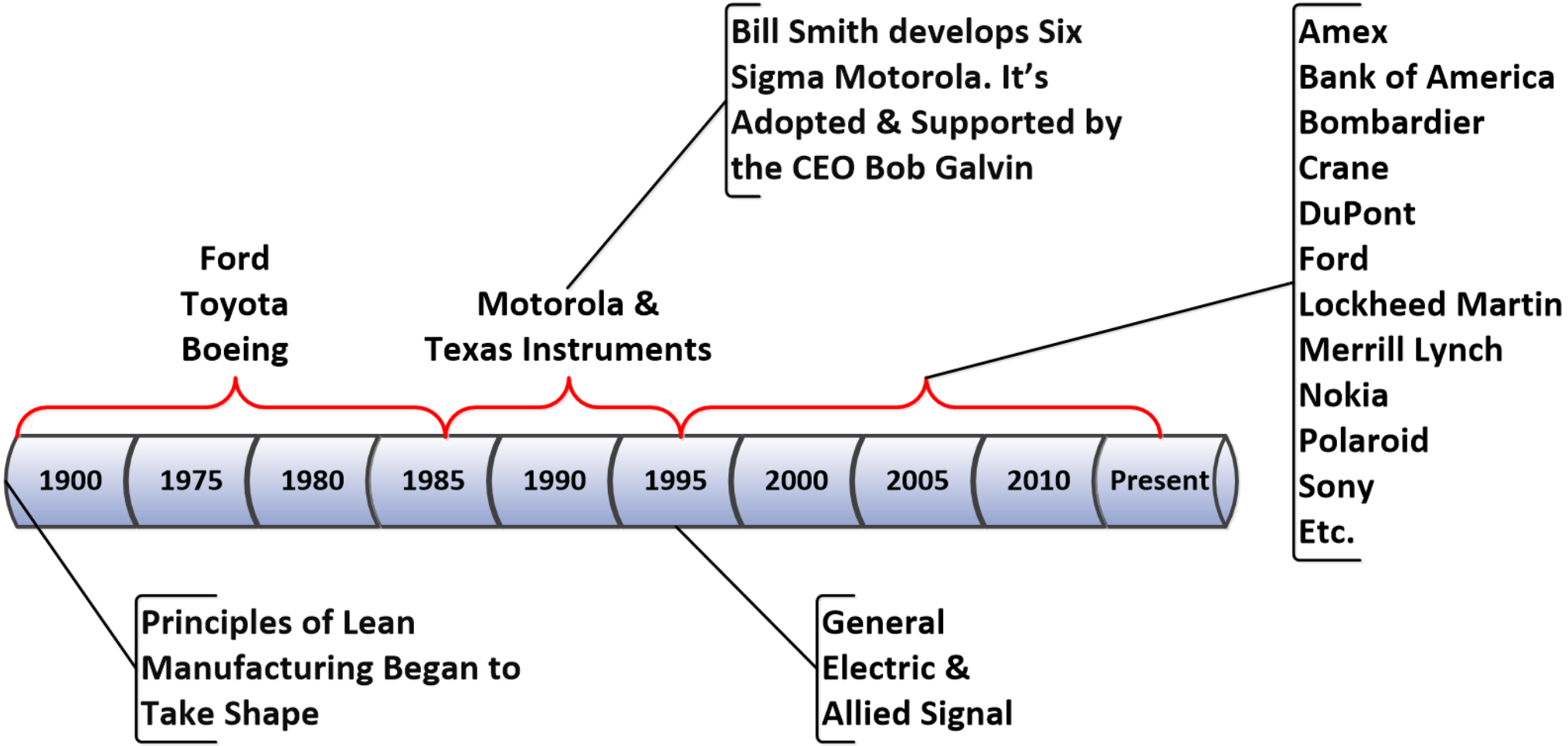
Lean Six Sigma: Historical Snapshot



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1.1.4 Roles & Responsibilities

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Six Sigma Roles

Each of the four Six Sigma “belts” represents a different level of expertise in the field of Six Sigma:

- Six Sigma Master Black Belt (MBB)
- Six Sigma Black Belt (BB)
- Six Sigma Green Belt (GB)
- Six Sigma Yellow Belt (YB)

In addition to Belts, there are other critical and complementary roles:

- Champions
- Sponsors
- Stakeholders
- Subject Matter Experts (SMEs)

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Organization Level

- The Six Sigma discipline is organization wide
- The **most important** part of a Six Sigma project's success is leadership support

➢Leadership by example: "walk the talk"

- Encourage and reward individual initiative
- Align incentive systems to support desired behaviors
- Eliminate functional barriers
- Embrace "systems" thinking
- Balance standardization with flexibility

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Executive Leadership

The Executive Leadership **owns the vision for the business**, they provide **sponsorship** and set **expectations** for the results from Six Sigma.

They **enable** the organization to apply Six Sigma then monitor the progress against expectations

- Makes decision to implement the Six Sigma initiative and **develop accountability** method
- Sets meaningful **goals** and **objectives** for the corporation
- Sets performance **expectations** for the corporation
- Ensures continuous improvement in the process
- Eliminates barriers

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Champions & Sponsors

- Champions identify and **select** the most meaningful **projects** to work on, they provide **guidance** to the Six Sigma belt and open the doors for the belts to apply the process improvement technologies

- **Own Project selection**
  - ➢Obtain needed project resources and eliminates roadblocks
  - ➢Participate in all project reviews
  - ➢One to three hours per week commitment

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Champions & Sponsors

- Champions and sponsors play critical roles in the successful deployment of Six Sigma
- **Strong endorsement of Six Sigma from the leadership team is critical for success!**
  - Typical Responsibilities of a Champion/Sponsor include:

- Maintains a strategic oversight
- Establishes strategy and direction for a portfolio of projects
- Clearly defines success
- Assists in project team formation

- Provides resolution for issues such as resources or politics
- Establishes routine tollgates or project reviews
- Clears the path for solution implementation

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Master Black Belt (MBB)

- The Master Black Belt (MBB) is the most experienced, educated, and capable Six Sigma expert

- The MBB can simultaneously lead multiple Six Sigma Belt projects while mentoring and certifying Black Belt and Green Belt candidates
  - ➢A typical MBB has managed dozens of Black Belt level projects

- The MBB typically works with high-level operations directors, senior executives, and business managers to help with assessing and planning business strategies and tactics

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MBB Responsibilities

- Reports-out/updates stakeholders and executives
- Establishes organization's Six Sigma strategy/roadmap
- Leads the implementation of Six Sigma
- Provides advice and counsel to Executive Staff
- On site mentoring
- Facilitates cultural change

- Identifies and defines the portfolio of projects required to support a business strategy
- Establishes scope, goals, timelines, and milestones
- Assigns and marshals resources
- Trains and mentors Green Belts, Black Belts, and others
- Facilitates tollgates or checkpoints for Belt candidates

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Black Belt (BB)

The Black Belt is well rounded in terms of project management, statistical analysis, financial analysis, meeting facilitation, prioritization, and a range of other value-added capabilities. BB's project scope is enterprise wide and results in a higher level of cost savings

- A typical BB:

  - Facilitates DMAIC teams in applying Six Sigma methods to solve problems
  - Led multiple projects
  - Trained and mentored various Green Belts candidates
  - Understood how to define a problem and drive effective solution

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BB Responsibilities

Project Management

- Defines projects, scope, teams etc.
- Marshals resources
- Establishes goals, timelines, and milestones
- Provides reports and/or updates to stakeholders and executives

Team Management

- Chooses or recommends team members
- Defines ground rules for the project team
- Coaches, mentors, and directs project team
- Coaches other Six Sigma Belts
- Manages the team's organizational interfaces

Task Management

- Establishes the team's Lean Sigma Roadmap
- Plans and implements the use of Lean Sigma tools
- Facilitates project meetings
- Does Project Management of the team's work
- Manages progress toward objectives

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Green Belt (GB)

Green Belts are practitioners of Six Sigma Methodology and typically work within their functional areas (Line of Business LOB) or support larger Black Belt Projects

A GB is exposed to all the comprehensive aspects of Six Sigma with less focus on the statistical theories and some other advanced analytical methodologies such as Design of Experiment (DOE)

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Green Belt (GB)

- When it comes to project management, a GB has almost the same responsibilities as a BB
- A GB responsibilities differ in:
  - **Scope**: Intra-departmental, Line-of-Business (LOB), Functional department
  - **Financial Benefits**: Usually less than $200,000 in cost savings
  - **Time**: Projects usually last less than 90 days

- Well versed in the definition & measurement of critical processes
- Creating Process Control Systems
- Involved in identifying improvement opportunities
- Involved in continuous improvement efforts
- Supporting projects with process knowledge & data collection

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Yellow Belt (YB)

- The Yellow Belt (YB) understands the basic objectives and methods of a Six Sigma project
- YB has an elementary understanding about what other Six Sigma Belts (GB, BB, MBB) are doing to help them succeed
- In a Six Sigma project, YB usually serves as a subject matter expert regarding some aspects of the process or project
- Supervisors, managers, directors, and sometimes executives are usually trained at the YB level

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YB Responsibilities

- Helps define process scope and parameters
- Contributes to team selection process
- Assists in information and data collection
- Participates in experiential analysis sessions (FMEA, Process Mapping, Cause and Effect etc.)
- Assists in assessing and developing solutions
- Yellow Belts apply the Six Sigma Methodology to their own work responsibilities and serve on project teams
- Provide support to Black Belts and Green Belts as needed
- May be team members on DMAIC teams

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Stakeholders

- Those that are affected by, or perceive themselves to be affected by, a project's or organization's activities
  - Stakeholders are usually the recipients or beneficiaries of the success of a Six Sigma project
  - Does not include competitors
  - Stakeholders are individuals owning the process, function, or production/service line that a Six Sigma Belt focuses on improving the performance of
- BBs and GBs need to keep strong working relationships with stakeholders; because, without their support, it would be extremely difficult to make the Six Sigma project a success

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Subject Matter Experts (SME's)

- Subject Matter Experts (SME's) are commonly known as the experts of the process or subject matter
  - Six Sigma Belts should proactively look to key SMEs to round out their working project team

- SMEs play critical roles to the success of a project
  - Based on SMEs' extensive knowledge about the process, they have the experience to identify which solutions can work and which cannot work
  - SMEs who simply do not speak up can hurt the chances of the process' success
  - SMEs are also the same people who prefer to keep the status quo. Six Sigma Belts may find many of them unwilling to help implement the changes

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1.1.5 Project Team Selection

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Project Team Selection

Six Sigma project team selection is the cornerstone of a successful Six Sigma project

What makes a team successful:

- Shared goals
- Commitment
- Leadership
- Respect
- Effective communication
- Autonomy
- Diversity (capabilities, knowledge, skills, experience etc.)
- Adequate resources (4 – 8 Team members)

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Project Team Selection

- Team Selection
  - Selected teammates have proper skills and knowledge
  - Adequately engaged management
- Successful teams use reliable methods
  - Follow the prescribed DMAIC methodology
- Successful teams always have exceeded players
  - Winning teams typically:
    - Have unusually high standards
    - Have greater expectations of themselves and each other
    - Do not settle for average or even above average results

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Team Development Patterns

FORMING Stage

- Roles and responsibilities are unclear
- Process and procedures are ignored
- Scope and parameter setting is loosely attempted
- Discussions are vague and frustrating
- There is a high dependence on leadership for guidance

STORMING Stage

- Attempts to skip the research and jump to solutions
- Impatience for some team members regarding lack of progress
- Arguments about decisions and actions of the team
- Team members establish their position
- Subgroups or small teams form
- Power struggles exist and resistance is present

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Team Development Patterns

NORMING Stage

- Agreement and consensus start to form
- Roles and responsibilities are accepted
- Team members' engagement increases
- Social relationships begin to form
- The leader becomes more enabling and shares authority

PERFORMING Stage

- Team is directionally aware and agrees on objectives
- Team is autonomous
- Disagreements are resolved within the team
- Team forms above average expectations of performance

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1.2 Six Sigma: The Fundamentals
1.2.1 Voice of the Customer (VOC)

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What is a Customer?

There are two primary classifications of customers, Internal and External. Each customer type has different needs and thus a different “voice.” Understanding who they are will help in identifying their needs

External

- **Direct**: those who receive the output of your services, they generally are the source of your revenue
- **Indirect**: those who do not receive or pay for the output of your services but have a vested interest in what you do (government agencies), society, etc.

Internal

- Those within your organization who receive the output of your work and other processes. The Output of one becomes the Input to another

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Voice of the Customer (VOC)

Clearly understanding the customer's requirements in terms of quantifiable specifics is critical to Six Sigma projects

- VOC is the primary tool used to identify these requirements
- Note: VOC also plays a critical role in other Six Sigma methodologies and tools, such as:
  - Design for Six Sigma (DFSS)
  - Quality Function Deployment (QFD)
  - House of Quality (HOQ)

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Gathering VOC

The two most popular methods of collecting VOC are

Indirect

Indirect data collection for VOC involves passive information exchange:

- Warranty claims
- Customer complaints/compliments
- Service calls
- Sales reports

Direct

Indirect methods are less effective, sometimes dated, require heavy interpretation, and are also more difficult to confirm

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Gathering VOC

Direct data collection methods for VOC are active and planned customer engagements:

- Conducting interviews
- Conducting customer surveys
- Conducting market research
- Hosting focus groups

Direct methods are typically more effective for several reasons:

- Less need to interpret meaning
- Researchers can go a little deeper when interacting with customers
- Customers are aware of their participation and will respond better upon follow-up
- Researchers can properly plan engagements (questions, sample size, information collection techniques etc.)

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VOC Translation

- When receiving the “raw” commentary and feedback from the customer, it is important to ensure a complete understanding has been captured and confirmed by the customer
  - ➢Customers may reply with opinions, biased observations, emotion, and anecdotes
- We can use the VOC approach to translate, sort, and classify the feedback
- The VOC approach involves asking clarifying questions that will result in specific information about the customer’s requirements

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VOC Translation

- Analysis questions used to translate the voice of the customer into critical information include the basic questions of:

  - **Who** (customer demographic)
  - **What** (customer purpose)
  - **When** (customer timeframe)
  - **Where** (customer location)
  - **Why** (customer emotional driver)
  - **How** (how the organization currently handles the requirement)

- Record answers, with measurable components, in a table

  - Recommended: Place these terms as column headers to aid in remembering and to better establish consistency

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Multiple “Voices”

Note: Though the current focus is on the “Voice of the Customer”, there are additional “voices” that need to be considered. Following are some of these “voices”:

- VOC: Voice of the **Customer** (is Customer Driven)
- VOB: Voice of the **Business** (is Profit Driven)
- VOE: Voice of the **Employee** (is Process/Operations Driven)
- VOP: Voice of the **Process** (is Performance/Results Driven)

- Though VOC is called out specifically here, various aspects of the other voices will be referenced throughout the course

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1.2.2 Critical to Quality (CTQ)

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Developing CTQ's

CTQ stands for Critical to Quality and consists of the translated data gathered via VOC

- CTQs are the quantifiable, measureable, and meaningful translations of the VOC
- Once translated and defined, the CTQ's need to be confirmed with the customer, then prioritized
- Confirming CTQ's can be accomplished by conducting surveys through one or more of the following methods:

  - Group sessions
  - One-on-one meetings
  - Phone interviews
  - Electronic means (chat, email, social media etc.)
  - Physical mail

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Translating CTQs to Requirements

- CTQs must be transformed into specifics that can be built upon in a process
- Some of the tools used to identify, define, and prioritize CTQ's include: Tree diagrams, Affinity diagrams, and Pareto analysis (covered in the next section)
- A requirements tree (a repurposed decision tree) helps in facilitating the translation of general CTQ categories into meaningful and measureable requirements for production processes and products

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CTQ Specifics

A fully developed CTQ has the following 4 components:

- “Y” characteristic
- Measureable Metric
- Target value (associated with the metric)
- Tolerance range (upper/lower) of the metric

  - These are the customer defined Specification Limits (Spec Limits) that are used when determining if a product meets customer requirements and used when performing various analyses (yield, defects, and etc.)

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1.2.3
Affinity Diagrams & Pareto Analysis (80:20)

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Affinity Diagram

- Affinity diagrams are ideal tools for large amounts of soft data resulting from brainstorming sessions or surveys
- One effective way to organize the VOC data is to group or bucket it using an affinity diagram
- Like most Six Sigma tools, affinity diagrams are not exclusive to the Define phase or this application

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Pareto Principle

The Pareto principle helps us to prioritize/focus on the vital few items (vs. the trivial many) that have the greatest impact

- It is also known as the 80/20 rule
  - ➢ It means that the majority (approximately 80%) of effects come from a few (approximately 20%) of the causes
  - ➢ Example of applying the Pareto principle:
    - 80% of the defects of a process come from 20% of the causes
    - 80% of sales come from 20% of customers
- Originally based upon the works of Wilfried (Vilfredo) Fritz Pareto, the 80/20 Pareto rule and references are capitalized

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Examples of the Pareto Principle

- 80% of your phone calls are made by 20% of the employees
- 20% of the airports handle 80% of the flights
- 80% of the food ordered in a restaurant come from 20% of the menu
- 80% of the information is in the first 20% of the report
- 20% of the employees cause 80% of the issues
- 20% of the features of software are used 80% of the time

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Pareto Chart

A Pareto chart is a modified histogram / bar chart where the data has been sorted in descending order then displayed as bars; overlaid by an ascending, cumulative line on the top and values displayed on the secondary Y-axis (right side)

- **Percent to Total**: A Pareto chart shows the percentage to the total for individual bars.
- **Cumulative Percentage**: A Pareto chart also shows the cumulative percentage of each additional bar. The data points of all cumulative percentages are connected into an ascending line on the top of all bars, reaching 100% at the end

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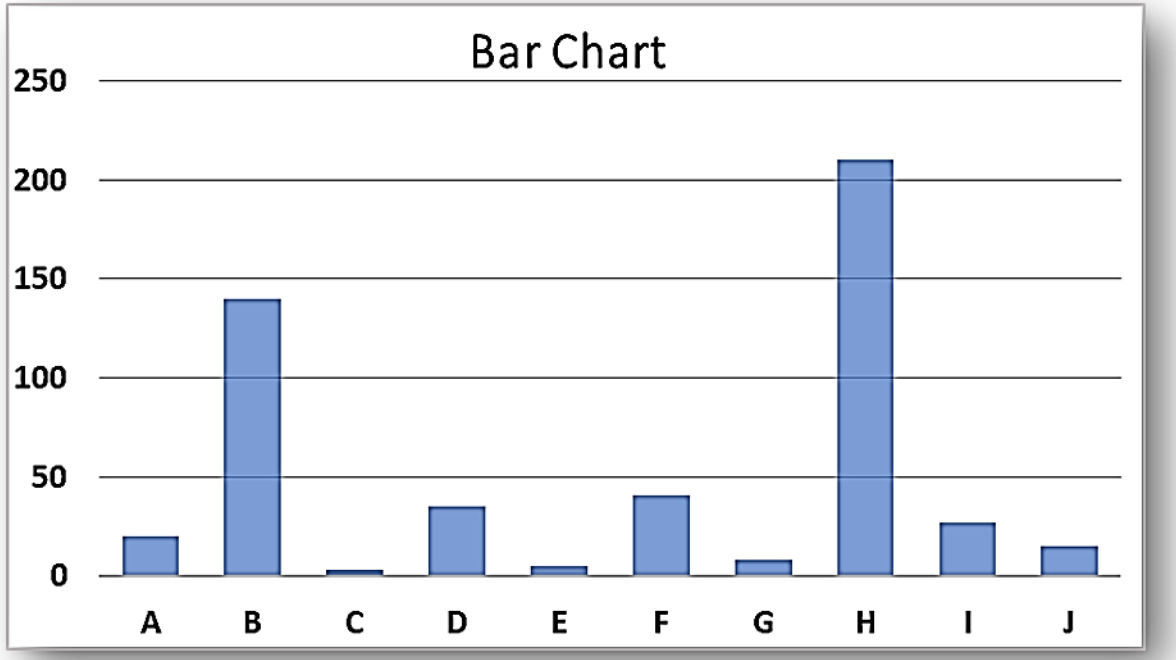
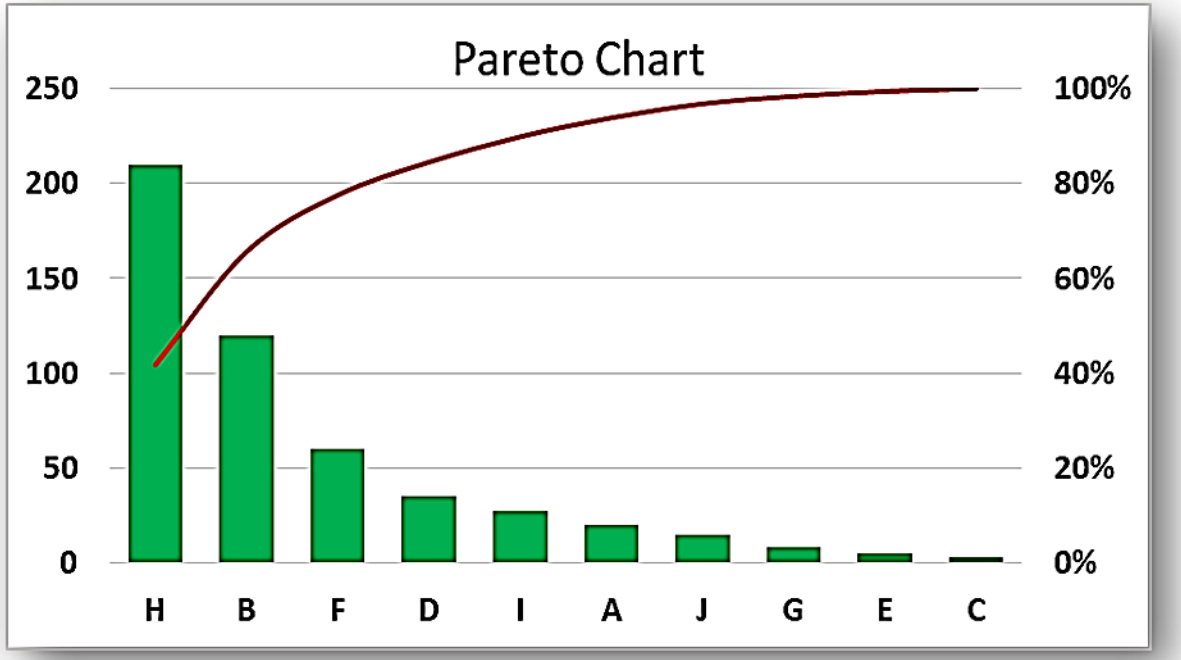
Bar vs. Pareto Chart





- A **bar** chart visually displays the corresponding value of each type of collected data

- A **Pareto** chart is a **histogram** of descending bars with an ascending cumulative line over the top

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1.2.4 Cost of Poor Quality (COPQ)

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Cost of Poor Quality

- Cost of Poor Quality (COPQ) is the expense incurred due to waste, inefficiencies, and defects
  - ➢ The COPQ has been proven to range from 5% to 30% of gross sales for most companies
  - ➢ COPQ helps us understand the financial impact of problems created by defects
- Understanding COPQ and where to look for it will help uncover process inefficiencies, and defects, within your business

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The Essence of COPQ

- COPQ is a **symptom**, not a defect
  - Projects fix defects with the intent of improving symptoms
- The concepts of traditional Cost of Quality are the foundation for COPQ
  - **External, Internal, Prevention, Appraisal**
- A significant portion of COPQ from any defect comes from effects that are difficult to quantify and must be **estimated**
  - This project COPQ represents the **financial opportunity** of your team's improvement effort
  - Calculating COPQ is **iterative** and **will change** as you learn more about the process

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COPQ: Types of Cost

There are two types of costs to be considered when determining COPQ

- **Hard Costs**
  - ➢Tangible costs that can be traced to the income statement
- **Soft Costs**
  - ➢Intangible costs: avoidance, opportunity costs, lost revenue etc.

Calculating the COPQ

- Determine the types of waste that are present in your process
- Estimate the frequency of waste that occurs
- Estimate the cost per event, item, or time frame
- Do the math

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COPQ: Categories

Prevention

- Error Proofing Devices
- Supplier Certification
- Design for Six Sigma
- Etc.

Detection / Appraisal

- Supplier Audits
- Sorting Incoming Parts
- Repaired Material
- Etc.

Internal COPQ

- Quality Control Department
- Inspection
- Quarantined Inventory
- Etc.

External COPQ

- Warranty
- Customer Complaint Related Travel
- Customer Charge Back Costs
- Etc.

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COPQ: Waste

- The Cost of Poor Quality is commonly driven by “waste”

➢There are 7 general forms of “waste” that are often referred to as the “7 deadly muda”

- Technically, there are more than 7 forms of waste but if you can remember these you will capture over 90% of your waste

➢These muda will be addressed in greater detail later

- Defects
- Overproduction
- Over-Processing
- Inventory
- Motion
- Transportation
- Waiting

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COPQ: Prevention Costs

- Costs related to the prevention of muda are those associated with trying to reduce or eliminate any of the “7 deadly muda”
  - Costs for error proofing methods or devices
  - Costs for process improvement and quality programs
  - Costs for training and certifications
  - Etc.
- Any costs directly associated with the prevention of waste and defects should be included in the COPQ calculation

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COPQ: Detection Costs

- **Costs** related to the detection of muda are those associated with trying to find or observe any of the “7 deadly muda”
  - Costs for sampling
  - Costs for quality control check points
  - Costs for inspection costs
  - Costs for cycle counts or inventory accuracy inspections
  - etc.
- Any costs directly associated with the detection of waste and defects should be included in the COPQ calculation

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COPQ: Internal Costs

- **Internal** Costs are related to the auditing and quality control procedures and its outcome, are those associated with trying to confirm that no waste has occurred prior to distribution to customers

  - Costs for maintaining a post-production, quality check team
  - Costs for time and tools expended in auditing products
  - Costs associated with testing; including destructive testing
  - etc.

- Any costs directly associated with the identification of waste and defects, before a product is shipped to a customer, should be included in the COPQ calculation

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COPQ: External Costs

- **External** Costs are related to obligation are those associated with addressing the muda that reaches a customer. These types of costs are the most expensive in terms of both hard and soft costs

  - Repair costs
  - Warranty costs
  - Replacement costs
  - Customer returns and customer service overhead
  - Etc.

- Any costs directly associated with customer obligations should be included in the COPQ calculation

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COPQ: Hard vs. Soft Savings

Cost savings occur when these poor-quality related costs are addressed. While hard savings are always more desirable because they are easier to quantify, it is also necessary to think about soft savings

- COPQ – **Hard Savings**
  - Labor Savings
  - Cycle Time Improvements
  - Scrap Reductions
  - Hidden Factory Costs
  - Inventory Carrying Cost

- COPQ – **Soft Savings**
  - Gaining Lost Sales
  - Missed Opportunities
  - Customer Loyalty
  - Strategic Savings
  - Preventing Regulatory Fines

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1.2.5 Six Sigma Projects: DMAIC

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Six Sigma's DMAIC Methodology

Six Sigma follows an organized, specific, repeatable methodology as a means of assessing and resolving challenges

- This five-phased methodology is entitled "DMAIC"
- Each phase has a specific purpose and specific tools and techniques that aid in achieving the phase objectives

➢ The 5 phases of DMAIC:

- Define
- Measure
- Analyze
- Improve
- Control

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5 Phases of DMAIC

DEFINE:	Define and quantify the problem and objective
MEASURE:	Baseline process, validate measures, and identify all possible inputs (x's) and dependent outputs (Y's)
ANALYZE:	Analyze and validate causes identifying critical factors
IMPROVE:	Design, Test, and Implement Solutions
CONTROL:	Implement and sustain solutions

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Define Phase Goals

- The goal of the **Define** phase is to establish a solid foundation and business case for a Six Sigma project
  - ➢Define is arguably the most important aspect of any Six Sigma project
- All successful projects start with a current state challenge or problem that can be articulated in a quantifiable manner
  - ➢It is not enough to just know the problem, you must quantify it and also determine the goal

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Define Phase Tools & Deliverables

Following are some of the tools and deliverables of the **Define** phase. Some of the shared tools will be covered in other phases

- Business Case
- Project Charter – Establish the:
  - ➢Problem Statement
  - ➢Project Objective
  - ➢Project Scope
  - ➢Project Timeline
  - ➢Project Team
- Stakeholder Assessment
- High-Level Pareto Chart Analysis
- High-Level Process Map
- VOC/VOB and CTQs Identified and Defined
- Financial Assessment

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Measure Phase Goals

The goal of the **Measure** phase is to gather baseline information about the process (process performance, inputs, measurements, customer expectations etc.)

- Throughout the Measure phase you will seek to achieve a few important objectives:

- Gather All Possible x's
- Assess Measurement Systems
- Determine Data Collection Requirements
- Validate Assumptions
- Validate Improvement Goals

- Determine COPQ (Cost of Poor Quality)
- Refine Process Understanding
- Determine Process Stability
- Determine Process Capability

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Measure Phase Tools & Deliverables

Following are some of the tools and deliverables of the **Measure** phase. Some of the shared tools will be covered in other phases

- Process Maps, SIPOC, Value Stream Maps
- Failure Modes and Effects Analysis (FMEA)
- Cause-and-Effect Diagram
- XY Matrix
- Six Sigma Statistics
  - Basic Statistics
  - Descriptive Statistics
- Measurement Systems Analysis
  - Variable and/or Attribute Gage R&R
  - Gage Linearity and Accuracy or Stability
- Basic Control Charts
- Process Capability (C_{pk}, P_{pk}) and Sigma Levels
- Data Collection Plan

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Analyze Phase Goals

The **Analyze** phase is all about establishing verified inputs (X's) that affect and control the outcome of various processes

- In the DMAIC methodology, the Analyze phase uses statistics and higher-order analytics to discover the relationships between process performance and process inputs (in other words, what are the root causes or drivers of the improvement effort)

- Ultimately, the Analyze phase establishes a reliable hypothesis for improvement solutions

  - ➢ Establish the Transfer Function Y = f(x)
  - ➢ Validate the List of Critical x's and Impacts
  - ➢ Create a Beta Improvement Plan (e.g., pilot plan)

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Analyze Phase Tools & Deliverables

Following are some of the tools and deliverables of the **Analyze** phase. Some of the shared tools will be covered in other phases

- Hypothesis Testing
  - Parametric and Non-Parametric
- Regression Analysis
  - Simple Linear Regression (SLR)
  - Multiple Linear Regression (MLR)

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Improve Phase Goals

- The goal of the **Improve** phase is about designing, testing, and implementing your solution

- To this point the problem and objective of the project has been defined, possible x's brainstormed, analyzed and verified critical x's

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Improve Phase Tools & Deliverables

Following are some of the tools and deliverables of the **Improve** phase.
Some of the shared tools will be covered in other phases

- Pilot or Planned Study Using:
  - Hypothesis Testing
  - Valid Measurement Systems
- Implementation Plan
- Any other tools from other phases
- Design of Experiments (DOE)

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Control Phase Goals

The goal of the **Control** phase is to establish automated and managed mechanisms to maintain and sustain your improvement

- A successful control plan also establishes a reaction and mitigation plan as well as an accountability structure

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Control Phase Tools & Deliverables

Following are some of the tools and deliverables of the **Control** phase. Some of the shared tools will be covered in other phases

- Statistical Process Control (SPC/Control Charts)
  - IMR, XbarS, XbarR, P, NP, U, C, etc.
- Control Plan Documents
  - Control Plan
  - Training Plan
  - Communication Plan
  - Audit Checklist
- Lean Control Methods
  - Poka-Yoke
  - Five-S
  - Kanban

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1.2.6 Basic Six Sigma Metrics

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Six Sigma Metrics

- There are many Six Sigma metrics and/or measures of performance used by Six Sigma practitioners

- In addition to the ones we will cover here, several others (Sigma level, C_p, C_{pk}, P_p, P_{pk}, etc.) will be covered in other modules throughout this training

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Some Key Terms

TERM	DEFINITION
Quality Product	A good unit. One that passes inspection. No defects
Scrap	Output units that are defective AND cannot be reworked. Truly defective or failed units / opportunities
Rework	Output units that are initially unacceptable (defective), but that become acceptable once reworked (by definition, NOT counted in a “First” pass-through calculation)

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Additional Key Terms

TERM	DEFINITION
Opportunity	Any point that can be inspected for Pass / Fail in a unit or process. There can be multiple opportunities per unit / process. The total number of possible defects
Unit	A deliverable (product, service, or result) produced by a process
Defect	Any opportunity that is unacceptable / unusable
Defective	A unit that has at least one defect, rendering it unacceptable (to customer requirements)

Remember: Defect means that part of a unit is bad; Defective means that the whole unit is bad

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Defects Per Unit

- **DPU** = Defect / Unit: The average number of defects observed when sampling a population

- **DPU** (TDPU) = Total # of Defects / Total population

  - Defects / Unit = D / U
  - For example: For a process step that produces an average of 65 defects for every 598 units, then that process produces a DPU = 65/598 = 0.109

    - The process produces 0.109 defects for every unit produced

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Defects per Million Opportunities (DPMO)

Defects per Million Opportunities (DPMO) is also known as **Non-Conformities per Million Opportunities** (NPMO)

- **Total number of opportunities** = Units * Number of opportunities per unit
- “Non-Conformity” = “Defect” at the Opportunity level
- Is a measure of process performance
- It standardizes the number of defects at the opportunity level
- It allows comparison of processes with different complexities

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Calculating DPMO

- Calculating Defects per Million Opportunities
- The equation is DPMO = (D/(U * O)) * 1,000,000
  - Example: Let us assume:
    - There are 6 opportunities (Pass / Fail) per unit
    - There are an average of 4 defects every 100 units
    - Total Opportunities = 6 * 100 = 600
    - Defect rate = 4/600 = 0.00667
    - DPMO = 4/600 * 1,000,000 = 6,667

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Yield & First Time Yield (FTY)

Yield: The percentage of units, of a process, that is free of defects

- Also a percentage of met commitments (total of defect free events) over the total number of opportunities

First Time Yield (FTY): Total number of quality units produced (after all process steps and including any reworks) divided by the number of total units going into the process

- (**Total Units**) – (**Scrap**) = (**Good Units**)
- **Good Units** includes **Rework**
- Yield rate = (**Good Units**) / (**Total Units**)

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First Time Yield (FTY) : Example

You have a process that is divided into four sub-processes: A, B, C and D

Assume that you have 100 units entering process A. To calculate first time yield (FTY) you would:

1. Calculate the yield (number out of step/number into step) of each step
2. Multiply these together

For Example: (# units leaving the process as good parts) / (# units put into the process) = FTY

Phase	Units Entering	Rework	Scrap	Good Units	FTY
A	100	5	10	90	90 / 100 = .9000
B	90	0	10	80	80 / 90 = .8889
C	80	10	5	75	75 / 80 = .9375
D	75	8	5	70	70 / 75 = .9333
					FTY = .7000

(Cont.)

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First Time Yield (FTY) : Example

Phase	Units Entering	Rework	Scrap	Good Units	FTY
A	100	5	10	90	90 / 100 = .9000
B	90	0	10	80	80 / 90 = .8889
C	80	10	5	75	75 / 80 = .9375
D	75	8	5	70	70 / 75 = .9333
					FTY = .7000

The total first-time yield is equal to FTY_A * FTY_B * FTY_C * FTY_D or .9000 * .8889 * .9375 * .9333 = **.7000**

You can also get the total process yield for the entire process by simply dividing the number of good units produced by the number going into the start of the process. In this case, **70**/100 = **.70 or 70% yield**

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Throughput Yield (TPY)

The number of acceptable pieces at the end of a process divided by the number of starting pieces, excluding scrap and rework

- TPY is used to measure a **single** process only
- Place multiple processes end-to-end, measuring TPY at each point, will give us the First Pass Yield (FPY)
- **When a single process is involved, TPY = FPY**

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First Pass Yield (FPY)

Also referred to as the quality rate, the percentage of units that completes a process and meets quality guidelines without being scrapped, rerun, retested, returned or diverted into an offline repair area. Used in place of TPY for multiple processes

- FPY is calculated by dividing the units entering the process minus the defective units by the total number of units entering the process. (ASQ.org)
- Does **NOT** include Rework
- (**Total Units**) – (**Scrap + Rework**) = (**Good Units**)
- **FPY** = (**Good Units**) / (**Total Units**)

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Rolled Throughput Yield (RTY)

- Rolled Throughput Yield (RTY) is the **probability** that a **single unit** can pass through a series of process steps defect free
- It is important as a metric when a process has excessive rework
- RTY is the product of each process's First Pass Yield (FPY) when the defectives are known
  - Reminder: FPY does not include rework

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Rolled Throughput Yield (RTY) : Example

You have a process that is divided into four sub-processes: A, B, C and D.

Assume that you have 100 units entering process A. To calculate Rolled Throughput Yield you would:

1. Calculate the First Pass Yield of each step.
2. Multiply these together.

Phase	Units Entering	Rework	Scrap	Good Units	FPY	FTY
A	100	5	10	90	85 / 100 = .8500	90 / 100 = .9000
B	90	0	10	80	80 / 90 = .8889	80 / 90 = .8889
C	80	10	5	75	65 / 80 = .8125	75 / 80 = .9375
D	75	8	5	70	62 / 75 = .8267	70 / 75 = .9333
					RTY = .5075	**FTY = .7000**

(Cont.)

Rework from prior process included

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Rolled Throughput Yield (RTY) : Example

Phase	Units Entering	Rework	Scrap	Good Units	FPY	FTY
A	100	5	10	90	85 / 100 = .8500	90 / 100 = .9000
B	90	0	10	80	80 / 90 = .8889	80 / 90 = .8889
C	80	10	5	75	65 / 80 = .8125	75 / 80 = .9375
D	75	8	5	70	62 / 75 = .8267	70 / 75 = .9333
					RTY = .5075	**FTY = .7000**

Rework from prior process included

The RTY of the set of processes is equal to FPY_A * FPY_B * FPY_C * FPY_D = .8500 * .8889 * .8125 * .8267 = **.5075**

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Yield & DPU

The following yield values can also be calculated using a known DPU, and vice-versa

In cases of multiple processes, utilize the Total DPU (TDPU) to calculate the RTY

- **$RTY = e^{-TDPU}$**
- **$TDPU = -\ln(RTY)$**

- **$TPY = e^{-DPU}$**
- **$DPU = -\ln(TPY)$**

  - DPU = .47
  - TPY = $e^{-x} = 1/e^{x}$
    - $= 1/ e^{0.47} = 0.625$

  - TPY = .625
  - DPU = -ln(TPY)
    - $= -[\ln(0.625)] = 0.47$

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Additional Process Terms

TERM	DEFINITION
Cycle Time	The time required to complete one cycle of an operation
Takt Time	The available production time divided by the rate of customer demand (the required pace of production) • e.g. 60 min to create 5 units > 60/5 = 12min/unit
Single-Piece-Flow	One product proceeds through various operations without interruptions or scrap
Standard Work	A precise description of each work activity, specifying cycle time, takt time, the work sequence of specific tasks, and the minimum inventory

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Six Sigma Performance Measurements

LT Sigma Level	ST Sigma Level	C_p	DPMO	LT Sigma Level	ST Sigma Level	C_p	DPMO	LT Sigma Level	ST Sigma Level	C_p	DPMO
4.5	**6**	**2.00**	3.4	2.8	4.3	1.43	2,555	1.1	2.6	0.87	135,666
4.4	5.9	1.97	5	2.7	4.2	1.40	3,467	1	2.5	0.83	158,655
4.3	5.8	1.93	9	2.6	4.1	1.37	4,661	0.9	2.4	0.80	184,060
4.2	5.7	1.90	13	2.5	**4**	**1.33**	6,210	0.8	2.3	0.77	211,855
4.1	5.6	1.87	21	2.4	3.9	1.30	8,198	0.7	2.2	0.73	241,964
4	5.5	1.83	32	2.3	3.8	1.27	10,724	0.6	2.1	0.70	274,253
3.9	5.4	1.80	48	2.2	3.7	1.23	13,903	0.5	**2**	**0.67**	308,538
3.8	5.3	1.77	72	2.1	3.6	1.20	17,864	0.4	1.9	0.63	344,578
3.7	5.2	1.73	108	2	3.5	1.17	22,750	0.3	1.8	0.60	382,089
3.6	5.1	1.70	159	1.9	3.4	1.13	28,717	0.2	1.7	0.57	420,740
3.5	**5**	**1.67**	233	1.8	3.3	1.10	35,930	0.1	1.6	0.53	460,172
3.4	4.9	1.63	337	1.7	3.2	1.07	44,565	0	1.5	0.50	500,000
3.3	4.8	1.60	483	1.6	3.1	1.03	54,799	-0.1	1.4	0.47	539,828
3.2	4.7	1.57	687	1.5	**3**	**1.00**	66,807	-0.2	1.3	0.43	579,260
3.1	4.6	1.53	969	1.4	2.9	0.97	80,757	-0.3	1.2	0.40	617,911
3	4.5	1.50	1,350	1.3	2.8	0.93	96,800	-0.4	1.1	0.37	655,422
2.9	4.4	1.47	1,866	1.2	2.7	0.90	115,070	-0.5	**1**	**0.33**	691,462

- This table shows each sigma level's corresponding DPMO (defects per million opportunities) and short-term Capability Index (C_p)
- The higher the sigma level, the lower the defect rate and DPMO

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1.3 Project Management
1.3.1 Change

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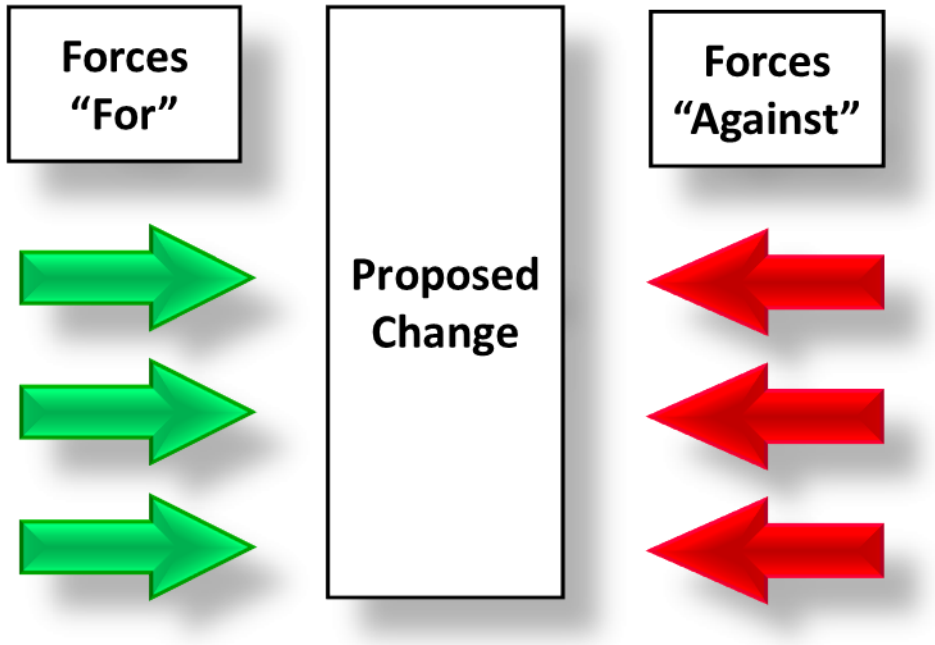
Force Field Analysis

- Created by socio-psychologist Kurt Lewin in the 1940

- Based on the assumption that any situation is the result of forces For and Against the current state being in equilibrium

  - To induce change:
    - Counter opposing forces
    - Increase Favorable



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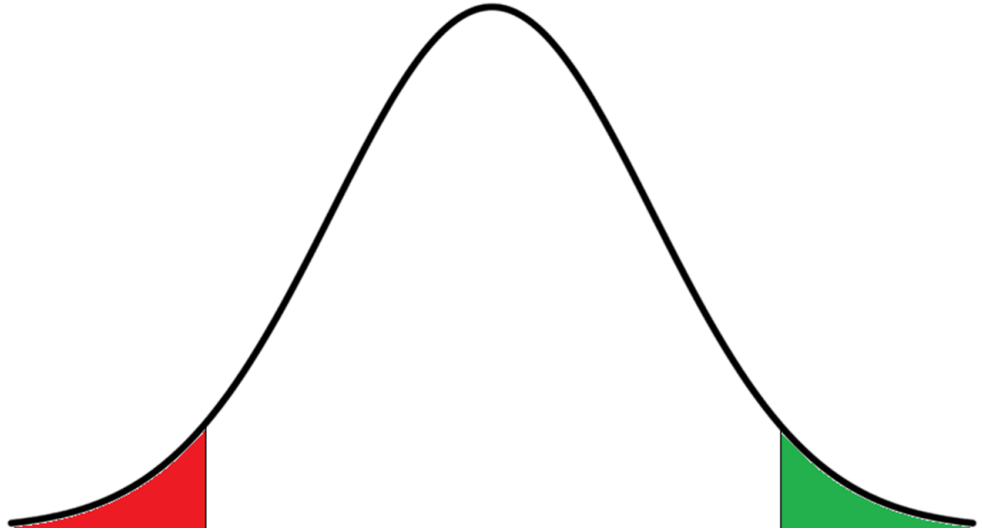
Change Resistance Curve

- Based upon the observation that: 15% Supportive, 70% Neutral, and 15% Resistant to change
- Message: Focus on the 70% Neutral to create support and momentum



➢ The 15% Supportive will support change either way

➢ The 15% Resistant will resist change regardless

- If the leadership becomes distracted by focusing on those Resistant to change, the 70% Neutral majority may feel that leadership has lost focus or commitment

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1.3.2 Project Management Basics

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Project Management

Project Management is the process of defining, planning, organizing, managing, leading, securing, and optimizing the project resources to complete the necessary activities (tasks) to produce the required deliverables (generally a product, service, or result)

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Project Plan

A project plan is a critical component of managing projects. It is a formal, approved document used to guide and execute project tasks, and measure performance

- It is comprised of a series of subsidiary plans and baselines (e.g. cost, schedule, scope, etc.) aggregated into one master plan
- It provides an overall framework for managing project tasks, schedules, and costs
- A project plan is a coordinating tool and communication device that helps define the crucial aspects of a project or program

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Statement of Work (SOW)

A formal document often accompanying a contract that outlines specific expectations, limitations, resources and work guidelines intended to define the work or project

- An SOW:
  - Defines the scope of the project
  - Establishes expectations and parameters of the project
  - Identifies technical requirements for the project
  - Provides guidance on materials to be used
  - Establishes timeline expectations
  - Is an input and reference for other steps in planning the project, running operations, and invoicing

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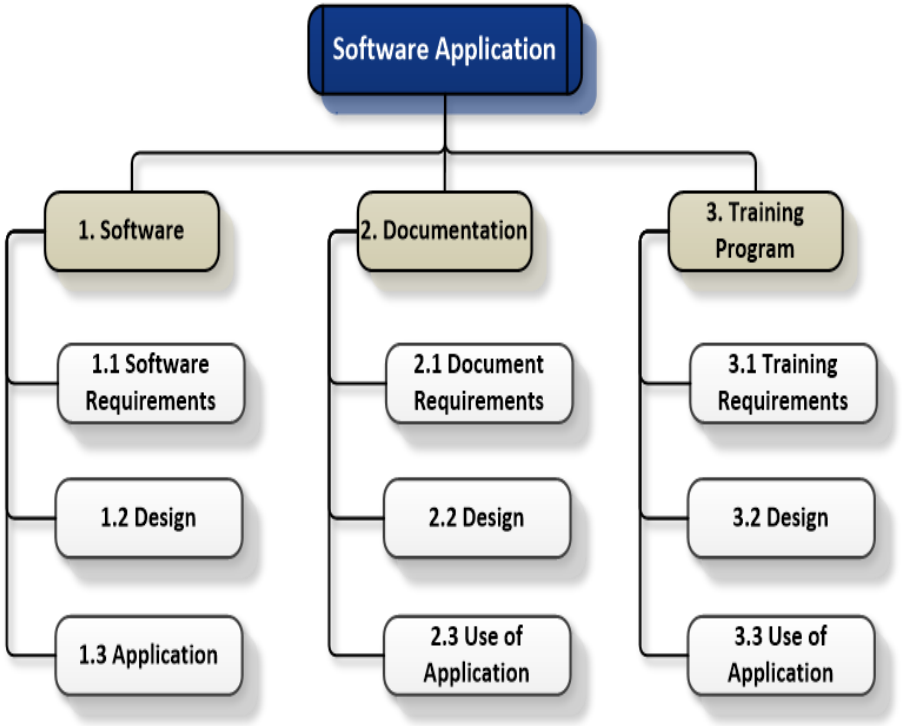
Work Breakdown Structure (WBS)

A decomposition of project components into small and logical bodies (work packages) of work or tasks

Work Breakdown Structures:

- Identify all required components of a project
- Cascades components into sub-components and tasks
- WBS's are not by themselves project plans or schedules but they are a necessary step to help establish project plans and timelines
- WBS's also enable logical reporting and summarizations of project progress



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Critical Path Method (CPM)

A schedule analysis technique used to identify the set of activities that are most influential on a project completion timeline. The Critical Path is the interconnected series of tasks that result in the longest duration to complete

- The project cannot complete any sooner than the Critical Path. Thus the Critical Path drives the project's completion date
- The longest path (Critical Path) is the shortest duration that a project can be completed
- Project timelines cannot be shortened without shortening the tasks or activities that are identified as critical path items

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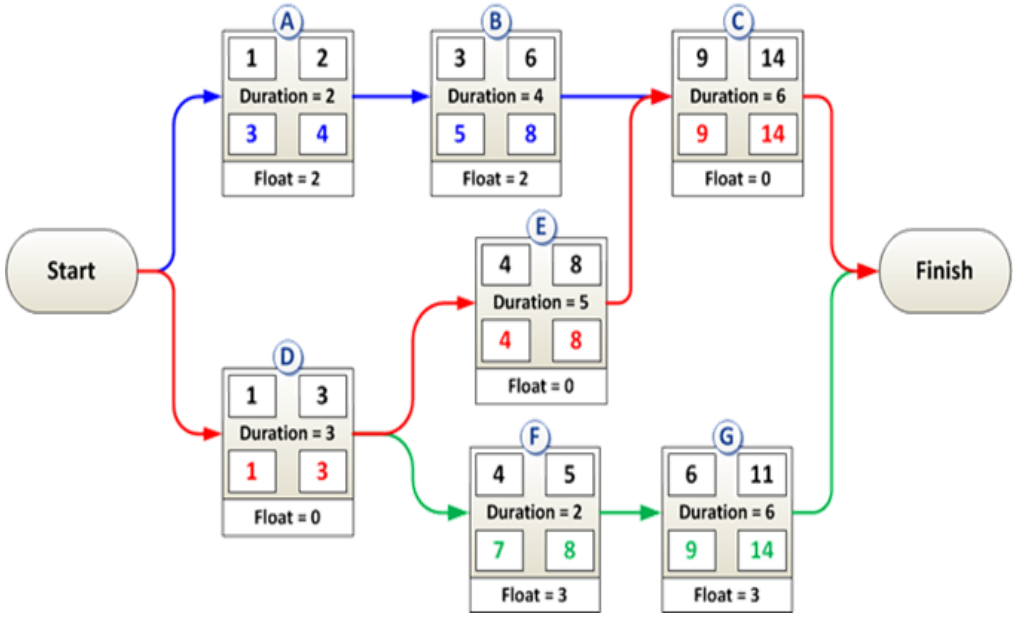
Critical Path Method (CPM)

Steps to implementing the Critical Path Method:

1. List all activities necessary to complete the project
   - ➢ Use Your Work Breakdown Structure (WBS)
2. Determine the time or duration of each activity
3. Identify the dependencies between the activities
4. Calculate the duration for each series of interconnected tasks (path)



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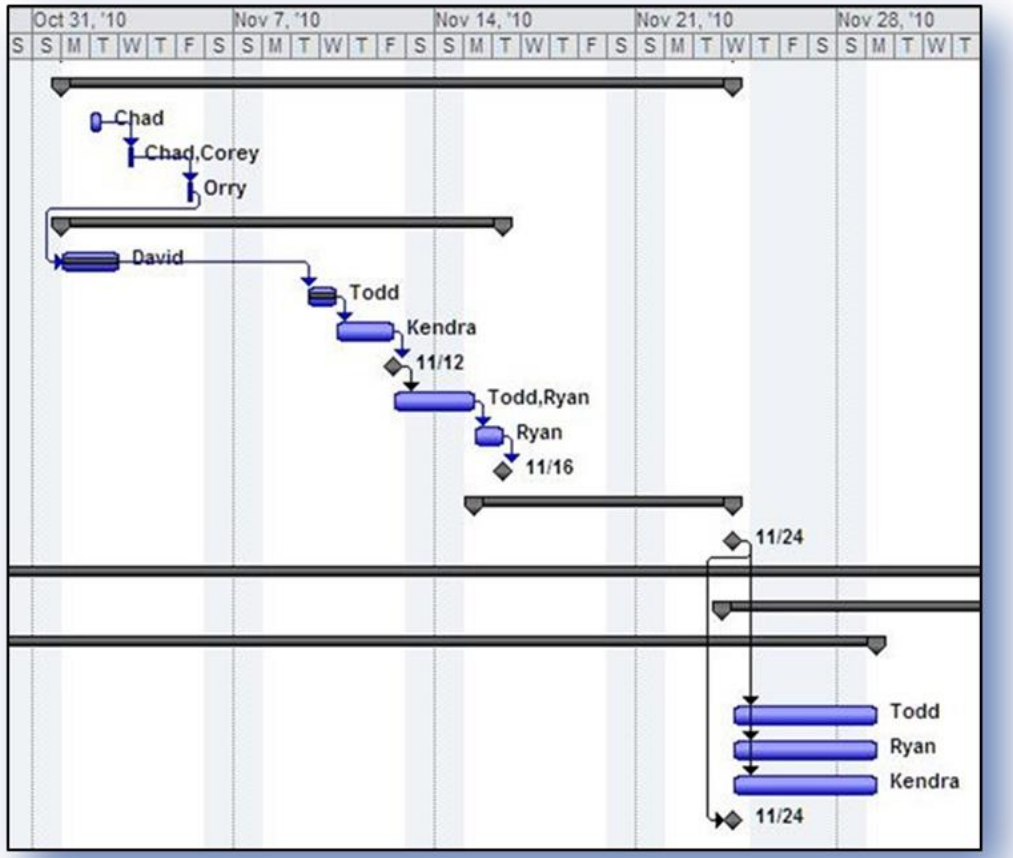
Project Schedule

- The most common tool used to display the results of a project's schedule is a Gantt chart
- The advantage of a Gantt chart is its ability to display the status of each task/activity at a glance
- Because it is a graphic representation, it is easy to demonstrate the schedule of tasks and timelines to all the stakeholders



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Program Evaluation & Review Technique

The Program Evaluation & Review Technique (**PERT**) is a method of evaluation that can be applied to **time and cost estimates**

- PERT provides a **weighted assessment** of time or cost
- PERT uses 3 parameters for estimation:
  - **Optimistic** or Best Case Scenario, represented by 'O'
  - **Pessimistic** or Worst Case Scenario, represented by 'P'
  - **Most Likely** Scenario, represented by 'M'
- PERT Equation: **(O + (4*M) + P) / 6**
  - The PERT equation provides for heavier weighting of the most likely scenario but also considers the best and worst cases. In the event that a best- or worst-case scenario is an extreme situation, the PERT will account for it in a weighted manner

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Cost Management Plan

Also known as the project budget or a project's financial plan. It will include the following items:

- The result is a time-phased, project budget
  - Planned **expenses** based upon the estimated and scheduled resources needed to complete the tasks outlined in the project schedule
  - A Six Sigma project will include estimations on **cost savings** (one of the objectives of a Six Sigma project)
  - May or may not include impacts on **revenues** or **profits** (depends on the purpose of the project
    - **Revenues – Cost = Profit**

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Communication Plan

Identify the communication needs of the stakeholders and project team members; then establish communication procedures among management, team members, and relevant stakeholders.

- Determine the communication schedule
- Define the acceptable modes of communication

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Risk Management Plan

- Identify the sources of project risks, then analyze them
  - ➢Risks might arise from new technology, availability of resources, lack of inputs from customers, business risks etc.
- A more detailed discussion of Risk is covered later in this presentation

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Project Planning Tools Advantages

- **Project planning tools** are very useful to organize and communicate project plans, status, and projections
- They help provide a holistic view of what needs to be accomplished
- They facilitate effective communication
- They allow for effective scope control and change management
- They establish project baselines for progress reviews and control points
- They act as an effective monitoring mechanism for the project

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1.3.3 Business Case & Project Charter

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Organizational Goals & Six Sigma Projects

Lean Six Sigma projects are tied to an organization's objectives

- They help an organization by addressing portions of the organization's strategic objectives
- The strategic objectives outline 'how' the organization's objectives will be met
- Lean Six Sigma projects are one of the ways that these two-tiered objectives are met

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Six Sigma Project Characteristics

Characteristics of a Six Sigma project include:

- Positively impacts profitability or other significant strategic value
- Addressing problems whose cause is unknown
- Addressing problems where the solution is not obvious or quickly solvable using traditional methods
- It is targeted to reduce the problem by 50-70% over existing performance levels

Focus is to solve a business problem that is:

- Inhibiting success
- Adding costs
- Negatively impacting employee or customer satisfaction
- Negatively impacting a customer (external and/or internal)

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What is a Business Case?

The Business Case communicates the justification, need for the project in terms of meeting business objectives

Components include:

- Identification of the output (goal) “Y” unit (product/service) for customers
- Identify the primary business metric used to measure project performance
- Baseline of the organization’s current performance relative to the metric
- Quantitatively establish the gap in baseline performance of primary business metric verses the organization’s objective
- Where possible, translate the ‘gap’ into monetary and/or time estimates

Example:

During FY 2017, the 1st Time Call Resolution Rate for New Customer Hardware Setup was 87%

This represents a gap of 9% from the industry standard of 96% that amounts to USD $3,000,000 of annualized cost impact

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Project-Based DMAIC

- As a methodology, Six Sigma's DMAIC refers to 5 phases of a project
  - ➢Define, Measure, Analyze, Improve, and Control

- Since DMAIC is project-based, we take the necessary steps to define and initiate the project. The first step in beginning a project is the creation of a Project Charter

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Project Charter

- The Project Charter is a document that:
  - Provides vital information about a project in a quick and easy-to-comprehend manner
  - Formally authorizes the project
  - Assigns the project manager with the authority to utilize organizational resources for the project
- It provides a preliminary delineation of:
  - Roles and responsibilities
  - Outlines the project objectives
  - Identifies the main stakeholders
  - Defines the authority of the project manager

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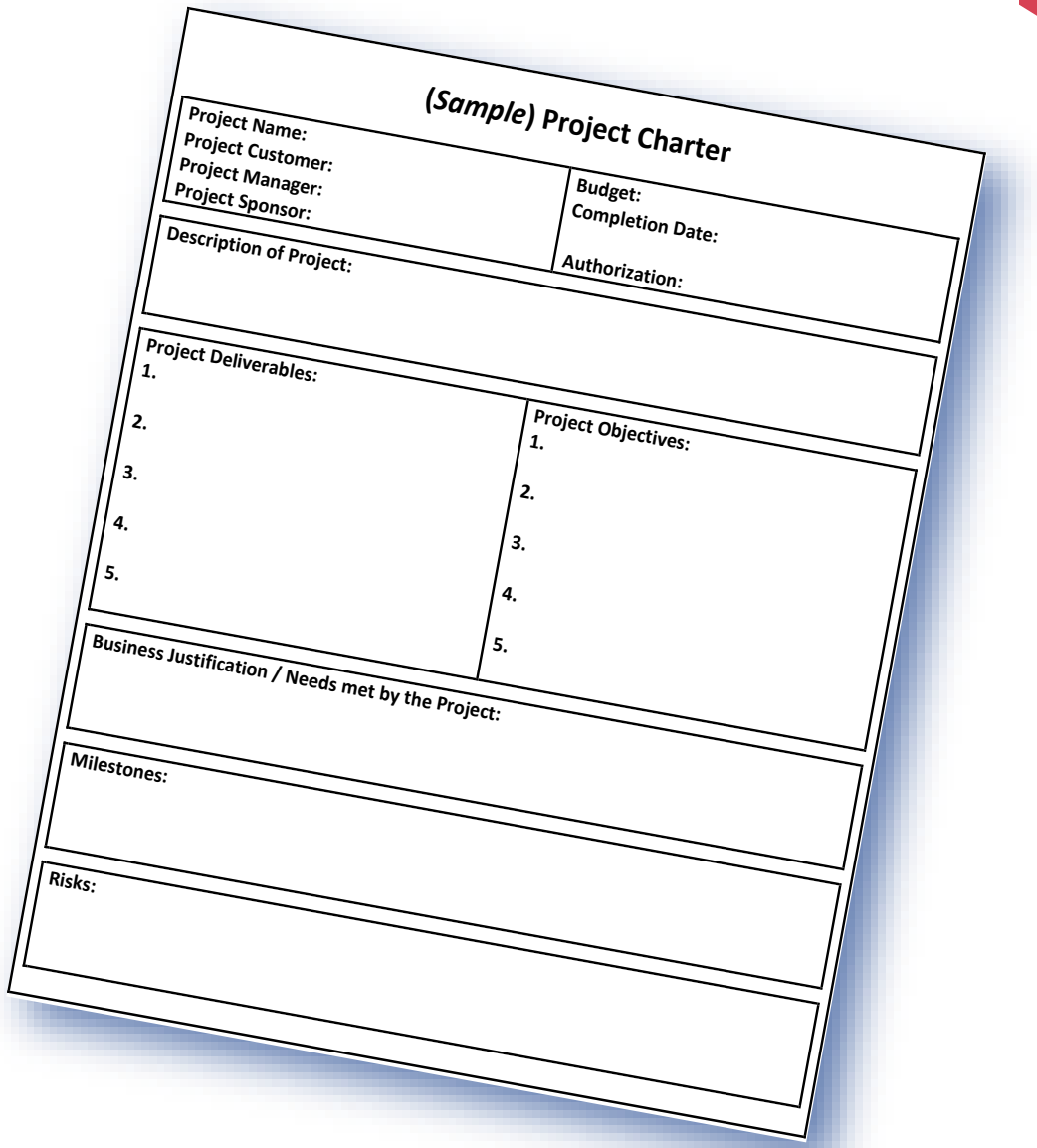
Project Charter Elements

Key Elements of Project Charters:

- Title
- Project Lead
- Problem Statement
- Project Objective
- Primary and Secondary Metrics
- Project Scope
- Project Timeline
- Project Team
- Stakeholders
- Approvers
- Constraints
- Dependencies
- Risks



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Problem Statement & Objective

Problem Statement:

Should call out the points identified in the Business Case, including the **current** state and the **ideal** (not goal) state, and the **gap** between the two

Objective:

Consists of the description of the **goal** state, which should reflect a **reduction** or **elimination** of the **gap**

Example:

Current process defect rates are 18% with an ideal rate of 3%. This represents a gap of 15%, costing the business $6.3 million dollars.

The goal of this project is to reduce this gap by 50% before Dec 2018 bringing process defect rates to 10.5% and saving $3.2 million dollars

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Metrics

- Are points where measurements are taken, and values monitored
  - E.g. Call time, count of sales, etc.
- Give clarity and the ability to quantify the work
- Establish how the initiative will be evaluated
- Provide points to establish a baseline or “starting point”
- Are mandatory for any Six Sigma project

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Primary Metric

The **Primary metric** (defined by the Champion, MBB, BB, or GB) is a generic term for a Six Sigma project's most important measure of success

- Good Primary metrics should be:
  - ➢Tied to the problem statement
  - ➢Measureable
  - ➢Expressed with an equation
  - ➢Aligned to business objectives
  - ➢Tracked at the proper frequency (hourly, daily, weekly, monthly etc.)
  - ➢Expressed pictorially over time with a run chart, time series, or control chart
  - ➢Validated with a Measurement Systems Analysis (MSA), covered later

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Project Charter – Secondary Metrics

The **Secondary metric(s)** are those things you do not want sacrificed on behalf of a primary improvement

- A secondary metric is one that makes sure problems are not just "changing forms" or "moving around“
- Measures positive & negative consequences

  - Example: If your primary metric is a cost or speed metric, then your secondary metric may be a quality measure
  - If your primary metric was energy consumption at a building, then you could shut off all the lights and the HVAC system and save a lot of energy; except secondary metrics are most likely comfort and functionality of the work environment

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1.3.4 Cost Benefit Analysis

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Cost-Benefit Analysis

The cost-benefit analysis is a comparison of financial costs and benefits in order to make viable economic decisions

- **Costs & Benefits** (e.g. cost savings, revenues, etc.) should be calculated on the baseline of key business process performance that relate to a business metric or “Key Performance Indicator” (KPI)
  - The Project’s Primary Metric must have a direct link between the process and its KPI’s
- A cost-benefit analysis is recommended to be done at the beginning of the project. The finance team may need to be involved

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Cost-Benefit Analysis

It is strongly recommended to update the cost-benefit analysis at each DMAIC phase of the project

- In the Define phase of the project, the cost-benefit analysis helps us understand the financial feasibility of the project
- In the middle phases of the project, updating and reviewing the cost-benefit analysis helps us compare potential solutions and make robust data-driven decisions
- In the Control phase of the project, the cost-benefit analysis helps us track the project's profitability

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Financial Analysis Approaches

Financial analysis approaches include:

- Time-value of Money (Present, Future, Net Present Value)
- Return on Investment (ROI)
- Internal Rate of Return (IRR)
- Breakeven and Payback Period
- Opportunity Cost
- Benefit over Cost
- Net Path Value
- Estimated Monetary Value (EMV)

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Project Cost-Benefit Estimation

Examples of **Benefits**:

- Direct revenue increase
- Waste reduction
- Operation cost reduction
- Quality and productivity improvement
- Market share increase
- Cost avoidance
- Customer satisfaction improvement
- Associate satisfaction improvement

Examples of **Costs**:

- Administration
- Asset
- Equipment
- Material
- Delivery
- Real estate
- Labor
- Training
- Consulting

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Return on Investment (ROI)

Return on Investment (ROI) is the ratio of the net (after removing costs) financial benefits (gain or loss) of a project or investment to the financial costs.

- The return on investment is used to evaluate the financial feasibility and profitability of a project or investment
- The higher the ROI, the more profitable the project

$$ROI = \frac{(\text{Total Benefits} - \text{Total Costs})}{\text{Total Costs}} \times 100\%$$

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Net Present Value (NPV)

The net present value (also called NPV, net present worth, or NPW) is the total present value of the cash flows calculated using a discount rate

- "*Net Cash Flow*$_t$" is the net cash flow happening at time "*t*"
- "*r*" is the discount rate
- "*t*" is the time of the cash flow

$$NPV = \frac{\text{Net Cash Flow}_t}{(1+r)^t}$$

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1.3.5 Project Risk Management

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Risk & Risk Management

- **Risk** is defined as a future event that may or may not occur that can impact the task/project if it occurs

  - **Negative** risks are also known as **Threats**
  - **Positive** risks are also known as **Opportunities**

- Risk management is concerned with Identifying, Assessing, and Monitoring project risks before they develop into issues and impact the project

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Project Risk Analysis Steps

The project risk analysis process consists of the following steps that are iterative and evolve through the life cycle of a project

STEP	DESCRIPTION
Risk Identification	Identify risks and risk categories, group risks, and define ownership
Risk Assessment & Prioritization	Evaluate and estimate the possible impacts and interactions of risks
Response Planning	Define mitigation and reaction plans
Mitigation Actions	Implement action plans and integrate them into the project
Tracking and Reporting	Provide visibility to all risks
Closing	Close the identified risk

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Risk Identification

The first action of risk management is the identification of individual events that the project may encounter during its lifecycle

- The **Identification** step consists of:
  - Identify the risks
  - Categorize the risks
  - Match the identified risks to categories
  - Define ownership for managing the risks

- **Risk categories may include:**
  - Human & Physical Resources
  - Technology
  - Suppliers
  - Customer
  - Security
  - Legal

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Risk Assessment

Risk Parameters for evaluating, categorizing, and prioritizing risks include the following:

- Risk likelihood (i.e., probability of risk occurrence)
- Define the various impacts (positive / negative) of each risk
- Rate each impact based on a logical severity level
- Sort and evaluate risks by severity level
- Determine if any controls already exist
- Define potential mitigation (for risks) and contingency actions (for issues)
- Thresholds to trigger contingency actions

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Risk Mitigation Planning & Actions

- Risk Mitigation involves planning and implementing the steps to reduce or eliminate the probability and/or impact of a risk

- The risk owners are responsible for planning and implementing mitigation actions with support from the project team

  - The project core team members are responsible for identifying an appropriate action owner for each identified risk
  - The risk owner must track all mitigation actions and expected completion dates

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Risk Occurrence & Contingency Plans

- A Risk that has occurred is known as an “**Issue**”
- A **Contingency** plan is designed to address Issues
  - ➢Whenever an issue arises, the project team should implement the predefined contingency plan to ensure that project deliverables can be met

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Risk Tracking & Reporting

- **Risk tracking** and reporting provides critical visibility to all risks
  - ➢Risk owners must report on the status of their mitigation and contingency actions
- Depending on the risk severity, project managers need to report the risk status of each category of risk to senior management
  - ➢The details of each risk and issue should be recorded in the risk register or other tracking tool maintained by the project manager and reviewed on a regular basis

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Risk Closure

- The risk owners are responsible for recommending the risk closure to the project manager
- A risk is closed only when the item is not considered a risk to the project anymore
- When a risk is closed, the project manager needs to update the risk status in the lessons learned document

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1.3.6 Failure Modes & Effects Analysis (FMEA)

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What is FMEA?

The FMEA (Failure Modes and Effects Analysis) is an analysis technique to identify, evaluate, and prioritize a potential deficiency in a process so that the project team can design action plans to reduce the probability of the failure/deficiency occurring

- An FMEA is completed in cross-functional, brainstorming sessions in which attendees have a good understanding of the entire process or of a segment of it

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Why Create an FMEA?

Failure Modes Effect Analysis (FMEA) is a structured approach to:

- Predict failures and prevent their occurrence in manufacturing and other functional areas which generate defects
- Identify the ways in which a process can fail to meet critical customer requirements (Y)
- Estimate the Severity, Occurrence and Detection (SOD) of defects
- Evaluate the current Control Plan for preventing these failures from occurring and escaping to the customer
- Prioritize the actions that should be taken to improve and control the process using a Risk Priority Number (RPN)

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Who Can Create an FMEA?

The team working on the project, and anyone who had/has a role in defining, executing, or changing the process would be candidates for assisting in creating the FMEA; including:

- Associates
- Technical Experts
- Supervisors
- Managers
- Etc.

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Types of FMEA's

Design DFMEA:

- Performed early in the **design** phase to analyze **product** fail modes before they are released to production

Equipment FMEA:

- Used to analyze failure modes in the equipment used in a process to detect or make the part

Process PFMEA:

- Performed in the early quality planning phase of manufacturing to analyze fail modes in **manufacturing and transactional processes**

System FMEA:

- Performed on a product or service product at the early concept/design level when various modules all tie together

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FMEA: Template

Example of an FMEA template

#	Process Function (Step)	Potential Failure Modes (process defects)	Potential Failure Effects (KPOVs)	SEV	Potential Causes of Failure (KPIVs)	OCC	Current Process Controls	DET	RPN	Recommend Actions	Responsible Person & Target Date	Taken Actions	SEV	OCC	DET	RPN
1																
2																
3																
4																
5																
6																
7																
8																

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FMEA: Severity Rating Scale

EFFECT	CRITERIA: Severity of Effect Defined	RANKING
Hazardous: Without Warning	May endanger operator. Failure mode affects safe vehicle operation and / or involves noncompliance with government regulation. Failure will occur WITHOUT warning.	10
Hazardous: With Warning	May endanger operator. Failure mode affects safe vehicle operation and / or involves noncompliance with government regulation. Failure will occur WITH warning.	9
Very High	Major disruption to production line. 100% of product may have to be scrapped. Vehicle / item inoperable, loss of primary function. Customer very dissatisfied.	8
High	Minor disruption to production line. Product may have to be sorted and a portion (less than 100%) scrapped. Vehicle operable, but at a reduced level of performance. Customer dissatisfied.	7
Moderate	Minor disruption to production line. A portion (less than 100%) may have to be scrapped (no sorting). Vehicle / item operable, but some comfort / convenience item(s) inoperable. Customers experience discomfort.	6
Low	Minor disruption to production line. 100% of product may have to be reworked. Vehicle / item operable, but some comfort / convenience item(s) operable at reduced level of performance. Customer experiences some dissatisfaction.	5
Very Low	Minor disruption to production line. The product may have to be sorted and a portion (less than 100%) reworked. Fit / finish / squeak / rattle item does not conform. Defect noticed by most customers.	4
Minor	Minor disruption to production line. A portion (less than 100%) of the product may have to be reworked on-line but out-of-station. Fit / finish / squeak / rattle item does not conform. Defect noticed by average customers.	3
Very Minor	Minor disruption to production line. A portion (less than 100%) of the product may have to be reworked on-line but in-station. Fit / finish / squeak / rattle item does not conform. Defect noticed by discriminating customers.	2
None	No effect.	1

*Potential Failure Mode and Effects Analysis (FMEA), Reference Manual, 2002. Pgs 29-45. Chrysler Corporation, Ford Motor Company, General Motors Corporation

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FMEA: Occurrence Rating Scale

PROBABILITY OF FAILURE	POSSIBLE FAILURE RATES	C_{pk}	RANKING
Very High:	³ 1 in 2	< 0.33	10
Failure is almost inevitable	1 in 3	³ 0.33	9
High: Generally associated with processes similar to previous	1 in 8	³ 0.51	8
processes that have often failed	1 in 20	³ 0.67	7
Moderate: Generally associated with processes similar to	1 in 80	³ 0.83	6
previous processes which have	1 in 400	³ 1.00	5
experienced occasional failures, but not in major proportions	1 in 2,000	³ 1.17	4
Low: Isolated failures associated with similar processes	1 in 15,000	³ 1.33	3
Very Low: Only isolated failures associated with almost identical processes	1 in 150,000	³ 1.5	2
Remote: Failure is unlikely. No failures ever associated with almost identical processes	1 in 1,500,000	³ 1.67	1

*Potential Failure Mode and Effects Analysis (FMEA), Reference Manual, 2002. Pgs 29-45. Chrysler Corporation, Ford Motor Company, General Motors Corporation

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FMEA: Detection Rating Scale

DETECTION	CRITERIA: Likelihood the existence of a defect will be detected by test content before product advances to next or subsequent process	RANKING
Almost Impossible	No known control(s) available to detect failure mode	10
Very Remote	Test content must detect 60 % of failures	9
Remote	Test content must detect 65 % of failures	8
Very Low	Test content must detect 70 % of failures	7
Low	Test content must detect 75 % of failures	6
Moderate	Test content must detect 80 % of failures	5
Moderately High	Test content must detect 85 % of failures	4
High	Test content must detect 90 % of failures	3
Very High	Test content must detect 95 % of failures	2
Almost Certain	Test content must detect 99.5 % of failures	1

*Potential Failure Mode and Effects Analysis (FMEA), Reference Manual, 2002. Pgs 29-45. Chrysler Corporation, Ford Motor Company, General Motors Corporation

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1.4 Lean
1.4.1 Understanding Lean

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The Lean Enterprise

“Lean Enterprise is based on the premise that anywhere work is being done, waste is being generated.

The Lean Enterprise seeks to organize its processes to the optimum level, through the continual focus on the identification and elimination of waste.”

- Barbara Wheat

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Lean

A lean enterprise intends to eliminate waste and allow only value to be pulled through its system

- Lean manufacturing is characterized by:
  - Identifying and driving value
  - Establishing flow and pull systems
  - Creating production availability and flexibility
  - Zero waste

- Waste Elimination
  - Waste identification and elimination is critical to any successful lean enterprise
  - Elimination of waste enables flow, drives value, cuts cost, and provides flexible and available production

The 5 principles of Lean, on the following slides, are taken from the book Lean Thinking (1996) by James P. Womack and Daniel T. Jones

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Lean: 5 Principles

LEAN 5 PRINCIPLES
1. Identify customers and specify value
2. Identify and map the value stream
3. Create flow through eliminating waste
4. Customer pull
5. Pursue perfection

Only a small fraction of the total **time** and **effort** spent in an organization actually adds value for the end customer. With a clear definition of **value (from the customer's perspective)**, it is much easier to identify where the waste is

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Lean: 5 Principles

LEAN 5 PRINCIPLES
1. Identify customers and specify value
2. Identify and map the value stream
3. Create flow through eliminating waste
4. Customer pull
5. Pursue perfection

The **value stream** is the entire set of **activities** across all parts of the organization involved in jointly delivering the product or service. It is the end-to-end process that **delivers the value to the customer**. As stated above, once you understand what determines value to the customer, it is easier to determine where the waste is

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Lean: 5 Principles

LEAN 5 PRINCIPLES
1. Identify customers and specify value
2. Identify and map the value stream
3. Create flow through eliminating waste
4. Customer pull
5. Pursue perfection

Typically you will find that only **5% of activities add value**, but the percentage can be as high as 45% in a service environment. **Eliminating this waste ensures that your product or service flows** better to the customer **without interruption or waiting**

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Lean: 5 Principles

LEAN 5 PRINCIPLES
1. Identify customers and specify value
2. Identify and map the value stream
3. Create flow through eliminating waste
4. Customer pull
5. Pursue perfection

When possible, understand the **customer demand** on your product or service, then create a process to respond to it.
In other words, **only produce what the customer wants when they want it**

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Lean: 5 Principles

LEAN 5 PRINCIPLES
1. Identify customers and specify value
2. Identify and map the value stream
3. Create flow through eliminating waste
4. Customer pull
5. Pursue perfection

At first reorganizing the process steps will create flow, and as this is done, the layers of waste become exposed and exploited. **Continuing the process of eliminating, reorganizing, and exploiting the waste will continue to drive down the number of steps and time needed to serve the customer**

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1.4.2 Lean: History

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Lean History

- Henry Ford established the first mass production system in 1913 by combining standard parts, conveyors, and workflow

- Decades later, Kiichiro Toyoda and Taiichi Ohno at Toyota improved and implemented various new concepts and tools (e.g., value stream, takt time, kanban etc.) based on Ford's effort

  - Toyota developed what is known today as the Toyota Production System (TPS) based on lean principles

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Lean History

- Starting in the mid 1990s, Lean became extensively recognized and implemented when more and more Fortune 100 companies began to adopt Lean and Six Sigma

- The term “Lean manufacturing” was introduced by James Womack in the 1990s

- Lean and Six Sigma share similar objectives, work hand in hand, and have benefited from one another in the past 30 years

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1.4.3 Lean vs. 7 Deadly Wastes

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Waste: The 7 Deadly “Muda”

- The Japanese word for **waste** is “**muda**”
- There are 7 commonly recognized forms of waste, often referred to as the “7 deadly muda”
  - **Transportation**
  - **Inventory**
  - **Motion**
  - **Waiting**
  - **Overproduction**
  - **Over-Processing**
  - **Defects**

- You can use the acronym “**Tim Wood**” to help in memorizing these 7 classifications
- 3 additional forms of muda include:
  - **Under utilized Skills**
  - **Lack of Safety**
  - **Poor performing Processes**

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Waste: Transportation

Transportation is considered wasteful because it does nothing to add value or transform the product

- In a similar way, the less transportation a product has to endure, the better. There would be fewer opportunities for delay, destruction, loss, damage etc.

Examples:

- Extra steps in the process
- Distance traveled
- Moving paper from place to place

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Waste: Inventory

Inventory is often an overlooked waste. Think of the time, materials, and logistics that went into establishing such an abundance of inventory

- Inventory is the liability of materials that are bought, invested in, and not immediately sold or used

Examples:

- Transactions not processed
- Over-ordering materials consumed in-house
- Over-ordering raw materials – just in case

* An evaluation of Inventory would need to consider an organization's customer base, nature of their industry, availability and pricing of raw materials, and etc.

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Waste: Motion

Motion is another form of waste often occurring as a result of poor setup, configuration, or operating procedures

Examples:

- Extra steps
- Extra data entry
- Having to look for something

- Wasted motion can be experienced by machines or humans
- Wasted motion is exaggerated by repetition or recurring tasks
- Wasted motion is very common with workers who are unaware of the impact of small unnecessary movements in repetitive tasks

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Waste: Waiting

Waiting is an obvious form of waste and is typically a symptom of an upstream problem causing a "bottleneck"

Examples:

- Processing once each month instead of as the work comes in
- Showing up on time for a meeting that starts late
- Delayed work due to lack of communication from another group

- Waiting is usually caused by inefficiency or poorly-designed workflows
- Waiting can also be caused by inefficient administration
- Reduction in waiting time will require thoughtful applications of lean and process improvement

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Waste: Overproduction

Overproduction is wasteful because your system expends energy and resources to produce more materials than the customer or next function requires.

- Overproduction is one of the most detrimental of the seven deadly muda because it leads to many others:
  - Inventory
  - Transportation
  - Waiting, etc.

Examples:

- Preparing extra reports
- Reports not acted upon or even read
- Multiple copies in data storage
- Over-ordering materials
- Duplication of effort/reports

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Waste: Over-processing

Over-processing occurs any time more work is done than is required by the next process step, operation, or consumer

- Relates to over-processing anything that may not be adding value in the eyes of the customer
- Over-processing is tasks, activities and materials that don't add value

Examples:

- Over-processing also includes being over capacity (scheduling more workers than required or having more machines than necessary).
- Sign-offs
- Reports that contain more information than the customer wants or needs
- Communications, reports, emails, contracts, etc that contain more than the necessary points (briefer is better)
- Voice mails that are too long

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Waste: Defects

Defects, or defectives, are an obvious waste for any working environment or production system

- Defects require rework during production and/or after the product is returned from an unhappy customer
- Some defects are difficult to solve and they create “workarounds” and hidden factories
- Eliminating defects is a sure way to improve product quality, customer satisfaction, and production costs

Examples:

- Incorrect data entry
- Paying the wrong vendor
- Misspelled words in communications
- Making bad product
- Materials or labor discarded during production

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1.4.4 Lean & Six Sigma

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Lean & Six Sigma

- Lean and Six Sigma both have the objectives of producing high value (quality) at lower costs (efficiency)
- They approach these objectives in somewhat different manners but in the end, both Lean and Six Sigma drive out waste, reduce defects, improve processes, and stabilize the production environment
- Lean and Six Sigma are a perfect combination of tools for improving quality and efficiency

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Lean & Six Sigma

Cost Savings for the Business & Value for the Customer

• LEAN	• SIX SIGMA
•Efficiency	•Effectiveness
•Eliminate Waste	•System Capability
•Pull Systems	•Pull Systems
•Process Flow	•Process Yield (Reduce Variation)
•Easy Visual Approach	•Rigorous Analytical Approach

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Define Phase: Roadblocks

Look for the potential roadblocks and plan to address them before they become problems:

- No historical data exists to support the project
- Team members do not have the time to collect data
- Data presented is the best guess by functional managers
- Data is communicated from poor systems
- The project is scoped too broadly
- The team creates the “ideal” Process Map rather than the “as is” current Process Map

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Thank You

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2.0 Measure

The "Measure" phase is about:

- ✓ **Understanding the current process**
- ✓ **How the process is measured and validity of data**
- ✓ **The process's historical and baseline performance**

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Measure Phase

Process Definition

- Process Mapping, SIPOC, Value Stream Map
- X-Y Matrix

Six Sigma Statistics

- Basic Statistics
- Data Collection
- Descriptive Statistics
  - Central Tendency
  - Dispersion
- Distributions & Normality
- Central Limit Theorem
- Graphical Analysis

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Measure Phase

Measurement System Analysis

- Precision & Accuracy
- Bias, Linearity & Stability
- Gage Repeatability & Reproducibility
- Attribute MSA
- Stability

Process Capability

- Capability Analysis
- Attribute & Discrete Capability

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2.1 Process Definition

2.1.1 Process Mapping, SIPOC, Value Stream Map

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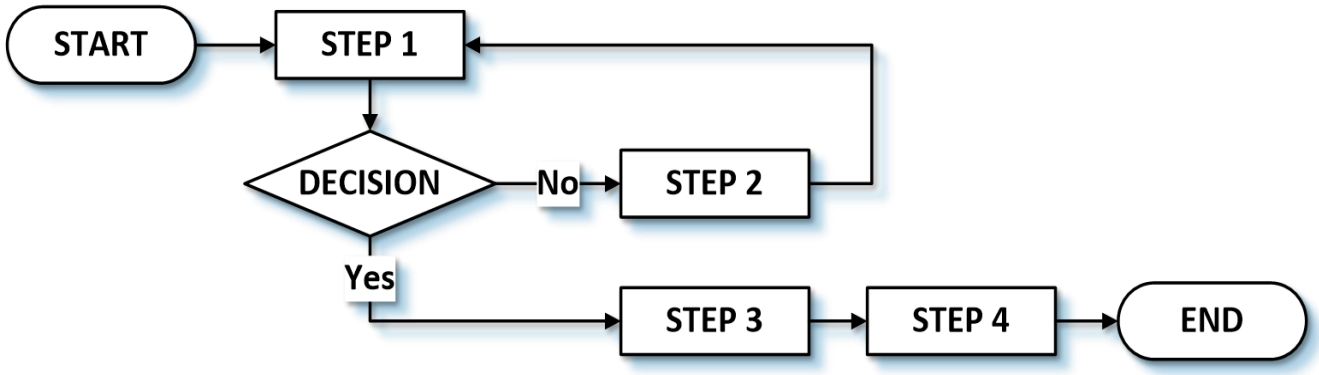
What is a Process Map?

- In the Measure phase, the project team should map the current state of the process instead of the ideal state



- What:
  - A visual representation of a process flow, or step by step procedures (also known as flow charts)
  - It describes how the information or materials sequentially flow from one business entity to the next

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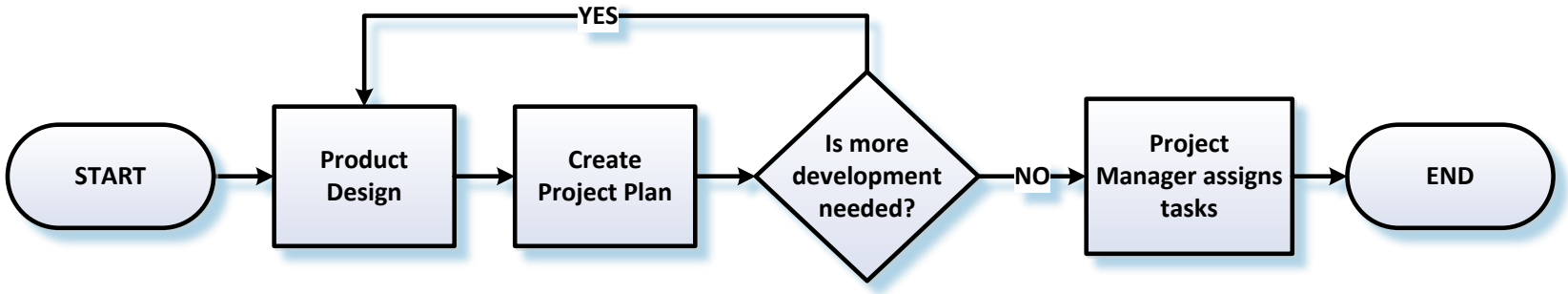
Process Maps

- Purpose:
  - ➢Identify the complexity of the process
  - ➢Communicate the focus of problem solving

- Living documents:
  - ➢They represent what is accurately occurring (present state)
  - ➢They should be created by the people who are closest to the process



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Process Map Benefits

By mapping processes we can identify many important characteristics and develop information for other analytical tools:

- Process inputs (X's)
- Supplier requirements
- Process outputs (Y's)
- Actual customer needs
- All value-added and non-value added process tasks and steps
- Data collection points
  - Cycle times
  - Defects
  - Inventory levels
  - Cost of poor quality, etc.
- Decision points
- Process control needs

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How to Create a Process Map

1. Identify and define the beginning and ending points
2. Consult with process owners and subject matter experts or observe the process in action
3. Record the process steps in order of their occurrence
4. Fill the step information into the appropriate process symbolic shapes and plot the diagram (see following slides for process map shapes)
5. Once completed, record the map using software such as Visio

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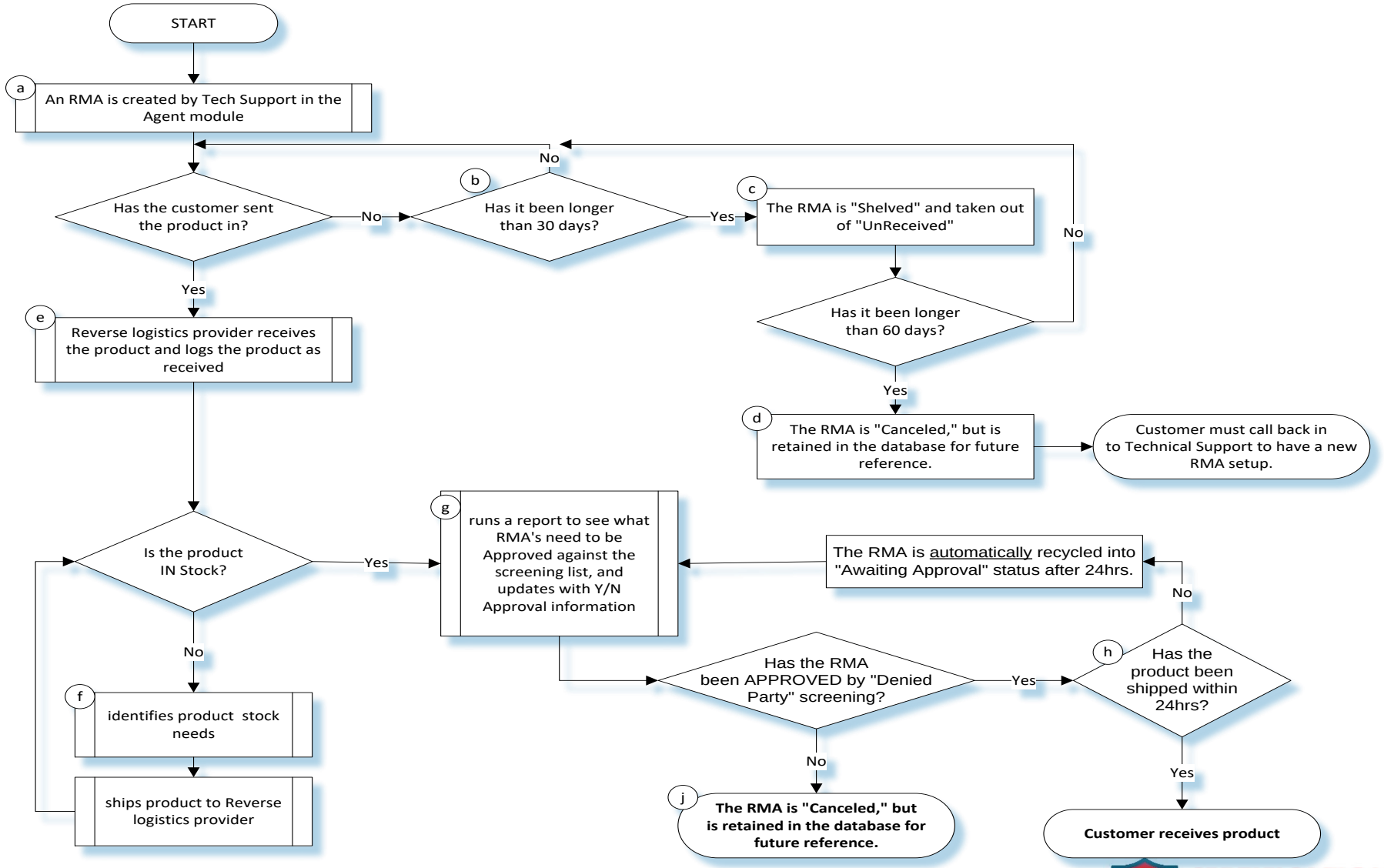
How to Create a Process Map

Note: If the process is too complicated to be covered in one single process map, you may create additional detailed sub-process maps for further information



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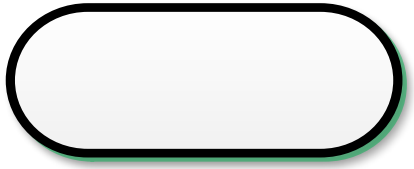
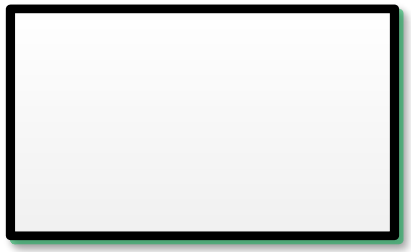
Process Map: Symbols

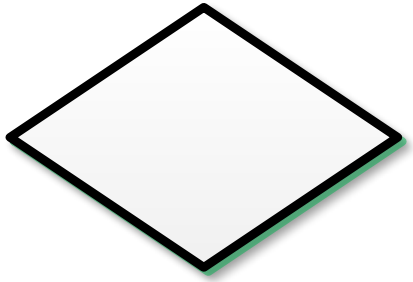
The following four symbols are the most commonly used symbols in a Process Map



Terminator (Oval): Shows the start/end points in the process.



Process (Rectangle): Indicates a single process step.



Decision (Diamond): Indicates a question with two choices (e.g., Yes/No).

Flow Line (Arrow): Shows the direction of the process flow.

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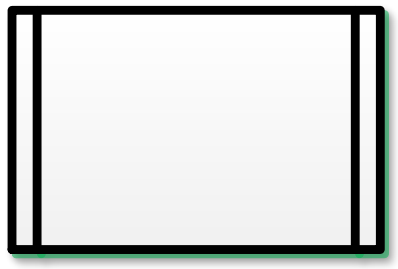
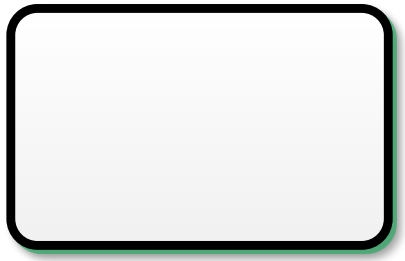
Process Map: Symbols



Predefined Process: Indicates a formally defined process step



Alternative Process: Indicates a process step as the alternate of a normal process step

Document: Indicates a process step that results in a document

Multi-Document: Indicates a process step that results in multiple documents

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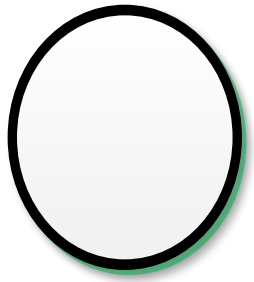
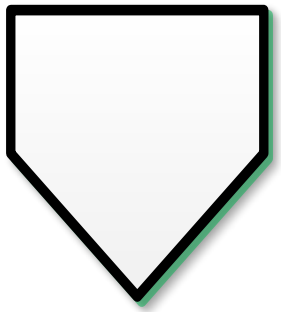
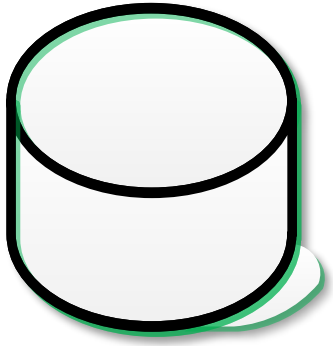
Process Map: Symbols



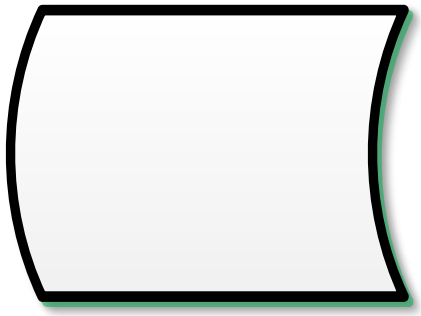
On-Page Connector: Indicates the process flow continues to another point on the same page



Off-Page Connector: Indicates the process flow continues onto another page



Database Indicates electronic storage of files, documents, and etc.



Stored Data: Indicates a process step that stores data

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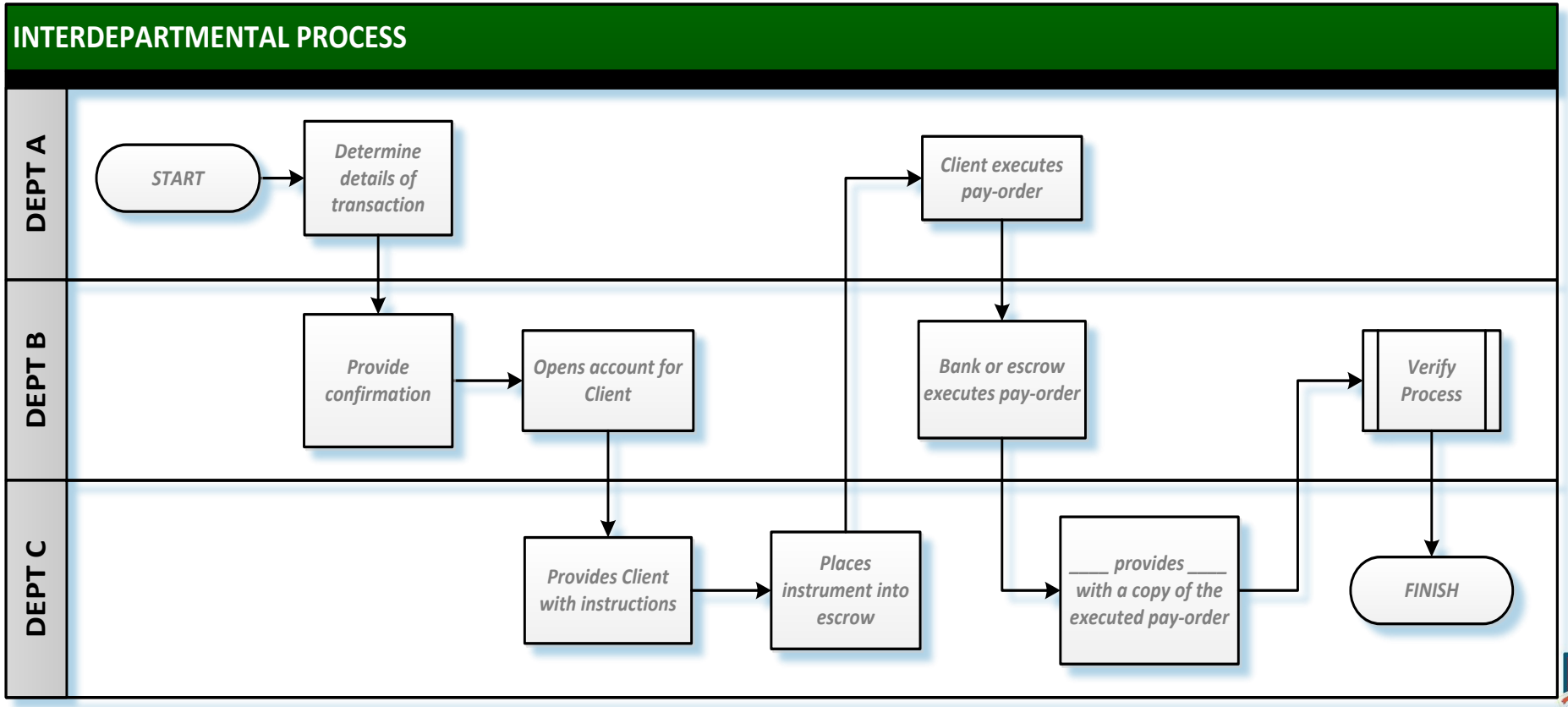
Types of Process Maps

- To illustrate the responsibility of different organizations involved in the process, use a Swim Lane Process Map
- The Swim Lane map shows who or which department is responsible for the steps in a process



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What is SIPOC?

- A SIPOC (Supplier-Input-Process-Output-Customer) is a high-level visualization tool to help identify and link the different components in a process

- It is usually applied in the Measure phase in order to better understand the current state of the process and define the scope of the project

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SIPOC Terms

TERM	DESCRIPTION
Suppliers	Vendors who provide the raw material, services, and information
Input (x)	The raw materials, information, equipment, services, people, environment involved in the process
Process f(x)	The high-level sequence of actions and decisions that transform inputs into outputs
Output (y)	The services or products delivered to the customers
Customers	The end users or recipients of the services or products

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SIPOC Benefits

- Visually communicate project scope
- Identify key inputs and outputs of a process
- Identify key suppliers and customers of a process
- Verify:
  - Inputs match outputs for upstream processes
  - Outputs match inputs for downstream processes
- If you have completed your high-level process map, follow the outlined steps (on the next series of slides) to create a SIPOC

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How to: SIPOC

1. Vertically List 4-6 High-Level Process Steps
2. List Process Outputs
3. List Process Customers
4. List Process Inputs
5. List Suppliers of Inputs

SUPPLIERS	INPUTS	PROCESS	OUTPUTS	CUSTOMERS
		Start		
Step 1: Suppliers	Step 1: Inputs	Step 1	Step 1: Outputs	Step 1: Customers
Step 2: Suppliers	Step 2: Inputs	Step 2	Step 2: Outputs	Step 2: Customers
Step 3: Suppliers	Step 3: Inputs	Step 3	Step 3: Outputs	Step 3: Customers
Step 4: Suppliers	Step 4: Inputs	Final Step	Step 4: Outputs	Step 4: Customers

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Additional Mapping Techniques

On the following series of slides we will see a few additional mapping techniques

- Value-Stream Map
- Spaghetti Chart
- Thought Process / Mind Map

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What is Value Stream Mapping?

Value is defined as ‘what a customer will pay for’

The objective of Value Stream Mapping is to:

- Analyze a stream of steps in a process to determine if they individually add “value” – is the customer willing to pay for the actions or results of this unique step?
- Customers pay for Value-Adding (VA) activities
- Eliminate or reduce the ‘waste’ produced by or inherent in activities that do not add value (NVA’s)

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Identifying Activity Disposition

Each process step, or activity, is evaluated to determine its value-add contribution

- A value stream map requires more detailed information for each step than the standard process map
  - Cycle time
  - Preparation time
  - Actual working time
  - Available time
  - Scrap rate
  - Rework rate
  - Number of operators
- Following are a couple of questions that you may ask in determining the disposition of an activity:
  - Is the form, fit, or function of the work item changed as a result of this activity?
  - Ultimately, is the customer willing to pay for this activity?

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Non-Value-Added Activities

- Non-Value-Adding (NVA) activities are activities in a process that do not provide “value” (per the customer)
- Some non-value-adding activities are required, but not by the customer, such as governmental or industrial inspections and approvals

- Examples of non-value-adding activities:
  - Rework
  - Overproduction
  - Excess transportation
  - Excess stock
  - Waiting
  - Unnecessary motion

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Spaghetti Chart

A spaghetti chart is a graphical tool to map out the physical flow of materials, information, and people involved in a process

- It can reflect the distances between multiple workstations the physical flow has been through
- A process that has not been streamlined has messy and wasteful movements of materials, information, and people, resembling a bowl of cooked spaghetti

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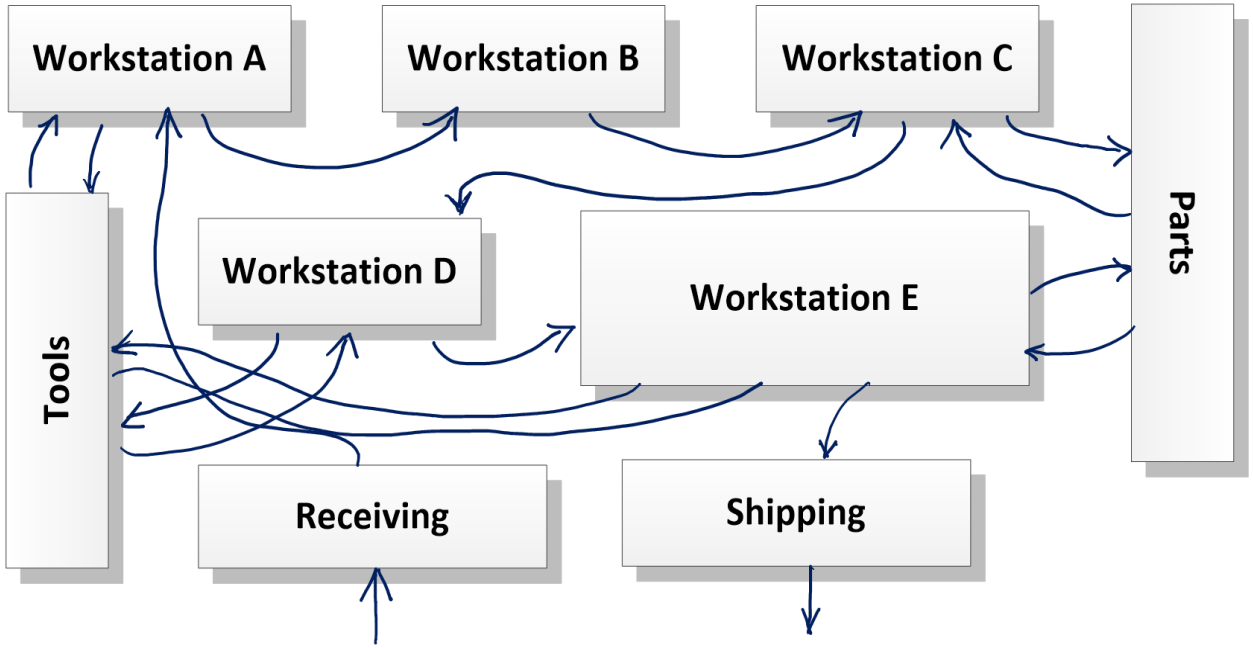
How to: Spaghetti Chart

1. Create a map of the work area layout
2. Observe the current workflow and draw the actual work path from the very beginning of work to the end when products exit the work area
3. Analyze the spaghetti chart and identify improvement opportunities



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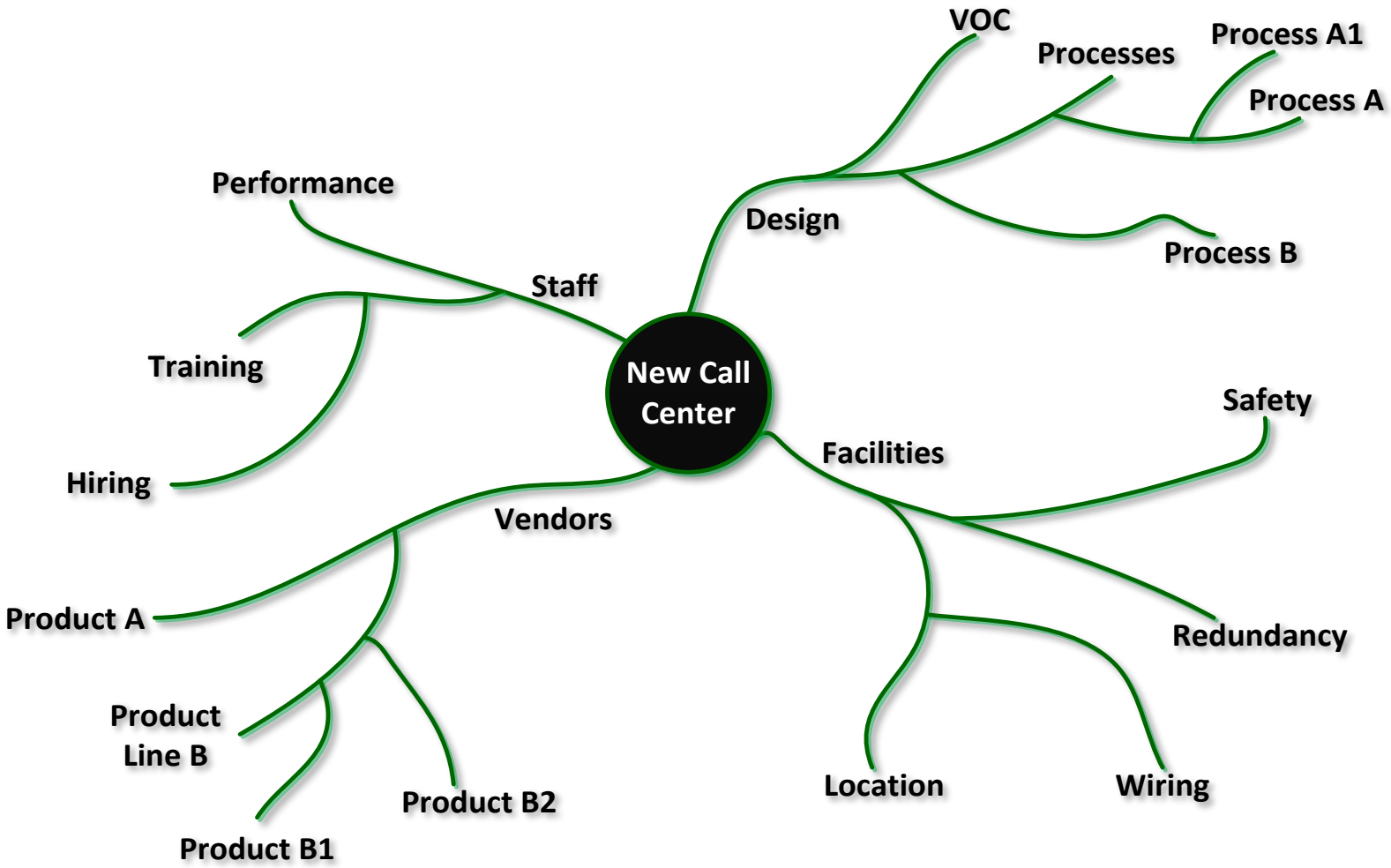
Thought Process / Mind Map

A thought process map is a graphical tool to help brainstorm, organize, and visualize the information, ideas, questions, or thoughts regarding reaching the project goal



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2.1.2 X-Y MATRIX

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What is an X-Y Matrix?

- The cause and effect matrix (X-Y Matrix) is a tool to help **subjectively quantify** the relationship of several X's to several Y's
  - ➢Among the Y's under consideration, two important ones should be the primary and secondary metrics of your Six Sigma project
- Once quantified, **prioritize** the X's that reflect the greatest impact on the Y's
- The goal is to **focus** in on the X's with the **greatest influence** on the desired Y's

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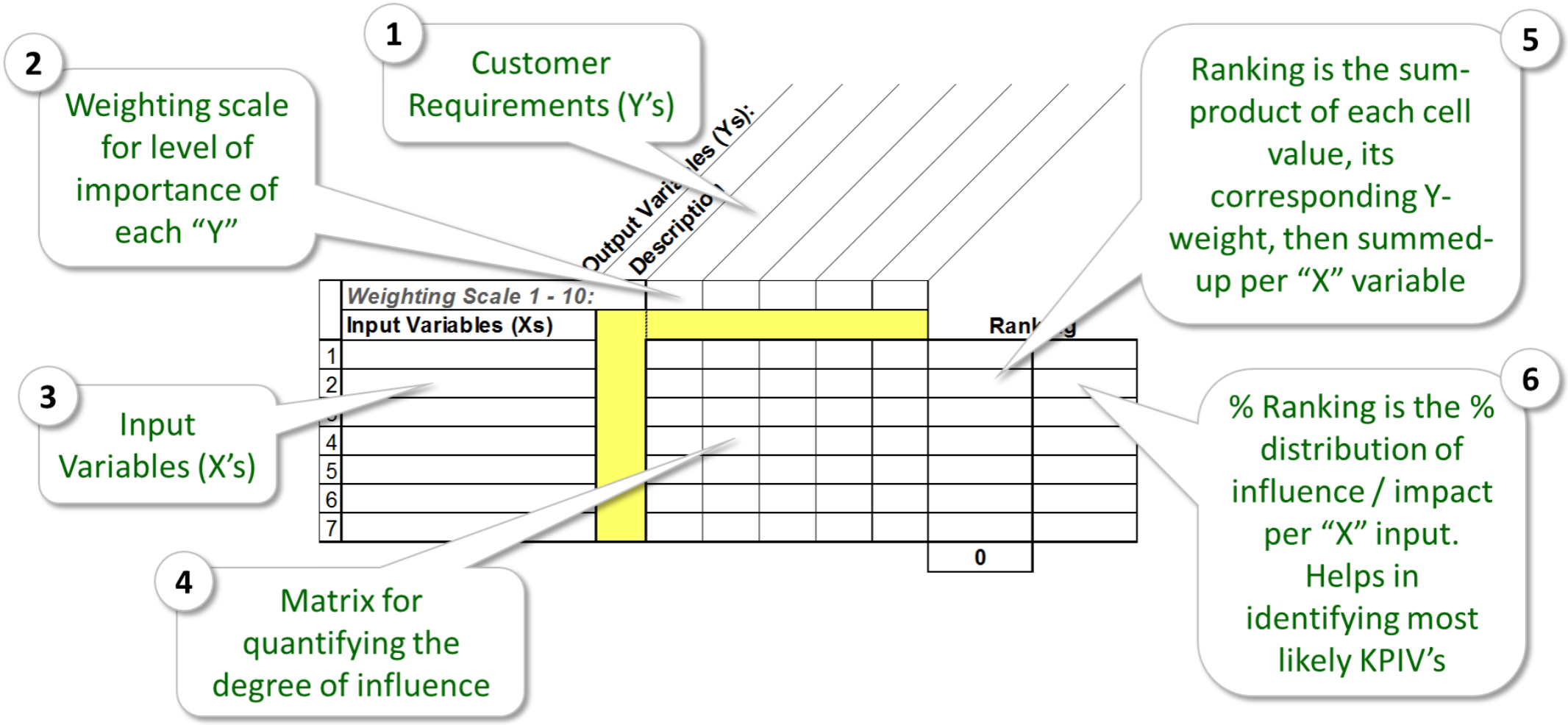
The “X-Y Matrix”



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X-Y Matrix

	Output Variables (Ys): / Description	Output "Y" Characteristic 1	Output "Y" Characteristic 2	Output "Y" Characteristic 3	Output "Y" Characteristic 4	Output "Y" Characteristic 5	Ranking	
	Weighting Scale 1 - 10:	10	6	8	7	9		
	Input Variables (Xs)							
1	Input Variable "X1"	3			9	2	111	15.08%
2	Input Variable "X2"	3	1		3	1	66	8.97%
3	Input Variable "X3	9	1		1	1	112	15.22%
4	Input Variable "X4"		9	1	9		125	16.98%
5	Input Variable "X5"	9	3	3			132	17.93%
6	Input Variable "X6"		1	9			78	10.60%
7	Input Variable "X7"		1	9	1	3	112	15.22%
							736	

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2.2 Six Sigma Statistics

2.2.1 Basic Statistics

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Data Types: Continuous vs. Discrete

CONTINUOUS

- Also known as "**Variable**" data
- Measureable on a continuum (**infinite**)
  - ➢ More precise and contains more information than counted data
- Answers questions such as: how long?, what volume?, or how far?
- Examples: Time, Pressure, Speed, Height, Weight

DISCRETE

- Includes "**Attribute**" data
- There is a **finite** number of values available
- Countable data
- Examples: how many?, how often?, or what kind?
- Pass/Fail, Yes/No

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Population & Parameter

- **Population**
  - All possible observations of similar items from which a sample is drawn

- **Parameter**
  - Arithmetic description of the true numeric population value
    - Often unknown, estimated by a statistic
    - For example: population mean and population standard deviation

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Sample & Statistic

- **Sample**
  - A randomly selected set of units, values, or items drawn from a population

- **Statistic**
  - A numerical value taken from a sample data set that may be used to make an inference about a population
  - For example: sample mean or sample standard deviation

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Refresher: Statistical Notation

SIGN / SYMBOL	DESCRIPTION	SIGN / SYMBOL	DESCRIPTION	SIGN / SYMBOL	DESCRIPTION
x	An individual value, an observation	s	The Standard Deviation of sample data	Σ	Summation
x_i	For each, all, individual values	σ	The Standard Deviation of population data	α	Alpha; risk of a Type 1 error
x_1	A specific (1st) individual value	s^2, V	The variance of sample data	β	Beta; risk of a Type 2 error
x-bar, $\bar{x}$	The Mean, average of sample data	σ^2, V	The variance of population data	k	Multi-purpose notation, i.e. # of subgroups, # of classes
x-double-bar, $\bar{\bar{x}}$	The grand Mean, grand average	R	The range of data	p	A proportion of sample data
μ	The Mean of population data	R-bar, $\bar{R}$	The average range of data	P	A proportion of population data
n	Sample size	N	Population (size)	λ	Lambda

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Data Collection Guidelines

Guidelines for collecting data:

- Clearly define **what** is to be measured
- Define the **characteristics** to be measured
- Select the right measurement **tools** and **technique**
- Establish **who** will **collect** the data
- Select an appropriate **sampling method**
- Decide **who** will **analyze** and **report** the results

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Sampling Techniques (Intro)

- **Random Sampling**
  - Requires giving every part an equal chance of being selected for the sample
  - Must be representative of the lot

- **Stratified Sampling**
  - Sampling from a subgroup of data that reflects a population parameter
  - For example: What is the mean height of children in 3rd grade?

Will be covered in greater detail under **"Analyze"**

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2.2.2 Descriptive Statistics

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Statistic Types: Descriptive & Inferential

DESCRIPTIVE	INFERENTIAL
• Describing what was going on • Enumerative (countable data) statistics • Includes organizing, summarizing, and presenting the data • Tools include histograms, pie charts, box plots, etc.	• Makes inferences about general conditions based upon the statistical analysis of a sample • Analytical statistics • Includes predicting and drawing conclusions • Tools include hypothesis testing, scattered diagram, etc.

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Basics of Descriptive Statistics

- Summarizes main features of the collected data in terms of their:
  - **Location**
  - **Spread**
  - **Shape**
- Presentation of data collected
- Does not provide any inferences about the population

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2.2.2.1 Descriptive Statistics:

Location – Measures of Central Tendency

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Location of the Data

- The location (i.e., central tendency) of the data describes the value where the data tend to cluster around (central value)

- Measurements to capture the location of the data:

  - **Mean**
  - **Median**
  - **Mode**

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Mean

- Sum total of all data values divided by the number of data points
  - ➢The arithmetic (calculated) mean (average) is the most widely used measure of central tendency
  - ➢Example for values: 2, 3, 5, 7, 11, 20, 20, 22, 36
    - (2+3+5+7+11+20+20+22+36) / 9 = 126/9 = **14**
    - Note: The **mean = 14**, median = 11, mode = 20

- **Sample**: **x-bar**, $\overline{x}$

- **Population**: **mu** μ

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Median

- The median (midpoint) is the **positional** mean

- It is the middle value when the data is arranged in numerical order

  - In cases of an even set of data, the median is the average of the middle two values

- Separates the set of data into two parts:

  - 50% of the values lie below the median and 50% above
  - Example for values: 2, 3, 5, 7, **11**, 20, 20, 22, 36
  - Note: The mean = 14, **median = 11**, mode = 20

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Mode

- The mode is the **frequency** mean
- The value that occurs most often
- Note: It is possible for groups of data to have more than one mode
  - Example for values: 2, 3, 5, 7, 11, **20, 20**, 22, 36
  - Note: The mean = 14, median = 11, **mode = 20**
- If no number is repeated, there is no mode for the data set

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2.2.2.2 Descriptive Statistics: Spread – Measures of Dispersion

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Spread of the Data

- The spread (i.e., variation) of the data describes the degree of data dispersion around the center value (location)

- There are multiple measurements to reflect the spread of the data:

  - **Range**
  - **Variance**
  - **Standard Deviation**

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Range (R)

- The range of a set of data is the **difference** between the **largest** and **smallest** values

  - Common uses in Six Sigma include determining the range between Specification limits and Control limits

- Example for values: 2, 3, 5, 7, 11, 20, 20, 22, 28

  - The Range = 28 – 2 = **26**

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Variance (V)

- A calculated value providing insight into the disbursement of data values
- The formula for a sample (n) is read as: “the **sum of squares** divided by the sample size minus one (n-1)”

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Population Variance

Differences between Population & Sample Variance are the mean and the denominator

$$s^2 = \frac{\sum (x_i - \overline{x})^2}{n-1}$$

Sample Variance

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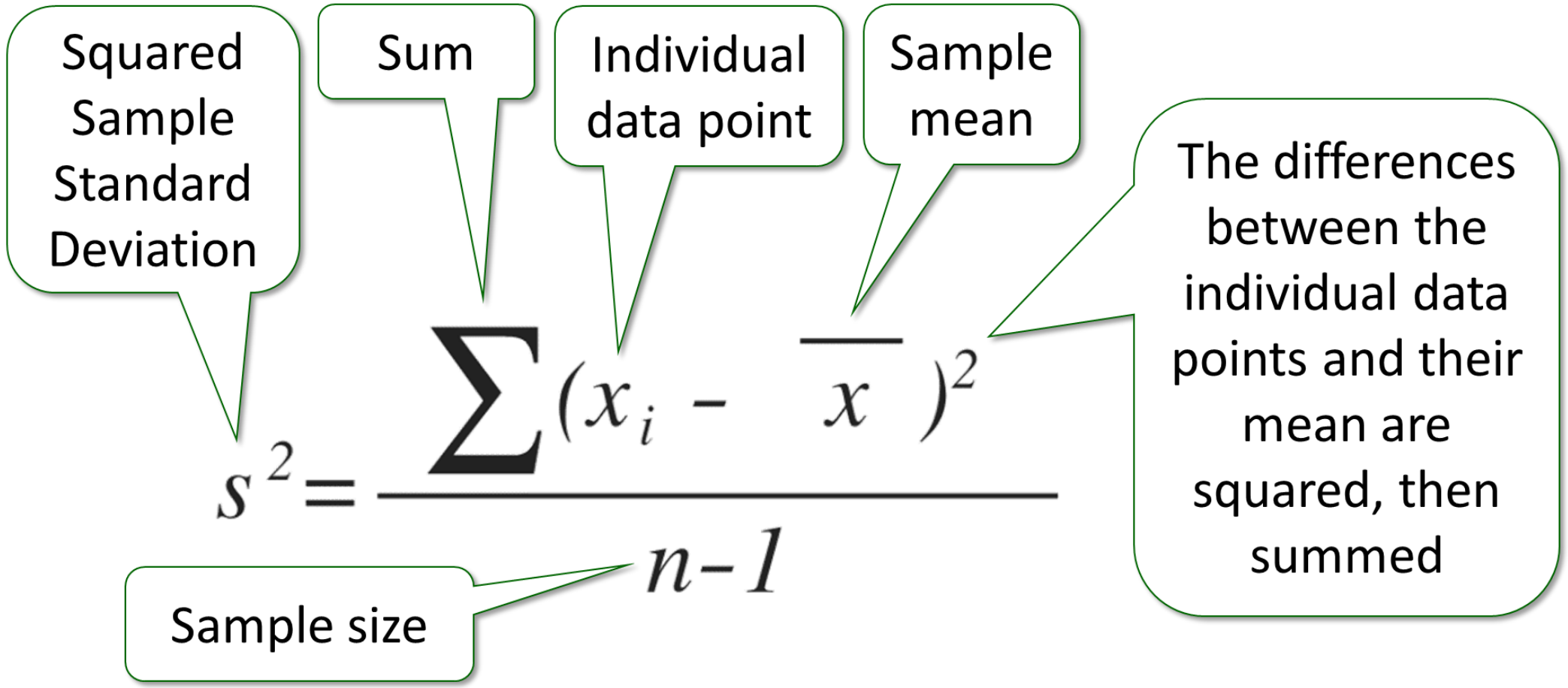
Variance (V)



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Standard Deviation (SD, s, σ)

- The square-root of the Variance
- A calculated unit of measure providing insight into the disbursement of data values
- Use when data are symmetrical

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Population Standard Deviation

Differences between Population & Sample Standard Deviation are the mean and the denominator

$$s = \sqrt{\frac{\sum (x_i - \overline{x})^2}{n-1}}$$

Sample Standard Deviation

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Variation = Bad

Variance away from a target value increases the chances of a product, service, etc. being rejected or not purchased by a customer

- The larger the standard deviation, the greater the chance of failing to meet customer requirements
- **Variation is the enemy of any quality process**
- We need to understand, manage, and minimize process variation

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Six Sigma Performance Measurements

LT Sigma Level	ST Sigma Level	C_p	DPMO	LT Sigma Level	ST Sigma Level	C_p	DPMO	LT Sigma Level	ST Sigma Level	C_p	DPMO
4.5	**6**	**2.00**	3.4	2.8	4.3	1.43	2,555	1.1	2.6	0.87	135,666
4.4	5.9	1.97	5	2.7	4.2	1.40	3,467	1	2.5	0.83	158,655
4.3	5.8	1.93	9	2.6	4.1	1.37	4,661	0.9	2.4	0.80	184,060
4.2	5.7	1.90	13	2.5	**4**	**1.33**	6,210	0.8	2.3	0.77	211,855
4.1	5.6	1.87	21	2.4	3.9	1.30	8,198	0.7	2.2	0.73	241,964
4	5.5	1.83	32	2.3	3.8	1.27	10,724	0.6	2.1	0.70	274,253
3.9	5.4	1.80	48	2.2	3.7	1.23	13,903	0.5	**2**	**0.67**	308,538
3.8	5.3	1.77	72	2.1	3.6	1.20	17,864	0.4	1.9	0.63	344,578
3.7	5.2	1.73	108	2	3.5	1.17	22,750	0.3	1.8	0.60	382,089
3.6	5.1	1.70	159	1.9	3.4	1.13	28,717	0.2	1.7	0.57	420,740
3.5	**5**	**1.67**	233	1.8	3.3	1.10	35,930	0.1	1.6	0.53	460,172
3.4	4.9	1.63	337	1.7	3.2	1.07	44,565	0	1.5	0.50	500,000
3.3	4.8	1.60	483	1.6	3.1	1.03	54,799	-0.1	1.4	0.47	539,828
3.2	4.7	1.57	687	1.5	**3**	**1.00**	66,807	-0.2	1.3	0.43	579,260
3.1	4.6	1.53	969	1.4	2.9	0.97	80,757	-0.3	1.2	0.40	617,911
3	4.5	1.50	1,350	1.3	2.8	0.93	96,800	-0.4	1.1	0.37	655,422
2.9	4.4	1.47	1,866	1.2	2.7	0.90	115,070	-0.5	**1**	**0.33**	691,462

- This table shows each sigma level's corresponding DPMO (defects per million opportunities) and short-term Capability Index (C_p)
- The higher the sigma level, the lower the defect rate and DPMO

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2.2.3 Distributions & Normality

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Shape of the Data

- The “Shape” of the sample data is presented in terms of its **distribution**

  - Also called **frequency distribution**

  - Summarizes the frequency (occurrence) of an individual value or a range of values of a variable

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Shape of the Data

Examples of **continuous** distribution

- Normal distribution
- t distribution
- Chi-square distribution
- F distribution

Examples of **discrete** distribution

- Binomial distribution
- Poisson distribution

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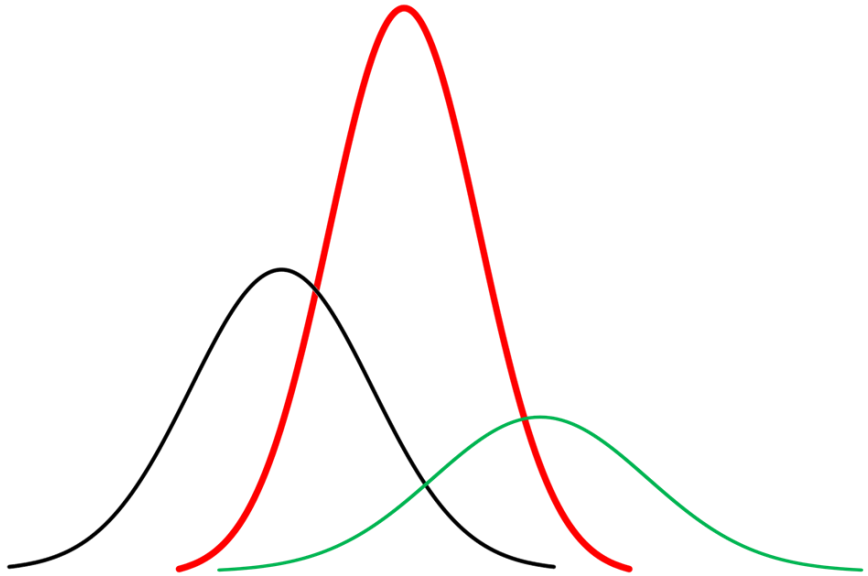
What is Normal Distribution?

The **normal** distribution is a probability distribution of a continuous random variable

- Values spread symmetrically about the mean
- A normal distribution can be described by using its mean (μ) and variance (σ2)
- Visually, the normal distribution creates a “bell” shape
- Also known as: Bell Curve, Gaussian, Parametric, Standard



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Z Distribution

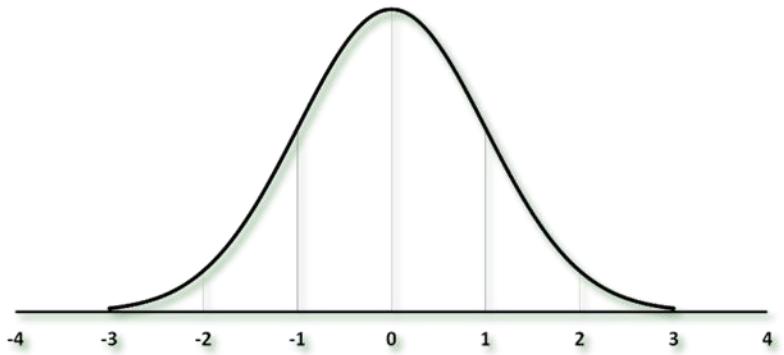
Z distribution is the simplest normal distribution with the mean equal to zero and the variance equal to one

- Any normal distribution can be transferred to a Z distribution (**Z score**) by applying:

$$z = \frac{x - \mu}{\sigma}$$

- The Z score is the measure of how many standard deviations an observation is above or below the mean

- Use lookup tables to find the proportion between μ and x



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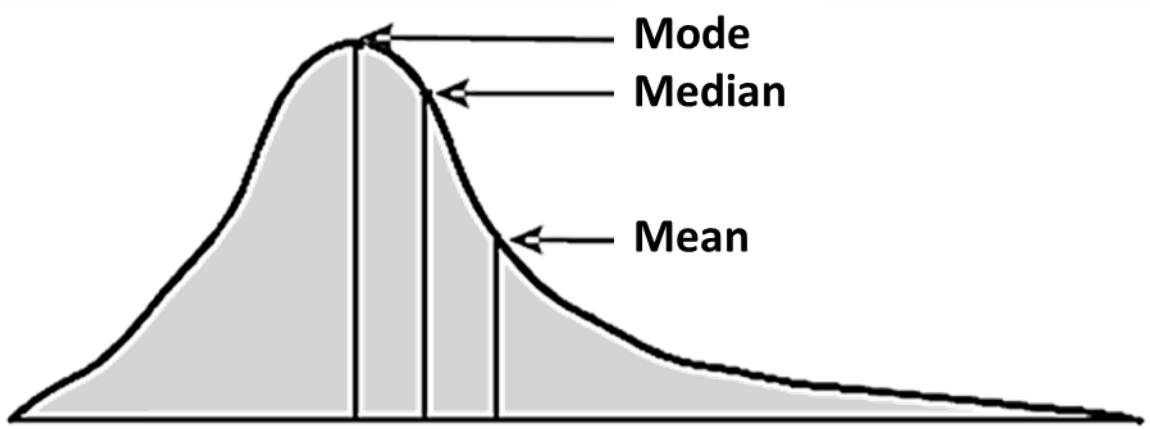
Location of Normal Distribution

- If a variable is normally distributed, the mean, the median, and the mode have the same value

Normal Distribution

Mean = Median = Mode

Right Skewed Distribution



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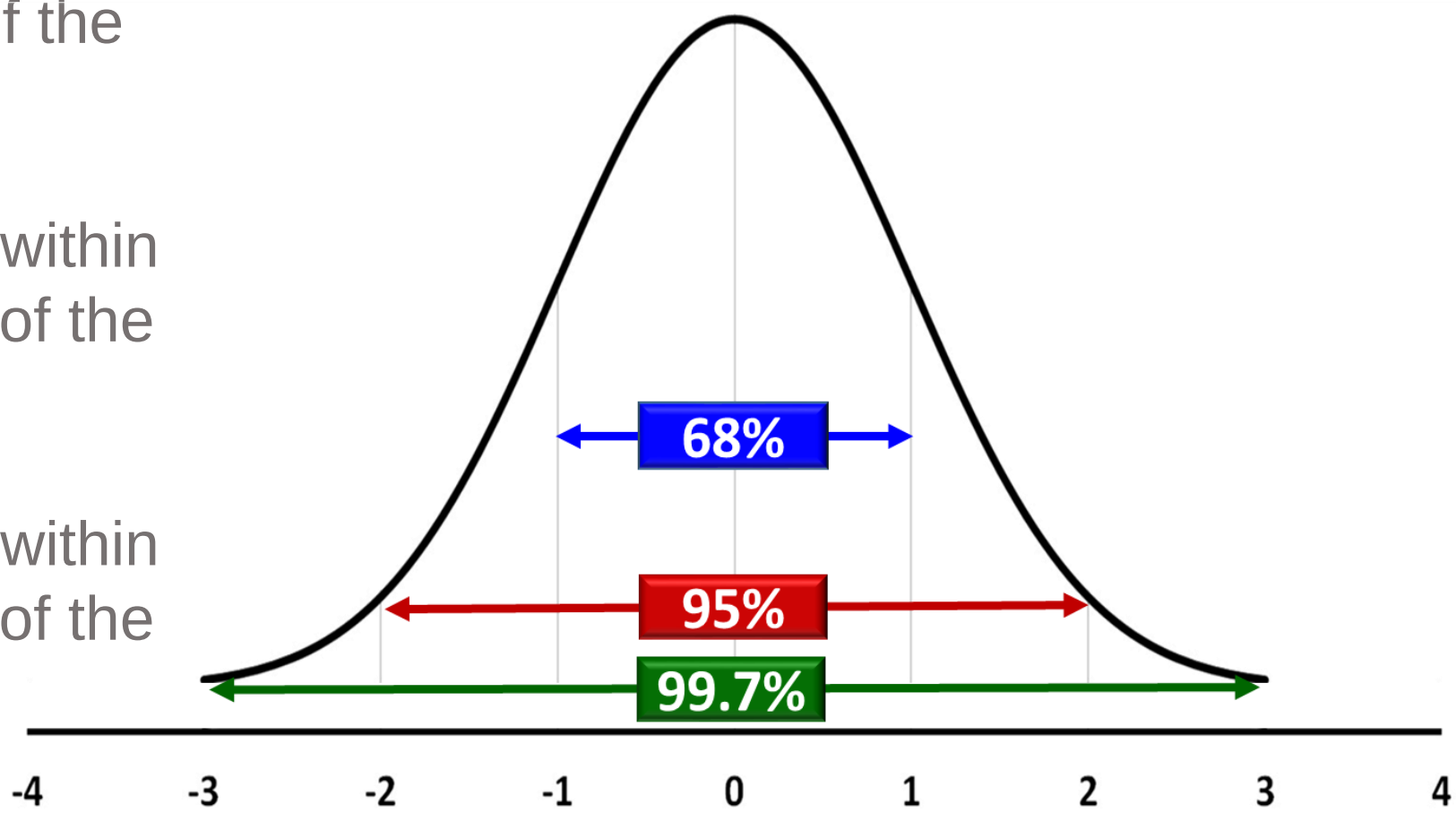
Empirical / 68-95-99.7 Rule

- **68.2 %** of the data will fall within +/- **1** Standard Deviation of the mean
- **95.4 %** of the data will fall within +/- **2** Standard Deviations of the mean
- **99.7** % of the data will fall within +/- **3** Standard Deviations of the mean



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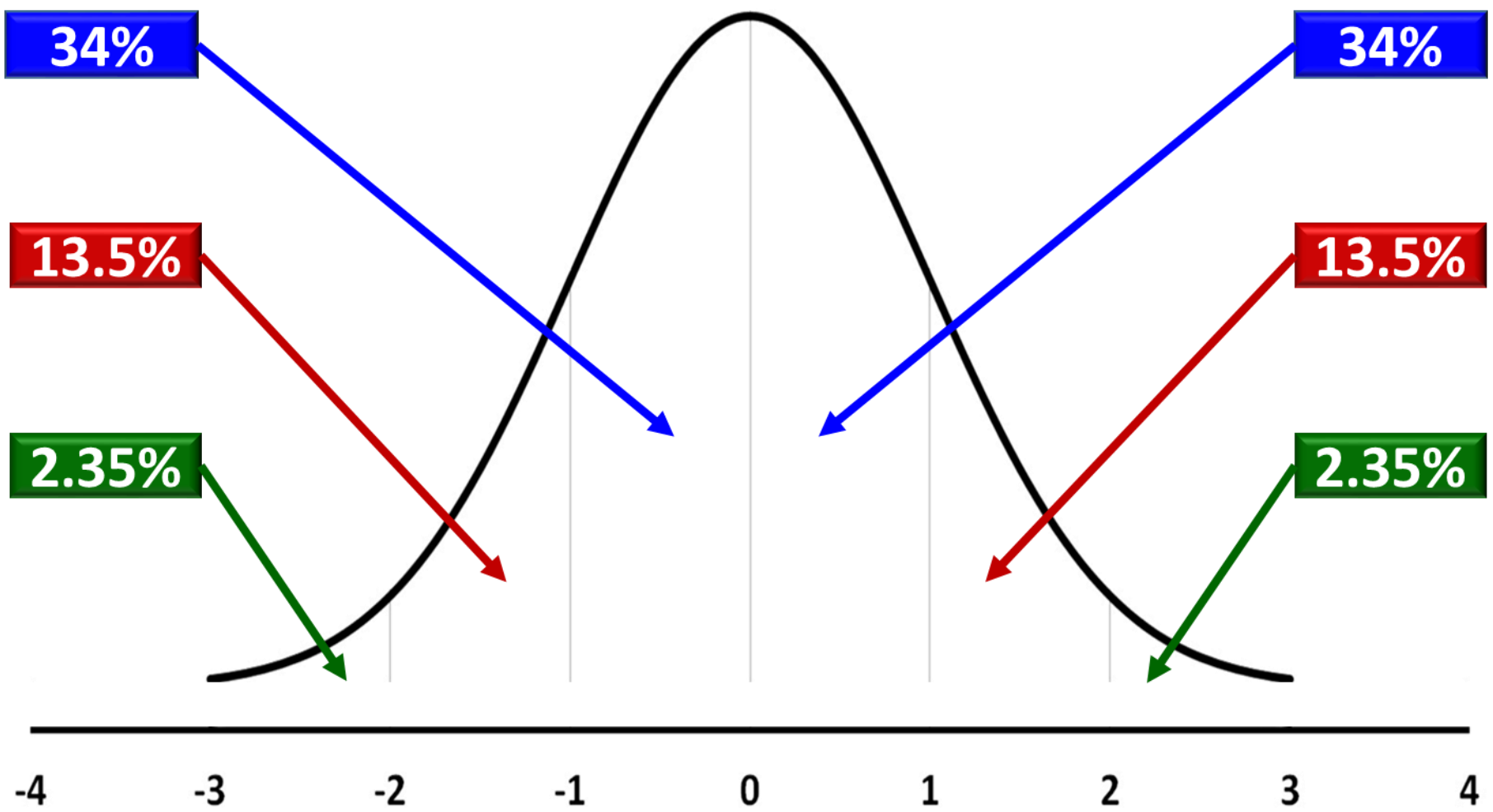
Empirical / 68-95-99.7 Rule



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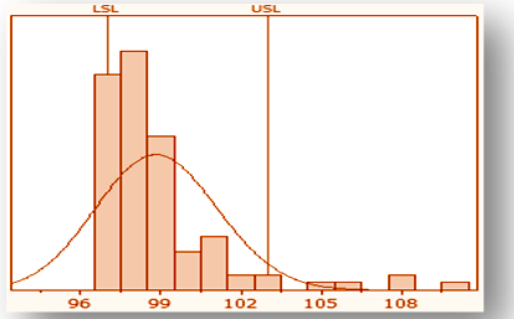
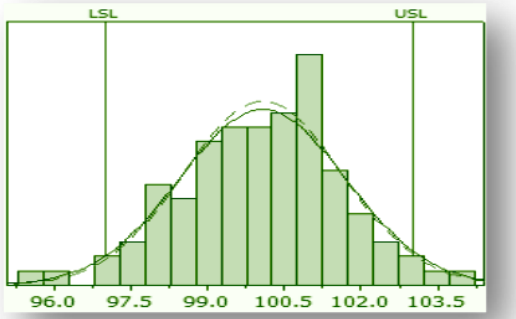
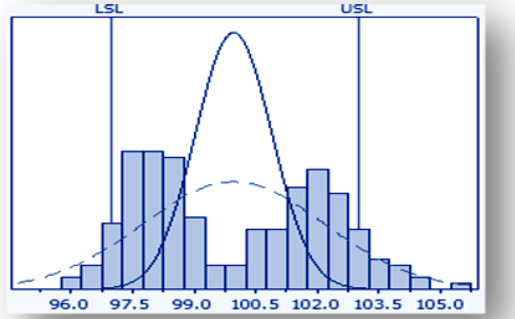
Why Assess Normality?







- There are many types of distributions
- Many processes have a **Normal Distribution**, but not all business functions do
  - ➢Example: One cannot have a phone call shorter than 0 seconds
- Knowing whether data are normally distributed will impact how it is analyzed and interpreted

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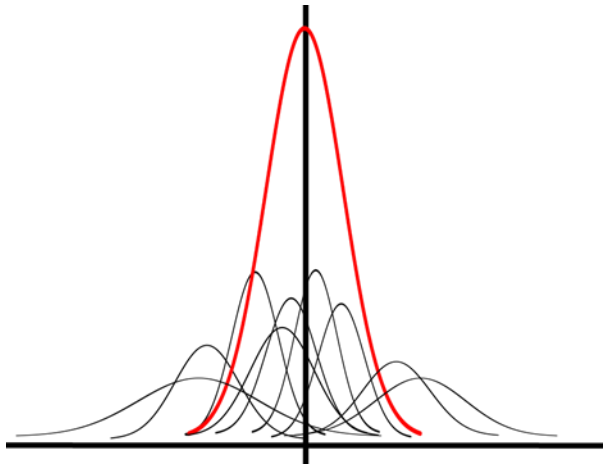
What is Central Limit Theorem?

- The Central Limit Theorem states that as the sample size increases data approaches a normal distribution

- Cumulatively, samples and sample means aggregate and shift (sample to sample) until combined they reflect the population's parameters

- Sample size threshold is when **n > 30**



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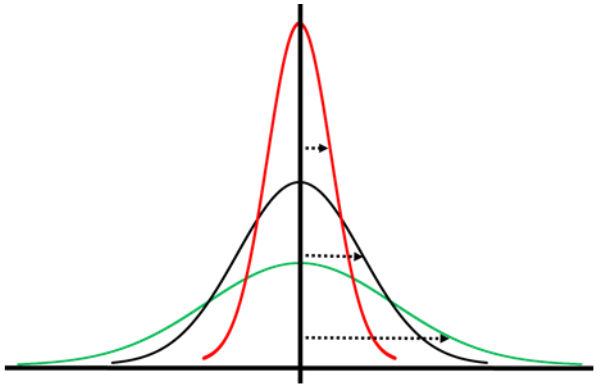
Central Limit Theorem Application

- Standard error (SE) of the mean to measure the standard deviation (s) of the sample mean's ($\overline{X}$) estimate of a population mean (μ)

- Standard deviation (σ) of the population mean (μ)



$$SE_{\overline{x}} = s_{\overline{x}} = \frac{s_x}{\sqrt{n}}$$

$$SD_{\overline{x}} = \sigma_{\overline{x}} = \frac{\sigma_x}{\sqrt{n}}$$

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2.2.4 GRAPHICAL ANALYSIS

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What is Graphical Analysis?

- Graphical analysis in statistics is a method to:
  - ➢Visualize the quantitative data
  - ➢Discover the structure and patterns in the data
- A complete statistical analysis includes both quantitative and graphical analysis

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Graphical Analysis Tools

There are various graphical analysis tools available. Following are four of the most commonly used:

- Box Plot
- Histogram
- Scatter Plot
- Run Chart
- Dot Plot

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Box Plot

A box plot is a **graphical method** to summarize a data set by visualizing the:

- Min & Max values
- 25^{th} percentile (1^{st} quartile)
- Median (50^{th} percentile, 2^{nd} quartile)
- 75^{th} percentile (3^{rd} quartile)
- Potential outliers

A percentile is the value below which a certain percentage of data fall. It is **not a calculated value**, it is **positional**

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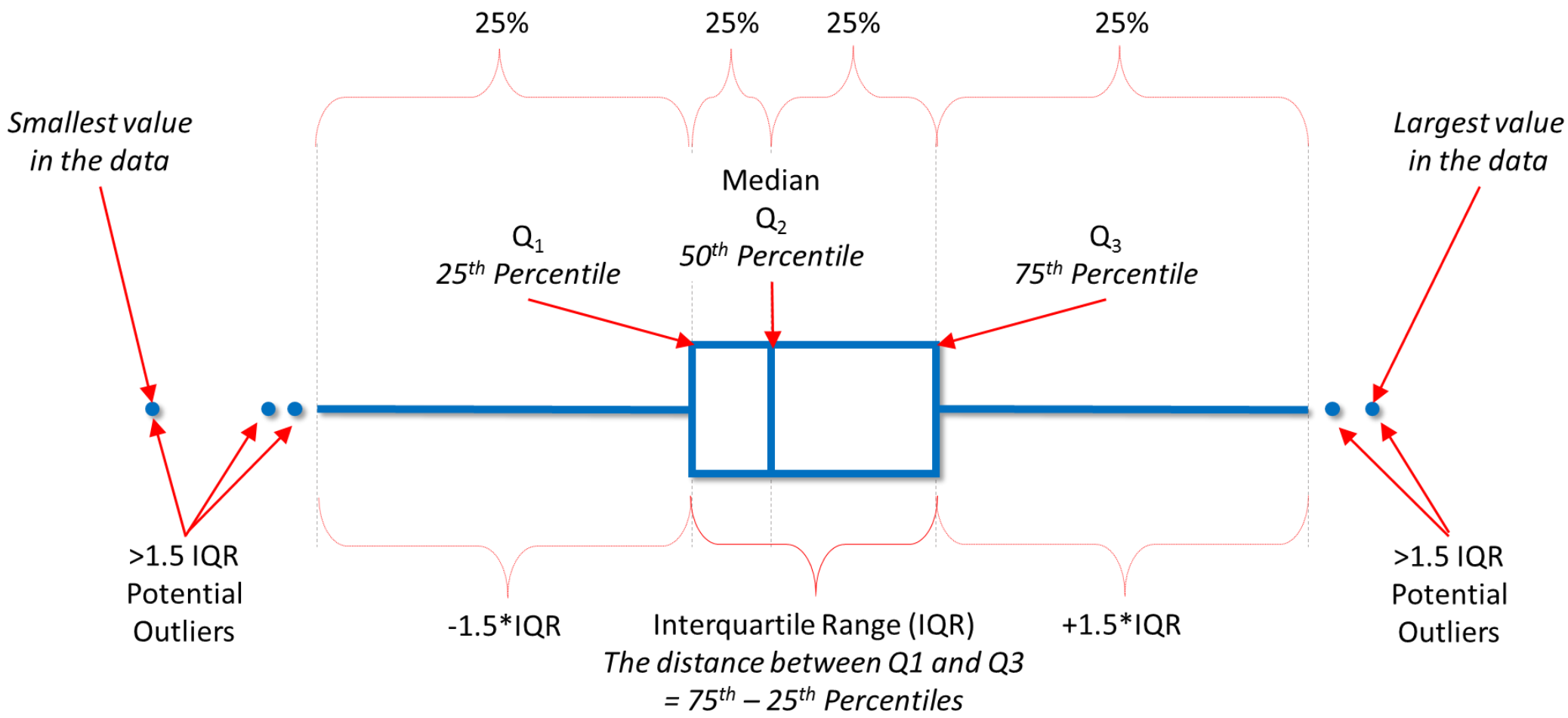
Box Plot



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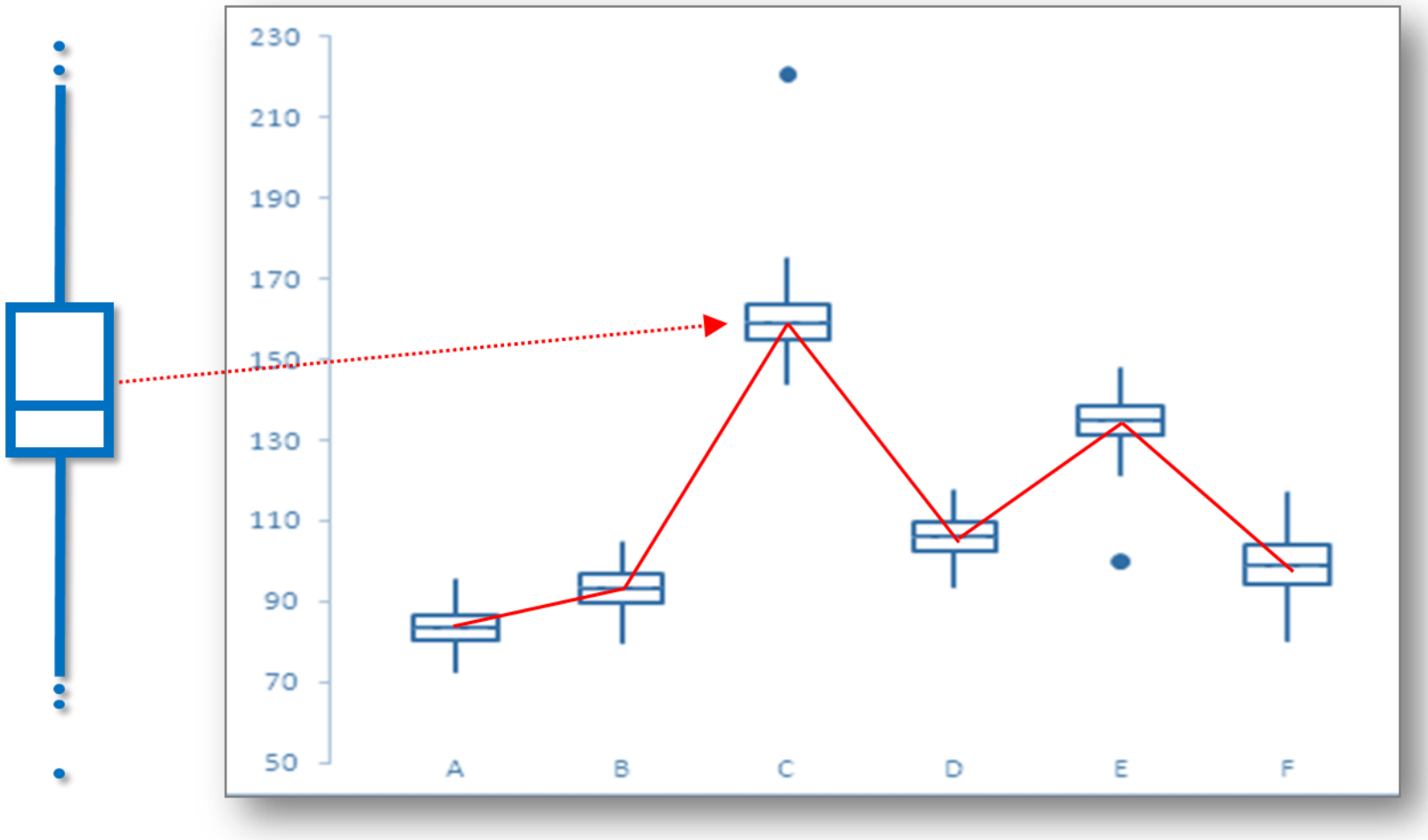
Box Plot Examples

What can you tell about the multiple sets of data expressed in these Box Plots?

- Because the boxes (IQR) do not overlap, there is significant variation



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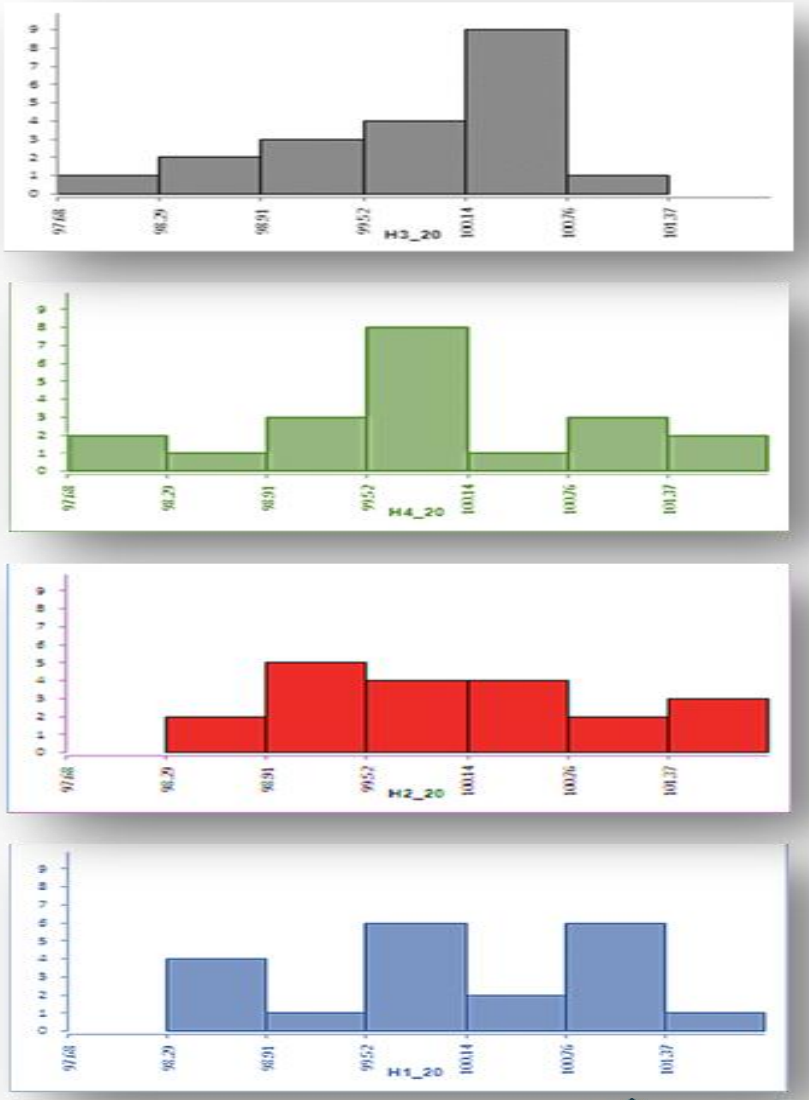
Histogram

- A **histogram** is used to graphically present the distribution of data

  - Total area of all the rectangles in a histogram is equal to 100% of all data values
  - Be careful not to determine Normality simply from a Histogram plot, if the sample size is low the data may not look very Normal



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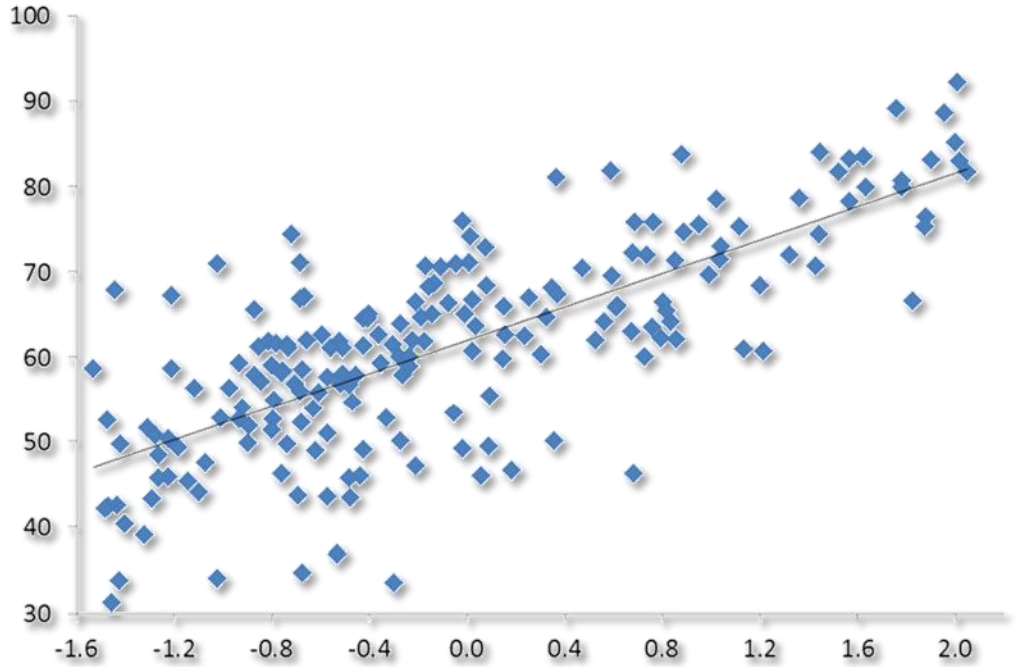
Scatter Plot

- On the scatter plot, a single observation is presented by a data point
- A scatter plot helps to understand:

  - Whether the two variables are related/correlated to each other or not
  - The strength of their relationship
  - What is the shape of their relationship
  - What is the direction of their relationship
  - Whether outliers are present



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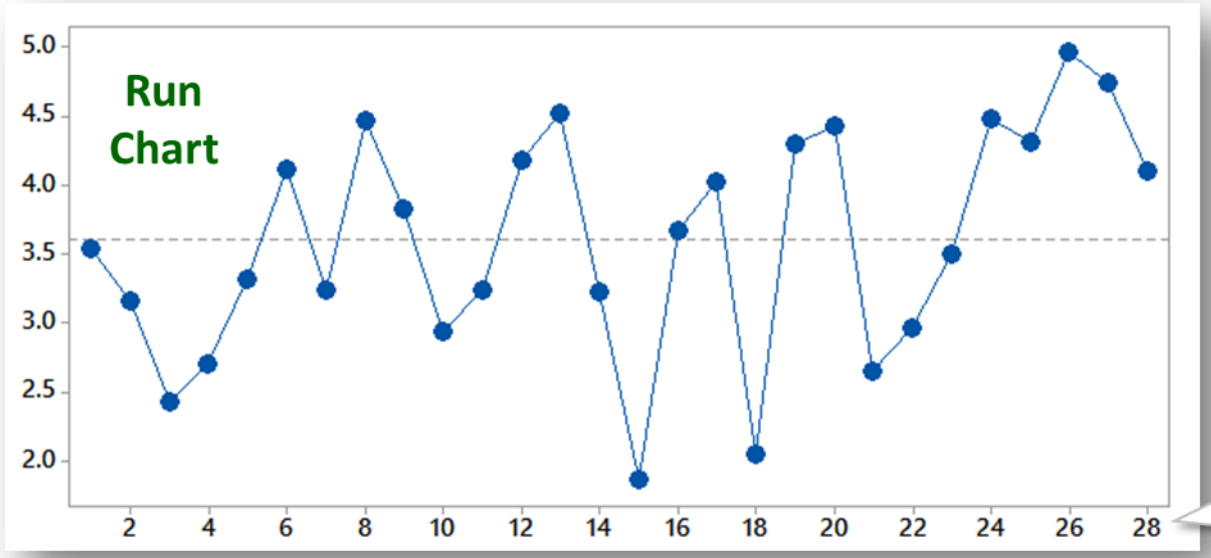
Run Chart

A run chart is a chart used to present the data in time order. It captures the process **performance over time**



The X axis of the run chart indicates the time and the Y axis indicates the observed values

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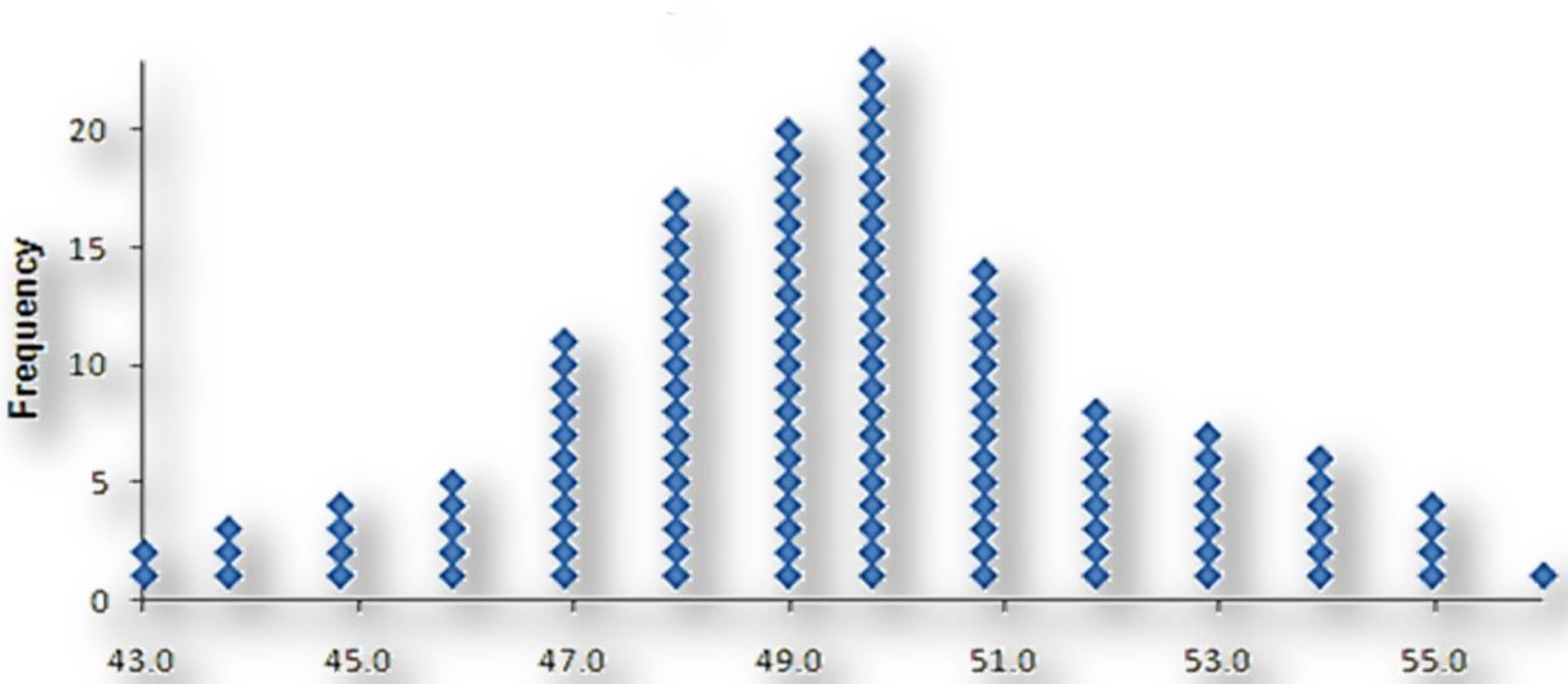
Dot Plot



The **Dot Plot** - an alternative to the Histogram, if you want to see the frequency of **individual values**

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2.3 Measurement Systems Analysis (MSA)

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Measurement Systems Analysis (MSA)

- Measurement Systems Analysis (MSA) precedes any further analysis conclusions from data

  - **Variable MSA** : for continuous measurements
  - **Attribute MSA** : for discrete measurements

- The goal of MSA is to qualify the measurement system by quantitatively analyzing its characteristics

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Measurement Errors

Observed Value = True Value + Measurement Error

- **True Value**
  - ➢The actual value we are interested to measure
- **Measurement Error**
  - ➢The errors brought in by measurement system
- **Observed Value**
  - ➢The observed/measured value obtained by the measurement system

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Measurement Errors

Common sources of measurement errors:

- Human
- Environment
- Equipment
- Sample
- Process
- Material
- Method

Note: the same general categories as found on Fishbone diagrams

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2.3.1 Precision & Accuracy

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Measurement System: Characteristics

Measurement systems can be characterized by two aspects:

- **Accuracy**
- **Precision**

A valid measurement system is both accurate and precise

- **Accurate ≠ Precise**
- **Precise ≠ Accurate**

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Accuracy vs. Precision

Accuracy (Location):

- [Average] observed value vs. the true value?
- **Bias**
- **Linearity**
- **Stability**

Precision (Variation):

- Consistency of the repeated measurement values under the same circumstances
- **Repeatability**
- **Reproducibility**

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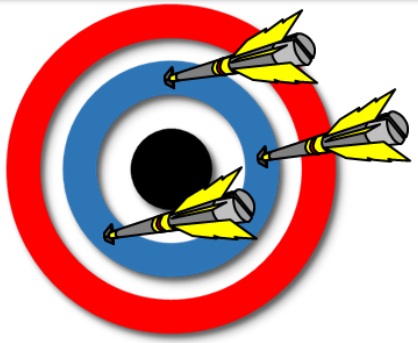
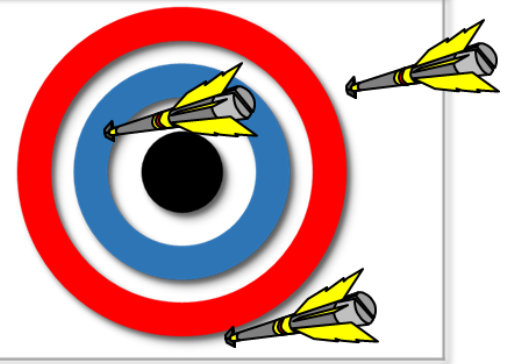
Accuracy vs. Precision

ACCURATE		PRECISE	
On average, the shots are in the center of the target but there is a lot of variability		Precision or repeatability is the ability to repeat the same measurement by the same operator at or near the same time	
ACCURATE & PRECISE		**NEITHER**	
Consistently on target with minimal variability		High level of variability and low level consistency	

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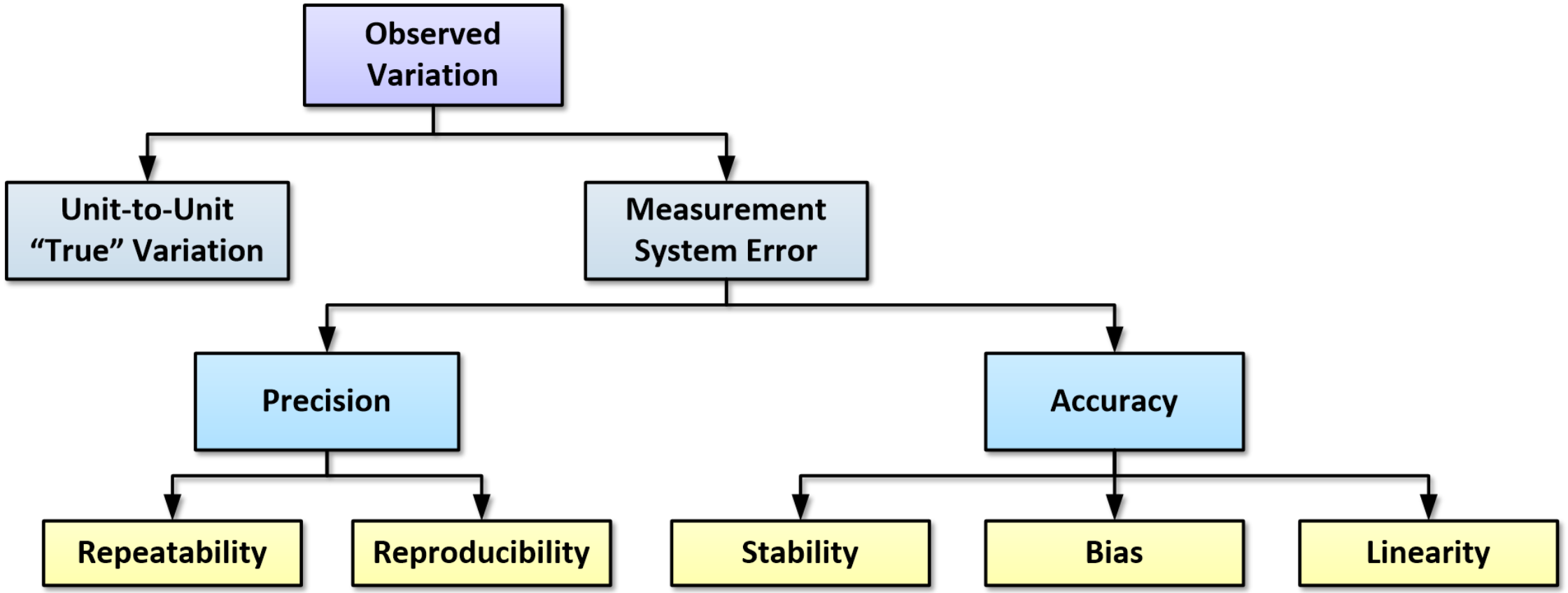
Measurement System & Variation



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Measurement System: Conclusions

The measurement system must be both accurate and precise, before we can start the data-based analysis or decision making

- If the measurement system is either not accurate or not precise:
  - ➢Identify and resolve the factor(s) affecting it
  - ➢Determine the measurement system's contribution to variation
  - ➢Calibrate the measurement system

May trigger a new project

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2.3.2 Accuracy: Bias, Linearity, & Stability

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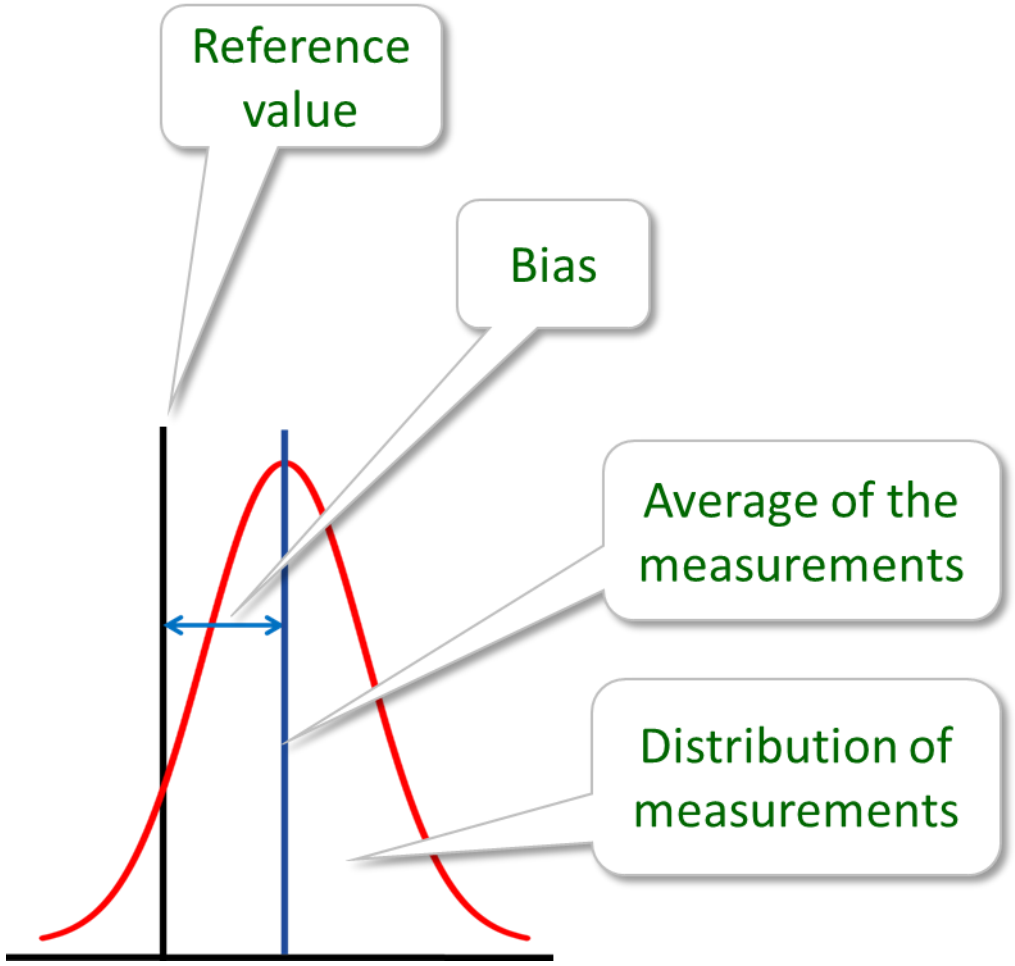
Bias

- **Bias** is the difference between the observed value and the true value
- Bias = Grand Mean – Reference Value



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Potential Sources of Bias

- Errors in measuring the reference value
- Lack of proper training for appraisers
- Damaged equipment or instrument
- Measurement instrument not calibrated precisely
- Appraisers read the data incorrectly

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Linearity

Linearity is the degree of the consistency of bias over the entire expected measurement range

- For example, a scale is off by 0.05 pounds when measuring an object of 20 pounds. However, it is off by 10 pounds when measuring an object of 110 pounds. The scale's bias is not linear

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Potential Sources of Linearity Errors:

- Errors in measuring the lower end or higher end of the reference value
- Lack of proper training for appraisers
- Damaged equipment or instrument
- Measurement instrument not calibrated correctly at the lower or higher end of the measurement scale
- Innate nature of the instrument

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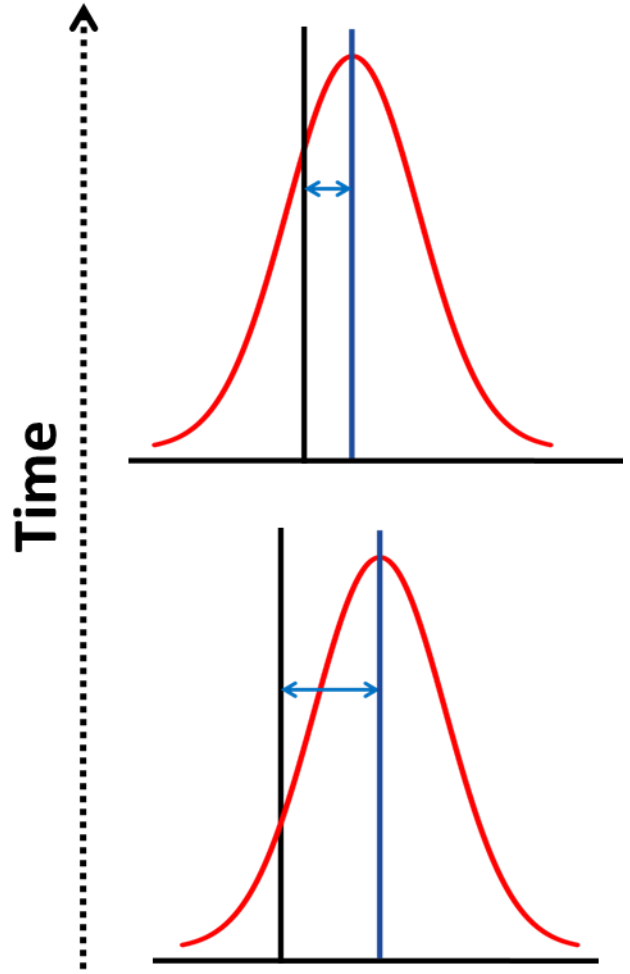
Stability

- Stability references the level of consistency of measurement values over an extended period of time
- A measurement system that cannot consistently perform well would not deliver reliable data
- Stability is evaluated using control charts



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Potential Sources of Instability

- Inconsistent training for appraisers
- Damaged equipment or instrument
- Worn equipment or instrument
- Measurement instrument not calibrated
- Appraisers do not follow the procedure consistently

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2.3.3 Variable MSA: Gage Repeatability & Reproducibility (GRR)

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Gage R&R

- Gage Repeatability & Reproducibility (Gage R&R, or GRR) is a method to analyze the variability of a measurement system

- Addresses the **precision** of a measurement system

- The potential sources of variance:

  - Equipment
  - Appraisers
  - Parts

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Gage Repeatability vs. Reproducibility

REPEATABILITY		REPRODUCIBILITY
Variation in measurements obtained when:		Variation in the average of measurements made by:
Same	**Operator**	**Different**
Same	**Gage**	**Same**
Same	**Characteristics**	**Same**
Same	**Part**	**Same**

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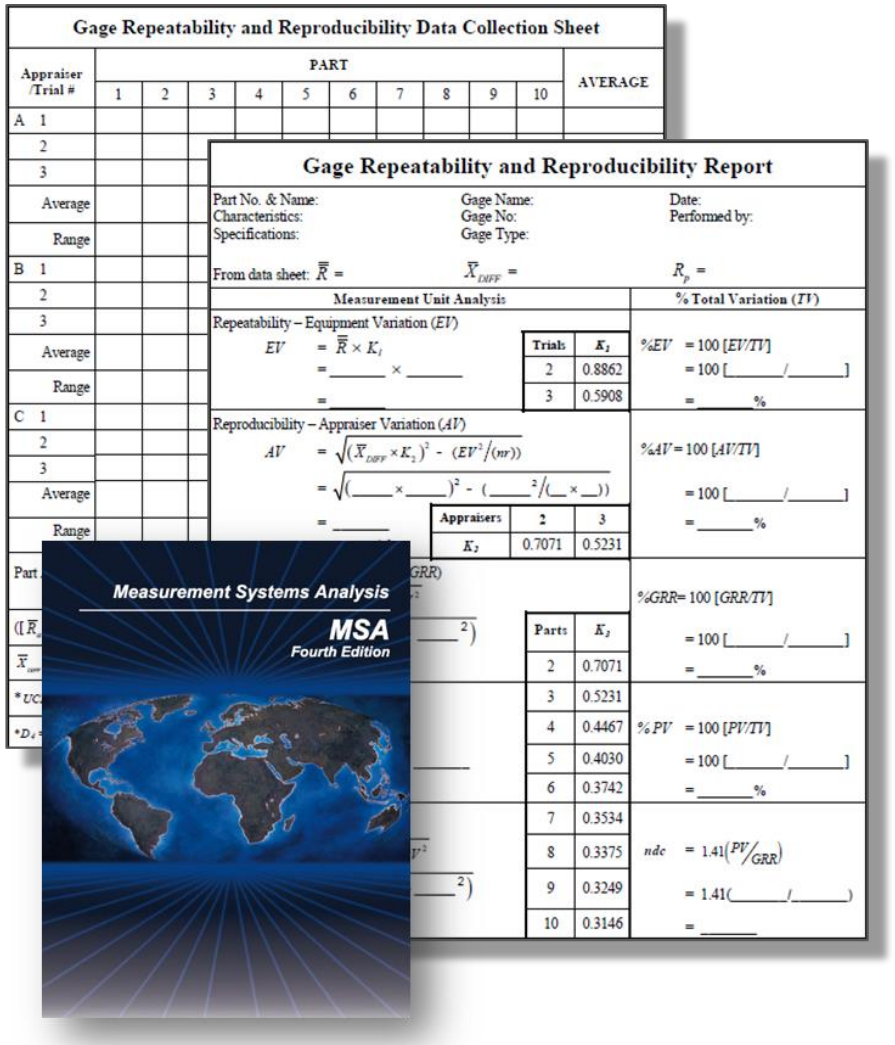
GRR Data Collection



- Let “k” appraisers
  - ➢(3 recommended)
- Measure “n” random samples independently
  - ➢(10 recommended)
- Repeat the process “p” times
  - ➢(3 recommended)
- Sequence of samples and appraisers is randomized

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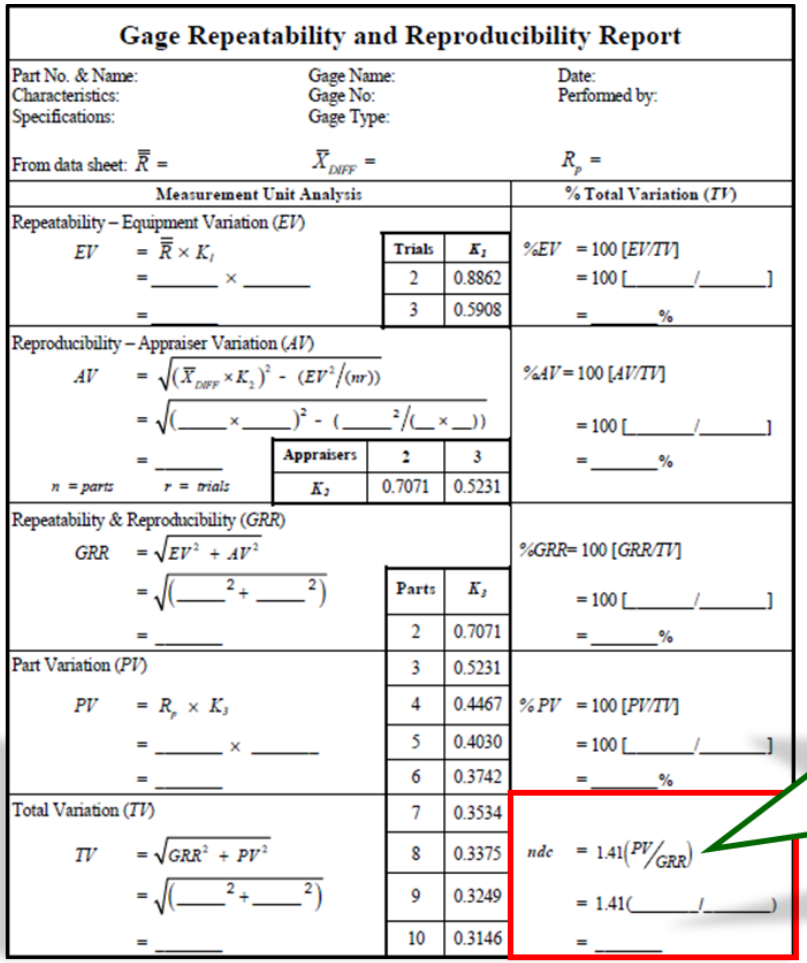
Guidelines for Distinct Categories (ndc)



Gage Repeatability and Reproducibility Report

Part No. & Name: | Gage Name: | Date:
Characteristics: | Gage No: | Performed by:
Specifications: | Gage Type:

From data sheet: $\bar{\bar{R}}$ = $\bar{X}_{DIFF}$ = R_p =

Measurement Unit Analysis | **% Total Variation (TV)**

Repeatability – Equipment Variation (*EV*)

$EV = \bar{\bar{R}} \times K_1$

= _____ × _____

= _____

Trials	K_1
2	0.8862
3	0.5908

%*EV* = 100 [*EV*/*TV*]

= 100 [_____/_____]

= _____ %

Reproducibility – Appraiser Variation (*AV*)

$AV = \sqrt{(\bar{X}_{DIFF} \times K_2)^2 - (EV^2/(nr))}$

$= \sqrt{(____ \times ____)^2 - (____^2/(__ \times __))}$

= _____

n = *parts* *r* = *trials*

Appraisers	2	3
K_2	0.7071	0.5231

%*AV* = 100 [*AV*/*TV*]

= 100 [_____/_____]

= _____ %

Repeatability & Reproducibility (*GRR*)

$GRR = \sqrt{EV^2 + AV^2}$

$= \sqrt{(____^2 + ____^2)}$

= _____

%*GRR* = 100 [*GRR*/*TV*]

= 100 [_____/_____]

= _____ %

Part Variation (*PV*)

$PV = R_p \times K_3$

= _____ × _____

= _____

Parts	K_3
2	0.7071
3	0.5231
4	0.4467
5	0.4030
6	0.3742
7	0.3534
8	0.3375
9	0.3249
10	0.3146

%*PV* = 100 [*PV*/*TV*]

= 100 [_____/_____]

= _____ %

Total Variation (*TV*)

$TV = \sqrt{GRR^2 + PV^2}$

$= \sqrt{(____^2 + ____^2)}$

= _____

ndc = $1.41(PV/GRR)$

= 1.41(_____/_____)

= _____

ndc	CONCLUSION
1	Measurement system cannot discriminate between parts
2	Measurement system can only distinguish between high/low or big/small
3 – 4	Measurement system is of little or no value
5+	According to AIAG, the measurement system can acceptably discriminate parts

The Number of Distinct Categories (**ndc**) is the number of categories of parts that your measurement system can distinguish

If it is below five, it is likely not able to distinguish between parts

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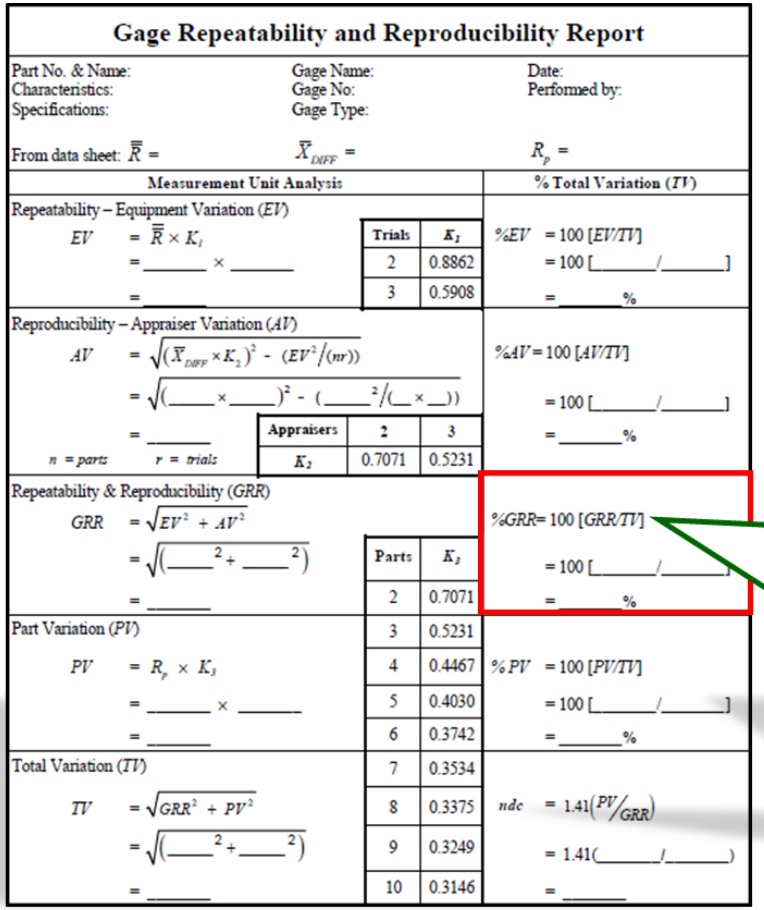
Gage Repeatability & Reproducibility



Gage Repeatability and Reproducibility Report

Part No. & Name: | Gage Name: | Date:
Characteristics: | Gage No: | Performed by:
Specifications: | Gage Type:

From data sheet: $\overline{\overline{R}}$ = | $\overline{X}_{DIFF}$ = | R_p =

Measurement Unit Analysis | **% Total Variation (*TV*)**

Repeatability – Equipment Variation (*EV*)

$EV = \overline{\overline{R}} \times K_1$
$= ____ \times ____$
$= ____$

Trials	K_1
2	0.8862
3	0.5908

$\%EV = 100\,[EV/TV]$
$= 100\,[____/____]$
$= ____\%$

Reproducibility – Appraiser Variation (*AV*)

$AV = \sqrt{(\overline{X}_{DIFF} \times K_2)^2 - (EV^2/(nr))}$
$= \sqrt{(____ \times ____)^2 - (____^2/(__ \times __))}$
$= ____$

n = parts *r* = trials

Appraisers	2	3
K_2	0.7071	0.5231

$\%AV = 100\,[AV/TV]$
$= 100\,[____/____]$
$= ____\%$

Repeatability & Reproducibility (*GRR*)

$GRR = \sqrt{EV^2 + AV^2}$
$= \sqrt{(____^2 + ____^2)}$
$= ____$

$\%GRR = 100\,[GRR/TV]$
$= 100\,[____/____]$
$= ____\%$

Parts	K_3
2	0.7071
3	0.5231
4	0.4467
5	0.4030
6	0.3742
7	0.3534
8	0.3375
9	0.3249
10	0.3146

Part Variation (*PV*)

$PV = R_p \times K_3$
$= ____ \times ____$
$= ____$

$\%PV = 100\,[PV/TV]$
$= 100\,[____/____]$
$= ____\%$

Total Variation (*TV*)

$TV = \sqrt{GRR^2 + PV^2}$
$= \sqrt{(____^2 + ____^2)}$
$= ____$

$ndc = 1.41(PV/GRR)$
$= 1.41(____/____)$
$= ____$

Gage R&R

The % of variation that R&R contributes to the Total Variation (TV) in the measurement system

% Total or % Study Variance	Measurement System Status
10% or less	Ideal
10% - 20%	Acceptable
20% - 30%	Marginal
30% or greater	Poor

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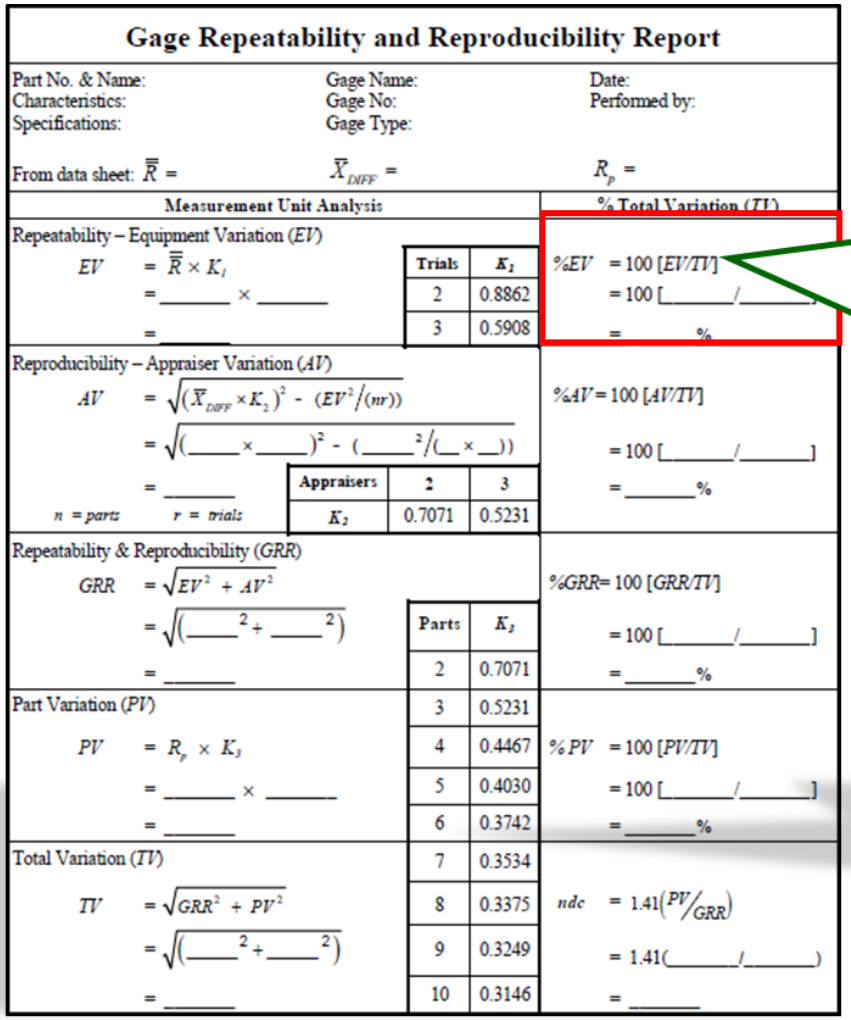
Repeatability: Equipment Variation (EV)



Gage Repeatability and Reproducibility Report

Part No. & Name: | Gage Name: | Date:
Characteristics: | Gage No: | Performed by:
Specifications: | Gage Type:

From data sheet: $\bar{\bar{R}}$ = $\bar{X}_{DIFF}$ = R_p =

Measurement Unit Analysis | **% Total Variation (TV)**

Repeatability – Equipment Variation (*EV*)

$EV = \bar{\bar{R}} \times K_1$
= _____ × _____
= _____

Trials	K_1
2	0.8862
3	0.5908

$\%EV = 100\,[EV/TV]$
= 100 [_____/_____]
= _____ %

Reproducibility – Appraiser Variation (*AV*)

$AV = \sqrt{(\bar{X}_{DIFF} \times K_2)^2 - (EV^2/(nr))}$
= $\sqrt{(____ \times ____)^2 - (____^2/(__ \times __))}$
= _____

n = parts *r* = trials

Appraisers	2	3
K_2	0.7071	0.5231

$\%AV = 100\,[AV/TV]$
= 100 [_____/_____]
= _____ %

Repeatability & Reproducibility (*GRR*)

$GRR = \sqrt{EV^2 + AV^2}$
= $\sqrt{(____^2 + ____^2)}$
= _____

$\%GRR = 100\,[GRR/TV]$
= 100 [_____/_____]
= _____ %

Parts	K_3
2	0.7071
3	0.5231
4	0.4467
5	0.4030
6	0.3742
7	0.3534
8	0.3375
9	0.3249
10	0.3146

Part Variation (*PV*)

$PV = R_p \times K_3$
= _____ × _____
= _____

$\%PV = 100\,[PV/TV]$
= 100 [_____/_____]
= _____ %

Total Variation (*TV*)

$TV = \sqrt{GRR^2 + PV^2}$
= $\sqrt{(____^2 + ____^2)}$
= _____

$ndc = 1.41\left(PV/GRR\right)$
= 1.41(_____/_____)
= _____

Equipment Variation (EV)

Repeatability Error

The % of Total Variation (TV) among all of the measurements that is attributable to the measurement equipment

%EV = 100 (EV/TV)

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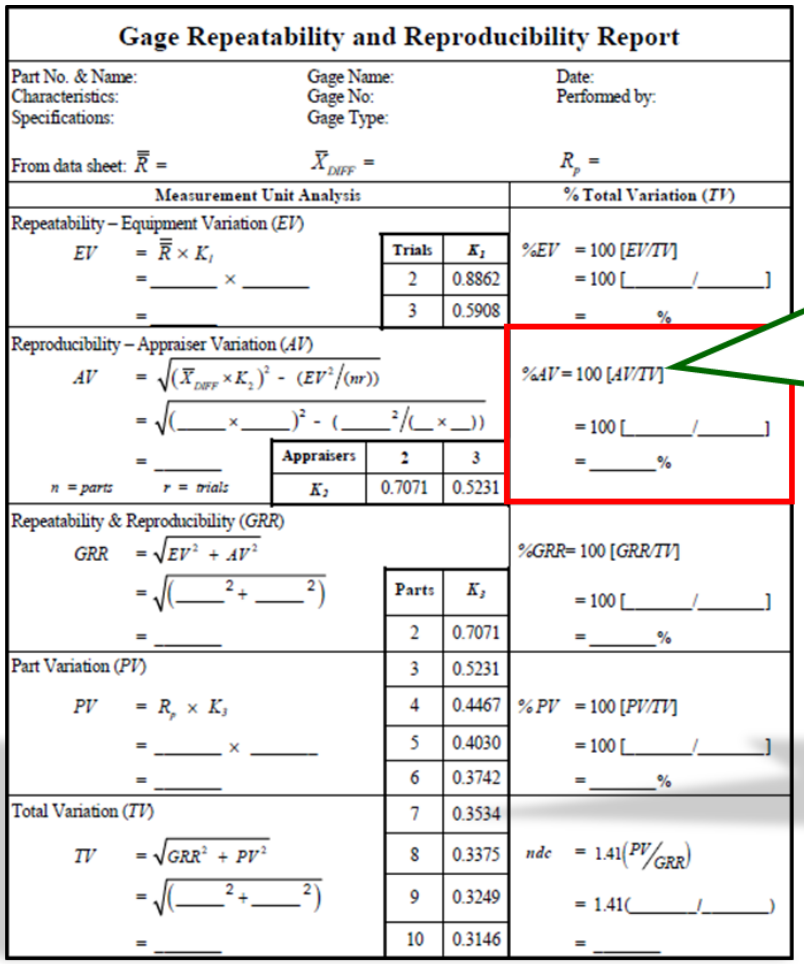
Reproducibility: Appraiser Variation (AV)



Gage Repeatability and Reproducibility Report

Part No. & Name: | Gage Name: | Date:
Characteristics: | Gage No: | Performed by:
Specifications: | Gage Type:

From data sheet: $\bar{\bar{R}}$ = $\bar{X}_{DIFF}$ = R_p =

Measurement Unit Analysis | **% Total Variation (TV)**

Repeatability – Equipment Variation (*EV*)

$EV = \bar{\bar{R}} \times K_1$

= _____ × _____

= _____

Trials	K_1
2	0.8862
3	0.5908

$\%EV = 100\ [EV/TV]$

= 100 [_____/_____]

= _____ %

Reproducibility – Appraiser Variation (*AV*)

$AV = \sqrt{(\bar{X}_{DIFF} \times K_2)^2 - (EV^2/(nr))}$

= $\sqrt{(____ \times ____)^2 - (____^2/(__ \times __))}$

= _____

n = parts *r* = trials

Appraisers	2	3
K_2	0.7071	0.5231

$\%AV = 100\ [AV/TV]$

= 100 [_____/_____]

= _____ %

Repeatability & Reproducibility (*GRR*)

$GRR = \sqrt{EV^2 + AV^2}$

= $\sqrt{(____^2 + ____^2)}$

= _____

$\%GRR = 100\ [GRR/TV]$

= 100 [_____/_____]

= _____ %

Parts	K_3
2	0.7071
3	0.5231
4	0.4467
5	0.4030
6	0.3742
7	0.3534
8	0.3375
9	0.3249
10	0.3146

Part Variation (*PV*)

$PV = R_p \times K_3$

= _____ × _____

= _____

$\%PV = 100\ [PV/TV]$

= 100 [_____/_____]

= _____ %

Total Variation (*TV*)

$TV = \sqrt{GRR^2 + PV^2}$

= $\sqrt{(____^2 + ____^2)}$

= _____

$ndc = 1.41\left(PV/GRR\right)$

= 1.41(_____/_____)

= _____

Appraiser Variation (AV)

Reproducibility Error

The % of Total Variation (TV) among all of the measurements that is attributable to the appraiser(s)

%AV = 100 (AV/TV)

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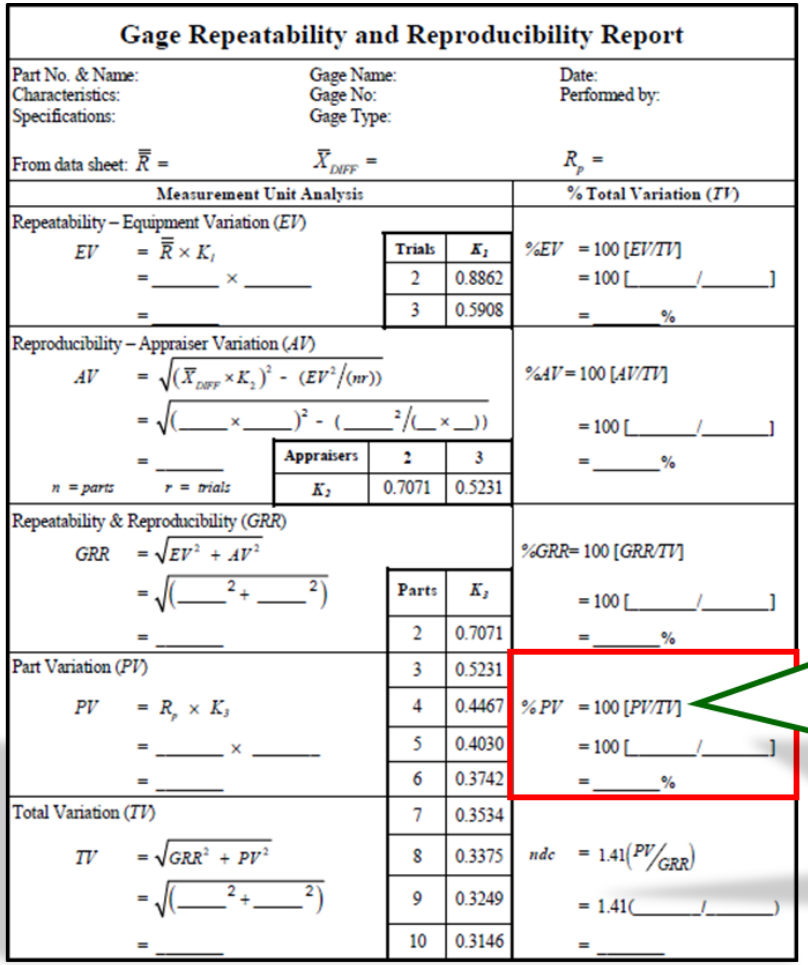
Part Variation (PV)



Gage Repeatability and Reproducibility Report

Part No. & Name: | Gage Name: | Date:
Characteristics: | Gage No: | Performed by:
Specifications: | Gage Type:

From data sheet: $\bar{\bar{R}}$ = $\bar{X}_{DIFF}$ = R_p =

Measurement Unit Analysis | **% Total Variation (TV)**

Repeatability – Equipment Variation (*EV*)

$EV = \bar{\bar{R}} \times K_1$

= _____ × _____

= _____

Trials	K_1
2	0.8862
3	0.5908

$\%EV = 100\ [EV/TV]$

= 100 [_____/_____]

= _____ %

Reproducibility – Appraiser Variation (*AV*)

$AV = \sqrt{(\bar{X}_{DIFF} \times K_2)^2 - (EV^2/(nr))}$

= $\sqrt{(____ \times ____)^2 - (____^2/(__ \times __))}$

= _____

n = *parts* *r* = *trials*

Appraisers	2	3
K_2	0.7071	0.5231

$\%AV = 100\ [AV/TV]$

= 100 [_____/_____]

= _____ %

Repeatability & Reproducibility (*GRR*)

$GRR = \sqrt{EV^2 + AV^2}$

= $\sqrt{(____^2 + ____^2)}$

= _____

$\%GRR = 100\ [GRR/TV]$

= 100 [_____/_____]

= _____ %

Parts	K_3
2	0.7071
3	0.5231
4	0.4467
5	0.4030
6	0.3742
7	0.3534
8	0.3375
9	0.3249
10	0.3146

Part Variation (*PV*)

$PV = R_p \times K_3$

= _____ × _____

= _____

$\%PV = 100\ [PV/TV]$

= 100 [_____/_____]

= _____ %

Total Variation (*TV*)

$TV = \sqrt{GRR^2 + PV^2}$

= $\sqrt{(____^2 + ____^2)}$

= _____

$ndc = 1.41\left(PV/GRR\right)$

= 1.41(_____/_____)

= _____

Part Variation (PV)

The % of Total Variation (TV) among all of the measurements that is attributable to the parts

Combined with GRR becomes the Measurement System's Total Variation

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R&R Problem Resolution

- Repeatability Problems / Equipment Variation:
  - ➢Calibrate or replace gage
  - ➢If only occurring with one operator, re-train
- Reproducibility Problems / Appraiser Variation:
  - ➢Training and skill level of the operators
  - ➢Ensure that standard procedures are followed

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Attribute MSA

- Assess Attribute Measurement Systems
- Attribute data are usually the result of human judgment (Pass / Fail)
  - For example: a Quality Team that monitors customer service representatives (CSR) in a call center and marks calls as Pass / Fail
- Situations where a continuous measure cannot be obtained

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Attribute MSA Purpose

- Use of the same criteria to determine “Pass” vs. “Fail”
- Inspection standards vs. customers’ requirements
- Inspector consistency
- Inspectors vs. a “known master”
- Discover areas where:
  - Training is required
  - Procedures must be developed
  - Standards are not available

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2.3.4 STABILITY

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What is Process Stability?

- A process is said to be stable when:
  - Within control limits
  - Consistent over time
  - Predictable (at least between some limits)
  - Variation is random and only due to chance or common causes
  - No trends, patterns, or outliers
- Control charts are used to analyze the stability of a process
  - Discussed in detail later, under Control

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Process Stability vs. Process Capability

- Stability is a prerequisite to determining process capability
  - ➢Stability ≠ Capability
- Process capability indicates how well a process meets customers' specifications

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2.4 PROCESS CAPABILITY

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Short-Term vs. Long-Term Data

SHORT-TERM (ST) DATA	LONG-TERM (LT) DATA
• Gathered and assessed in the near-term • Sample size (n) is typically smaller (n≤30) • May not align with the population parameters • Changes sample to sample	• Typically cumulative or aggregated over time • Sample size (n) is typically larger (n>30) • Calculated values are more stable • Reflective of the population

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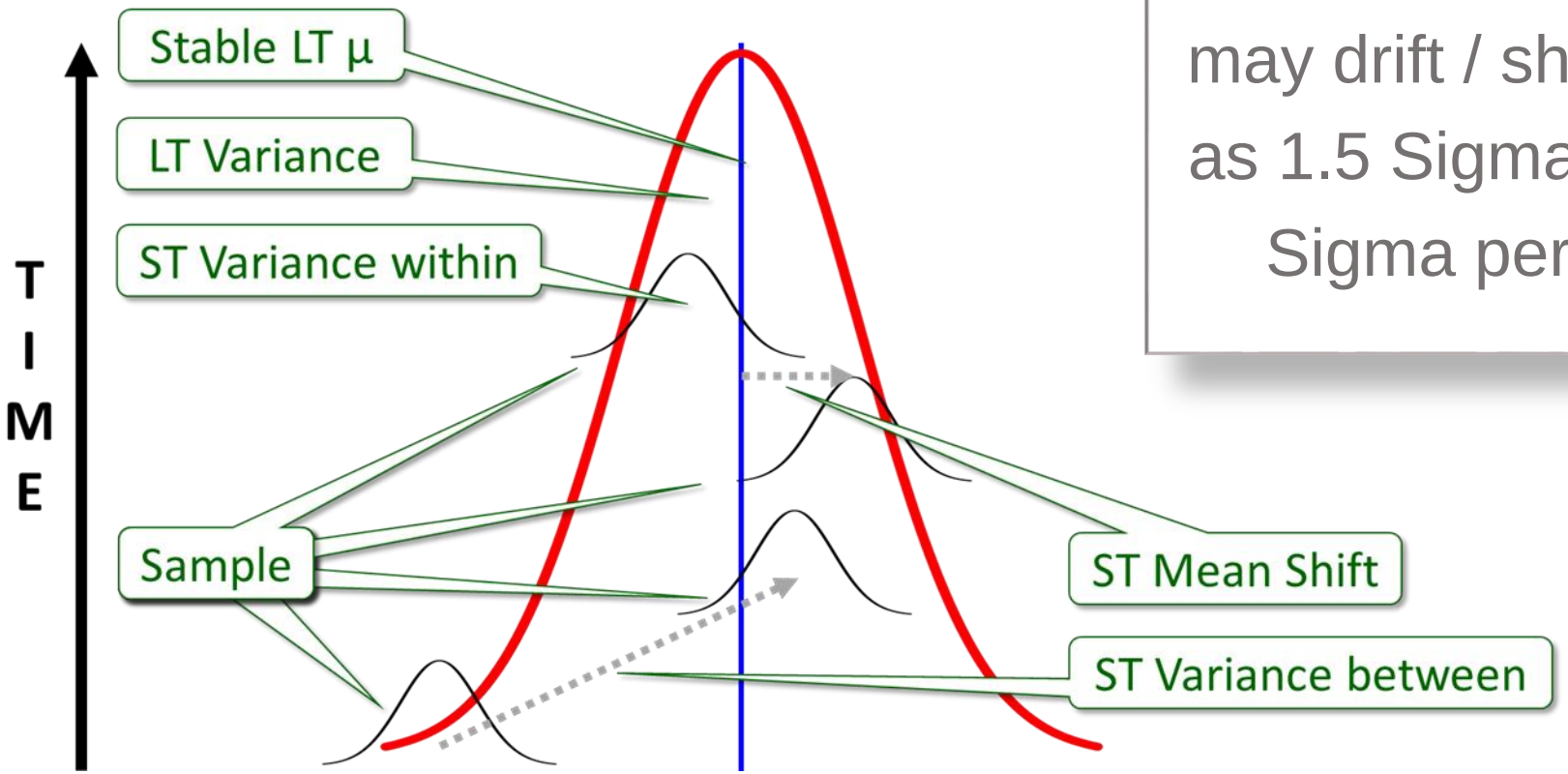
Sigma Shift



In the LT, processes perform at 4.5 Sigma, yet even stable processes may drift / shift over time by as much as 1.5 Sigma; resulting in a 4.5 + 1.5 Sigma performance of **6 Sigma**

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2.4.1 CAPABILITY ANALYSIS

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What is Process Capability?

- Inherent ability of a process to meet the customer requirements without any additional efforts
- Provides insight:
  - ➢Centering and / or Variation Issues
  - ➢Inappropriate specification limits
- Various indices are the measuring tools, including:
  - ➢For Normally Distributed Continuous Data: C_p , C_{pk} , P_p , P_{pk}
  - ➢For Discrete Data: DPMO, DPU

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Index: C_p

- C_p stands for **capability** of the process
- USL and LSL are the upper and lower specification limits
- C_p measures the process' **potential** capability
- High C_p indicates the small spread of the process within the SL's

$$C_p = \frac{USL - LSL}{6 \times \sigma_{within}}$$

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Index: C_{pk}

- C_{pk} measures the process' actual capability
  - Uses both the variation and average of the process (μ)
- Gives insight into how aligned the process is with the customer requirement

- Cpk for **lower** specification limit:

$$C_{pl} = \frac{\mu - LSL}{3 \times \sigma_{within}}$$

$$C_{pk} = \min(C_{pl}, C_{pu})$$

- Cpk for **upper** specification limit:

$$C_{pu} = \frac{USL - \mu}{3 \times \sigma_{within}}$$

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Index: P_p & P_{pk}

- Reference the **performance** of the process
- While C_p & C_{pk} consider the within-subgroup standard deviation, the P_p & P_{pk} indices consider the total standard deviation from the sample data

$$P_p = \frac{USL - LSL}{6 \times \sigma_{overall}}$$

$$P_{pk} = \min\left(\frac{USL - \mu}{3 \times \sigma_{overall}}, \frac{\mu - LSL}{3 \times \sigma_{overall}} \right)$$

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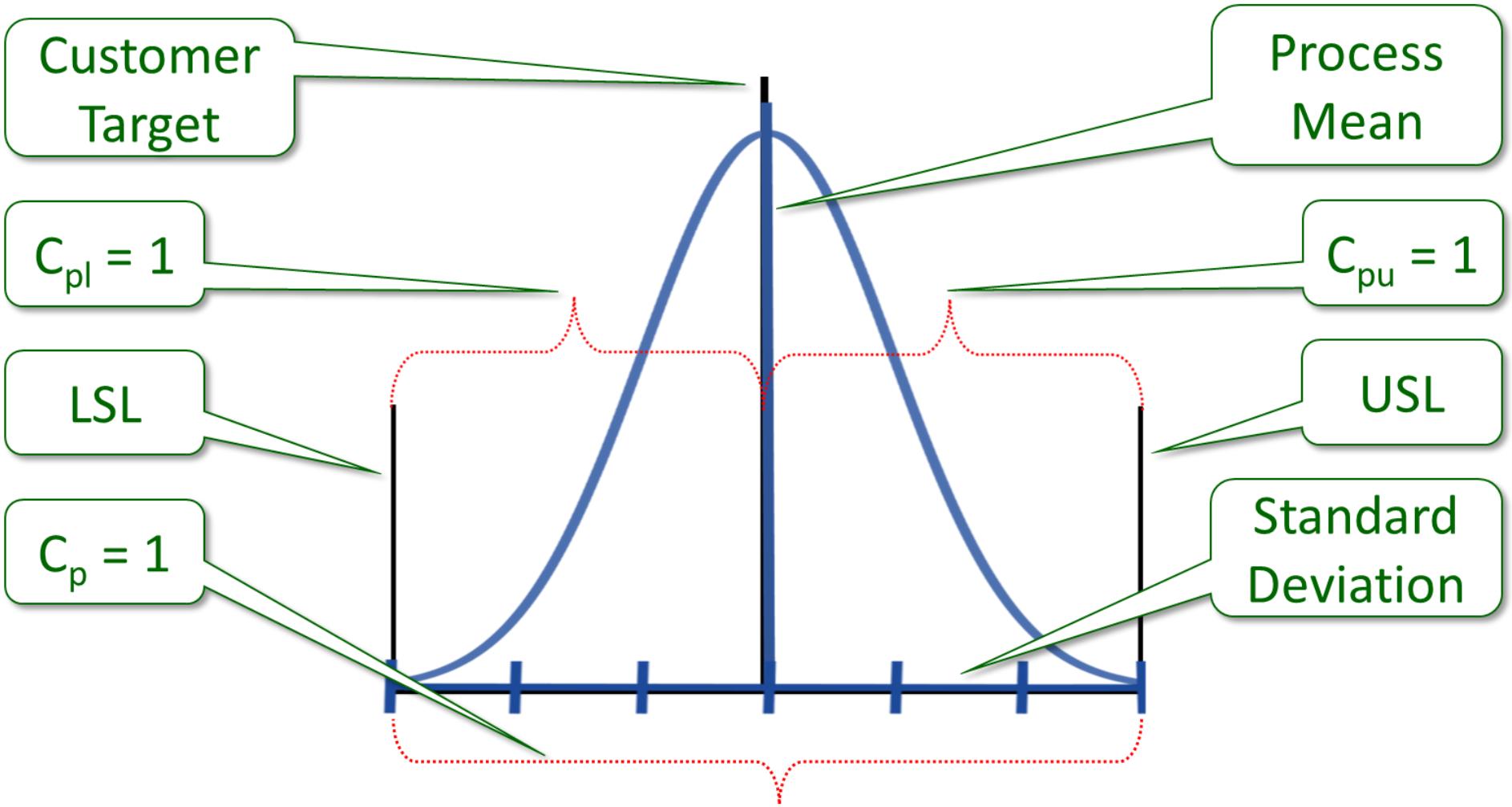
Visual: Process Capability



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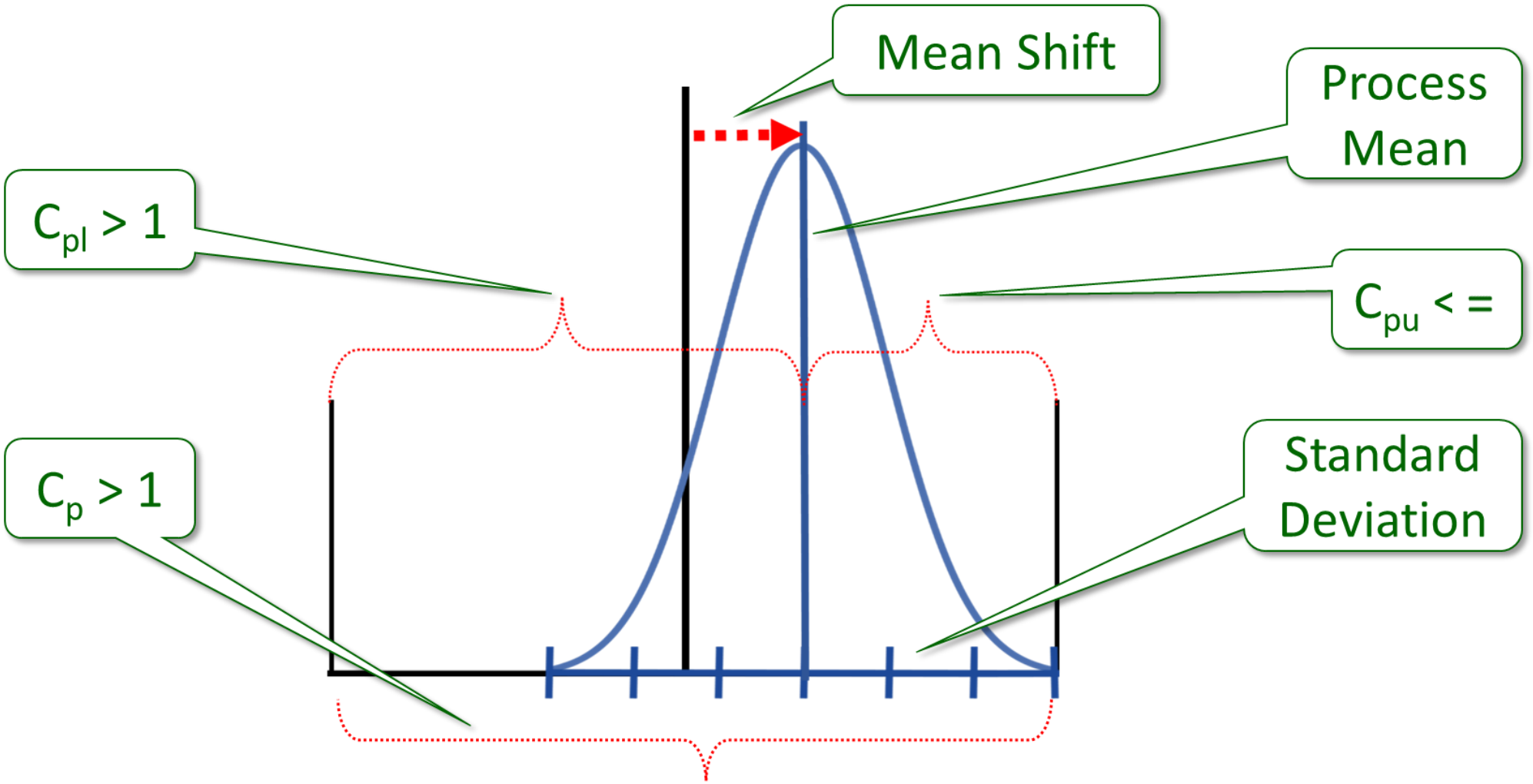
Visual: Process Capability



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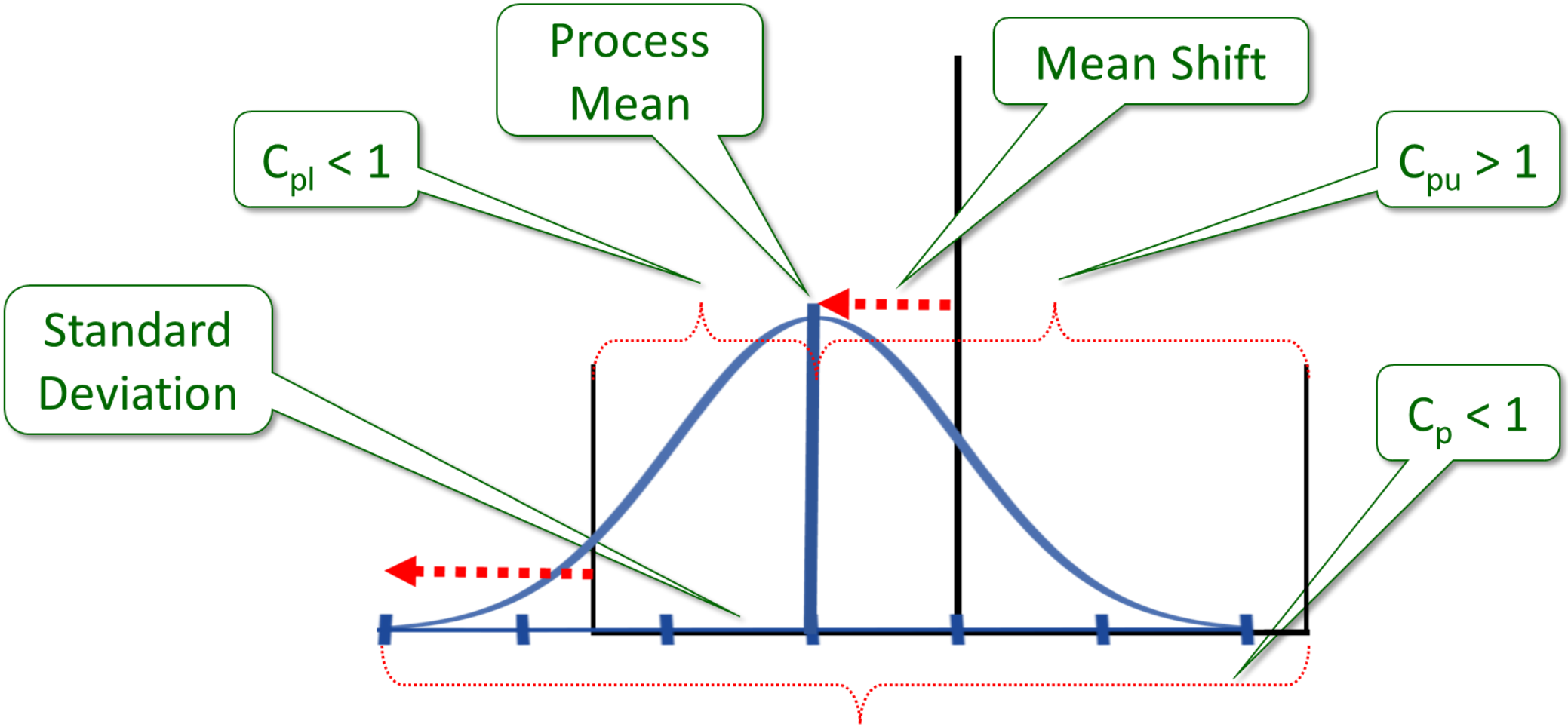
Visual: Process Capability



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2.4.2 Attribute & Discrete Capability

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Attribute & Discrete Capability

- For attribute or discrete data, process capability is determined by the mean rate of DPMO
- Mean and standard deviation have to be defined

For **Defectives**:

- $\overline{np}$ represents the mean when the sample size is constant
- $\overline{p}$ represents the mean when the sample size is variable
- Standard deviation: $\sqrt{\bar{p}\,(1-\bar{p})/n}$

For **Defects**:

- $\overline{c}$ represents the mean when the sample size is constant
- $\overline{u}$ represents the mean when the sample size is variable
- Standard deviation: $\sqrt{\frac{\bar{u}}{a}}$ and $\sqrt{\bar{c}}$

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Capability: Binomial vs. Poisson

BINOMIAL

Using count of **defectives** in each sample set to assess the process performance of meeting the customer specifications, use "**%Defective**" as the process capability index

$$\%\text{Defective} = \frac{N_{defectives}}{N_{overall}}$$

POISSON

Using count of **defects** in each sample set to assess the process performance of meeting the customer specifications, use Mean **DPU** as the process capability index

$$DPU = \frac{N_{defects}}{N_{overall}}$$

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Capability & Monitoring

- In the **Measure** phase:

  - Process stability and capability analysis are used to baseline the performance of current process

- In the **Control** phase:

  - Process stability and capability analysis are combined to monitor whether the improved process is maintained consistently as expected

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THANK YOU

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3.0 Analyze

The “Analyze” phase is about:

- ✓ Finding the top 1-3 causes of the problem being addressed
  - Phase starts with team input regarding potential causes
  - Uses statistical methods to isolate the top 1 to 3 causes

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Analyze Phase

Patterns of Variation

- Multi-Vari Analysis
- Classes of Distributions

Inferential Statistics

- Understanding Inference
- Sampling Techniques & Uses

Hypothesis Testing

- General Concepts & Goals of Hypothesis Testing
- Significance: Practical vs. Statistical
- Risk: Alpha & Beta
- Hypothesis Testing: Types & Decisions
- Reading Tables

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Analyze Phase

Hypothesis Testing, cont.

- Testing for Normality
- Hypothesis Testing with Normal Data
- Tests of Equal Variance
- Z Tests

- 1 Sample t-test
- 2 Sample t-test
- 1 Sample Variance: Chi-Square
- One-Way ANOVA

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Analyze Phase

Hypothesis Testing with Non-Normal Data

- Mann-Whitney
- Kruskal-Wallis
- Mood's Median
- Friedman
- 1 Sample Sign
- 1 Sample Wilcoxon

- One and Two Sample Proportion
- Chi-Squared (Contingency Tables)

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3.1 Patterns of Variation
3.1.1 Multi-Vari Analysis

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Multi-Vari Analysis

- As a tool, Multi-Vari graphically displays patterns of variation
- Families of variation can exist within a subgroup, between subgroups, or over time
- Variation source classifications:
  - **Positional**
  - **Cyclical**
  - **Temporal**

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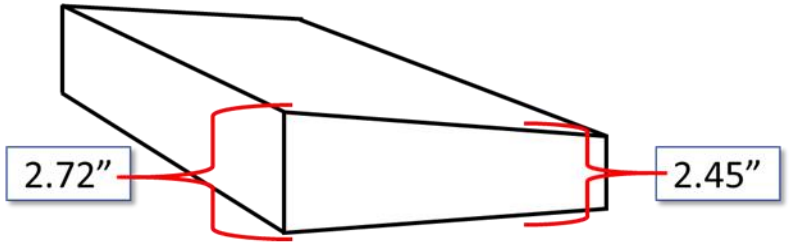
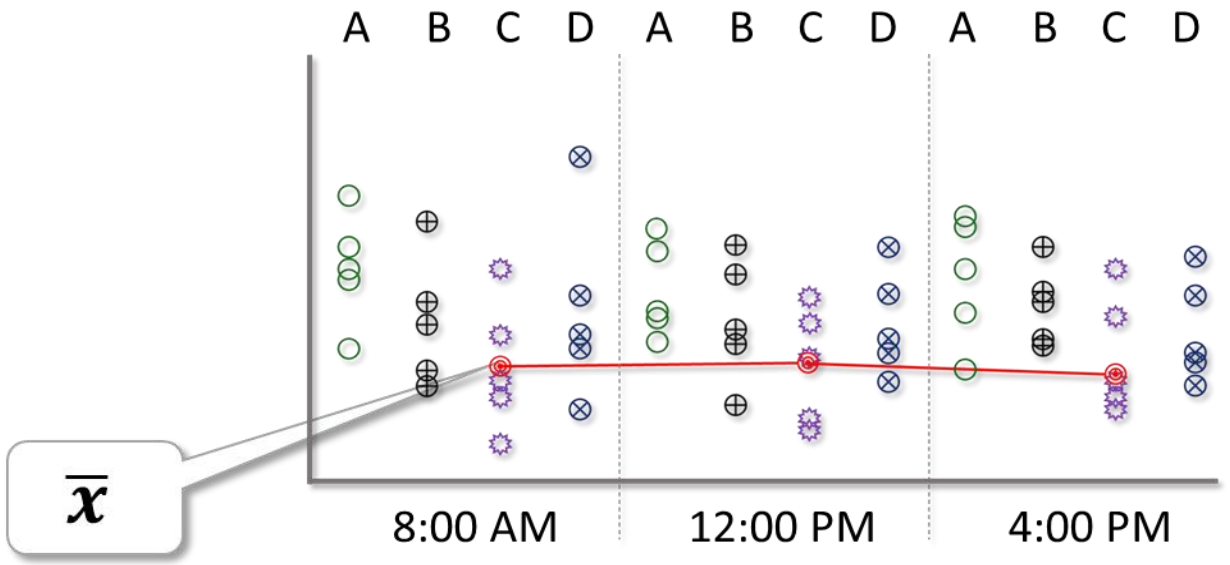
Sources of Variation

Positional: Within Unit

- Within piece
  - ➢Related to part geometry
- Single unit
  - ➢Containing many individual parts
- Location in a batch process
  - ➢Such as plating





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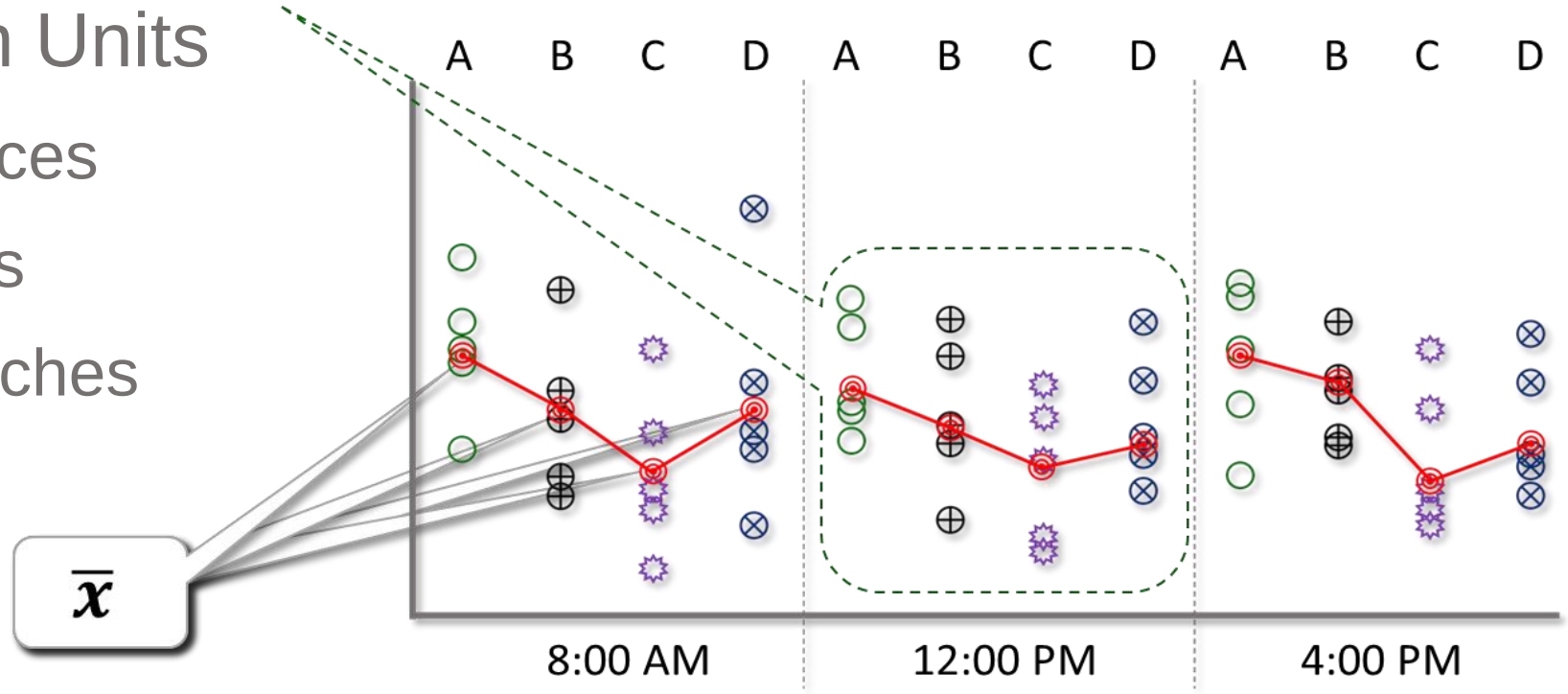
Sources of Variation

Cyclical: Between Units

- Consecutive pieces
- Groups of pieces
- Consecutive batches



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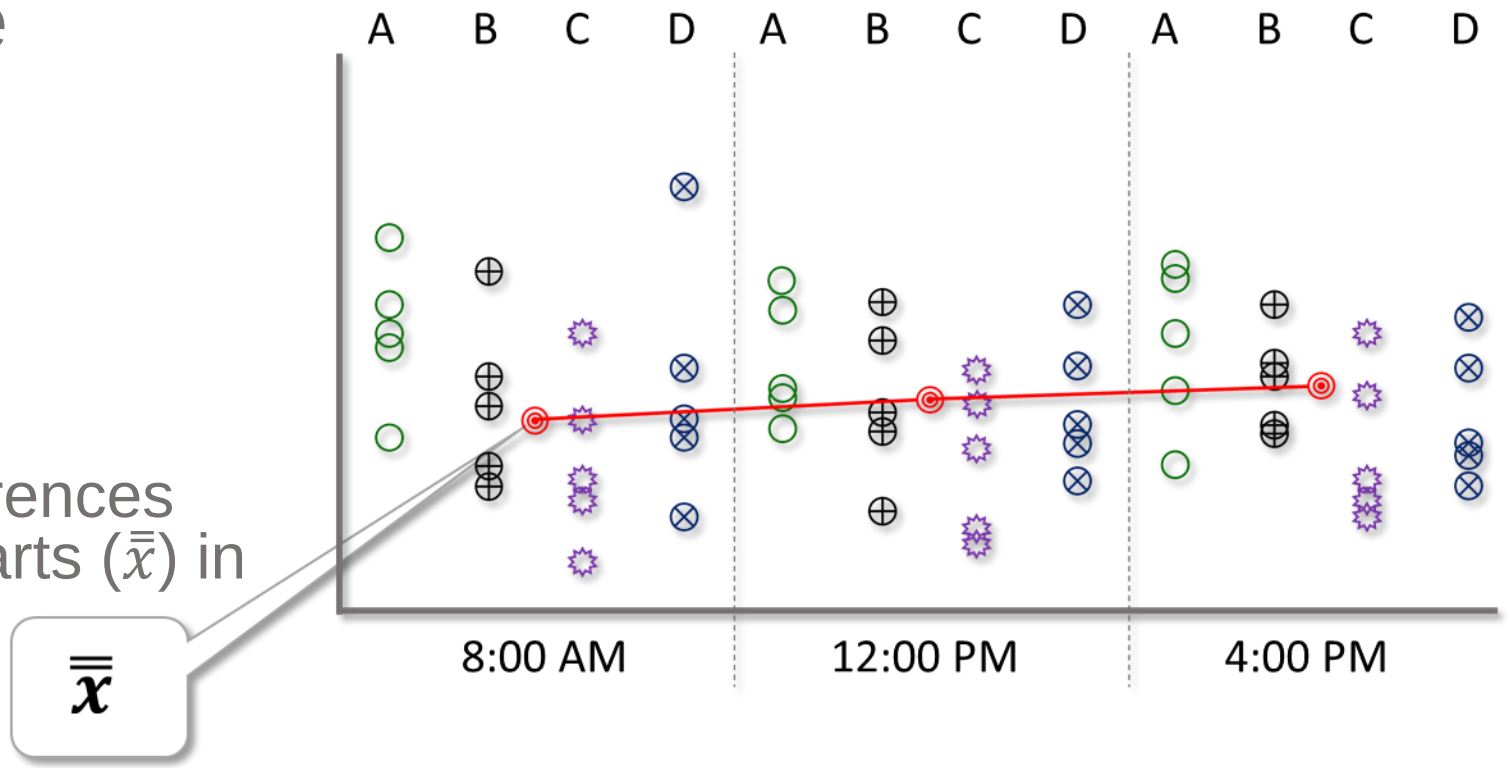
Sources of Variation

Temporal: Over time

- Shift-to-Shift
- Day-to-Day
- Week-to-Week

  - Note: Image references the mean of all parts ($\bar{\bar{x}}$) in a time period



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3.1.2 Classes of Distributions

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Distributions

Most practitioners generally stay within these following distributions:

- Binomial
- Poisson
- Student’s t
- Normal
- F distribution
- Chi-square

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Distributions: Binomial & Poisson

Binomial Distribution

- The binomial distribution is used to model discrete data
- It applies when the population is large (N > 50) and the
- Sample size (n) is small compared to the population
- It is appropriate to use when proportion **defective** is equal to or greater than 0.1
- Excel function: **BINOM.DIST**

Poisson Distribution

- The Poisson is used as a distribution for **defect** counts and can be used as an approximation to the binomial
- The Poisson distribution is closely related to the exponential distribution
- Excel function: **POISSON.DIST**

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Distribution: Normal & Student's T

Normal Distribution

- Total area under the curve is equal to 1 (100%)
- Shaped like a bell "curve"
- Extends indefinitely in both directions and is asymptotic
- Mean, Median, and Mode are the same
- All processes will exhibit a normal curve shape if only pure random variation exists

Student's t Distribution

- Same general shape as the Normal distribution
- Small sample size usually <30
- The t distribution is formed by combining a standard normal random variable and a chi-square random variable
- Population variance is unknown

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Distributions: F & Chi-Square

F Distribution

- Used extensively to test for **equality of variances** from two normal populations
- The distribution of findings serves as a precursor for the two-sample t-test
- Very sensitive to non-normality

Chi-square Distribution

- Used for multiple hypothesis tests, such as:
  - ➢ Test the equality between the variance of a normally distributed population and a specified value
  - Contingency tables for:
    - ➢ Test equality of proportions, using 1 factor
    - ➢ Evaluating the independence of 2 factors

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3.2 Inferential Statistics
3.2.1 UNDERSTANDING INFERENCE

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Inference

- in·fer·ence (n.) “The act or process of **deriving logical conclusions** from premises known or assumed to be true. The act of reasoning from factual knowledge or evidence.” (Dictionary.com)

- **Inferential Statistics**: To draw inferences, or conclusions, about a process or population being studied by leveraging the findings and validation of samples analysis

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Inferential Statistics

- Statistical inference contains two elements:
  - The Inference (“claim”)
  - The Measure of Validity (“confidence”)

- Steps involved in statistical inference are:
  - Clearly **define** the objective
  - Decide if it will be evaluated by a **one or two tail test**
  - Formulate a **null** and an **alternate hypothesis**
  - Select a test **distribution** and a **critical value** of the test statistic
  - **Calculate a test statistic** from the sample
  - **Compare** the calculated value to the critical value and determine if the null hypothesis is to be **rejected or not**

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Outcome of Statistical Inference

Conclusion of statistical inference is a statistical claim about the population

- Examples of statistical propositions:
  - Estimation of a population parameter
  - Identification of an interval or range where the true population parameter would fall with some degree of certainty
  - Decide whether to reject a hypothesis about the population
  - Make predictions

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3.2.2 Sampling Techniques & Uses

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Statistical Notation

SIGN / SYMBOL	DESCRIPTION	SIGN / SYMBOL	DESCRIPTION	SIGN / SYMBOL	DESCRIPTION
x	An individual value, an observation	s	The Standard Deviation of sample data	Σ	Summation
x_i	For each, all, individual values	σ	The Standard Deviation of population data	α	Alpha; risk of a Type 1 error
x_1	A specific (1st) individual value	s^2, V	The variance of sample data	β	Beta; risk of a Type 2 error
x-bar, $\bar{x}$	The Mean, average of sample data	σ^2, V	The variance of population data	k	Multi-purpose notation, i.e. # of subgroups, # of classes
x-double-bar, $\bar{\bar{x}}$	The grand Mean, grand average	R	The range of data	p	A proportion of sample data
μ	The Mean of population data	R-bar, $\bar{R}$	The average range of data	P	A proportion of population data
n	Sample size	N	Population (size)	λ	Lambda

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Population (N)

In statistics, Population is the entire (100%) set of objects or observations in which one is interested in drawing conclusions about or making generalizations based upon representative sample data

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What is Sampling?

- Sampling is the process of creating a sample, by selecting objects or observations from a population of interest

- Used extensively to collect information about a population

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Advantages of Sampling

- Usually **impractical** or **impossible** to collect the data of an entire population:
  - ➢Dynamic nature of the population
  - ➢Unavailability of records
  - ➢Time consuming
  - ➢High Cost

- **Advantages** of sampling a representative subset of the population:
  - ➢Faster data collection
  - ➢Easier to work with
  - ➢Lower cost

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Uses of Sample

Two primary uses of samples in making statistical inference:

- **Estimation**: estimating the population parameters using the sample statistics
- **Hypothesis testing**: testing a statement about the population characteristics using sample data

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Basic Sampling Strategies

- Random Sampling
  - Simple
  - Matched
- Stratified Sampling
- Systematic Sampling
- Proportional to Size Sampling
- Cluster Sampling

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Simple Random Sampling

Simple random samples are selected in such a way that each item in the population has an equal chance of being selected

- Basic sampling strategy
- No bias
- Minimizes the variation between samples and population

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Matched Random Sampling

Matched random samples are samples randomly selected in pairs, each of which has the same attribute

Examples:

- Height of twins
- Patients' blood pressure before and after medication

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Stratified Sampling

- Population can be grouped or “stratified” into distinct and independent categories or sub-populations
- Stratified samples are randomly selected in each category of the population
- Assumes advanced knowledge of population characteristics
  - Examples: Product mix, gender, region, income, or etc.
  - Specific: 60% of total apples come from Washington, thus 60% of the sample should consist of Washington apples

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Systematic Sampling

- Selected at regular intervals

- Random start and then every ‘ith’ item is selected going forward
  - Example: Sampling the every 5th unit produced on the production line

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Cluster Sampling

- Samples are only selected from certain clusters or groups of the population
- Reduces the time and cost spent on the sampling
- Increases risk that selected clusters are biased
  - Example: Selecting samples from a region where researchers are located so that the cost and time spent on travelling is reduced

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Sample Size

Sample size is a critical element that can influence the results of statistical inference

- Smaller the sample size = higher the risk that the sample statistic will not reflect the true population parameter

- Greater the sample size = more time and money spent on collecting the samples

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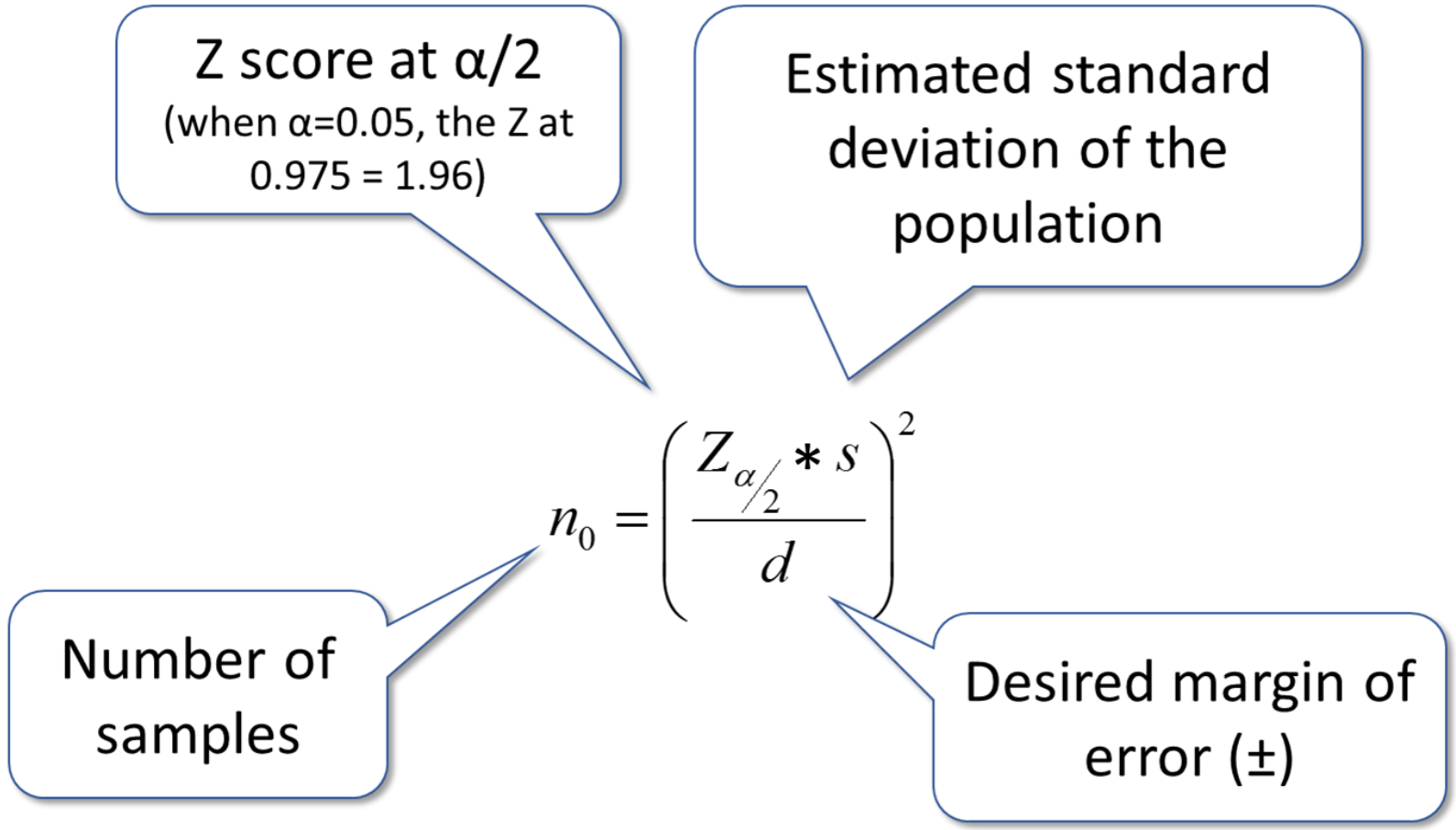
Sample Size: Continuous Data

- Equation for sample size of **continuous** data
- Capability Analysis – a minimum of **30** samples



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Sample Size: Continuous Data, cont.

- If calculated sample size exceeds 5% of the population, discount using the following equation

$$n = \frac{n_0}{\left(1 + \frac{n_0}{N}\right)}$$

$$n_0 = \left(\frac{Z_{\alpha/2} * s}{d}\right)^2$$

Population size

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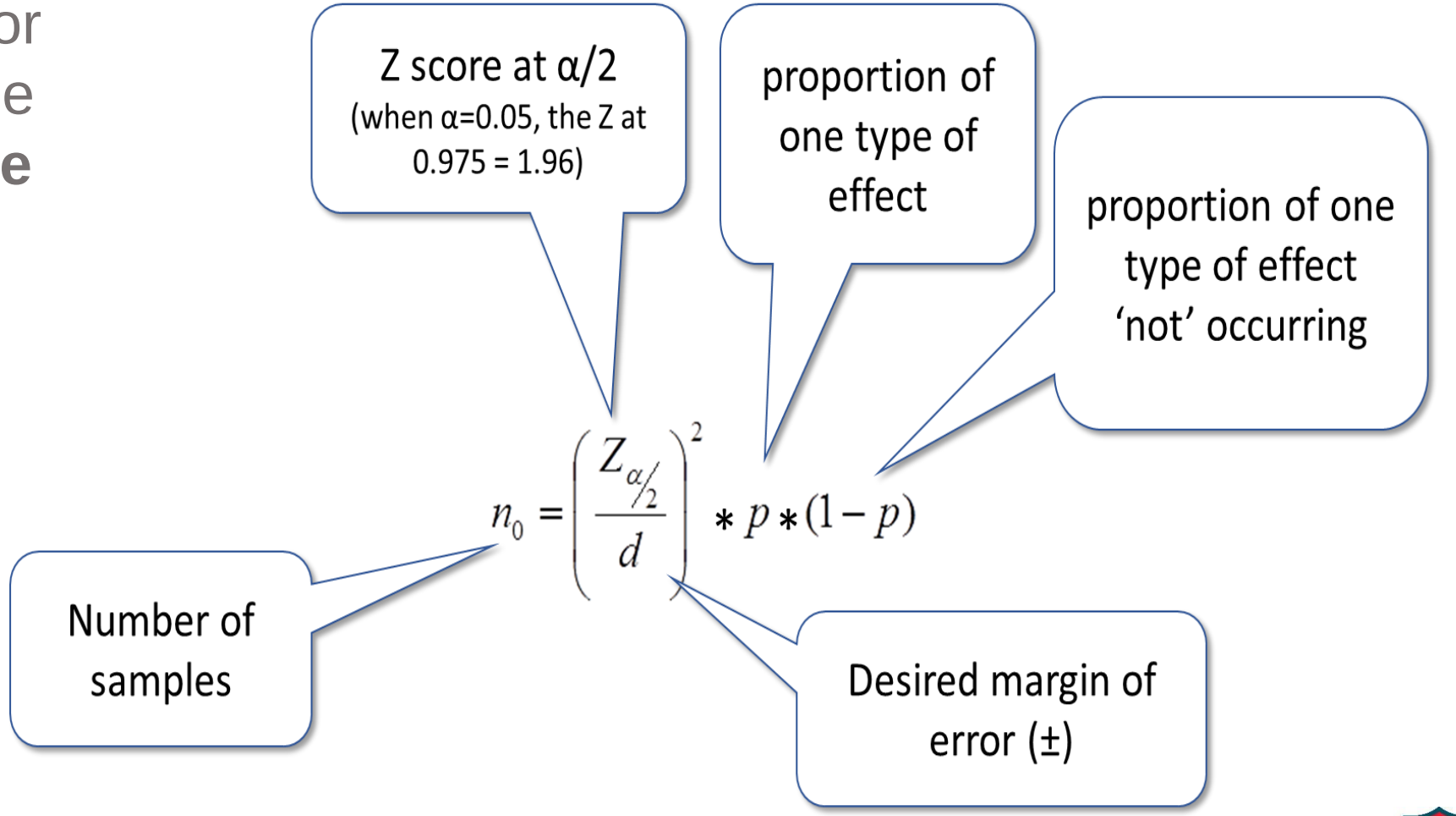
Sample Size: Discrete Data

- Equation for sample size of **discrete** data



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Sample Size: Discrete Data, cont.

- If calculated sample size exceeds 5% of the population, discount using the following equation

$$n_0 = \left(\frac{Z_{\alpha/2}}{d}\right)^2 * p * (1-p)$$

$$n = \frac{n_0}{\left(1 + \frac{n_0}{N}\right)}$$

Population size

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Sampling Errors

- Despite best efforts, some errors are anticipated:

- Random Sampling Error
  - ➢Random variation due to random selection
  - ➢Inherent in the sampling process
  - ➢Beyond one's control

- Selection Bias
  - ➢Non-random variation because of inadequate design of sampling
  - ➢Improve by adjusting the sampling size and strategy

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3.3 HYPOTHESIS TESTING
3.3.1 GENERAL CONCEPTS & GOALS OF HYPOTHESIS TESTING

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What is Hypothesis Testing?

- A hypothesis test is a statistical method in which:
  - A specific hypothesis (claim) is formulated about a population
  - It is tested for validity

- A statistical hypothesis is formulated in pairs:
  - Null Hypothesis "H_0"
  - Alternative Hypothesis "H_a"

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Hypothesis Testing: Why?

- **Hypothesis tests attempt to answer whether there is a:**
  - **Significant difference between groups**
  - **Significant difference from an expected value**
  - **Significant change occurring**

- Following are questions that may trigger a hypothesis test:
  - Are the average SAT scores of freshman entering school A and B the same?
  - Is the production rate of vendor A higher than that of vendor B?

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Null & Alternative Hypotheses

Following are examples of how hypotheses are approached and their general disposition:

Null Hypothesis (H_0):

- There is no difference in the measurement of different groups
- No change has occurred
- Variation is due to random causes
- Variables are independent

Alternative Hypothesis (H_a or H_1):

- There is a difference in the measurement of different groups
- Some change has occurred
- Variation is due to non-random causes
- Variables are not independent

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Hypothesis Testing: Conclusions

- There are two possible conclusions of hypothesis testing:
  - **Reject the null**
  - **Fail to reject the null**
- If there is significant evidence, based on the sample information to prove the alternative hypothesis, we reject the null
- If there is not significant evidence, or the sample information is not sufficiently persuasive, we fail to reject the null

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3.3.2 SIGNIFICANCE: PRACTICAL VS. STATISTICAL

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Practical Significance

- Observed differences are practically significant when they are large enough to make a practical difference
  - In some business situations, statistical differences can have little to no meaning because the difference is not large enough to be practical for a business to act upon

- At what point, or level of [statistical] significance, does it make sense (practical significance)?

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Statistical Significance

- **In statistics, an observed difference is statistically significant if it is unlikely that the difference occurred by pure (random or common) chance**
  - ➢ Statistical significance indicates that there are some non-random factors (special cause) causing the result to take place

- Statistical significance level in hypothesis testing indicates: The amount of evidence which is sufficiently persuasive to prove that a difference between groups exists **not** due to random chance alone
  - ➢ The amount of gathered evidence (analyzed sample) is compared to given a predetermined probability threshold (statistical significance level)

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Probability

- Probability refers to the likelihood, chance, or possibility of an occurrence
- Probability of any event (E) lies between 0 and 1
  - 0% < P(E) < 100%
  - Sum of the probabilities of all possible events (E) in a sample space = 1 = 100%
- Probability (P) is always a ratio
  - Ratio of the chances of event success vs. the total number of ways an event can occur (success + failure)

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Probability: Dice Example

- Die has 6 sides = Sample space of 6
- Each face is unique, thus each time a die is rolled there is a 1 out of 6 chances that it will land on any specific number
  - Ways to state "1 out of 6" = 1:6 = 1/6 = 0.1667 = 16.67%
- Probability of 5 one's in a row?
  - P(1 one) = 0.1667; P(5 ones) = $(0.1667)^5$ = 0.00013 (scientific notation: 1.3e-4)
  - Approximately 1.3 chances out of 10,000 that one could have gotten 5 ones (with a non-loaded die)
- Calculated probability allows the establishment of "acceptable" risk; thus "gambling"

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P-value

- Differences between observed data vs. claims made under H_0 may be real or due to chance
- **P-value**: Probability of these differences occurring solely due to chance (random)
- The alpha level (level of significance) of a test represents the threshold against which P-values are measured
  - ➢"alpha level" or "α" is addressed shortly
- H_0 is rejected if the p-value < α
  - ➢The most commonly used levels are 5%, 10%, and 1%

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3.3.3 Risk:Alpha & Beta

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Type I Error

- Type I error is also known as a false positive, false alarm, or alpha (α) error

- Type I error is associated with the risk of accepting false positives
  - ➢Believing there is a difference between groups but in fact there is none
  - ➢Example: telling a patient she is pregnant, when in fact she is not

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Alpha “α” & Confidence Interval (CI)

- α indicates the probability of making a type I error
  - Level of significance
  - Ranges from $0 < \alpha < 1$
- 5% (0.05) is the most commonly used “α” value
  - Other common values include 1% and 10%
- Confidence interval is the compliment to “α” (100% – α)
  - Used to calculate the confidence intervals (values at the bookends of the level of confidence)
  - The line(s) distinguishing the threshold between confidence and significance

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Hypothesis Decision Rules

- Hypothesis decision rules; Compare the p-value against α
  - ➢If p-value < α, reject the null
  - ➢If p-value > α, fail to reject the null

- To reduce the α risk, decrease the α value to which the p-value is compared

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Type II Error

- Type II error is also known as a false negative, oversight, or beta (β) error

- Type II error is associated with the risk of accepting false negatives
  - ➢Believe there is not any difference between groups but in fact there is
  - ➢Not rejecting the null when, in fact, it should be
  - ➢Example: telling a patient she is not pregnant, when in fact she is

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Beta “β”

- β indicates the probability of making a type II error
  - ➢It ranges from 0 < β < 1
- 10% is the most commonly used “β” value
- The compliment of is known as the power (100% – β)
  - ➢The probability of detecting a difference between groups when in fact the difference truly exists
- Increase the sample size to reduce the β risk
- Assuming other factors constant, β is inversely related to α

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Summary: Errors in Hypothesis Testing

	H_0 ACTUALITY	
HYPOTHESIS	**FALSE**	**TRUE**
Null Hypothesis: H_0 (Fail to reject the H_0)	• **False Negative** • **Type 2 Error** • Risk = **β** = Typically set at 10% (0.1) • "Consumer's Risk" • e.g. Not Convicted | Guilty	• Conclusion is correct • Have not rejected the H_0, when in fact, it is True • Risk = (1-α) • e.g. Not Convicted| Innocent
Alternate Hypothesis: H_A (Reject the H_0)	• Correct, reject H_0 when it is in fact false • Risk = (1- β) • e.g. Convicted | Guilty	• **False Positive** • **Type 1 Error** • Risk = **α** = Typically set at 5% (0.05) • "Producer's Risk" • e.g. Convicted | Innocent

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Confidence Interval

- Confidence interval is the range where the true population parameter would fall, within a certain level of confidence

  - Used to describe the reliability of a statistical inference
  - 95% is the most commonly used confidence level vs. 5% for α

- A 95% confidence interval indicates that:

  - The population parameter would fall in that region 95% of the time
  - We are 95% confident that the population parameter would fall in that region

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Confidence Interval: Influencers

The width of a confidence interval depends on:

- **Confidence level**
  - ➢The higher the confidence level, the wider the confidence interval
- **Sample size**
  - ➢The smaller the sample size, the wider the confidence interval
- **Variability in the data**
  - ➢The more variability, the wider the confidence interval

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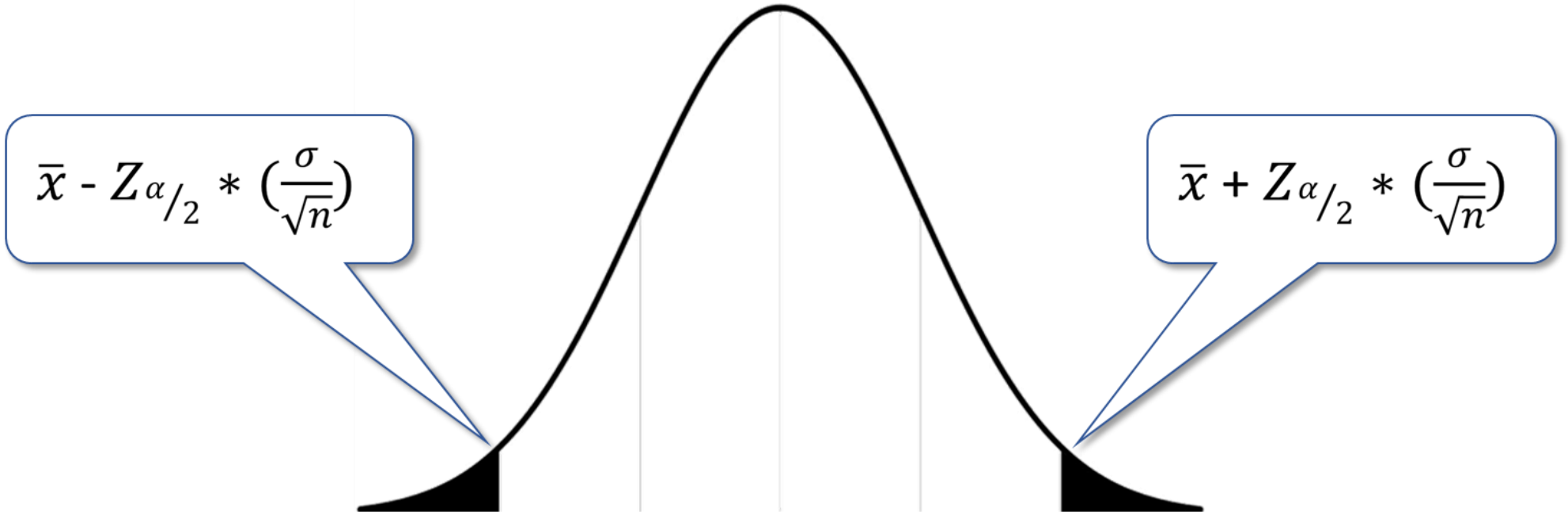
Confidence Interval: Calculation

- Confidence interval calculations are built into most software. The following is for reference
- Further analysis of this image is discussed shortly



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Comments About Sample Size

- Sample sizes are influenced by availability of units, cost, type of testing (e.g. destructive), and etc.
- One of the messages of the Central Limit theorem is that as sample size increases, standard error decreases
- Discussed later: Hypothesis testing and Statistical Process Control take sample sizes into account

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3.3.4 Hypothesis Testing: Types & Decisions

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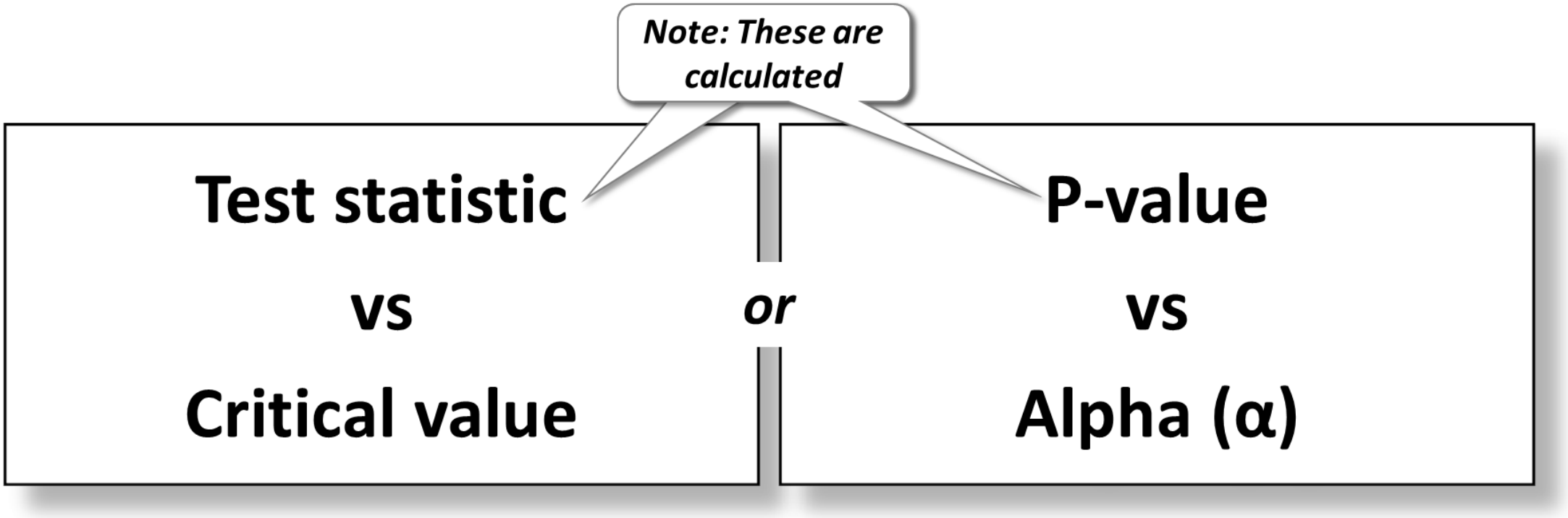
Important

As hypothesis tests are explored, it is important to keep the following as a reference. This information will be explained and expanded upon through the remainder of this module



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Hypothesis Testing: Test Statistic

Test Statistic

- In hypothesis testing, it is a value calculated using a function of the sample
  - ➢Different hypothesis tests have different formulas

- Test statistics are considered the sample data's numerical summary that can be used in hypothesis testing

- Also denoted as: "*Test*"$_{calc}$ or "*Test*"$_{stat}$
  - ➢Where "*Test*" is the particular value being calculated

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Hypothesis Testing: Critical Value

Critical Value

- In hypothesis testing, it is a threshold value to which the test statistic is compared in order to determine whether the null hypothesis is rejected

- The critical value is obtained from statistical tables
  - ➢Different hypothesis tests need different statistical tables

- Also denoted as: "CV" or "*Test*"$_{crit}$ or "*Test*"$_{CV}$
  - ➢Where "*Test*" is the particular value being calculated

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Hypothesis Testing: Decision Rules

- When the **test statistic falls into the acceptance region**:
  - Fail to reject the null (H_0)
  - Claim that there is no statistically significant difference between the groups
- When the **test statistic falls into the critical (significance or rejection) region:**
  - Reject the null (H_0)
  - Claim that there is a statistically significant difference between the groups

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Hypothesis Testing: Decision Rules

- When the **p-value is greater than (>) the α level**:
  - Fail to reject the null (H_0)
  - Claim that there is no statistically significant difference between the groups
- When the **p-value is less than (<) the α level**:
  - Reject the null (H_0)
  - Claim that there is a statistically significant difference between the groups

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Analysis Pitfalls

When Hypothesis Testing:

- The null hypothesis (H_0) is the item being evaluated
- The **decision** is about H_0 and **not** H_A
  - It is just a decision, and nothing has been proven
  - Example: "Reject H_0"
- The **conclusion** statement is whether the counter-claim of H_A was upheld
  - Example of when the H_0 is rejected: "There is sufficient evidence, at the α level of significance, to reject the H_0 in lieu of the H_A"

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Two Types of Hypothesis Tests

Two-tailed hypothesis test

- It is a statistical hypothesis test in which the critical region is split into two equal parts; each of which stays on one side of the sampling distribution

One-tailed hypothesis test

- It is a statistical hypothesis test in which the critical region is only on one side of the sampling distribution
  - ➢ It is also known as a one-sided hypothesis test

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Two-Tailed Hypothesis Test

- A two-tailed hypothesis test is the most commonly used hypothesis test
- A two-tailed hypothesis test is used when:

  - ➢ The interest is whether there is a significant difference between groups
  - ➢ The direction of the difference (higher or lower) is not important
  - ➢ Example of two-tailed hypothesis tests:
    - H0: $\mu = \mu_0$
    - HA: $\mu \neq \mu_0$
      - HA may include any of the following: ≠, <, >

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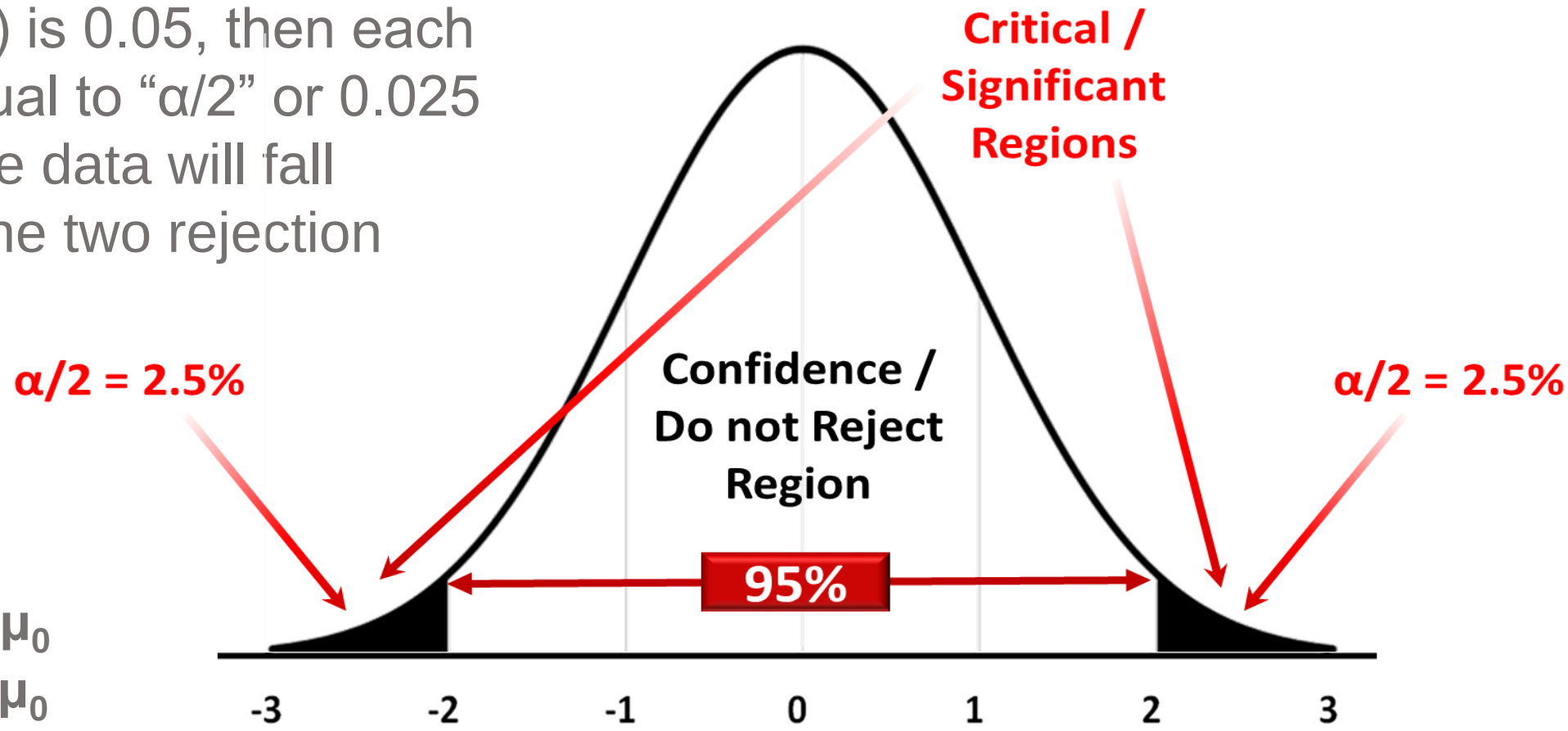
Two-Tail Hypothesis Test

- If alpha (α) is 0.05, then each "tail" is equal to "α/2" or 0.025
- **95 %** of the data will fall between the two rejection regions



- H_0: $\mu = \mu_0$
- H_a: $\mu \neq \mu_0$

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One-Tailed Hypothesis Test

- A one-tailed hypothesis test is used when the direction of the difference between groups is important
- The null hypothesis is rejected when the test statistic:

➢ falls into the critical region which only exits on the right side of the sampling distribution

➢ is sufficiently large

➢ is greater than the critical value

- H_0: $\mu = \mu_0$
- H_a: $\mu > \mu_0$

➢ falls into the critical region which only exits on the left side of the sampling distribution

➢ is sufficiently small

➢ is smaller than the critical value.

- H_0: $\mu = \mu_0$
- H_a: $\mu < \mu_0$

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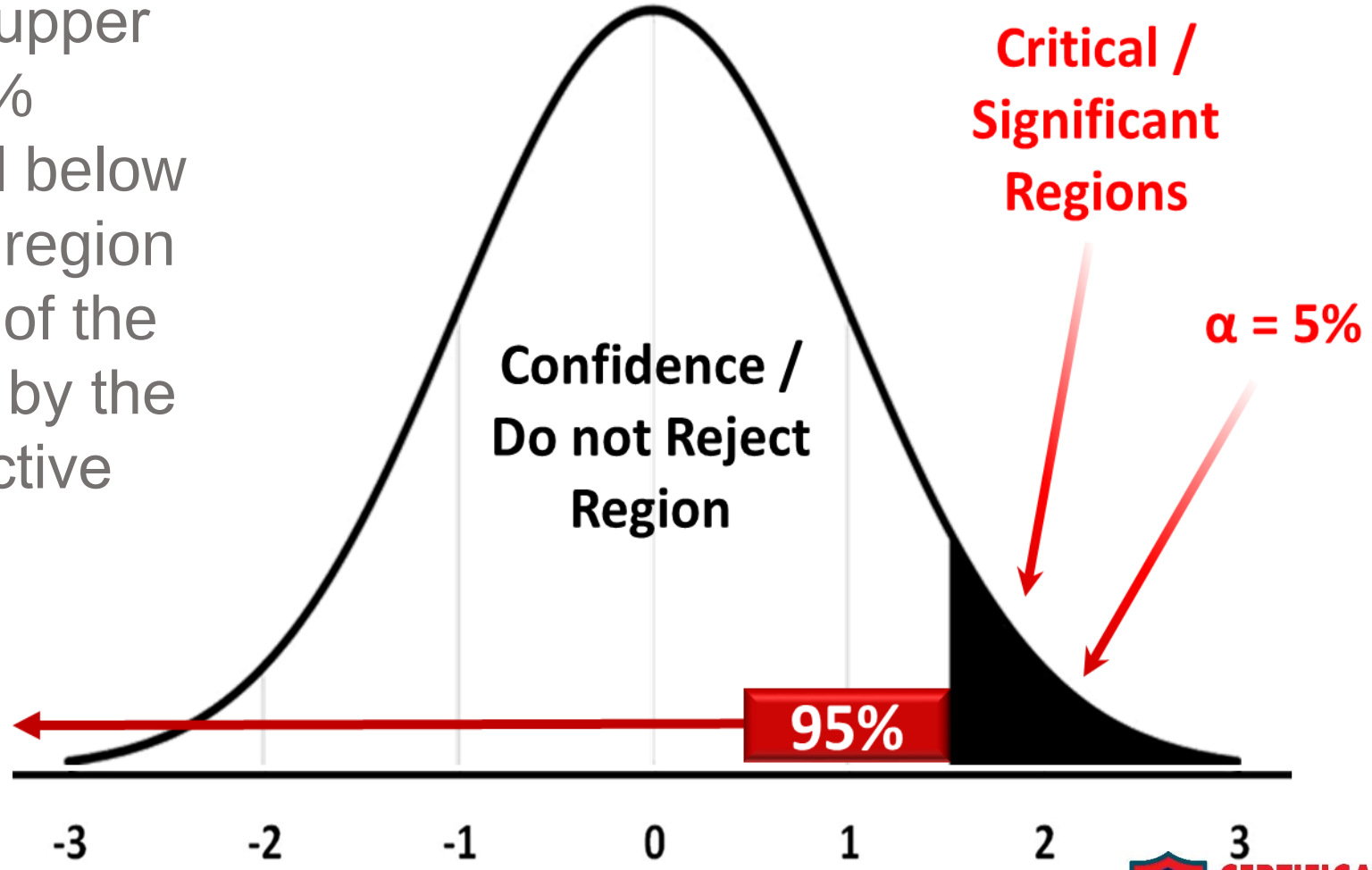
Upper-Tail Test

- If alpha (α) is 0.05, the upper portion will consist of 5%
- **95 %** of the data will fall below the upper-tail, rejection region
- The exact critical value of the alpha (α) is determined by the hypothesis test's respective parameters and the corresponding table
  - ➢ H_0: $\mu = \mu_0$
  - ➢ H_a: $\mu > \mu_0$



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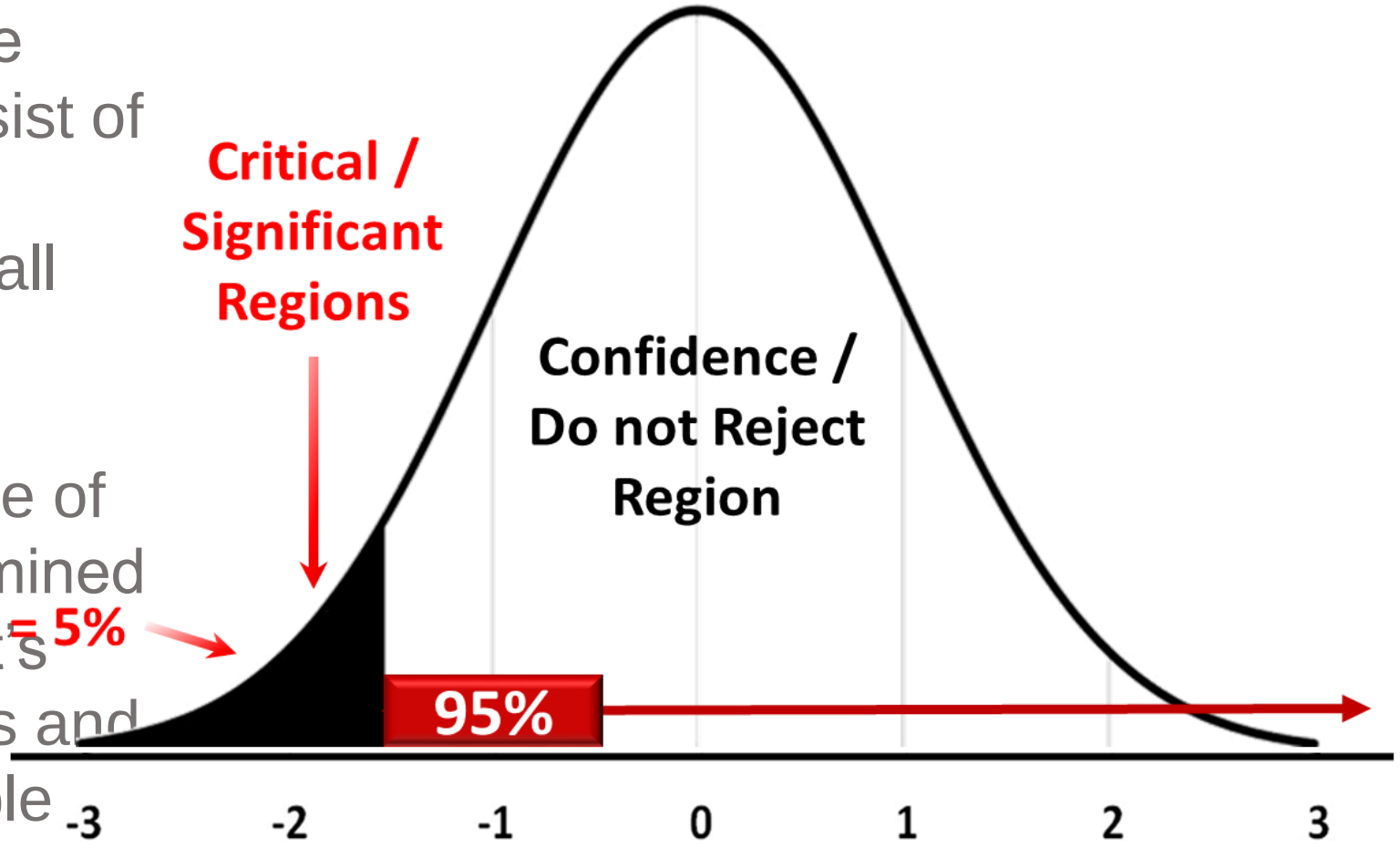
Lower-Tail Test

- If alpha (α) is 0.05, the lower portion will consist of 5%
- **95 %** of the data will fall above the lower-tail, rejection region
- The exact critical value of the alpha (α) is determined by the hypothesis test's respective parameters and the corresponding table
  - ➢ H_0: $\mu = \mu_0$
  - ➢ H_a: $\mu < \mu_0$



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3.3.5 READING TABLES

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Things to Know Before Using Tables

- Distribution classification
- What Alpha (α) / Significance level
- Sample size
- Degrees of freedom (df)
  - ➢ The number of unrestricted values (free to vary) in the calculation of a statistic (default: n-1)
- Right-tail vs. Left Tail test

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Z-Table: How to Read

The values in the table reference the shaded area in the curve. Always identify which Z table you are using!

Rows are labeled with the Z score's number and first decimal position
Z = 0.63

Z score / Z_{crit}

Columns are labeled with the Z score's second decimal value
Z = 0.63

% of values that sit in the shaded area for a given Z value
73.57% of values sit to the left of Z = 0.63

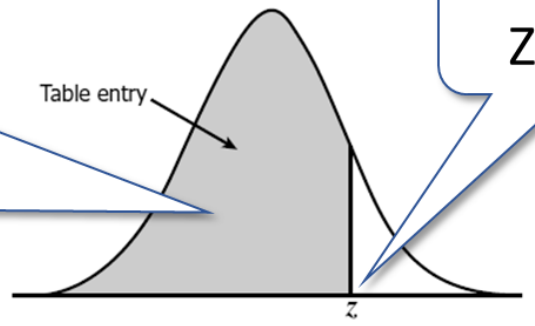

Table entry f
to the l

z	.00	.01	.02	.03	.04	.05
0.0	.5000	.5040	.5080	.5120	.5160	.519
0.1	.5398	.5438	.5478	.5517	.5557	.559
0.2	.5793	.5832	.5871	.5910	.5948	.598
0.3	.6179	.6217	.6255	.6293	.6331	.636
0.4	.6554	.6591	.6628	.6664	.6700	.673
0.5	.6915	.6950	.6985	.7019	.7054	.708
0.6	.7257	.7291	.7324	.7357	.7389	.742
0.7	.7580	.7611	.7642	.7673	.77	
0.8	.7881	.7910	.7939	.7967	.7995	
0.9	.8159	.8186	.8212	.8238	.8264	.82
1.0	.8413	.8438	.8461	.8485	.8508	.853
1.1	.8643	.8665	.8686	.8708	.8729	.874

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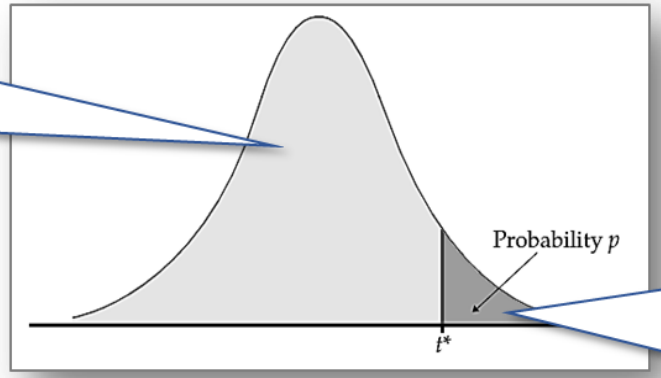
T-Table: How to Read

General shape of a t distribution



Shaded area represents probability or the significance region, depending on test

Identify the degrees of freedom (df)

df	.25	.20	.15	.10	.05	.025	.02	
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	3
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2

Alpha (α)

t_{crit}

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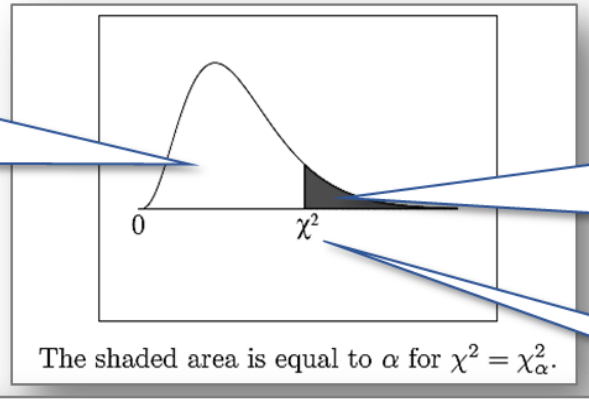
Chi-Square Table: How to Read

General shape of a χ^2 distribution



Shaded area is the significance or rejection region

χ^2_{crit}

Identify the degrees of freedom (df)

Alpha (α)

df	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	2.706	3.841	5.024	6.635	7.879
2	4.605	5.991	7.378	9.210	10.597
3	6.251	7.815	9.348	11.345	12.838
4	7.779	9.488	11.143	13.277	14.860
5	9.236	11.070	12.833	15.086	16.750
6	10.645	12.592	14.449	16.812	18.548
7	12.017	14.067	16.013	18.475	20.278
8	13.362	15.507	17.535	20.090	21.955
9	14.684	16.919	19.023	21.666	23.589

χ^2_{crit} value

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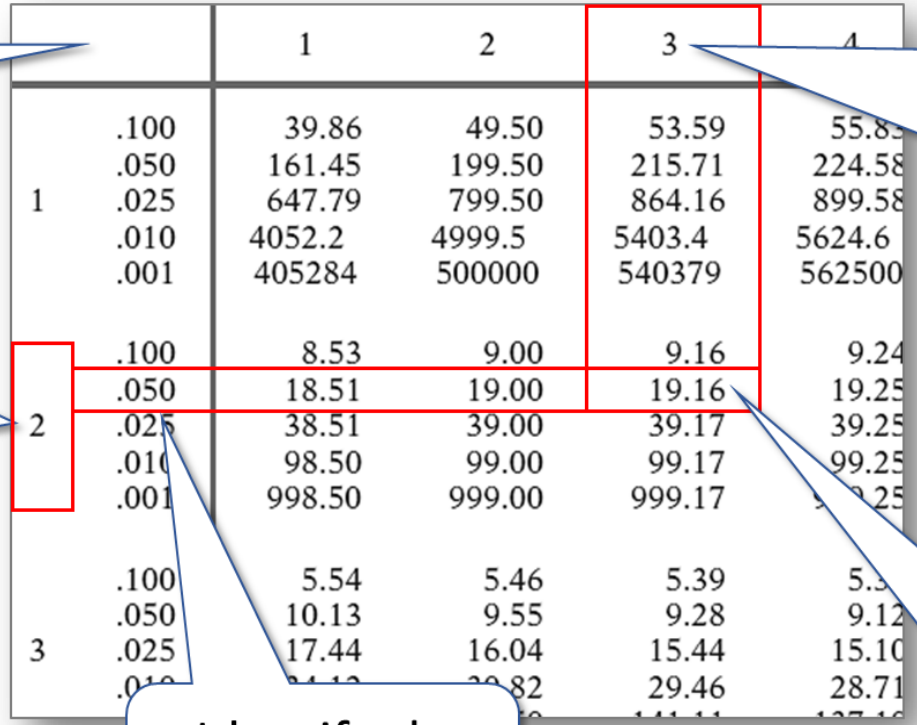
F-Table: How to Read

A portion of an F-table

In the rows, locate the value of the df in the denominator



		1	2	3	4
1	.100	39.86	49.50	53.59	55.8
	.050	161.45	199.50	215.71	224.58
	.025	647.79	799.50	864.16	899.58
	.010	4052.2	4999.5	5403.4	5624.6
	.001	405284	500000	540379	562500
2	.100	8.53	9.00	9.16	9.24
	.050	18.51	19.00	19.16	19.25
	.025	38.51	39.00	39.17	39.25
	.010	98.50	99.00	99.17	99.25
	.001	998.50	999.00	999.17	.25
3	.100	5.54	5.46	5.39	5.3
	.050	10.13	9.55	9.28	9.12
	.025	17.44	16.04	15.44	15.10
	.010		82	29.46	28.71

Identify the alpha (α)

In the column headers, locate the degrees of freedom (df) in the numerator

Cross-reference the various sets of values to find the F_{crit} value

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3.3.6 EXCEL: DATA-ANALYSIS TOOLPAK

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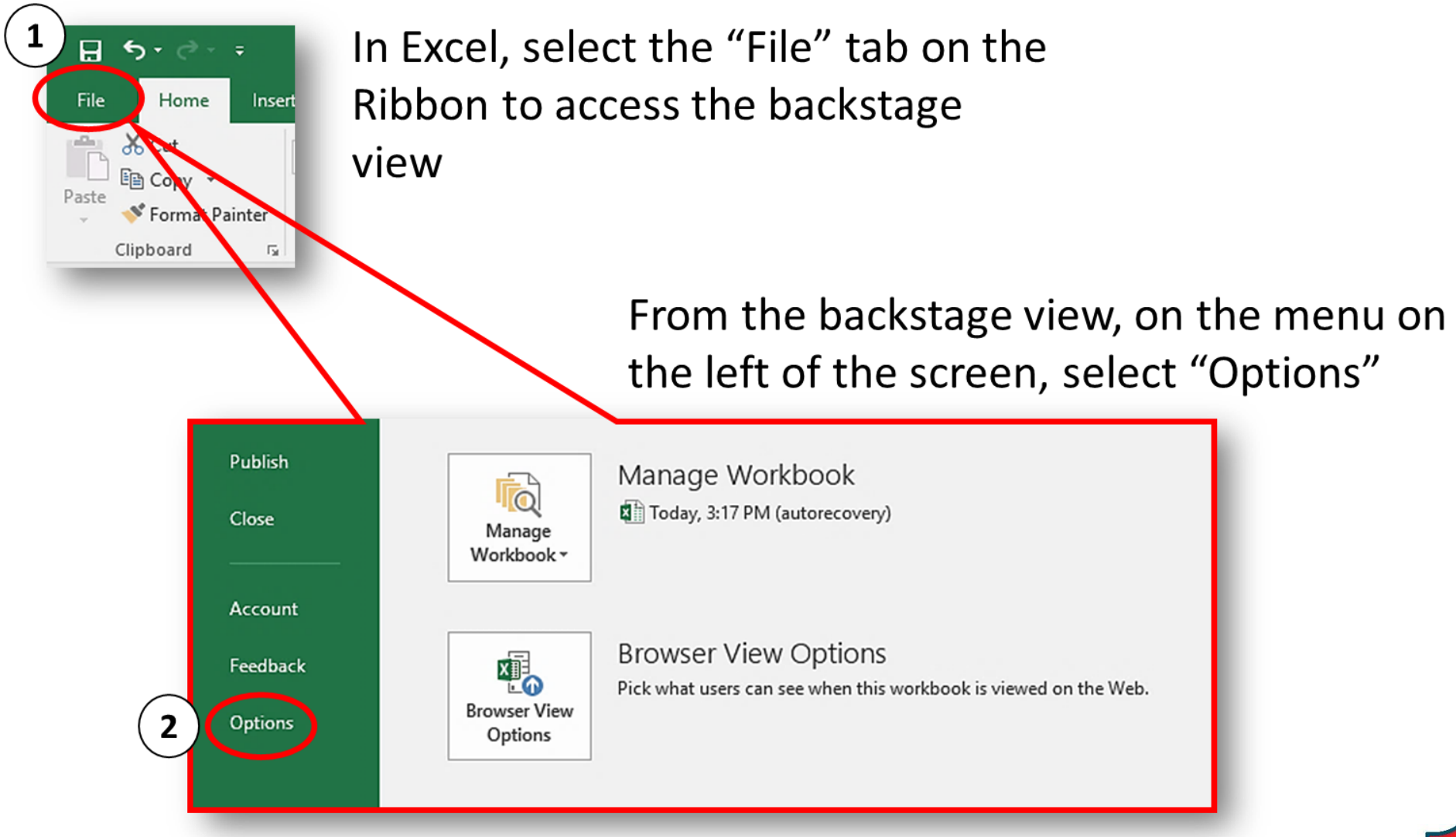
Excel: Data-Analysis ToolPak Install



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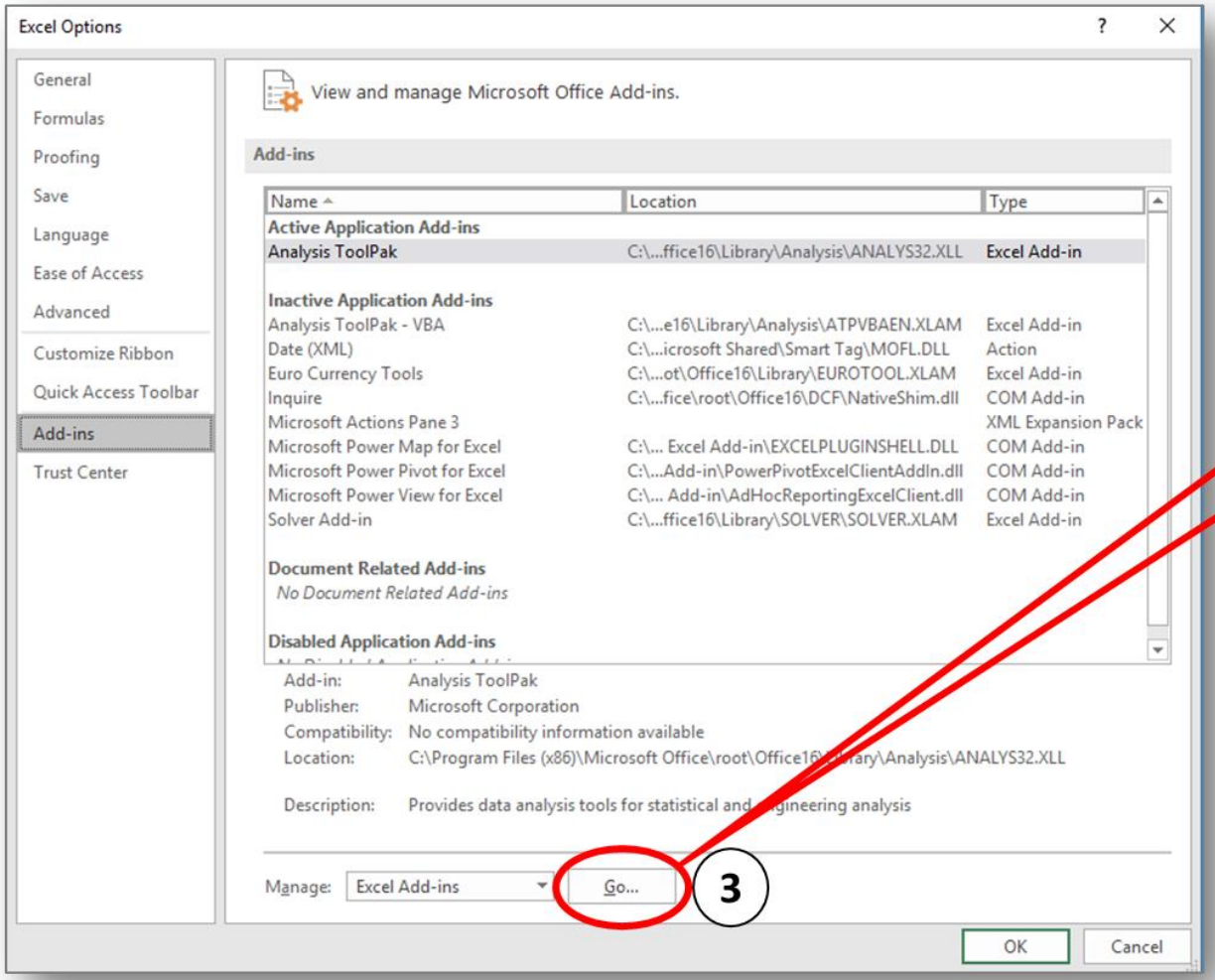
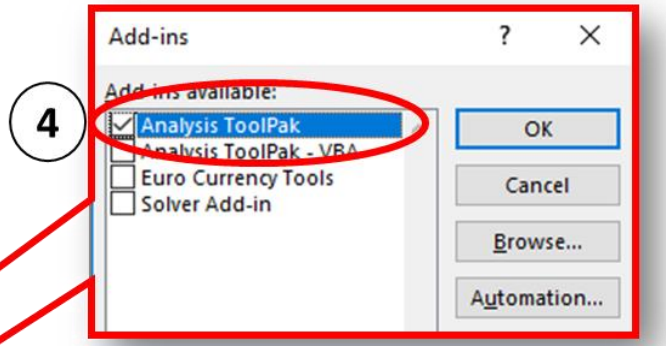
Excel: Data-Analysis ToolPak Install, cont.





From "Add-ins," select "Analysis ToolPak," then select, "OK" to activate

From the "Excel Options" window that appears, select "Add-ins," then select "Go" to manage the Excel Add-ins

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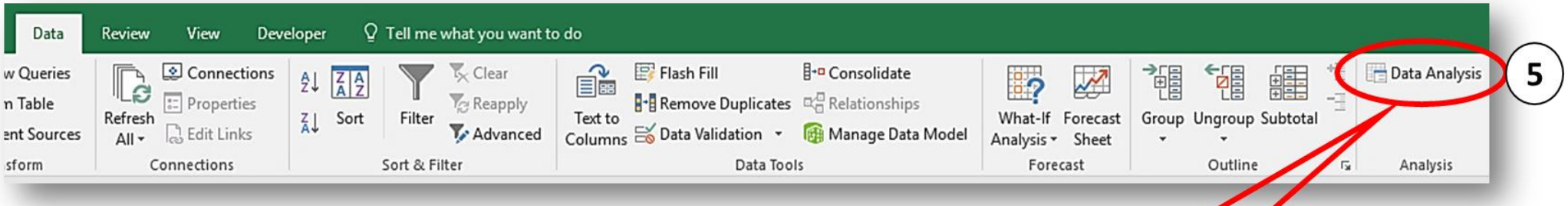
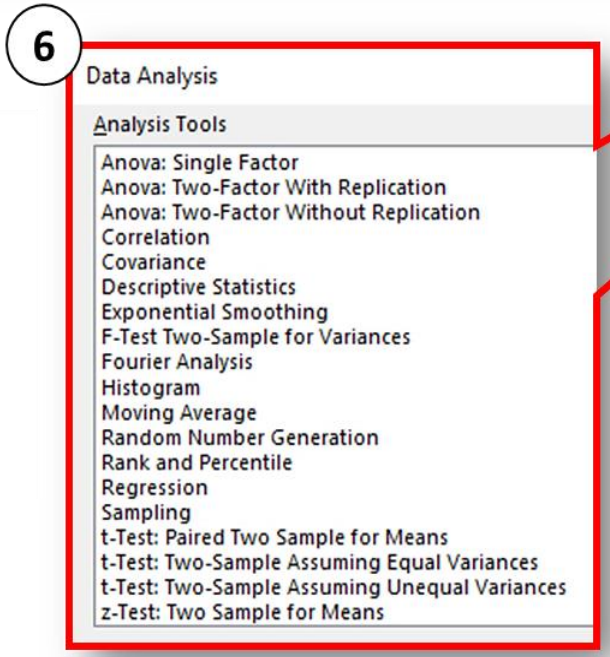
Excel: Data-Analysis ToolPak Install, cont.



When selected, the "Data Analysis" toolpak will appear, with the following options…



Now, under the "Data" tab there will be a "Data Analysis" option

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3.3.7 TESTING FOR NORMALITY

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Normality Testing: Why?

- One of the first questions in Hypothesis testing is whether the data are Normally distributed
- Normality will impact how it is analyzed and interpreted
- Quick reminder: Introduction to Normal distribution was covered under the Measure phase
- Normality is not a given. There are many functional or physical limitations that restrict the ability of normality
  - ➢Example: A call cannot last less than 0 seconds

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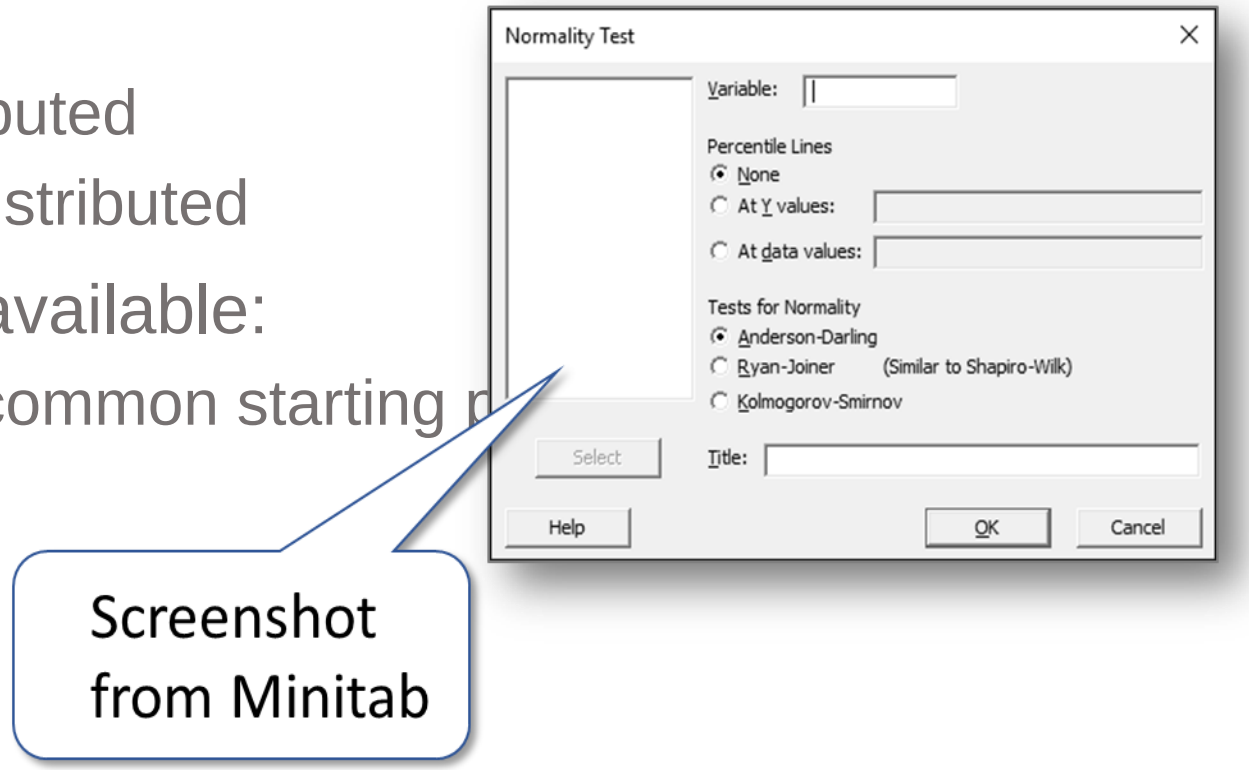
Normality Testing: Guidelines

- Framing the hypothesis:
  - H_0: The data are normally distributed
  - H_A: The data are not normally distributed
- There are many normality tests available:
  - Anderson-Darling (AD) – most common starting p
  - Sharpiro-Wilk
  - Jarque-Bera
  - Ryan-Joiner
  - etc.



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Normality Testing: Guidelines

- **While you can take time to built a Normality testing tool in Excel or download one from the Internet, it is MUCH easier (but pricier) to utilize statistical software, such as: Minitab, SigmaXL, or etc.

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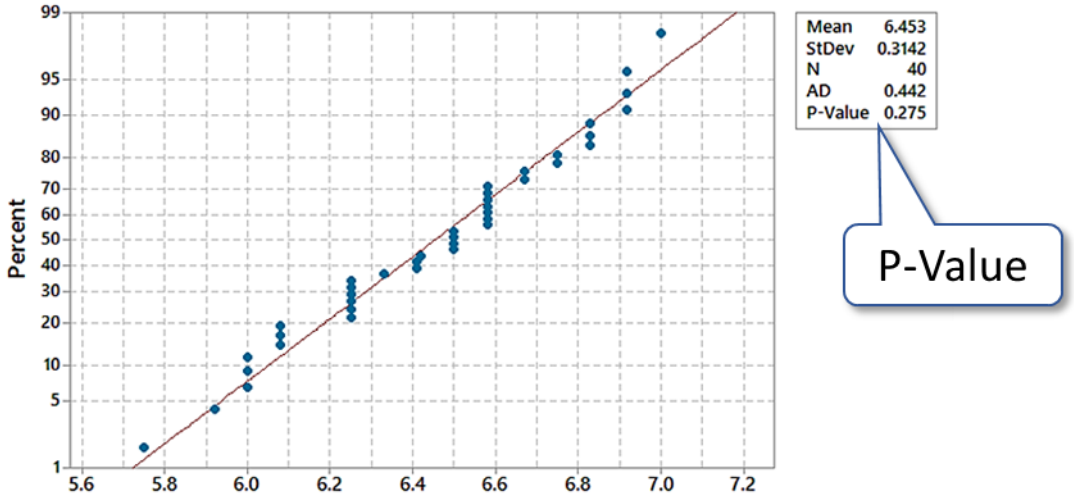
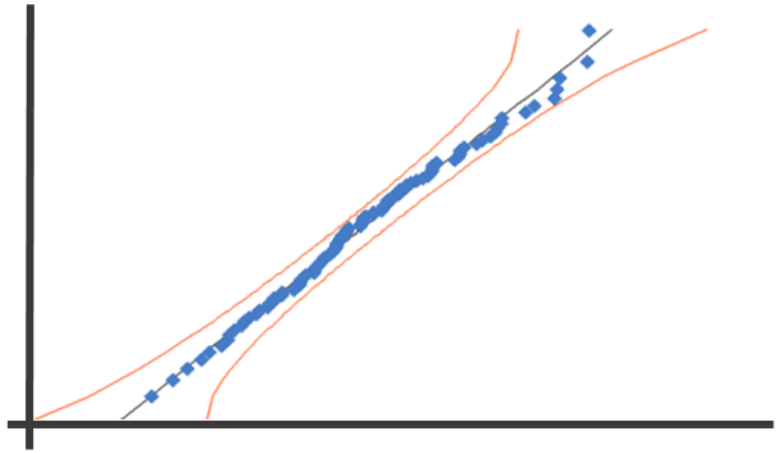
Normality Testing

Run a **normality test** on the data to determine if they are normally distributed or not

- Comparing the “P-Value” vs the Alpha level (addressed later under Risk)
- Normally distributed data will lie either on the line (top image) or within boundaries (bottom image)





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3.4 Hypothesis Testing with Normal Data

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Hypothesis Testing: Steps

- **Step 1:** State the null and alternative hypothesis
- **Step 2:** Determine α level
- **Step 3:** Collect sample data
- **Step 4:** Select and run the appropriate hypothesis test
  - ➢Though the presentation reviews formulas, using software is much easier!
  - ➢The calculations, findings, and interpretations are exactly the same!
- **Step 5:** Determine whether to reject the null and state the conclusion

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Normal Data: Testing Roadmap

DATA TYPE	DISTRIBUTION	PURPOSE	SAMPLES	N PARAMETERS 1	N PARAMETERS 2	N PARAMETERS 3	HYPOTHESIS TEST
Continuous	Normal	Mean	> 2 Samples		Equal Variances	1 Factor	**1-Way ANOVA**
Continuous	Normal	Mean	> 2 Samples		Equal Variances	>1 Factor	**2-Way ANOVA**
Continuous	Normal	Mean	1 Sample	"N" Standard Deviation Known			**Z Test**
Continuous	Normal	Mean	2 Samples	"N" Standard Deviation Known			**2 Sample Z Test**
Continuous	Normal	Mean	1 Sample	"N" Standard Deviation Not Known			**1 Sample t-Test**
Continuous	Normal	Mean	2 Samples	"N" Standard Deviation Not Known	Dependent / Pairs	Before / After	**Paired t-Test**
Continuous	Normal	Mean	2 Samples	"N" Standard Deviation Not Known	Equal Variances		**2 Sample t-Test**
Continuous	Normal	Mean	2 Samples	"N" Standard Deviation Not Known	Unequal Variances		**2 Sample t-Test**
Continuous	Normal	Variance	1 Sample			1 Factor	**Chi-square Test for Variance**
Continuous	Normal	Variance	>= 2 Samples			2+ Factors	**Chi-square Test for Variance**
Continuous	Normal	Variance	2 Samples				**F-Test**
Continuous	Normal	Variance	> 2 Samples				**Bartlett's test**

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3.4.1 Tests of Equal Variance

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What are Tests of Equal Variance?

- Tests of equal variance are a family of hypothesis tests used to check whether there is a statistically significant difference between the variances of two or more populations
  - H_0: $\sigma_1^2 = \sigma_2^2 = \ldots = \sigma_k^2$
  - H_A: At least the variance of one population is different from others
  - k is the number of populations of interest. $k \geq 2$
- The variance test may be used as a prerequisite for other statistics tests
  - Often, the validity of a test rests upon whether the variance of the populations are the same

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F-Test

Criteria:

- Continuous Data
- Variance
- Normality ‘Not Required’
- 2 Samples

- An F-test can be used to compare the variances of two samples
  - ➢ Having inferred that the samples are representative of their respective population
    - H_0: $s_1^2 = s_2^2$
    - H_A: $s_1^2 \neq s_2^2$
  - ➢ F-test is known to be extremely sensitive to non-normality
    - When any one of the two populations is not normal, use the Brown-Forsythe test for checking the equality of variances

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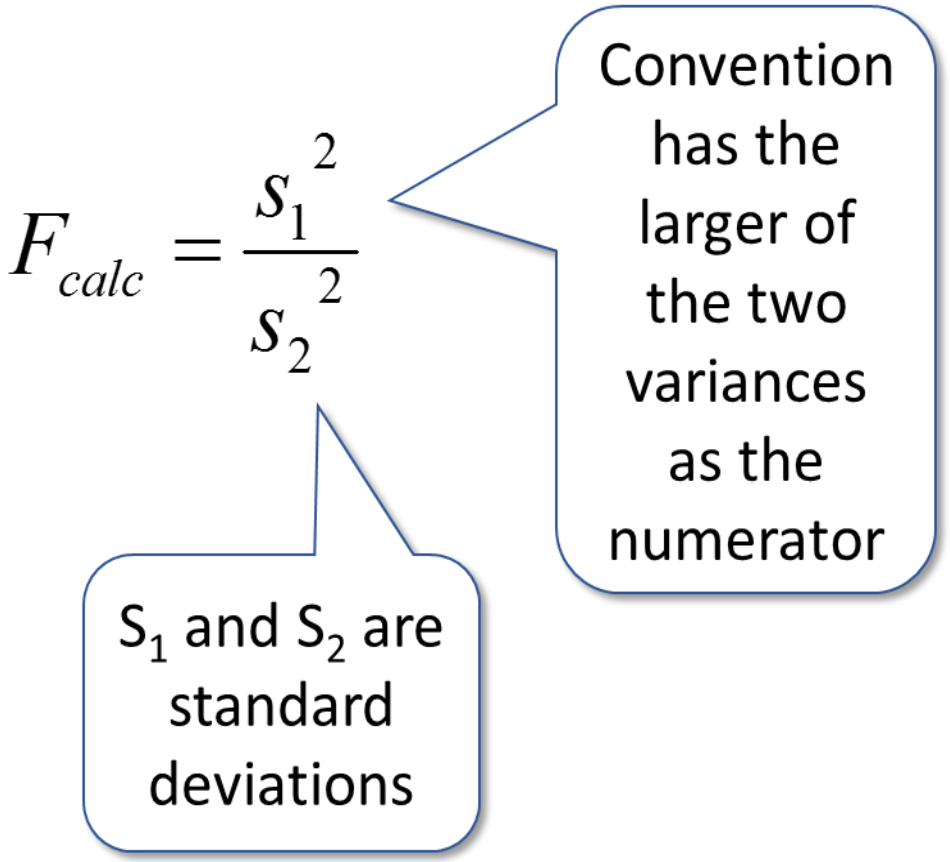
F-Test: Test Statistic & Decisions

- The sampling distribution of the test statistic follows F distribution when the null is true
- Compare test statistic to the critical value from an F-table
- $F_{calc} > F_{crit}$, we reject the null and claim there is a statistically significant difference between the variances of the two populations



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F-Test: Test Statistic & Decisions

- $F_{calc} < F_{crit}$, we fail to reject the null and claim there is not any statistically significant difference between the variances of the two populations

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Additional Tests for Equal Variance

Bartlett's test

- Used to test if k samples are from **populations with equal variances**
- Some statistical tests, for example the analysis of variance (ANOVA), assume that variances are equal across groups or samples. The Bartlett test can be used to verify that assumption
- Very sensitive to Non-Normal data

Levene's test

- An inferential statistic used to assess the **equality of variances for a variable calculated for two or more groups**
- Levene's test is often used before a comparison of means

Brown–Forsythe's test

- A statistical test for the equality of **non-normal, group variances**
- The Brown–Forsythe test statistic is the F statistic resulting from an ordinary one-way analysis of variance (ANOVA) on the absolute deviations from the median, vs. the mean for normal data

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Unequal Variance

- Unequal variances are usually the result of differences in the shape of the distribution
  - Extreme tails
  - Outliers
  - Multiple modes
  - Etc.
- Should be explored through data demographics

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3.4.2 Z Tests

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Reminder: Z Distribution

- **Z distribution** is the simplest normal distribution with the mean equal to zero and the variance equal to one

- The Z score is the measure of how many standard deviations an observation is above or below the mean

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Z Test – 1 or 2 Samples

Criteria:

- Continuous Data
- Mean
- Normally Distributed
- 1 or 2 Samples
- Large sample size (≥ 30)
- Standard Deviation Known

Z Test is based upon the Z distribution (basic Normal distribution) referenced in the Measure phase

- Where μ is the mean of the population of interest and μ_0 is the mean of the target
  - H0: $\mu = \mu_0$
  - HA: $\mu \neq \mu_0$

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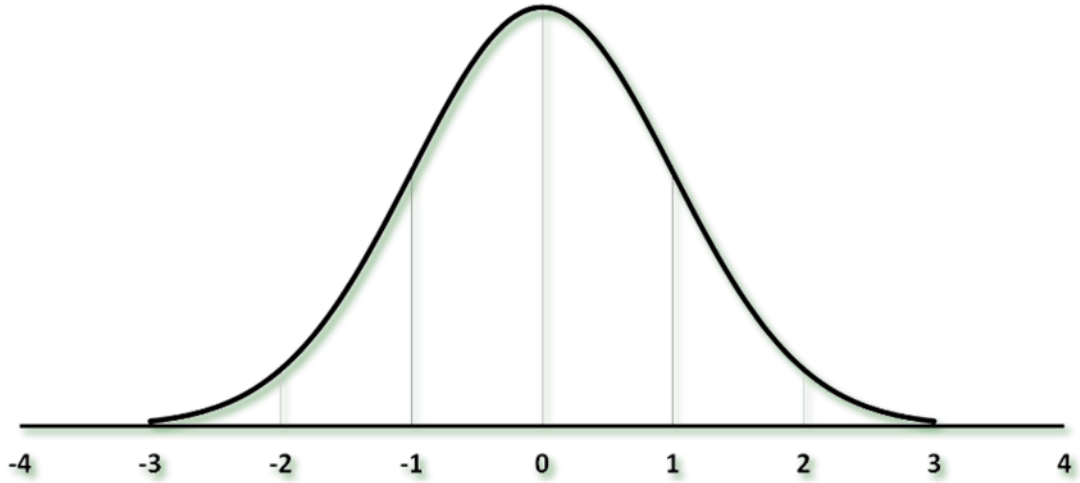
Z Test Statistic: “Z Score”

- The Z Score effectively transforms the actual data into standard normal units (not values)

- Not only can the Z be calculated, any of the values may be calculated algebraically if the others are known



$$z = \frac{x - \mu}{\sigma}$$

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Z-Test : Decision Rules

$|z_{calc}|$ vs. z_{crit}

- Identify the z_{crit} value from the z-table
- If $|z_{calc}| > z_{crit}$, we reject the null and claim there is a statistically significant difference between the population mean μ and the specified value μ_0
- If $|z_{calc}| < z_{crit}$, we fail to reject the null and claim there is no statistically significant difference between the population mean μ and the specified value μ_0

p-value vs. α

- Assuming $\alpha = 0.05$
- If p-value > 0.05 fail to reject H_0
- If p-value < 0.05 reject H_0

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3.4.3 1 SAMPLE T-TEST

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Test of Means (t-tests)

- t-tests are used when comparing:
  - The mean of a single sample against a target
  - One piece of equipment to another
  - Supplier A's quality to Supplier B's quality
  - Performance or products Before and After a given process
- Also known as the "Student's t-test"

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1 Sample : T-Test

Criteria:

- Continuous Data
- Normally Distributed
- 1 Sample
- Value vs. Mean
- Small sample size (<30)
- Population Standard Deviation (σ) and Variance Not Known
- Sample Standard Deviation(s) and Variance are Used
- Preferred over the Z-test when sample size is <30

One sample t-test is a hypothesis test to study whether there is a statistically significant difference between a population mean and a specified value

- Where μ is the mean of the population of interest and μ_0 is the mean of the target
  - H_0: $\mu = \mu_0$
  - H_A: $\mu \neq \mu_0$

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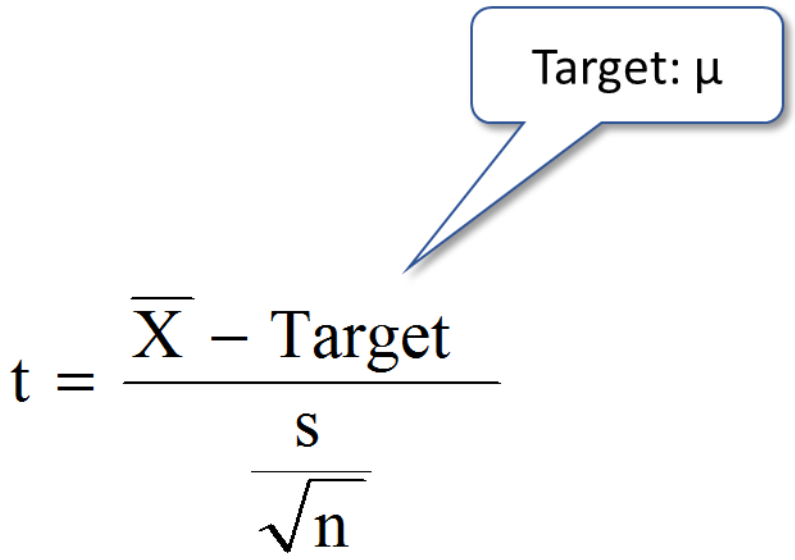
1 Sample T-Test : Test Statistic & Crit Value

Critical Value

- t_{crit} is the t-value in a Student t distribution with the predetermined significance level α and degrees of freedom (n –1)
- t_{crit} values for a two-sided and a one-sided hypothesis test with the same significance level α and degrees of freedom (n – 1) are different

Test Statistic



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1 & 2 Sample T-Test : Decision Rules

$|t_{calc}|$ vs. t_{crit}

- Identify the t_{crit} value from the t-table
- If $|t_{calc}| > t_{crit}$, we reject the null and claim there is a statistically significant difference between the population mean μ and the specified value μ_0
- If $|t_{calc}| < t_{crit}$, we fail to reject the null and claim there is no statistically significant difference between the population mean μ and the specified value μ_0

p-value vs. α

- Assuming $\alpha = 0.05$
- If p-value > 0.05 fail to reject H_0
- If p-value < 0.05 reject H_0

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3.4.4 2 SAMPLE T-TEST

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Types of Two Sample T-Tests

Two sample t-tests when comparing two populations:

- Dependency exists (e.g. Before vs. After)
  - ➢ Paired t-test
- Variances are unknown but (assumed) equal
  - ➢ Two sample t-test (when $\sigma_1^2 = \sigma_2^2$)
- Variances are unknown but (assumed) unequal
  - ➢ Two sample t-test (when $\sigma_1^2 \neq \sigma_2^2$)

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2 Sample : T-Test

Criteria:

- Continuous Data
- Normally Distributed
- 2 Samples
- Value vs. Mean
- Small sample sizes (<30)
- Population Standard Deviation (σ) and Variance Not Known
- Sample Standard Deviation(s) and Variance are Used
- Preferred over the Z-test when sample size is <30

Two sample t-test is a hypothesis test to study whether there is a statistically significant difference between the means of two populations

- Where μ_1 is the mean of one population and μ_2 is the mean of the other population of our interest
  - H_0: $\mu_1 = \mu_2$
  - H_A: $\mu_1 \neq \mu_2$
- Focus is on unknown, unequal variances

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2 Sample Paired T-Test

Criteria:

- Continuous Data
- Mean
- Normally Distributed
- 2 Samples
- Standard Deviation Not Known
- Dependent / Paired Samples

Two sample t-test is a hypothesis test to study whether there is a statistically significant difference between the means of the same sample (e.g. Before vs. After)

- Where μ_1 is the mean of one population and μ_2 is the mean of the other, dependent population of our interest
  - H_0: $\mu_1 = \mu_2$
  - H_A: $\mu_1 \neq \mu_2$

$$t_{calc} = \frac{\bar{d}}{s_d / \sqrt{n}}$$

Average difference between each pair of data

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3.4.5 1 Sample Variance: Chi-Square

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Chi-square Test for Variance

Criteria:

- Continuous Data
- Variance
- Normally Distributed
- 1 Sample

A chi-square test is a statistical hypothesis test in which the test statistic follows a chi-square distribution when the null hypothesis is true

- Can be used to test the equality between the variance of a normally distributed population and a specified value
- The null hypothesis states there is no difference between the standard deviations or variances
  - H_0: $\sigma_1 = \sigma_2 = \ldots = \sigma_k$
  - H_A = at least on is different

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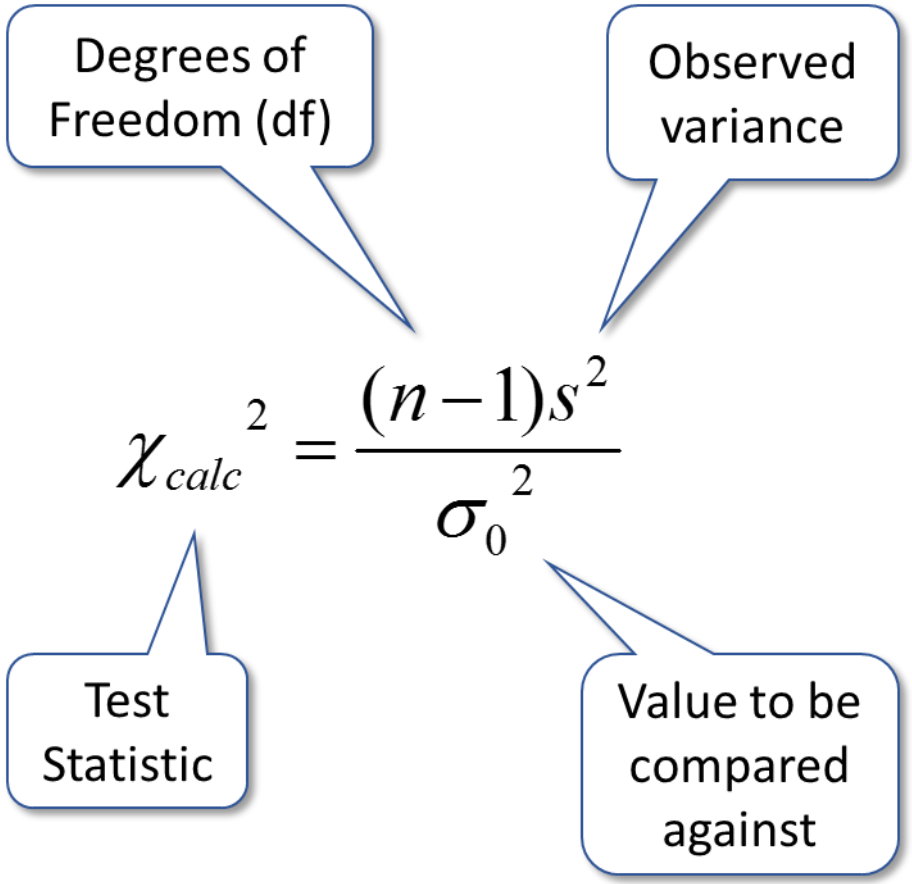
Chi-Square: Decision Rules

- χ_{crit}^2 is the χ^2 value in a χ^2 distribution with the predetermined significance level α and degrees of freedom (n – 1)
- If $\left|\chi_{calc}^2\right| > \chi_{crit}^2$, we reject the null and claim there is a statistically significant difference between the population variance and the specified value
- If $\left|\chi_{calc}^2\right| < \chi_{crit}^2$, we fail to reject the null and claim there is not any statistically significant difference between the population variance and specified value



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3.4.6 One-Way ANOVA

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One-Way ANOVA

Criteria:

- Continuous Data
- Mean
- Normally Distributed
- ≥ 2 Means
- Equal Variances
- 1 Factor

One-way ANOVA (one-way analysis of variance) **compares the means** of different groups **by analyzing the variances** between and within groups

- It is a generalized form of the two-sample t-test
- One-way ANOVA compares k population means, where k ≥ 2
  - H_0: $\mu_1 = \mu_2 = ... = \mu_k$
  - H_A: at least one μ_i is different

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Sources of Variation

- Based on sample data, the means of the three populations might look different because of two variation sources:

  - Variation between groups
    - There are non-random factors leading to the variation between groups
    - The desired value is the variation between groups; because, the interest is whether the groups are statistically different from each other

(Cont.)

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Sources of Variation

- Variation within groups
  - There are random errors resulting in the variation within each individual group
  - Variation within groups is the noise which corrupts the Variation between groups signal

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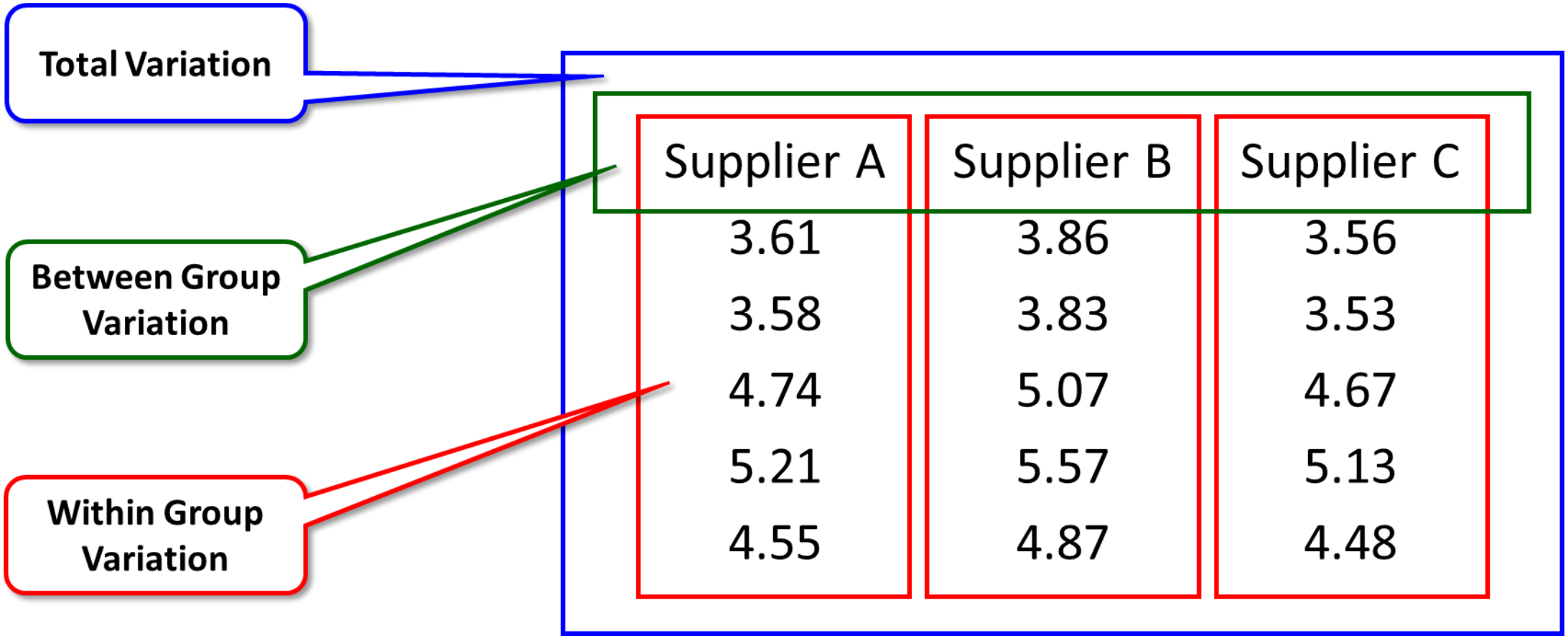
ANOVA: Variation Components



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ANOVA: Variation Components, Cont.

- Total Variation = Variation Between Groups + Variation Within Groups

$$\sum_{j=1}^{k}\sum_{i=1}^{n_j}(Y_{ij}-\bar{\bar{Y}})^2 = \sum_{j=1}^{k} n_j(\bar{Y}_j-\bar{\bar{Y}})^2 + \sum_{j=1}^{k}\sum_{i=1}^{n_j}(Y_{ij}-\bar{Y}_j)^2$$

Total Variation = sums of squares of the vertical difference between the individual observation and the grand mean

Variation Between Groups = sums of squares of the vertical difference between the group mean and the grand mean

Variation Within Groups = sum of squares of the vertical difference between the individual observation and the group mean

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ANOVA: Decision Rules

- ANOVA uses the F-test to compare the means of different groups
- The F-ratio calculated below is the test statistic F_{calc}

$$F_{calc} = \frac{MS_{between}}{MS_{within}} = \frac{SS_{between} / DF_{between}}{SS_{within} / DF_{within}} = \frac{\sum_{j=1}^{k} (\bar{Y}_j - \bar{\bar{Y}})^2 \Big/ (k-1)}{\sum_{j=1}^{k} \sum_{i=1}^{n_j} \left(Y_{ij} - \bar{Y}_j\right)^2 \Big/ (n-k)}$$

- If $|F_{calc}| < F_{crit}$, we fail to reject the null and claim that the means of all the populations of our interest are the same
- If $|F_{calc}| > F_{crit}$, we reject the null and claim that there is at least one mean different from the others

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ANOVA Validation

- To make sure the conclusions made in ANOVA are reliable, we need to perform residuals analysis

  - Good residuals:
    - Have a mean of zero
    - Are normally distributed
    - Are independent of each other
    - Have equal variance

- Residual analysis is further addressed in the Improve module

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Using Software to Calculate ANOVA

- Use software to dramatically, and realistically, speed up the ANOVA test
- Software will calculate both the F-test and the p-value
- Easiest to compare the p-value to the alpha level (0.05) and reject the null if the p-value is less than alpha
- If the null is rejected, conclude that the at least one of the means of the groups is different from others
- Software applications include: Excel, Minitab, SigmaXL, etc.

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ANOVA: Follow-up Comments

- If the variances are not all equal, use other hypothesis testing methods other than one-way ANOVA

- Two-way ANOVA is available for data that fits the following criteria:
  - ➢Continuous Data
  - ➢Mean
  - ➢Normally Distributed
  - ➢> 2 Means
  - ➢Equal Variances
  - ➢> 1 Factor

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3.5 Hypothesis Testing with Non-Normal Data

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Non-Normal Data: Testing Roadmap

DATA TYPE	DISTRIBUTION	PURPOSE	SAMPLES	N PARAMETERS 1	N PARAMETERS 2	N PARAMETERS 3	HYPOTHESIS TEST
Continuous	Non-Parametric	Median	1 Sample	Symmetrical			**1-Sample Wilcoxon**
Continuous	Non-Parametric	Median	1 Sample	Not Symmetrical			**1-Sample Sign**
Continuous	Non-Parametric	Median	2 Sample	Matched samples			**Friedman**
Continuous	Non-Parametric	Median	2 Sample	Not Matched samples	Equal Variances		**Mann-Whitney**
Continuous	Non-Parametric	Median	2 Sample	Not Matched samples	Unequal Variances	Outliers	**Moods Median**
Continuous	Non-Parametric	Median	2 Sample	Not Matched samples	Unequal Variances	No Outliers	**Kruskall Wallis**
Continuous	Non-Parametric	Median	> 2 Sample			Outliers	**Moods Median**
Continuous	Non-Parametric	Median	> 2 Sample			No Outliers	**Kruskall Wallis**
Discrete	Non-Parametric	Proportion	1 Sample			1 Factor	**1 Sample Proportion**
Discrete	Non-Parametric	Proportion	2 Samples			1 Factor	**2 Sample Proportion**
Discrete	Non-Parametric	Proportion	> 2 Samples			1 Factor	**Chi-square Test (Contingency Table)**
Discrete	Non-Parametric	Proportion	n/a			2 Factor	**Chi-square Test (Contingency Table)**
Continuous	Non-Parametric	Variance	2 Samples				**F-Test**

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Non-Normal Data: General Comments

- Non-Normal data is classified as Non-Parametric
- The four primary causes for Non-Normal Data:

➢Skewness – Natural and Artificial Limits
  - **Focus of the hypothesis tests in this module**
  - Any data that has natural or artificial limits typically exhibits a Skewed Distribution when it is operating near the limit
    - Example: Call times in a call center have a natural limit of 0

➢Mixed Distributions – Multiple Modes

➢Kurtosis

➢Granularity

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Non-Parametric Tests

- A non-parametric test makes no assumptions about Normality
- For a Skewed Distribution:
  - The central tendency is the Median
- In cases of multiple distributions, if just one distribution is non-normal, a non-parametric test should be used

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3.5.1 MANN-WHITNEY

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Mann-Whitney

Criteria:

- Continuous Data
- Not Normally Distributed
- Median
- 2 Samples
- Not Symmetrical
- Not Matched samples
- Equal Variances

The Mann-Whitney test (also called Mann-Whitney U test or Wilcoxon rank-sum test) is a statistical hypothesis test to compare the medians of two populations that are not normally distributed

- H_0: $\eta_1 = \eta_2$
- H_A: $\eta_1 \neq \eta_2$
  - where η_1 is the median of one population and η_2 is the median of the other population

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Mann-Whitney Test: Assumptions

- Mann-Whitney test:
  - ➢Preferential for populations that are non-normally distributed
  - ➢Can be used when shapes of the two populations' distributions are different

- Two populations are independent to each other

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Mann-Whitney: Steps to the Test Statistic

1. Gather two samples (n_1 and n_2) from two populations (N_1 from N_2)
2. Group the two samples into a single data set
3. Sort the data in ascending order, ranked from 1 to n
    - Where n is the total number of observations ($n_1 + n_2$)
4. Sum the ranks (not values) for all the observations from n_1 and call it R_1
5. Sum the ranks (not values) for all the observations from n_2 and call it R_2

(Cont.)

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Mann-Whitney: Calculating Test Statistic

6. Calculate the test statistics

   Where: $$U = \min(U_1, U_2)$$

7. Decide whether to reject the null hypothesis
   - H0: η1 = η2
   - HA: η1 ≠ η2

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

Sample sizes

Sum of ranks for observations from n_1 and n_2, respectively

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Mann-Whitney: Samples < 10

- When both of the sample sizes are smaller than 10, the distribution of U under the null is tabulated
  - ➢The test statistic is U and, use the Mann-Whitney table, find the p-value
    - If the p-value < alpha level (0.05), reject the null hypothesis
    - If the p-value > alpha level (0.05), fail to reject the null hypothesis
- See following slide for sample sizes > 10

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Mann-Whitney: Samples > 10

- If both sample sizes are greater than 10, the distribution of U can be approximated by a normal distribution
- In other words, $\frac{U-\mu}{\sigma}$ follows a standard normal distribution

$$U = \min(U_1, U_2)$$

$$Z_{calc} = \frac{U-\mu}{\sigma}$$

$$\mu = \frac{n_1 n_2}{2}$$

$$\sigma = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

- Where $|Z_{calc}| > Z$ value at $\alpha/2$ level reject the null hypothesis
  - Example: When $\alpha = 5\%$, the z value we compare $|Z_{calc}|$ to is 1.96

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3.5.2 KRUSKAL-WALLIS

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Kruskall Wallis

Criteria:

- Continuous Data
- Not Normally Distributed
- Median
- 2 or More Samples
- Not Symmetrical
- Not Matched samples
- Unequal Variances
- No Outliers

The Kruskal-Wallis one-way analysis of variance is a statistical hypothesis test to compare the medians among more than two groups

- The Kruskal-Wallis test is robust for the non-normally distributed population
- It is an extension of Mann-Whitney test
  - H_0: $\eta_1 = \eta_2 = \ldots = \eta_k$
  - H_A: At least one of the medians is different from others

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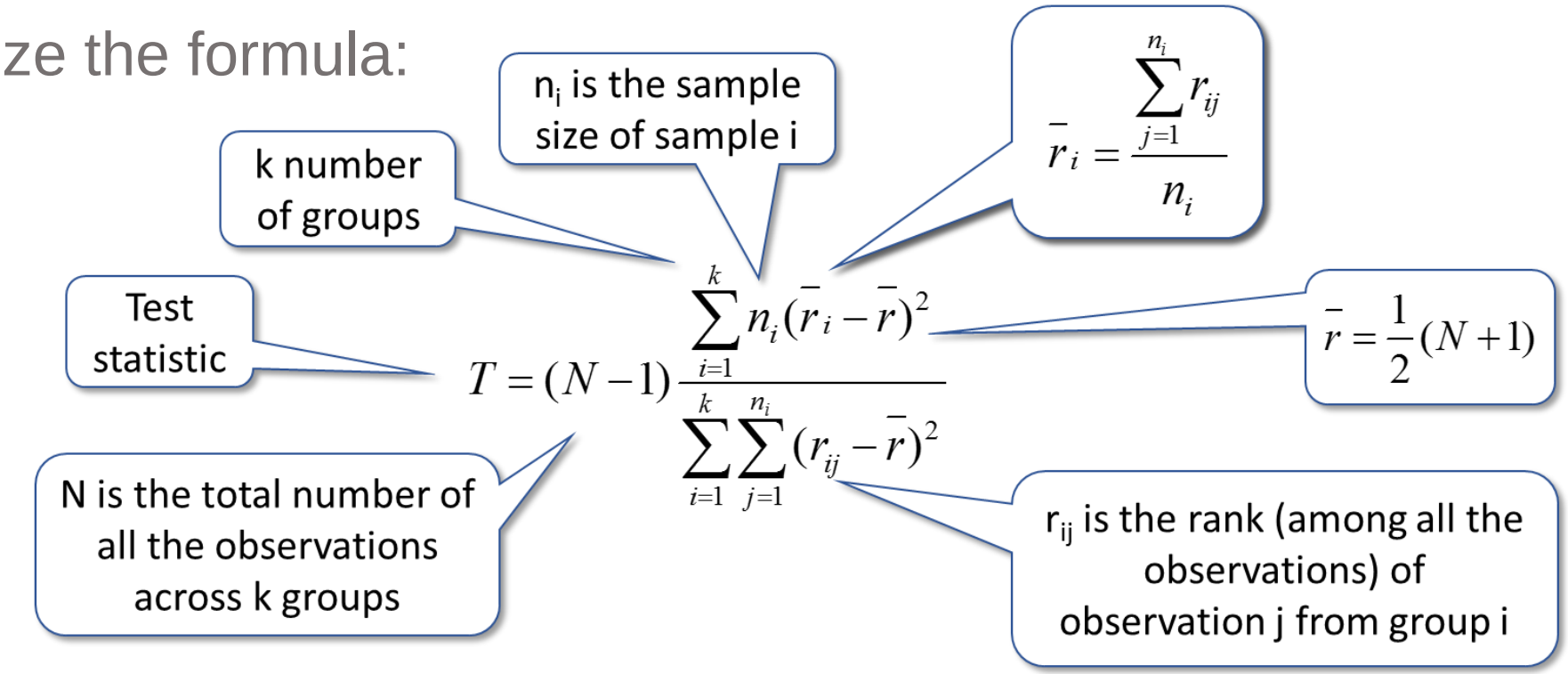
Kruskal-Wallis: Steps to the Test Statistic

- To calculate the Kruskal-Wallis test statistic, utilize the same steps as the Mann-Whitney test, but for “k” samples
- Utilize the formula:



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Kruskal-Wallis: Decision Rules

- The test statistic follows **chi-square** distribution when the null hypothesis is true
  - If T is greater than χ^2_{k-1} , we reject the null and claim there is at least one median statistically different from other medians
  - If T is smaller than χ^2_{k-1} , we fail to reject the null and claim the medians of k groups are equal
- May also utilize the p-value to make the decision as well
  - Best to use software to run the p-value calculation
- If the distributions of k populations are not identically shaped or there are outliers in the distribution, Mood's median test is a more robust than Kruskal-Wallis

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3.5.3 Mood's Median

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Moods Median

Criteria:

- Continuous Data
- Not Normally Distributed
- Median
- ≥ 2 Samples
- Not Symmetrical
- Not Matched samples
- Unequal Variances
- Outliers

Mood's median test is a statistical test to compare the medians of two or more populations

- Mood's median is an alternative to Kruskal-Wallis
- For data with outliers, Mood's median test is more robust than the Kruskal-Wallis test
- It is the extension of one sample sign test
  - H_0: $\eta_1 = \ldots = \eta_k$
  - H_A: At least one of the medians is different from the others

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Mood's Median: Calculating Test Statistic

- Group the k samples from k populations into one single data set
- Find the median of the combined data set
- Separate the data in each sample into two groups
  - ➢ One consists of all the observations with values higher than the grand median
  - ➢ The other consists of all the observations with values lower than the grand median

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Mood's Median: Decision Rules

- Use Pearson's chi-square test to compare the actual number of observations less than and greater than the grand median to what is expected if the populations were the same
  - Run a Pearson's chi-square test to determine whether to reject the null hypothesis
    - H_0: $\eta_1 = \ldots = \eta_k$
    - H_A: At least one of the medians is different from the others
  - May also utilize the p-value to make the decision as well
    - Best to use software to run the p-value calculation
- If the distributions of k populations are not identically shaped or there are outliers in the distribution, Mood's median test is a more robust than Kruskal-Wallis

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3.5.4 FRIEDMAN

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Friedman

Criteria:

- Continuous Data
- Not Normally Distributed
- Median
- 2 Samples
- Not Symmetrical
- Matched samples

The Friedman test is a hypothesis test used to detect the differences in various groups across multiple attempts (e.g., trials)

- A unique grouping of attempts and the corresponding values are known as a block
- Results in one block are independent of the results in another
  - H_0: The treatments have identical effects (i.e., $\eta_1 = \ldots = \eta_k$)
  - H_A: At least one of the treatments has different effects or median (η) that is statistically different from others

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Friedman: Formula for Test Statistic

- Formula for the Friedman test statistic (Q)
  - ➢Use software
- Standard hypothesis decision rules
  - ➢Calc vs. Crit
  - ➢P-value vs. α

$$\bar{r}_j = \frac{1}{n}\sum_{i=1}^{n} r_{ij}$$

$$\bar{r} = \frac{1}{nk}\sum_{i=1}^{n}\sum_{j=1}^{k} r_{ij}$$

$$Q = \frac{n\sum_{j=1}^{k}(\bar{r}_j - \bar{r})^2}{\frac{1}{n(k-1)}\sum_{i=1}^{n}\sum_{j=1}^{k}(r_{ij} - \bar{r})^2}$$

n : # of blocks

r_{ij} : specific value where a block and treatment intersect

j : specific treatment

k : # of treatments

i : specific block

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3.5.5 1 SAMPLE SIGN

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1-Sample Sign

Criteria:

- Continuous Data
- Not Normally Distributed
- Median
- 1 Sample
- Not Symmetrical

It is an alternative test of one sample t-test when the distribution of the data is non-normal

- The one sample sign test is a hypothesis test to compare the median of a population with a specified value
  - Null Hypothesis (H0): $\eta = \eta 0$
  - Alternative Hypothesis (Ha): $\eta \neq \eta 0$

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One Sample Sign: Test Statistic Steps

- Separate the sample set of data into two groups
  - One with values greater than the hypothesized median η_0
  - The other with values less than the hypothesized median η_0
- Count the number of observations in each group
- Calculate the test statistic
  - If the null is true, the number of observations in each group should not be significantly different from half of the total sample size
  - The test statistic follows a binomial distribution when the null is true
  - When n is large, use the normal distribution to approximate the binomial distribution

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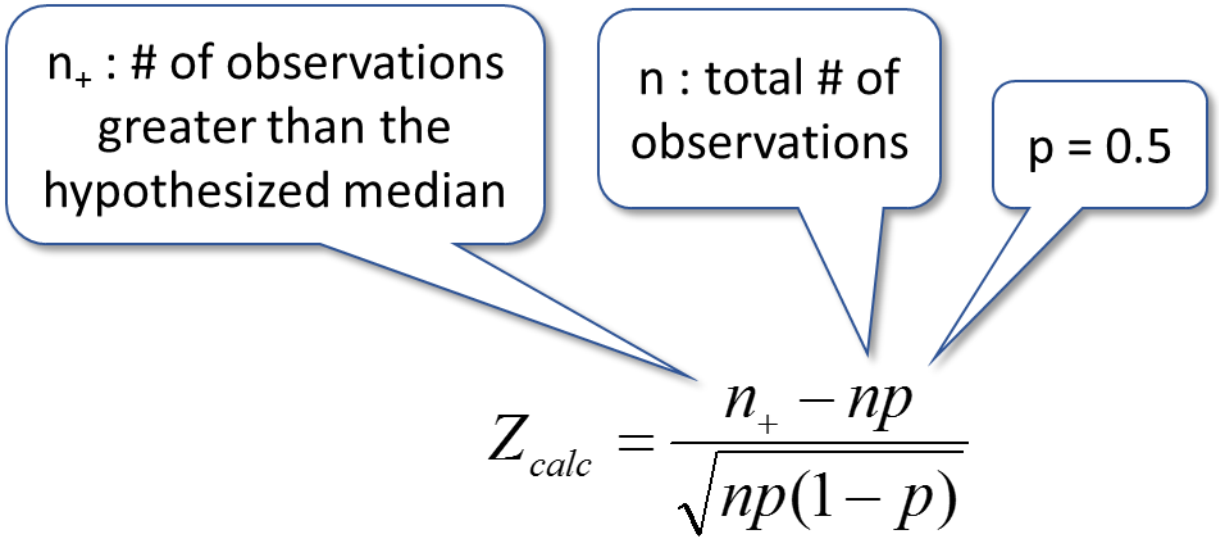
One Sample Sign: Calculating Test Statistic

- Test Statistic (Z)
- Standard hypothesis decision rules
  - Calc vs. Crit
  - P-value vs. α



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3.5.6 1 SAMPLE WILCOXON

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1-Sample Wilcoxon

Criteria:

- Continuous Data
- Not Normally Distributed
- Median
- 1 Sample
- Symmetrical

It is an alternative test of one sample t-test when the distribution of the data is non-normal

- It is more powerful than one sample sign test, but it assumes the distribution of the data is symmetric
- The one sample Wilcoxon test is a hypothesis test to compare the median of one population with a specified value
  - ➢ H_0: $\eta = \eta_0$
  - ➢ H_A: $\eta \neq \eta_0$

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One Sample Wilcoxon: Test Statistic

Create the following columns one by one:

- Column 1: all the raw observations (X)
- Column 2: the differences between each observation value and the hypothesized median ($X - \eta_0$)
- Column 3: the signs (+ or –) of column 2
- Column 4: the absolute value of column 2
- Column 5: the ranks of each item in column 4 in ascending order
- Column 6: the product of column 3 and column 5

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One Sample Wilcoxon: Decision Rules

- Calculate the test statistic W_{calc} , which is the sum of all the non-negative values in column 6

- Use the table of critical values for the Wilcoxon test to get the W_{crit}

- Utilize standard hypothesis decision rules for rejecting or failing to reject the null
  - ➢Calc vs. Crit
  - ➢P-value vs. α

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3.5.7 One & Two Sample Proportion

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One Sample Proportion

Criteria:

- Discrete Data
- 1 Factor
- Not Normally Distributed
- 1 Sample

One sample proportion test is a hypothesis test to compare the proportion of one certain outcome occurring in a population with a specified proportion

- H_0: $p = p_0$
- H_A: $p \neq p_0$
- Population follows a binomial distribution

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One Sample Proportion: Test Statistic

- Test Statistic: Z
  - When $np \geq 5$ and $np(1-p) \geq 5$, we use normal distribution to approximate the underlying binomial distribution of the population
  - Standard hypothesis decision rules
    - Calc vs. Crit
    - P-value vs. α

The observed probability of one certain outcome occurring

hypothesized probability

$$Z_{calc} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

number of trials

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Two Sample Proportion

Criteria:

- Discrete Data
- 1 Factor
- Not Normally Distributed
- 2 Sample

The two-sample proportion test is a hypothesis test to compare the probability of a certain event occurring in two populations

- There are only two possible outcomes in each trial for both populations: success/failure, yes/no, defective/non-defective, etc.
  - H0: p1 = p2
  - HA: p1 ≠ p2
- The underlying distributions of both populations are binomial distribution

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One Sample Proportion: Test Statistic

- Test Statistic: Z
  - When $np \geq 5$ and $np(1-p) \geq 5$, we use normal distribution to approximate the underlying binomial distribution of the population
  - Standard hypothesis decision rules
    - Calc vs. Crit
    - P-value vs. α

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_0\left(1-\hat{p}_0\right)\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$

observed proportions of events in the two samples

$$\hat{p}_0 = \frac{x_1 + x_2}{n_1 + n_2}$$

number of trials in the two samples respectively

number of events in the two samples respectively

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3.5.8 Chi-Squared (Contingency Tables)

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Chi-square Test (Contingency Table)

Criteria:

- Discrete Data
- 1 or 2 Factors
- Not Normally Distributed
- > 2 Samples

A chi-square test is a hypothesis test in which the sampling distribution of the test statistic follows a chi-square distribution when the null hypothesis is true

- Different than the chi-square test used for testing variances
- There are multiple chi-square tests (contingency tables) available
  - One tests equality of proportions, using 1 factor
  - Another use for this test is evaluating the independence of 2 factors

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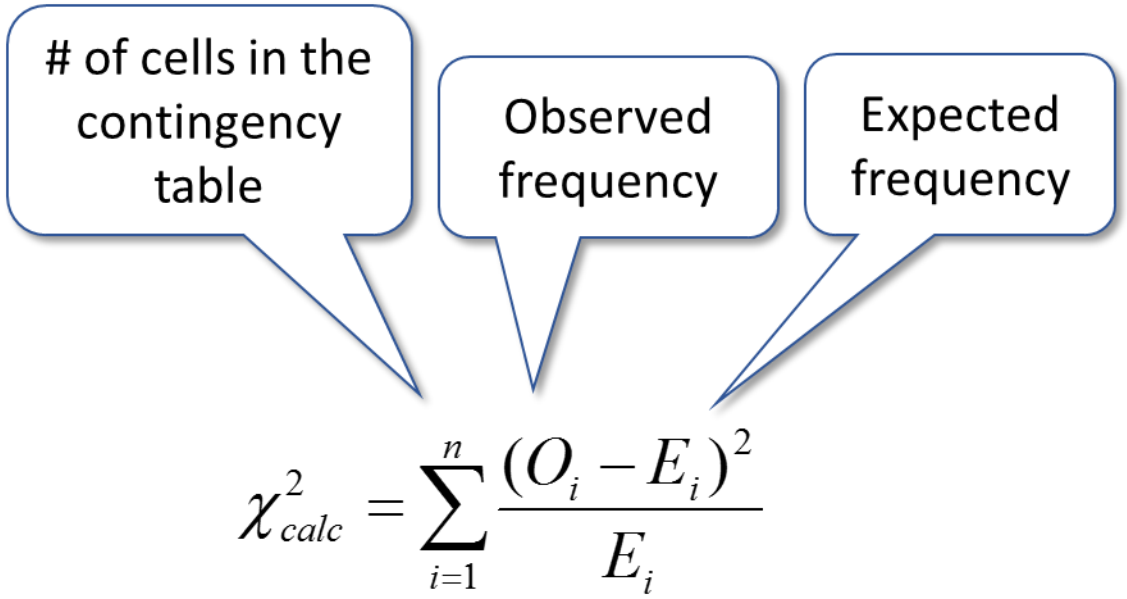
Chi-Square: Test Statistic

- Test Statistic: χ^2
- Both objectives, 1 and 2 factors, use the same general test statistic formula
- Standard hypothesis decision rules
  - Calc vs. Crit
  - P-value vs. α



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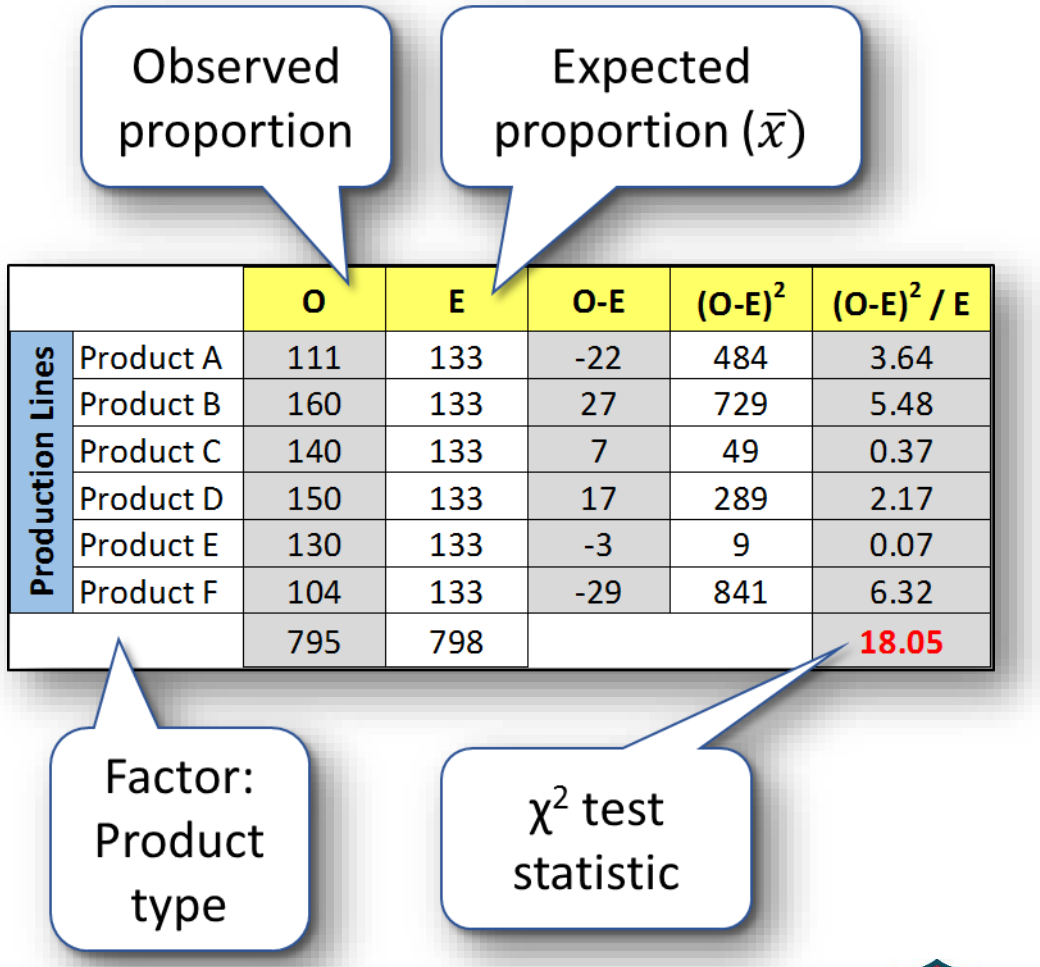
Contingency Table: 1 Factor

Testing for **equal proportions** using the example of:

- Product type is the 1 factor
- Different product types have the same yield (the proportion “p”)
- H_0: $p_1 = p_2 = \ldots = p_k$
- H_A: At least one of the proportions is different from others



		O	E	O-E	$(O-E)^2$	$(O-E)^2 / E$
Production Lines	Product A	111	133	-22	484	3.64
	Product B	160	133	27	729	5.48
	Product C	140	133	7	49	0.37
	Product D	150	133	17	289	2.17
	Product E	130	133	-3	9	0.07
	Product F	104	133	-29	841	6.32
		795	798			18.05

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Contingency Table: 2 Factors

2 Factor: Testing for **independence** of two factors (meaning there is no relationship between them)

- In other words, it can be used to test whether there is any statistically significant relationship between two discrete factors
- Example: Facility has no impact on product line yield (facility and product are the 2 factors)
- H_0: Factor 1 is independent of factor 2
- H_A: Factor 1 is not independent of factor 2

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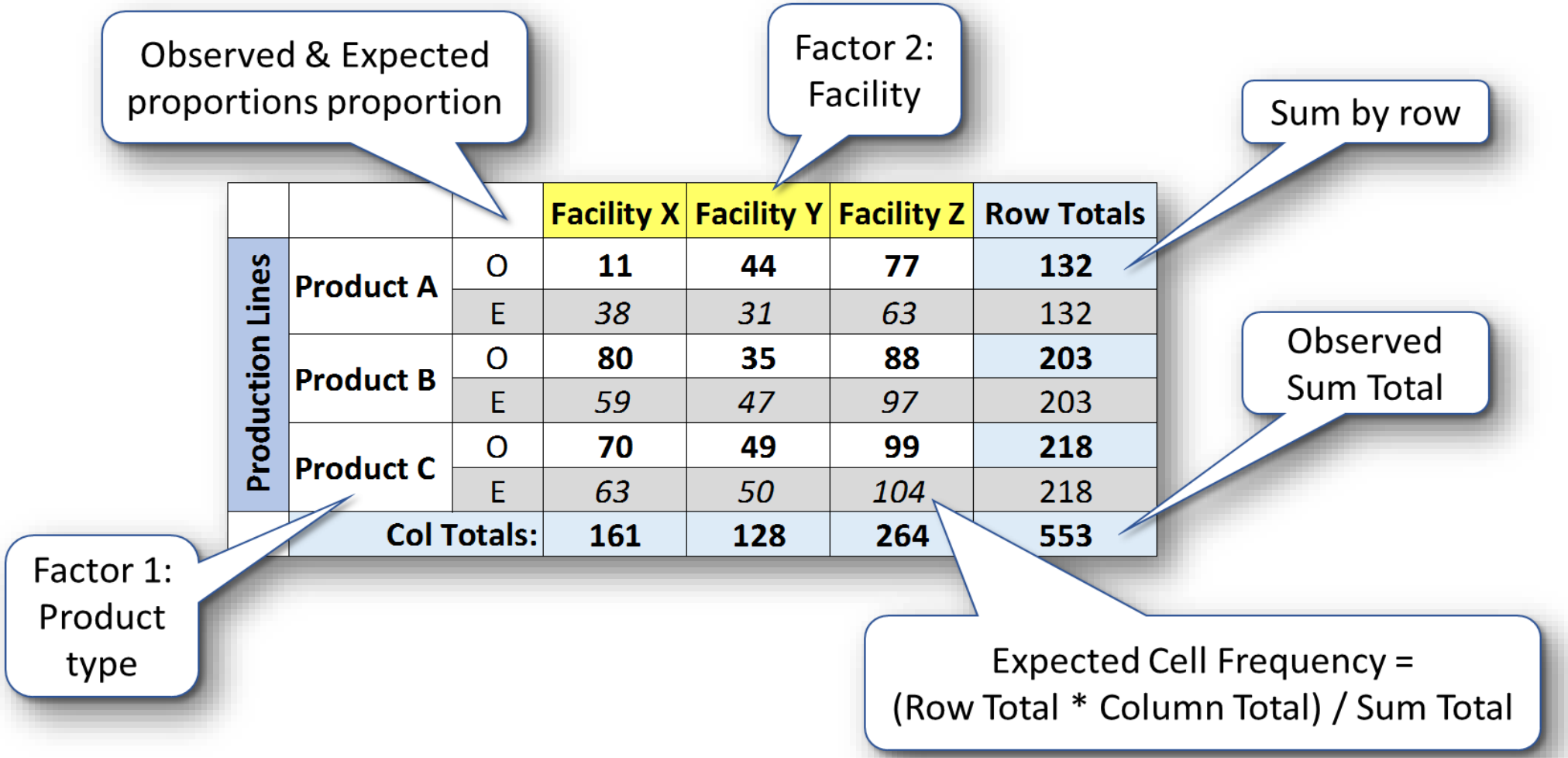
Contingency Table: 2 Factors



			Facility X	Facility Y	Facility Z	Row Totals
Production Lines	Product A	O	11	44	77	132
		E	*38*	*31*	*63*	132
	Product B	O	80	35	88	203
		E	*59*	*47*	*97*	203
	Product C	O	70	49	99	218
		E	*63*	*50*	*104*	218
	Col Totals:		161	128	264	553

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Chi-Square: 2 Factor Test Statistic

Observed proportion

Expected proportion

O	E	O-E	$(O-E)^2$	$(O-E)^2 / E$
11	38	-27.43	752.426	19.58
44	31	13.45	180.813	5.92
77	63	13.98	195.545	3.10
80	59	20.90	436.757	7.39
35	47	-11.99	143.696	3.06
88	97	-8.91	79.413	0.82
70	63	6.53	42.662	0.67
49	50	-1.46	2.130	0.04
99	104	-5.07	25.729	0.25
	553.00			**40.83**

χ^2 test statistic

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Chi-Square: Decision Rules

- To use the χ^2 table to identify the χ^2_{crit} , you need to know:
  - Alpha (α)
  - Degrees of Freedom (df)
    - For 1 Factor: **df = (k-1)**; where k = sample size or categories
    - For 2 Factor: **df = (# of rows - 1) * (# of columns -1)**
- Standard hypothesis decision rules apply to both χ^2 tests (1 & 2 Factors)
  - Calc vs. Crit
  - P-value vs. α

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THANK YOU

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4.0 Improve

The “Improve” phase is about:

- ✓ **Redesigning the process to achieve improved capability**
  - **Improvements stem from an increased focus on root causes and the application of various Lean tools**

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Improve Phase

Selecting Improvements

- Impact
- Cost
- Time
- Roadblocks

Root Cause Analysis

- Cause & Effect / Fishbone Diagram

Correlation & Linear Regression

- Correlation
- Scatter Plot / X-Y Diagram
- Regression Analysis
- Residuals Analysis
- Multiple Linear Regression
- Data Transformation / Box Cox

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Improve Phase

Design of Experiments (DOE)

- Terminology
- OFAT
- DOE Objectives, Methods
- Full Factorial Experiments
- Fractional Factorial Experiments

Lean Controls

- 5S
- Visual Factory
- Standardized Work
- Kaizen
- Kanban
- Poka-Yoke

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Selecting Improvements

- High-level objective evaluation
  - Impact
  - Cost
  - Time
- Balance desire with quantifiable evaluation
  - Examples: Gold standard, Inventory, Capacity

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Impact Considerations

- Long-term vs. Short-term effectiveness
  - Greatest urgency
  - Best solution or outcome for what's most important
- Effectiveness of the improvement types
  - Removing the Root Cause
  - Monitoring for the condition
  - Inspection
  - Training

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Cost Considerations

Costs to consider include:

- Initial
- Training
- Materials
- Resources
- Equipment
- Building space
- On-going

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Time Considerations

Time considerations include:

- Technical time constraints
  - Minimum time it would take
  - Time to build/create
  - Time to implement
- Competing priorities
- Waiting
  - To gain support or approval
  - Resource availability
  - Inspection

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Improve Phase: Roadblocks

- Potential roadblocks include:
  - Lack of data
  - Data is anecdotal
  - No time to collect data
  - Lack of participation in analysis planning
  - Lack of access to the process, etc.
- Plan to address roadblocks before they become problems

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4.1 Root Cause Analysis

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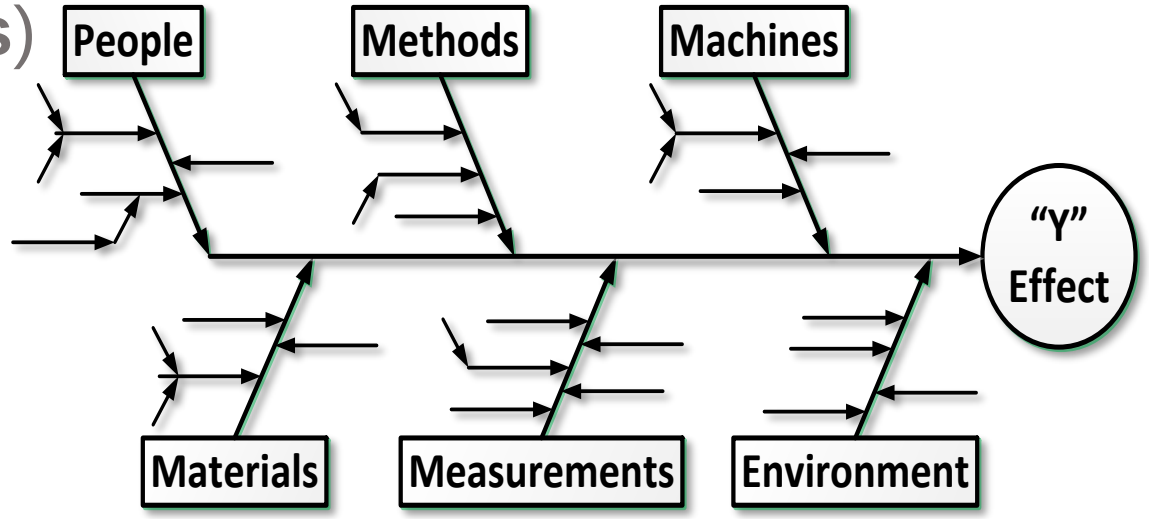
What is a Cause & Effect Diagram?

- A powerful brainstorming tool, the **cause and effect diagram** can be used to identify and organize possible causes of a specific effect or event
- It is sometimes referred to as a **fishbone** diagram or an **Ishikawa** diagram
- …with a finished layout that looks like a “fish skeleton”
- It shows the relationship between the **effect** (**output**) and **potential causes** (**inputs**)



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Major Categories of Potential Causes

- The potential, individual causes for an effect, may be organized into categories
- P4ME is an acronym that reflects the following set of categories:

People	People (Manpower) who are involved in the process
Methods	How the process is completed (e.g., procedures, policies, regulations, laws)
Machines	Equipment or tools needed to perform the process
Materials	Raw materials or information needed to do the job
Measurements	Data collected from the process for inspection or evaluation
Environment	Surroundings of the process (e.g., location, time, culture)

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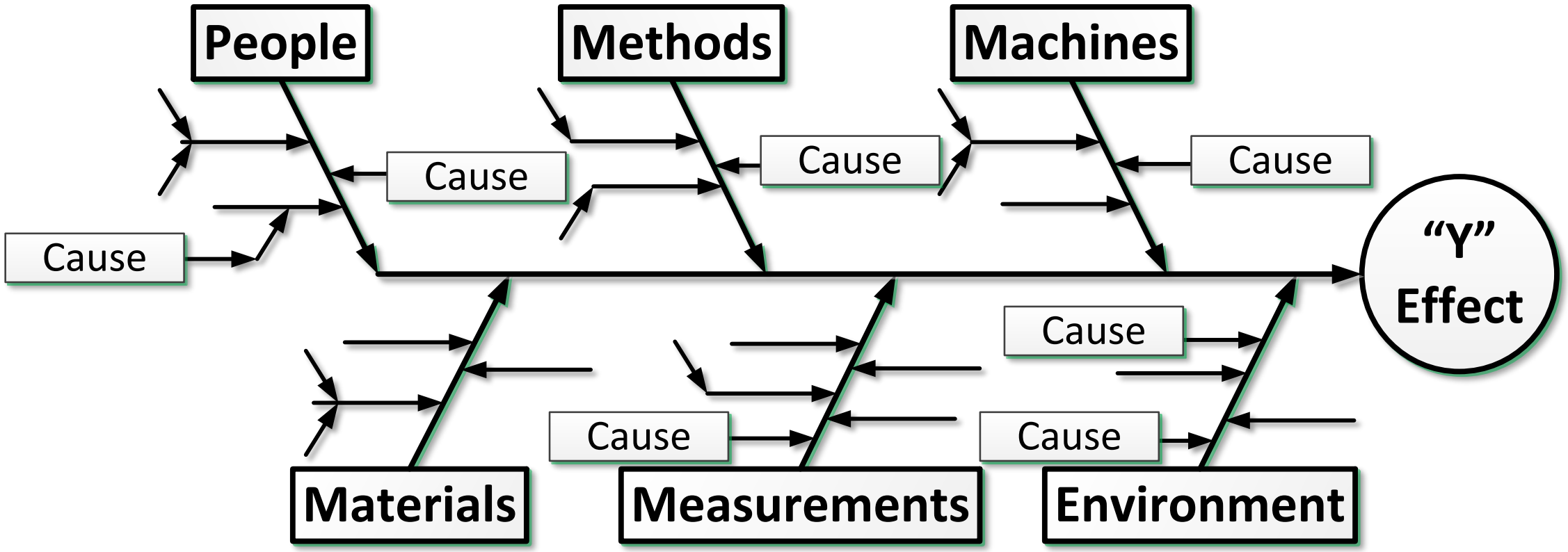
How to: Cause & Effect Diagram



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Analysis: Cause and Effect Diagram

- Use a Pareto chart to filter and focus on causes
- Identify causes with high impact that the team can take action upon
- Further statistical analysis is required to quantify the relationship between various factors and the effect and identify the root causes
- A cause and effect diagram only provides qualitative analysis of correlation between each cause and the effect

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4.2 Correlation & Linear regression
4.2.1 Correlation

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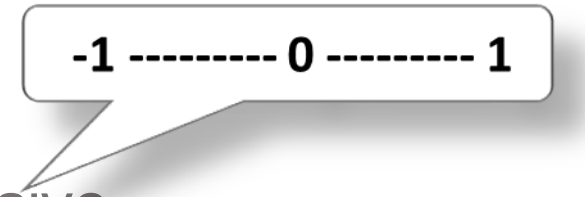
What is Correlation?

Pearson's correlation coefficient "r" is a statistical technique that helps describe:



- Inter-variable **linear relationship** (X and Y)
  - ➢Correlation Coefficient "r" range from **-1** to **1**, inclusive
- **Strength** / degree
  - ➢Absolute "|#|" value of "**r**" describes strength
- **Direction** / type
  - ➢Sign (+/-) of "r" indicates the direction
- (Suggests) whether one (x) can be used to **predict** another (Ŷ)

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Linear Relationship

- Pearson's correlation coefficient is only sensitive to the **linear** interaction
  - ➢Perfect, positive linear relationship: r = 1
  - ➢Perfect, negative linear relationship: r = -1
  - ➢No linear relationship: r = 0
- Common threshold for determining if a strong correlation exists between two variables is: **-.85 ≥ r ≥ .85**
- It is possible that two variables have a perfect non-linear relationship when the correlation coefficient is low

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Correlation vs. Causation

- Correlation does not imply causation!!

  - If variable A is highly correlated with variable B, it does not necessarily mean A **causes** B or vice versa

    - Possible that an unknown third variable C is causing both A and B to change
    - For example, if deaths increase as milk consumption increases. Where a separate, shared variable is the driving factor; such as population. Can be re-stated that as population increases so does milk consumption and the number of deaths

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Statistical Significance

- Perform a hypothesis test to determine whether the correlation coefficient is statistically different from zero
  - Hypothesis Test Statements:
    - **H_0: r = 0**: There is no significant correlation
    - **H_A: r ≠ 0**: There is a significant correlation
  - Analysis reminder:
    - p-value < 0.05, reject the null hypothesis and conclude that significant correlation exists
    - p-value > 0.05, fail to reject the null hypothesis and conclude that no significant correlation exists

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Calculating the Correlation Coefficient

- The formula "**CORREL**" in Excel calculates the sample correlation coefficient of two data sets

  - The correlation coefficient here shows a .98, a near perfect positive relationship

	A	B	C	D	E
1		KPIV	KPOV		
2		X	Y		
3		Distance	Speed		
4	Unit 1	2.2	9000		
5	Unit 2	1.81	7500		
6	Unit 3	1.9	8000		
7	Unit 4	1.44	5500		
8	Unit 5	1.8	7000		
9	Unit 6	1.4	5000		
10	Unit 7	1.3	4500		
11	Unit 8	1.77	6500		
12	Unit 9	2	8500		
13	Unit 10	1.5	6000		

=CORREL(B4:B13,C4:C13)

0.98

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Correlation Analysis: Next Steps

- **Scatter plot**, or **X-Y diagram**, can help reveal additional characteristics of the variables' relationship

  - Correlation coefficient does not replace the scatter plot in studying the variables' relationship

  - Correlation coefficient does not cover all dependency patterns

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4.2.2 Scatter Plot / X-Y Diagram

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X-Y Diagram / Scatter Plot

- X-Y diagrams are scatter plots qualitatively depicting the relationship between two variables
- Graphically displays the variables' relationship
- X is the horizontal axis; Y is the vertical axis
- Each point represents a corresponding pair of X and Y values
- Linear or Nonlinear correlation
- Strength and Direction
- Helps identify any outliers

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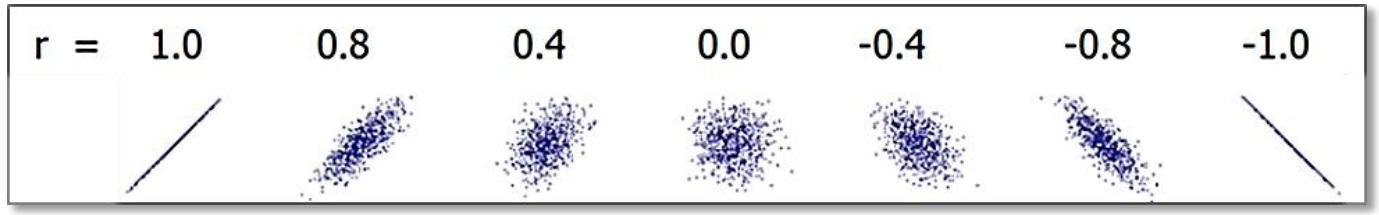
X-Y Diagram / Scatter Plot, cont.

- X = independent variable; Y = dependent variable
- Strong correlation: data points will be close together (tight)
- Weak correlation: data points are more spread apart (loose)
- The five types of correlation are:

  - Perfect positive
  - Moderate positive
  - None
  - Moderate negative
  - Perfect negative



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Limitations of the X-Y Diagram

- X-Y diagram does **not** provide any **quantitative** conclusions about the data
- Further statistical analysis, such as Pearson's "r," is needed to:

  - Quantitatively describe the pattern of the data
  - Assess whether the association between variables is statistically significant
  - Measure the strength of the relationship between variables
  - Determine whether outliers exist in the data

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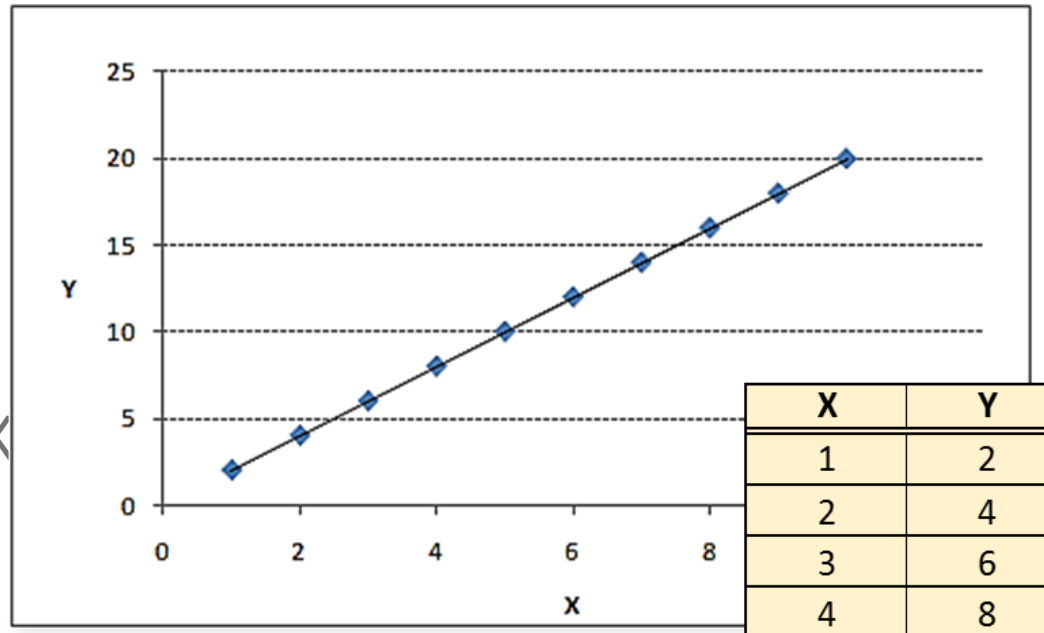
Perfect Linear Correlation

- This chart demonstrates a **perfect positive linear** correlation



X	Y
1	2
2	4
3	6
4	8
5	10
6	12
7	14
8	16
9	18
10	20

  - ➢ The relationship between X and Y can be perfectly "described" by a linear equation
  - ➢ Equation format is: **Y = a * X + b** (where a ≠ 0)

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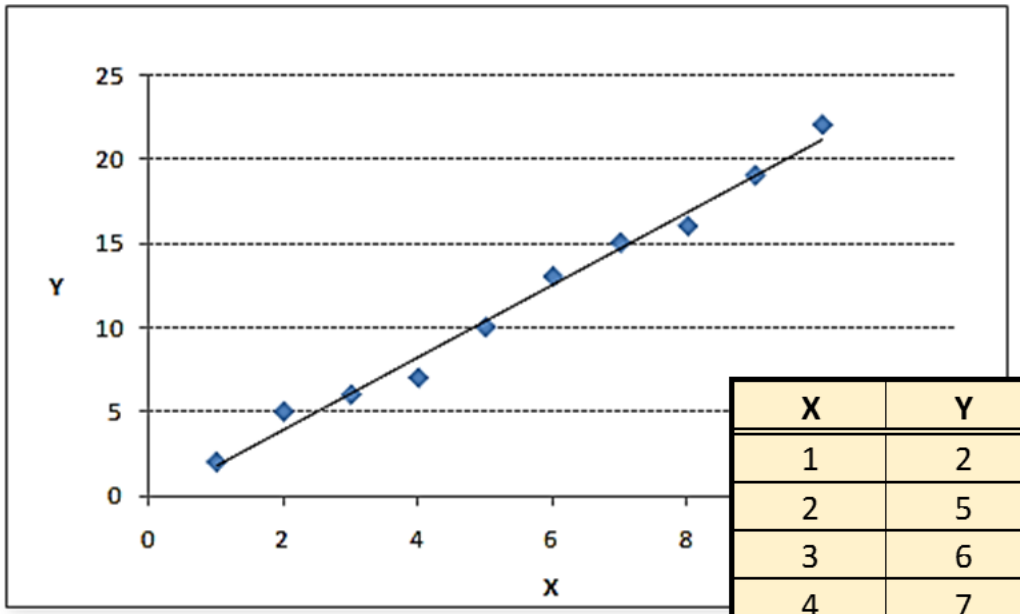
Strong Linear Correlation

- This chart demonstrates a **strong positive linear** correlation between X and Y

  - When X increases, Y increases accordingly
  - In this chart, the data points scatter closely around a straight line



X	Y
1	2
2	5
3	6
4	7
5	10
6	13
7	15
8	16
9	19
10	22

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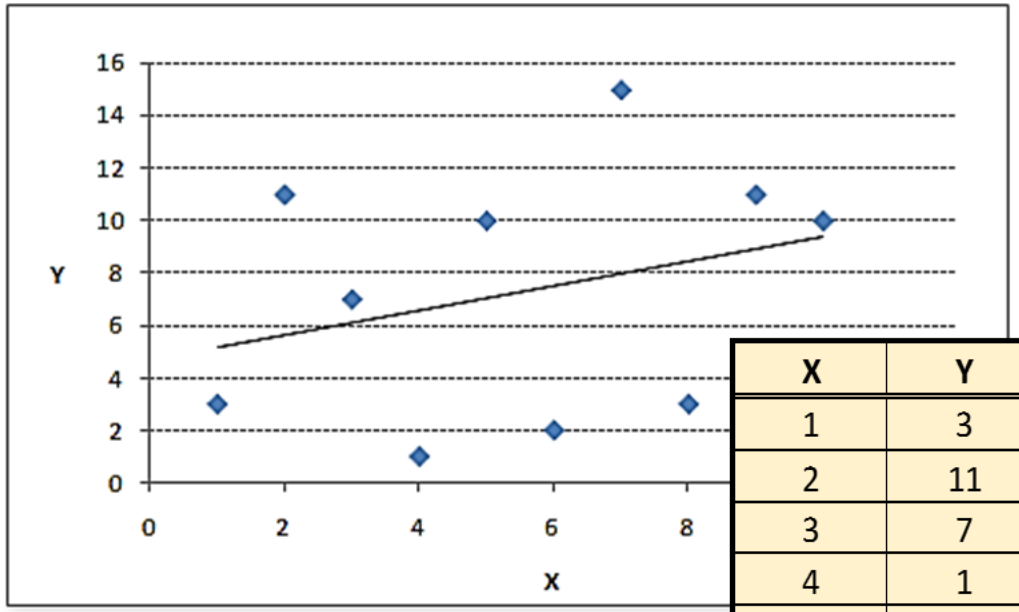
Weak Linear Correlation

- This chart demonstrates a **weak positive linear** correlation between X and Y

- The distance between the data points and the trend line is relatively far on average



X	Y
1	3
2	11
3	7
4	1
5	10
6	2
7	15
8	3
9	11
10	10

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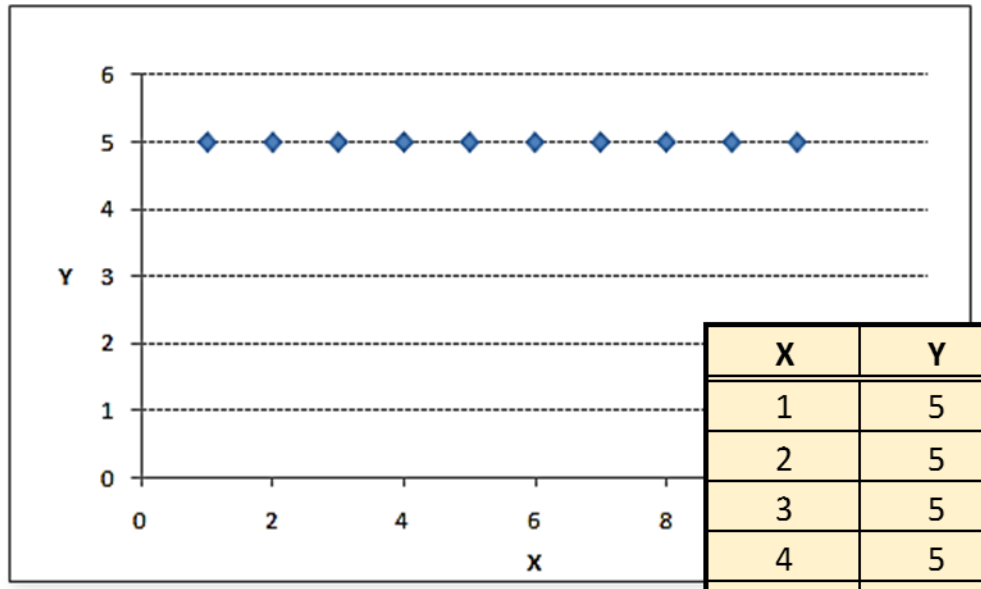
Uncorrelated

- In this chart, the Y value of each data point is a **constant** regardless of what the X value is

  - There is **no correlation** between X and Y
  - Changes in X do not show any relative impact on Y



X	Y
1	5
2	5
3	5
4	5
5	5
6	5
7	5
8	5
9	5
10	5

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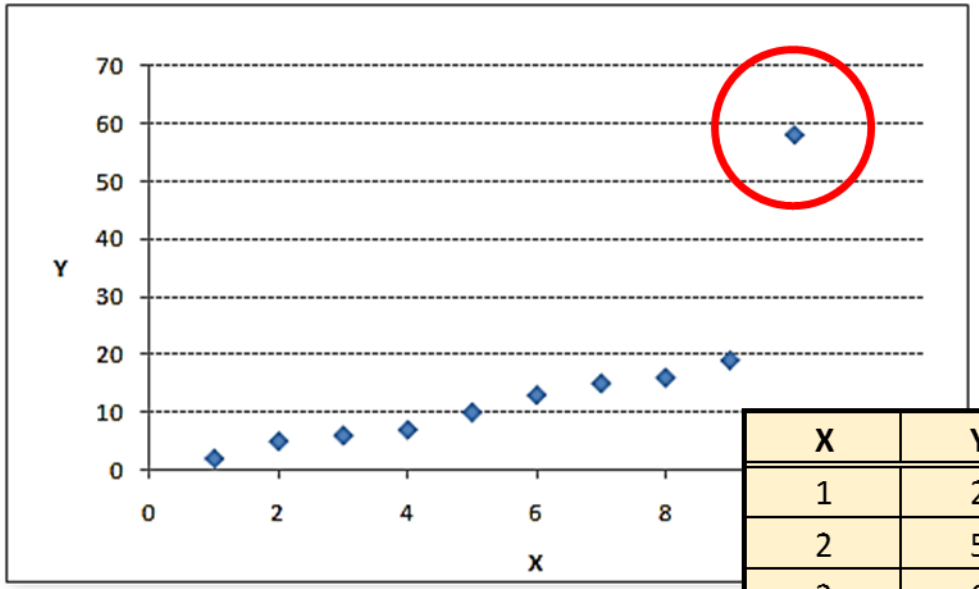
Outlier

- In this chart, the last data point does not seem to follow the trend of other data points

- Investigate that value further to determine if it is actually an **outlier**



X	Y
1	2
2	5
3	6
4	7
5	10
6	13
7	15
8	16
9	19
10	58

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4.2.3 Regression Analysis

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Simple Linear Regression (SLR)

- The correlation coefficient cannot answer the following questions:

  - **Degree** one variable changes when the other variable changes?
  - How to **obtain a "Y" target** value by manipulating the "X" input?
  - How to make "Y" output **predictions**?

- Simple linear regression (SLR) analysis helps to answer these questions

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What is Simple Linear Regression?

- Simple linear regression is a statistical technique to fit a straight line through the data points
  - ➢ “Simple” because only one predictor “X” variable is involved
  - ➢ “Line of best fit”
- “Models” the quantitative relationship between two continuous variables
  - ➢ Describes how one variable changes according to the change of another variable

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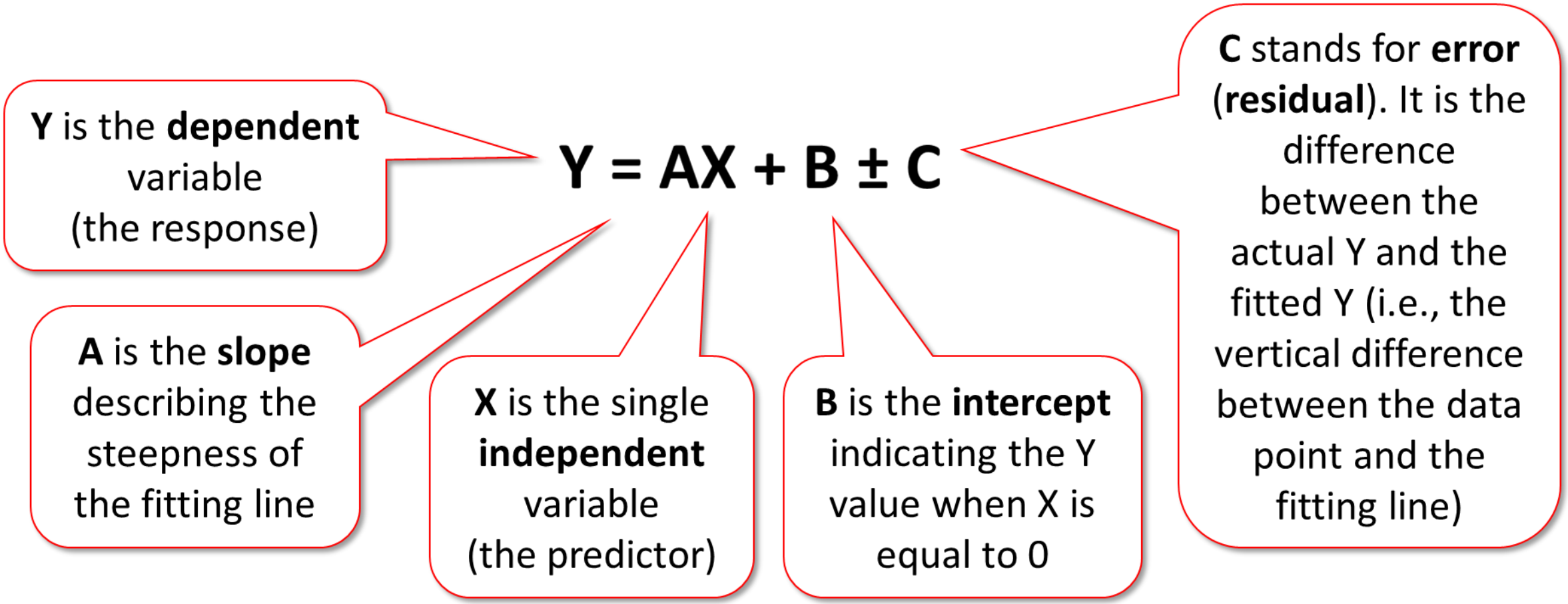
Simple Linear Regression Equation

Simple linear regression analysis fits the data to a regression equation in the form:



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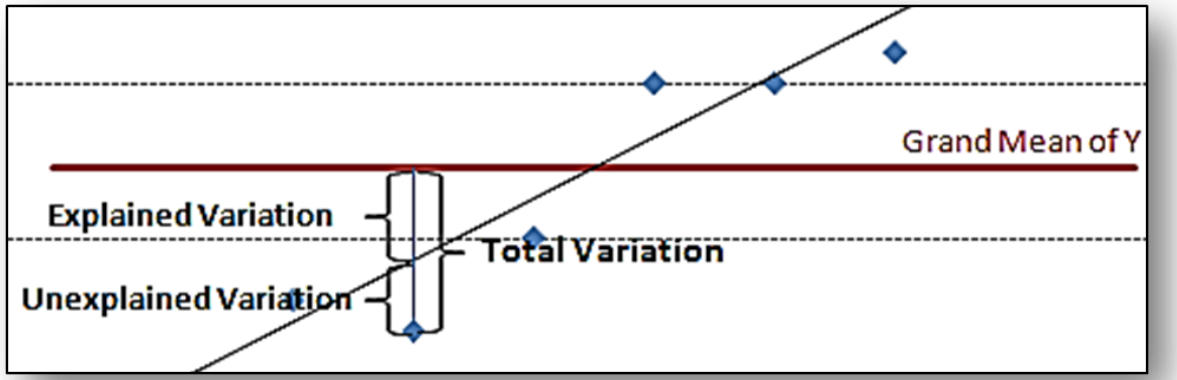
ANOVA in Simple Linear Regression

- **Total** Variation = Total Sums of Squares = SST $\sum_{i=1}^{n}(Y_i - \bar{Y})^2$
- **Explained** Variation = Regression Sums of Squares = SSR $\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2$
- **Unexplained** Variation = Error Sums of Squares = SSE $\sum_{i=1}^{n}(Y_i - \hat{Y})^2$



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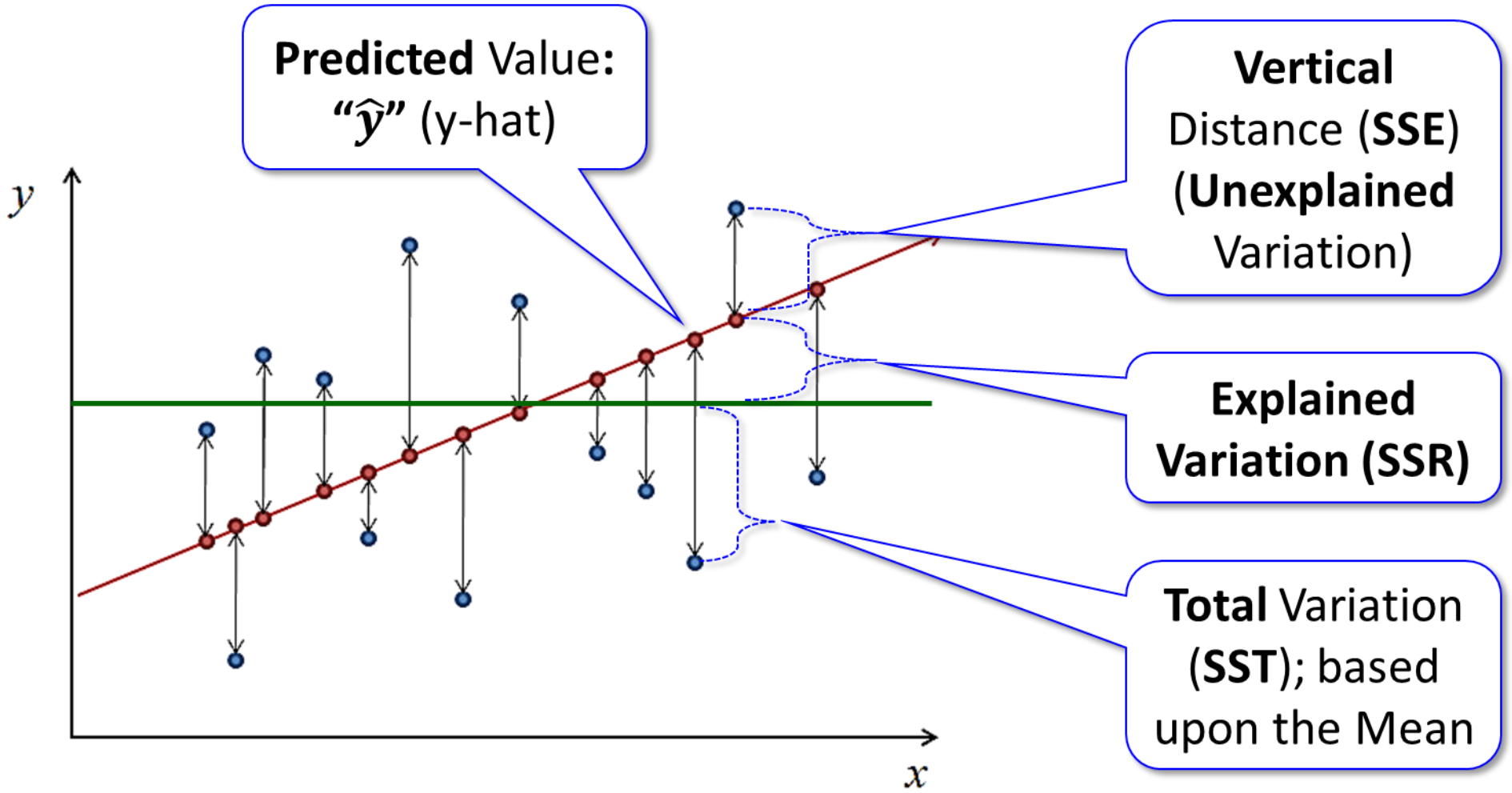
Refresher: Linear Regression



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Coefficient of Determination: R^2

- Coefficient of Determination "R^2" measures the proportion of variability in the data that can be explained by the model

- R^2 ranges from 0 to 1
  - Positive values because of the squaring
  - With the "R^2" target of .73, the corresponding "R" value is ±.85

- Higher R^2, the better the model (regression equation) can fit the actual data

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4.2.4 Residuals Analysis

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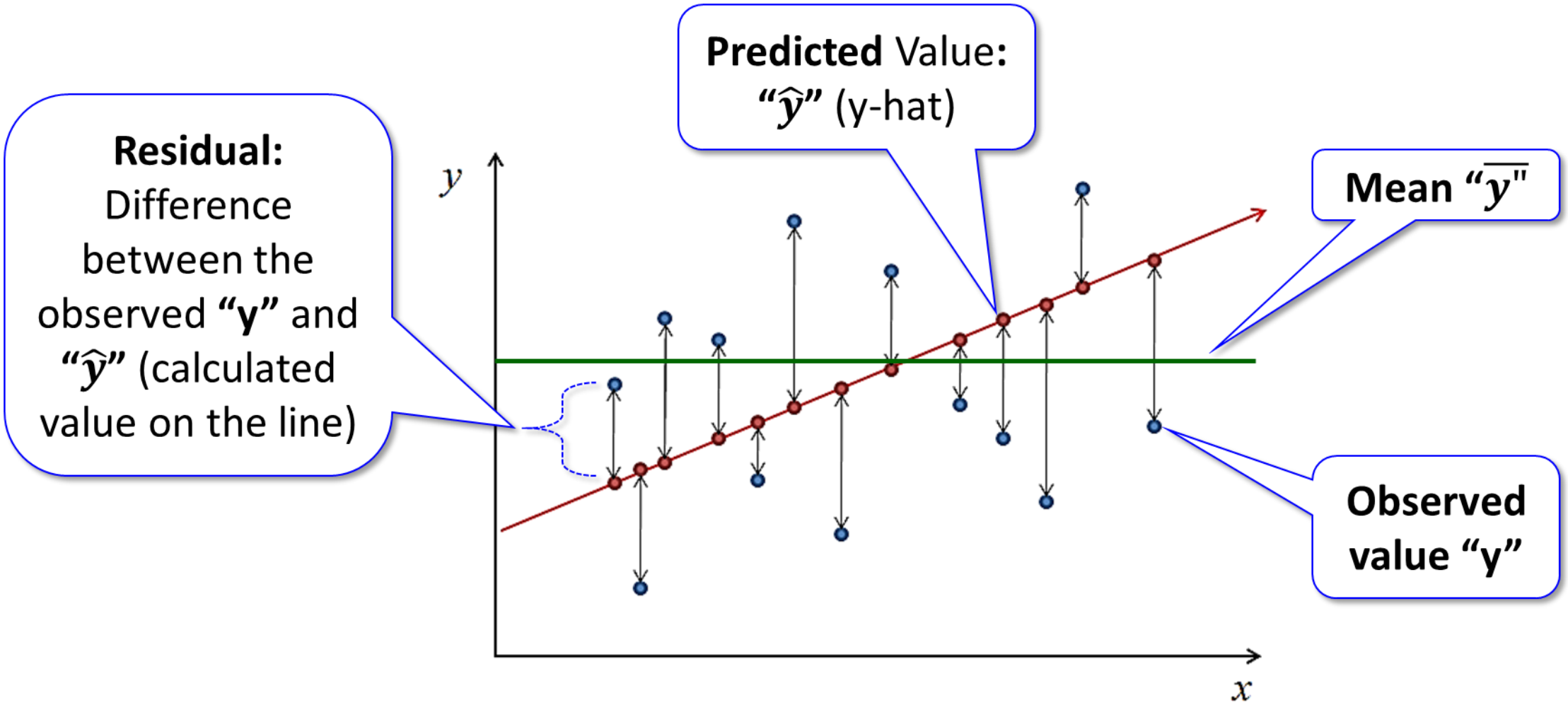
What Residuals Look Like



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Residual Formula

- Regression equation produces the “line of best fit” which consists of predicted values "$\hat{y}$” (y-hat)
- Differences between the predicted and observed values are known as residuals
- **Residual** = Observed (“**y**”) – Predicted ("$\hat{y}$”)

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Why Residuals Analysis?

- The quality of the linear regression model should be assessed by defining residuals and examining the residual plots
- Values of both sum and mean of the residuals are zero
- Indicates equally balanced / distributed on both sides of the regression line

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Residuals Requirements

- To ensure the reliability of the linear regression model, residuals (errors) must meet the following criteria:
  - Normally distributed, with the mean equal to zero
  - Independent
  - Constant variance
  - Underlying population relationship is linear
- If requirements are not met, a new model will need to be rebuilt
  - Replace predictors with new ones, adding new predictors, building non-linear models, etc.

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4.2.5 Multiple Linear Regression

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Multiple Linear Regression (MLR)

- Multiple Linear Regression evaluates multiple input (x_1, x_2, x_n) variable's effect on an output simultaneously
  - If R^2 is not as high as desired in the Simple Linear Regression
  - Process knowledge implies more than one input affects the output
- Simple Linear Regression assumptions stay the same for Multiple Linear Regression
  - One additional assumption for MLR is the independence of predictors (X's)

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4.2.6 Data Transformation, Box Cox

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Transforming Data

- Box Cox Transformation leverages a process to convert non-normal data into normally distributed data for the **sake of simplicity**
  - ➢Cases where data is Non-linear it is possible to perform Regression using transformed data
  - ➢Either the X or Y may be transformed
- This table consists of some ways in which data can be transformed

Exponent	Function
-2	$y^{-2} = 1/y^2$
-1	$y^{-1} = 1/y^1$
-0.5	$y^{0.5} = 1/y^{0.5} = 1/\sqrt{y}$
0	Log(y)
0.5	$y^{0.5} = \sqrt{y}$
1	$y^1 = y$ (no transformation)
2	y^2

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Transforming Rules

- Must preserve the relative order of the data
- Must be a smooth and continuous function
- Most often useful when the ratio of largest to smallest value is greater than two
- In most cases, the transform will have little effect when this rule is violated
- All external reference points (spec limits, etc.) must use the same transform

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Transforming Advantages vs. Disadvantages

- **Advantages** of transforming data:
  - Linear Regression is easier to visually understand and manage
  - Where required, Non-normal Data can be changed to resemble Normal Data for statistical analysis

- **Disadvantages** of transforming data:
  - Difficult to understand transformed units
  - Difficult without automation or computers

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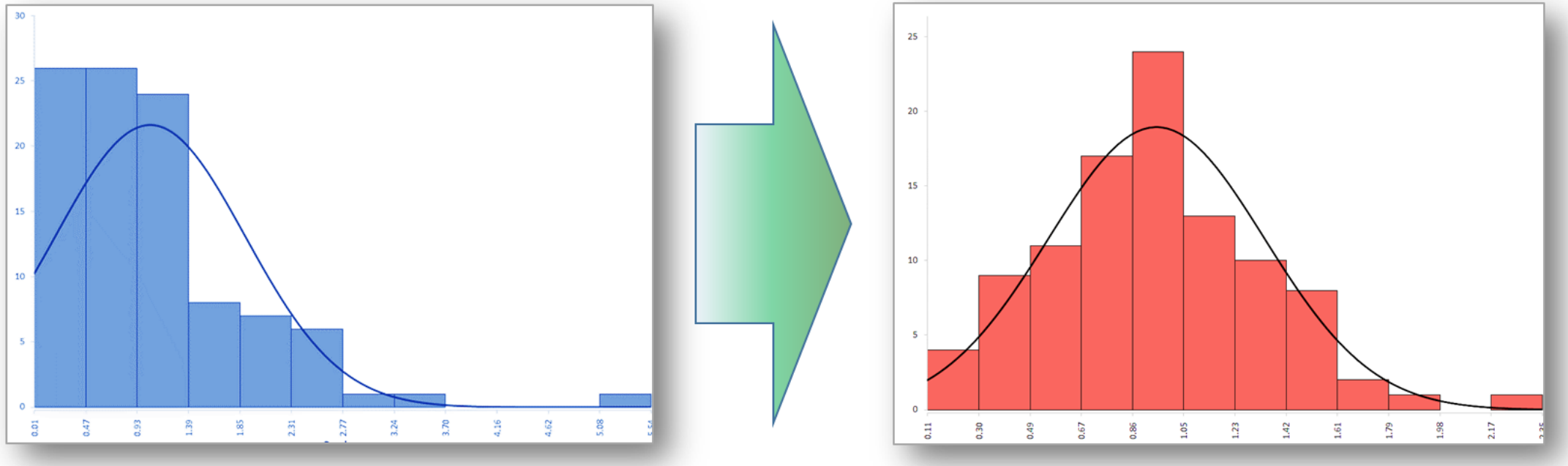
Effect of Transformation

- The transformed data now shows a Normal Distribution



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4.3 Design of Experiments (DOE)

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Terminology

TERM	DESCRIPTION
Independent variables	Factors intentionally varied by the experimenter.
Dependent variables	Responses that vary as a result of changes made to the independent variable.
Response	Outcome of an experimental treatment that varies as changes are made to levels and factors.
Factors	Independent variables changed during an experiment to validate their impact on the output.
Levels	Values, settings, or conditions of the factors that are tested during the experiment.
Treatment Combination	Combination of factor levels whose effect on the response variable is of interest.
Error	Variation in experimental units that have been exposed to the same treatment. This variability is due to uncontrollable factors.
Experimental unit	Quantity of material to which one trial of a single treatment is applied.
Confounded	Exists when the effects of two factors are not separable, or distinguishable
Interaction	Occurs when the effect of one input factor depends upon, or impacted by, the level of another input factor

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Terminology, cont.

TERM	DESCRIPTION
Blocking	Organizational method of grouping treatments at specific level settings to ensure all variations are tested, and tested as needed
Replication / Replicate	Entire experiment is repeated with a change in the setting of experimental conditions between trials; Shows long-term variability
Repetition	During one experimental setup, several samples are run without changing the setting; Shows short-term variability
Principle of Parsimony	Use the minimum amount of data to discover the maximum amount of information
Main Effect	Average change in the response resulting from the change in the levels of a factor
Interaction Effect	Average change in the response resulting from the change in the interaction of multiple factors
Run	Number of experiments in a DOE setting
Design Generator	A Design Generator is the mathematical definition for how to begin aliasing a Full Factorial to create a Fractional Factorial
Noise	Uncontrollable variations from the inputs

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Factors

- Factors are the independent inputs (X's) that are controlled or set directly
- Many factors (of input or process) can impact the output (Y response)
- Factors can be either qualitative or quantitative
- Quantitative factors are preferred, since they can be measured and set at specific levels (continuous data)
- Qualitative factors can only be rated (attribute data)

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Levels

- Values, settings, or conditions of the factors that are tested during the experiment
  - Example: specific temperature (factor) settings (level) on an oven, such as 100° F and 200° F
- Code of factor levels:
  - High vs. Low
  - (+1) vs. (-1)
  - (+) vs. (-)
  - e.g. 100° F would be the Low and 200° F the High

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One factor at a time (OFAT)

Changing just one factor at a time, has shortcomings:

- Too many experiments are necessary
- The optimum combination of all the variables may never be revealed
- The interaction between factors cannot be determined
- Conclusions may be wrong or misleading
- Many of the observed effects tend to be mysterious
- Time and effort may be wasted by studying the wrong variables or obtaining improper data

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4.3.1 Designed Experiments

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Design of Experiments (DOE)

- Design of Experiments (DOE, DOX, or experimental design) is a scientific method of planning and conducting an experiment that will yield the true cause-and-effect relationship between the X variables and the Y variables of interest
  - ➢ Objective is to find the set of factors and corresponding levels to produce the optimal output
- DOE allows the experimenter to study the effect of many input variables that may influence the product or process simultaneously, as well as possible interaction effects (for example synergistic effects).
- One Factor at a Time (OFAT) is an experimental style but not a planned experiment or DOE

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Advantages of DOE

- Many factors can be evaluated simultaneously, making the DOE process economical
- Input factors can be controlled to make the output insensitive to noise factors
- In-depth, statistical knowledge is not always necessary to get big benefit
- One can look at a process with few experiments

(Cont.)

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Advantages of DOE

- Since the designs are balanced, there is confidence in the conclusions
- If important factors are overlooked in an experiment, the results will indicate that fact
- Precise statistical analysis can be run using standard computer programs
- Results can be improved with low costs

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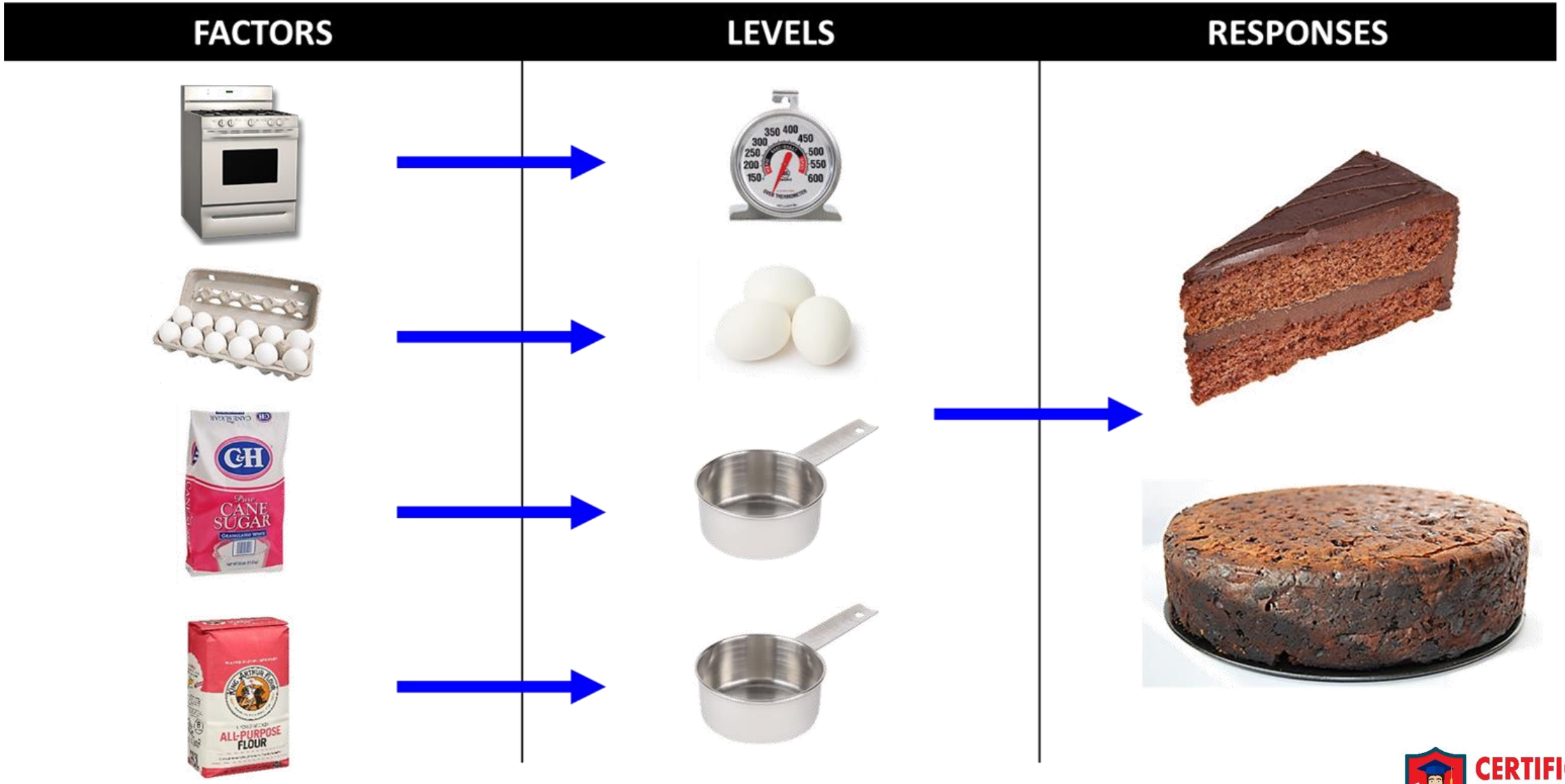
DOE Example



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Types of Designed Experiments

- Full Factorial
- Fractional Factorial
- Taguchi’s designs
- Plackett Burman’s designs
- Response surface designs

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Experiment Objectives

- **Active Learning:**
  - ➢Learn as much as possible with as little resources as possible (efficiency)

- **Identify Critical X's:**
  - ➢Identify the critical factors that drive the output or dependent variable ("Y")

(Cont.)

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Experiment Objectives

- **Quantify Relationships:**
  - Generate an equation that characterizes the relationship between the X's and the Y
  - The end result of a DOE is a mathematical function to describe the results of the experiment
- **Optimize:**
  - Determine the required settings of all the X's to achieve the optimal output or response.
- **Validate:**
  - Prove results through confirmation or pilot implementation before fully implementing solutions

Models typically reflect errors given by the "ε" in the following equation

$$Y = f (x_1 + x_2 + x_3 + \ldots + x_k + \varepsilon)$$

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DOE Methodology

1. Quantifiably define the problem
2. Establish the Experimental Objective
3. Select the Output (response) Variables “Y” : Primary Metric
4. Select the Input (independent) Variables “X’s” : KPIV
5. Choose the Levels for the Input Variables (Factors)

Preserve all the raw data - not just averages!

(Cont.)

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DOE Methodology

6. Select the specific Experimental Design
7. Plan and Execute the experiment and Collect Data
8. Analyze the data from the designed experiment and draw Statistical Conclusions
9. Draw Practical Solutions
10. Replicate or validate the experimental results
11. Implement Solutions

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4.3.2 Full Factorial Experiments

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Full Factorial Experiments

In a full factorial experiment, **all** the possible combinations of factors and levels are created and tested

- Full Factorial designs are used when:
  - There are five or fewer factors
  - You know the critical factors and need to explain interactions
  - Optimizing processes

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2^k Full Factorial Designs

- Two-Level is the most popular DOE
  - ➢When dealing with “k” factors, each with 2 levels, there are 2^k treatments
    - For example, for two-level design (i.e., each factor has two levels) with k factors, there are 2^k possible scenarios or treatments.
    - 2^k Full factorial – 2 factors are tested at k levels completely
      - 2^2 Full factorial – 2 factors are tested at 2 levels completely = $2^2 = 4$
    - 3^k Full factorial – 2 factors are tested at k levels completely
      - 3^2 Full factorial – 3 factors are tested at 2 levels completely = $3^2 = 9$
- Also known as “ L^F ” designs

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Linear & Quadratic Mathematical Models

Linear DOE

- If we assume the relationship between the response and the factors is linear, we use a two-level design experiment in which there are two settings for individual factors
- **Center Points**
  - A method to check for adequacy of linear model; e.g. if a line connecting factor settings passes through the center of the design, the model is adequate to predict within the inference space

Quadratic DOE

- To test the non-linear relationship between the response and the factors, we need at least a three- level design, i.e., quadratic design of experiment

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Balanced & Orthogonal Designs

*Keep experiments Balanced and Orthogonal:

Balanced Design

- Experiment design is balanced if individual treatments have the same count of observations
- Each level (high or low) of each factor is run the same number of times

Orthogonal Design

- Main effect of one factor can be evaluated independently of other factors
- Each level of a factor, the count of high setting and low setting in any other factors must be the same
- **How to check: the sum of products for each run should equal zero**

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4.3.3 Fractional Factorial Experiments

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Fractional Factorial Experiments

- A fractional factorial experiment is a subset of a full factorial experiment
- Fractional factorials use fewer treatment combinations and runs
- Fractional Factorials are often referenced as “screening experiments”
  - Fewer runs with larger number of factors
- Fractional Factorials are usually done in early stages of the improvement process
- Fractional factorials are less able to determine effects because of fewer degrees of freedom available to evaluate higher order interactions
- Fractional factorials can be used to screen a larger number of factors
- Fractional factorials can also be used for optimization

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Fractional Factorial Designs

2^{k-1} Fractional Factorial – k factors are tested at 2 levels at (k-1) combinations

- 2^4 Fractional Factorial – 4 factors are tested at 2 levels at 3 combinations
- 2^5 Fractional Factorial – 5 factors are tested at 2 levels at 4 and 3 combinations
- 2^{k-2} Quarter Fractional Factorial – k factors are tested at 2 levels at (k-2) combinations

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Alias & Design Generation

Run	Treatment	Factor			2-Way Interaction			3-Way Interaction
		A	B	C	AB	BC	AC	ABC
1	c	-1	-1	+1	+1	-1	-1	+1
2	a	+1	-1	-1	-1	+1	-1	+1
3	b	-1	+1	-1	-1	-1	+1	+1
4	abc	+1	+1	+1	+1	+1	+1	+1

Example: A has the same values (aliased) as BC, A is known as the alias of BC

Note: Design Generators can capitalize on aliases by turning Full Factorial experiments into Fractional Factorial experiments, thus reducing time and resources

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Confounding & Resolution

Confounding

- The probability of confounding increases with fractional factorial experiments because less data resulting from fewer runs
- When two input factors are aliases of each other, the effects they each have on the response cannot be separated
  - E.g. if A is the alias of BC, then is the effect because of A or the combination of BC?

Resolution

- Resolution is the measure or degree of confounding
- Higher resolution means less confounding, or merely confounding main effects with higher order interactions

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4.4 Lean Tools & Practices

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Lean Reminder: Elimination of Waste

- Waste can be a root cause of a Six Sigma project
- Muda specifics were discussed in the Define Phase
- The 7 basic elements of waste (muda) include:
  - ➢Transportation
  - ➢Inventory
  - ➢Motion
  - ➢Waiting
  - ➢Overproduction
  - ➢Over-Processing
  - ➢Defects

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The Vision of Lean Supporting Your Project

- Following are Lean concepts and tools used to Improve the overall success of the organization
- The Lean tools are most likely to succeed when the following, progressive model is in place:

  - 5S Workplace Organization
  - Visual Factory
  - Standardize Work
  - Kaizen
  - Kanban

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4.4.1 Lean: 5S, Visual Factory, & Standardized Work

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5S - Workplace Organization

- 5S is a systematic approach of cleaning and organizing the workplace
- 5S is the **starting point** for implementing improvements to a process
- Company and employee **commitment** is critical to maintaining improvements

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Lean: 5S

The “5” general categories are (**English** / Japanese):

- **Sort** / Seiri
- **Straighten** (Set in Order or Storage) / Seiton
- **Shine** (or Sweep) / Seiso (Seisou and Seis)
- **Standardize** (Systemize or Schedule) / Seiketsu
- **Sustain** (Self-Discipline) / Shisuke

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Sort

- **Review** all tools, parts, equipment, supplies, and material in the workplace
- **Categorize** items as necessary or unnecessary
- Store **frequently** used items at the work area
- Store **infrequently** used items away from the work area
- **Dispose** of items that are not needed
- Sort them in the order of **priority**

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Straighten (Set in Order or Storage)

- Create an **efficient** workflow
- **Arrange** work station so that all necessary equipment is near
- **Reduce** the **motion** and time required to locate items
- **Centralize** (location) sharable items (eliminated excess)
- Use labels, tape, floor markings, signs, and shadow outlines

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Shine (Sweep)

- **Clean** the workplace thoroughly
- Indicates a **quality** product and process
- Use as an **inspection** method
- Keep workplace **safe**
- **Prevent** machinery and equipment **deterioration**

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Sustain (Self-Discipline)

- **Commitment** and **discipline**
- Also translates as "do without being told"
- Develop **schedules**, **check lists**, **audits**
- **Training**
- **Practice** and **repeat** until it becomes a way of life
- Create the **culture** to adhere to the first four S’s consistently

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The Visual Factory

- **Visual notice of status**
- **Visual alert of variation**
- 5S standards are basis and foundation
- Provides clear indications of opportunities
- Examples: Sign boards, reader boards

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Standardized Work

- 5S and Visual Factory are pre-requisites
- **Standardized work provides consistency**
  - ➢Examples: Cycle times, assembly methods, batches of materials, etc.
- **Measurable**
  - ➢Consistency is required
- Required to sustain system benefits

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4.4.2 Lean:
Kaizen, Kanban, & Poka-Yoke

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Kaizen: Definition

Definition*: “The philosophy of **continuous improvement**, that every process can and should be continuously evaluated and improved in terms of time required, resources used, resultant quality and other aspects relevant to the process.”

* Note: Kaizen Definition from: All I Needed To Know About Manufacturing I Learned in Joe’s Garage. Miller and Schenk, Bayrock Press, 1996. Page 75.

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Kaizen & Kaizen Blitz

- Successful Kaizen efforts need the following:
  - ➢Management Support
  - ➢Measurable Process
  - ➢Analysis Tools
  - ➢Operator Support
- **Kaizen Blitz or Event** is the actual effort of Kaizen designs and plans

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What is Kanban?

- Japanese word “Kanban” means “signboard”
  - ➢Originally developed by Taiichi Ohno
- Production is [customer] demand driven
- Match production to demand
  - ➢Only meet the immediate demand
  - ➢No demand = No production
- Success requires: 5S, Visual Factory, standardized work, ongoing Kaizens

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Kanban Card

- Kanban card is the ticket / signal authorizing production or movement of materials
  - ➢The Kanban card can be a physical card or an electronic signal
- Initiated by the end customer
- Reflects movement of materials
  - ➢May follow material through an entire production cycle
- Upon receiving a Kanban card, goods are produced

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What is Poka-Yoke?

- Japanese term “poka-yoke” means “**mistake-proofing**”
  - Initially called “baka-yoke” (fool-proofing)
  - Originally developed by Shigeo Shingo
- Defect Prevention – The highest level of improvement
  - Minimize or eliminate the possibility of making a mistake
- Philosophy of “humans don’t make mistakes”
- Additional details under the Control phase

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Poka-Yoke: Examples

Examples of Poka-Yoke:

- Automated shutoffs on electric coffee pots
- Butane lighters with safety button
- Car engine warning lights
- Child proof caps on medications
- Computer device connections
- Confirmation of “Delete” of files
- Ground fault circuit breakers
- Pilotless gas ranges and hot water heaters
- Spell check in word processing software
- Tamper proof packaging

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Thank You

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5.0 Control

The “Control” phase is about:

✓ Controlling the “vital few” variables

- Typically the top 1 – 3, that were identified in the Analyze and Improve phases of the project

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Control Phase

- **Types of Variation**
- **SPC Data Collection**
- **Control Chart Structure & Rules**
  - I-MR Chart
  - Xbar-R Chart
  - Xbar-S Chart

- NP Chart
- P Chart
- C Chart
- U Chart
- CumSum Chart
- EWMA Chart

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Control Phase, cont.

Control Plans

- Control Plan
- Standard Operating Procedures (SOPs)
- Communication Plan
- Training Plan
- Audits
- Response Plan

Lean Controls

- 5S
- Visual Factory
- Standardized Work
- Kaizen
- Kanban
- Poka-Yoke
- Preventative Maintenance & Jidohka

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5.1 Statistical Process Control (SPC)

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What is SPC?

Statistical process control (SPC) is a statistical method to:

- Monitor the performance of a process using control charts
- Distinguish between special cause variation (SVC) and common cause variation (CCV)
- Present the voice of the process (VOP)

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Process Stability

A process is stable when:

- No special cause variation is involved
- The process is in statistical control
  - ➢No trends, unnatural patterns, or outliers
- Future performance of the process is predictable within certain limits
- Changes are all due to random, inherent variation

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Common Cause Variation (CCV)

- Common cause variation is the inherent natural variation in any processes
  - ➢Also known as chance or random variation
- Random background “noise”
  - ➢Cannot be controlled or eliminated from the process
- Presence is expected and acceptable
  - ➢Relatively small influence on the process

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Special Cause Variation (SCV)

- Special cause variation is the unnatural variation in the process
  - ➢Also known as assignable cause variation
- Cause of process instability
- Leads to defects
- Signal of unanticipated change (positive or negative)
- Possible to eliminate the special cause variation

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Variation Factors

- **Special Cause**: Variation is caused by **known** factors that result in a non-random distribution of output

  - Easier to identify
  - Special Causes = Opportunity

- **Common Cause**: Variation caused by **unknown** factors resulting in a steady but random distribution of output around the average of the data

  - More difficult to identify
  - Typically follows a Normal Distribution

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5.1.1 Data Collection for SPC

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Data Collection Considerations

Considerations when collecting data for plotting control charts:

- Metric of interest
- Discrete or continuous data
- Sampling strategy
- Number and frequency of samples
- When and where to sample
- Data and information formats

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Subgroups & Rational Subgrouping

- Randomly select a group of items (i.e., a subgroup) from the population of interest
- Subgroup (sample) size can be constant or variable
  - Number of subgroups, subgroup size, and frequency of sampling impact the quality of control charts
- Match control chart type to subgroup size
- Rational subgrouping is the basic sampling scheme in SPC

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Purpose for Rational Subgrouping

- Maximize the likelihood of **detecting** special cause variation
- Minimize the opportunity of having special cause variation **within** subgroups
- Assuming only random variation (background noise) within subgroups, all the special cause variation would be reflected **between** subgroups

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Size of Subgroups

Typical subgroup sizes are 3-12 for variable data

- If sampling is difficult or expensive, size may be smaller
- 3, 5, and 10 are the most common size
  - ➢Because of ease of calculations when done without computers

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Frequency of Sampling

- Objective: Have **sufficient data** to signal the changes in a process within reasonable **time** and **costs**
- Balance cost of sampling and testing vs. cost of not detecting significant variation
- Leverage the subject matter experts' knowledge

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5.1.2 Control Chart Structure & Rules

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Control Charts: Purpose

Control charts are graphical tools used to:

- **Present**, **Analyze**, and **Monitor** the process performance
- **Detect** special cause variation
- **Determine** whether the process is in statistical control (stable)

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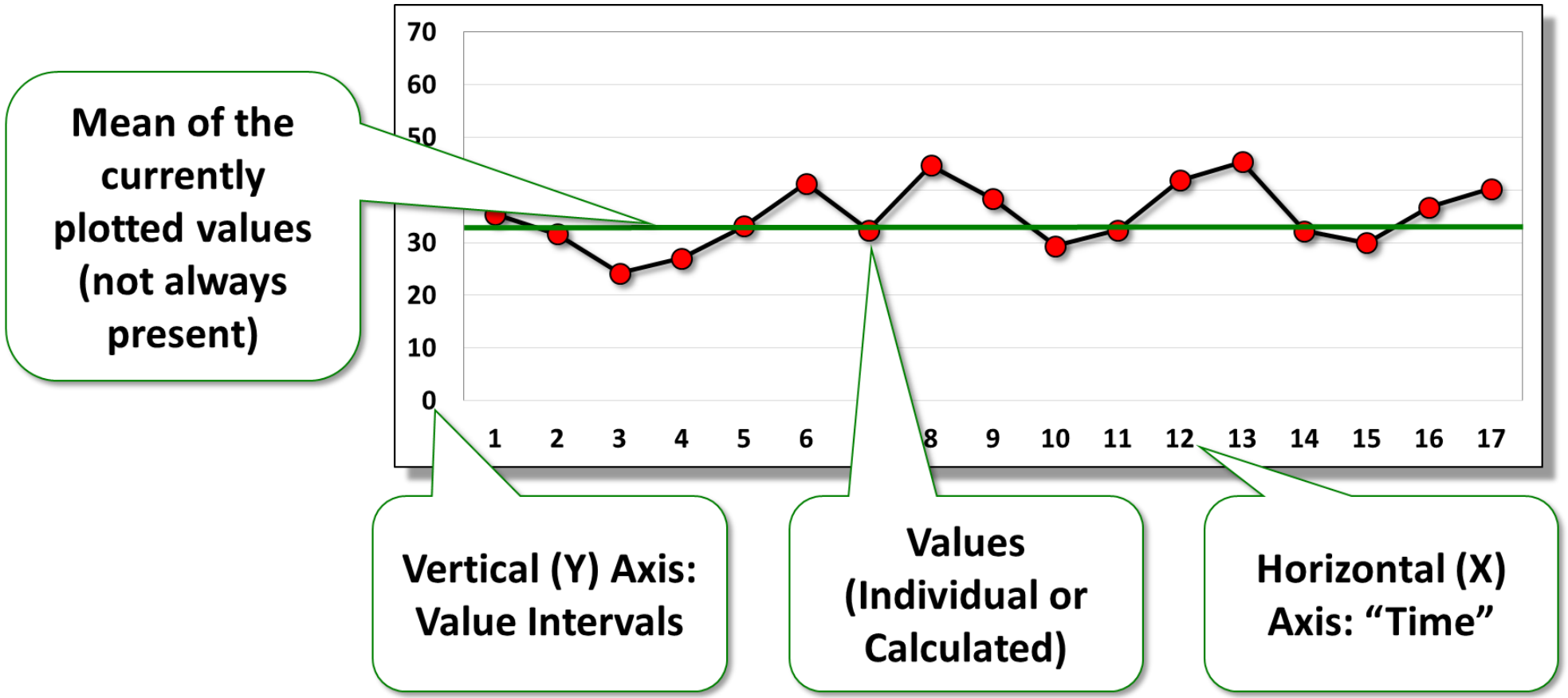
Run Chart with Mean



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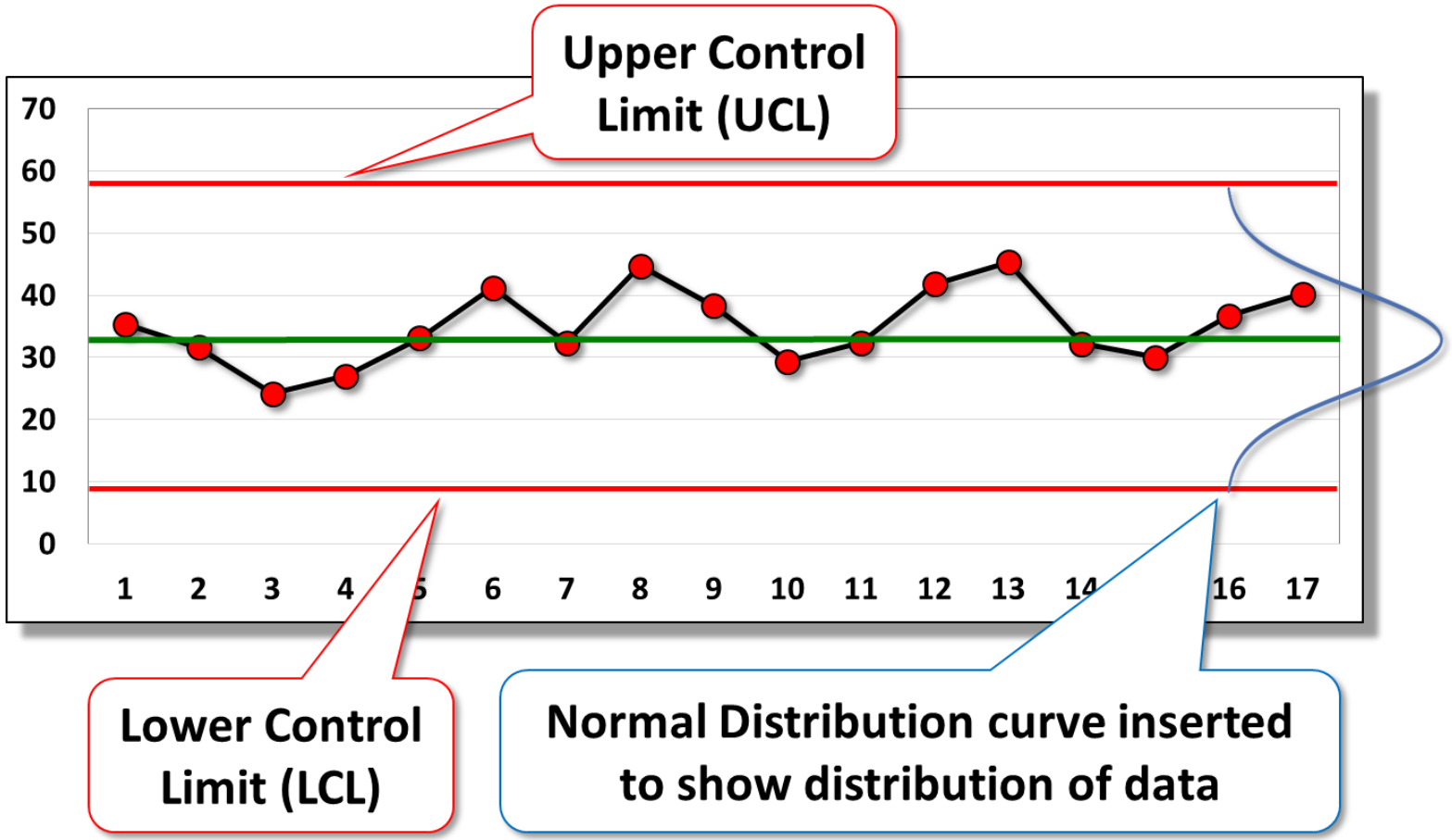
Control Chart Elements

- **Control limits are calculated** based upon the control chart type, sample size, etc.
- Closely align with ±3 sigma of the mean



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Western Electric Rules

Western Electric Rules are the most popular decision rules to detect unnatural patterns, or special causes, in the control charts

- Typical unnatural patterns in control charts:
  - Outliers
  - Trending
  - Cycling
  - Auto-correlative
  - Mixture

- Process is in control if all the data points on the control chart are randomly spread out within the control limits

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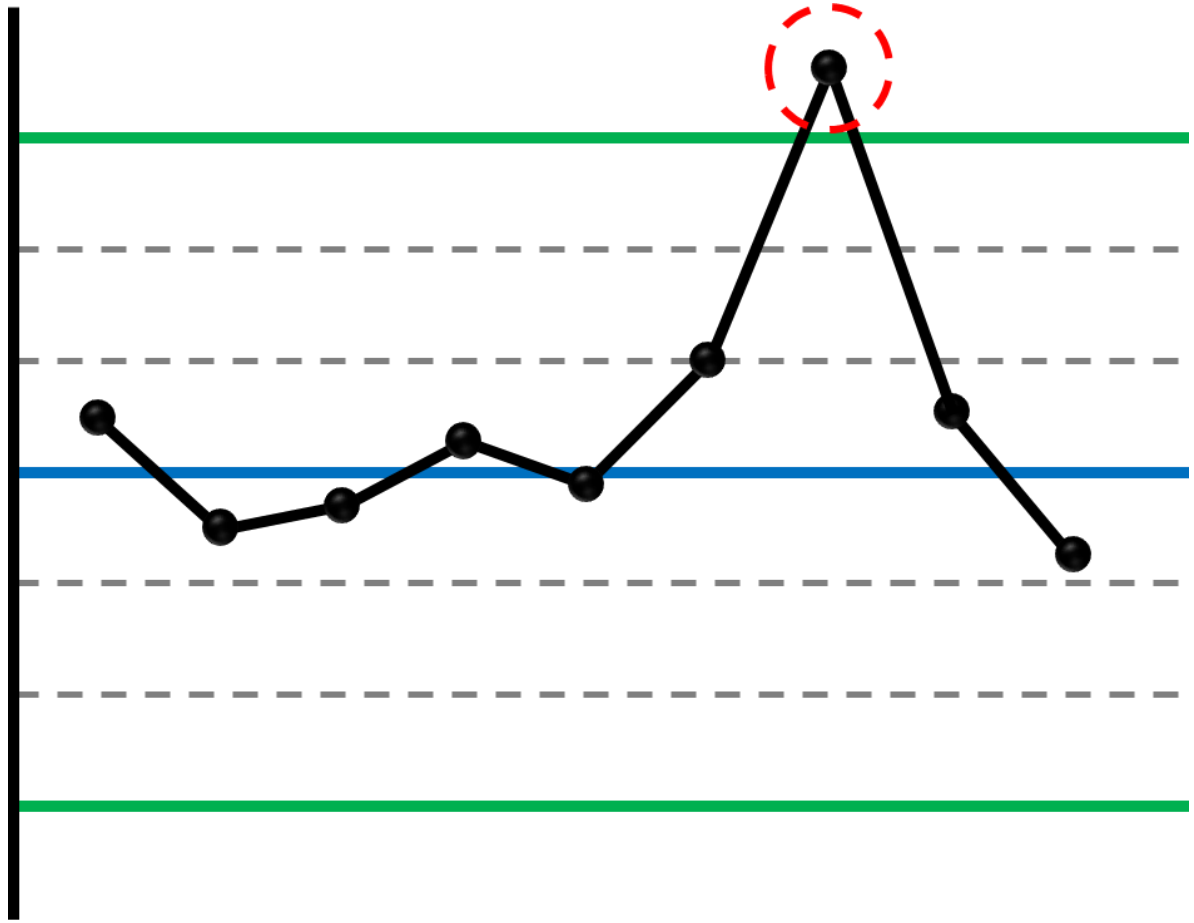
Western Electric Rules: Test 1

Test 1:

1 point outside the Control limits (more than 3 standard deviations from the mean)

- Typically known as an outlier



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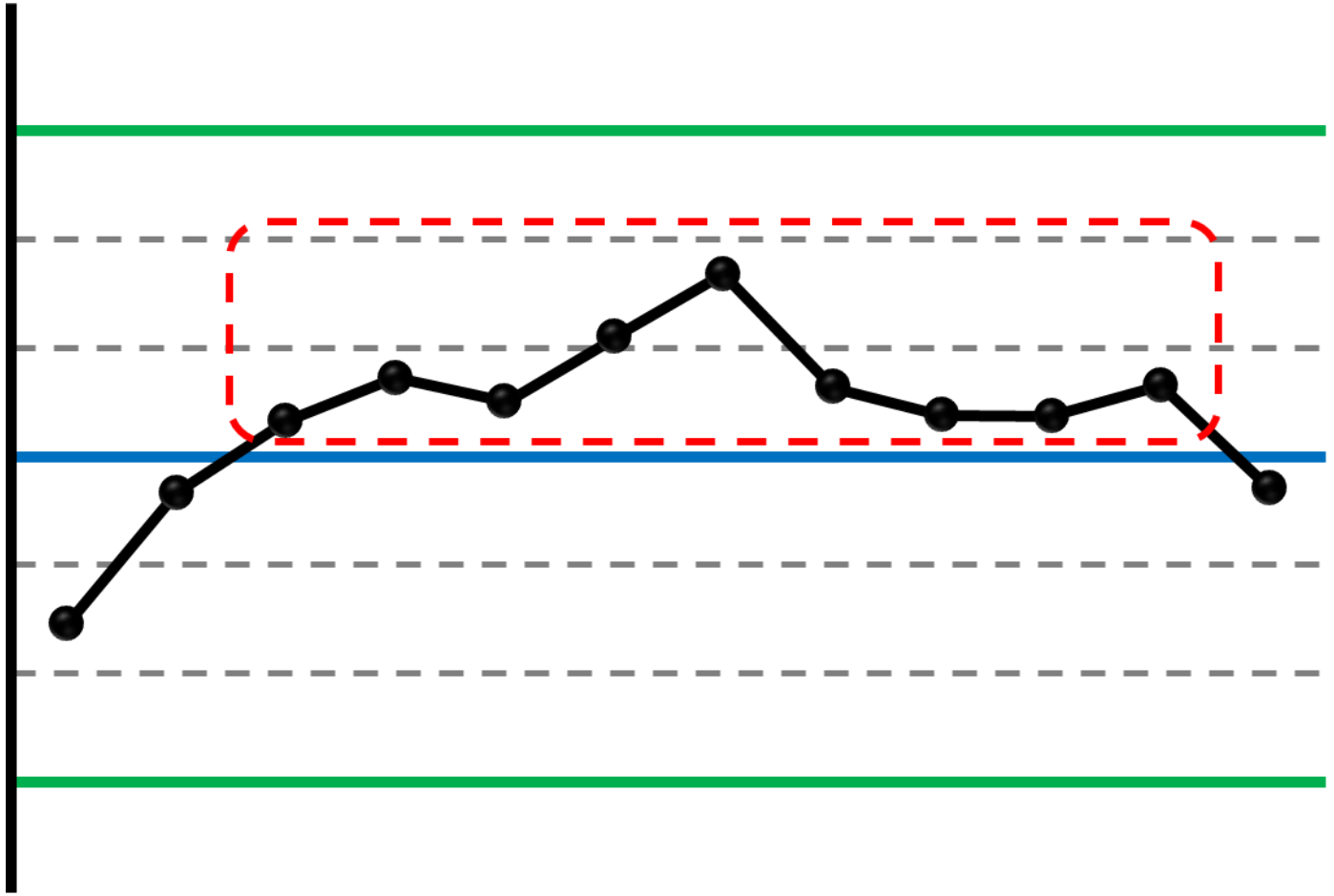
Western Electric Rules : Test 2

Test 2:

9 points in a row on the same side of the center line

- Identifies situations where the process mean has temporarily shifted in the process



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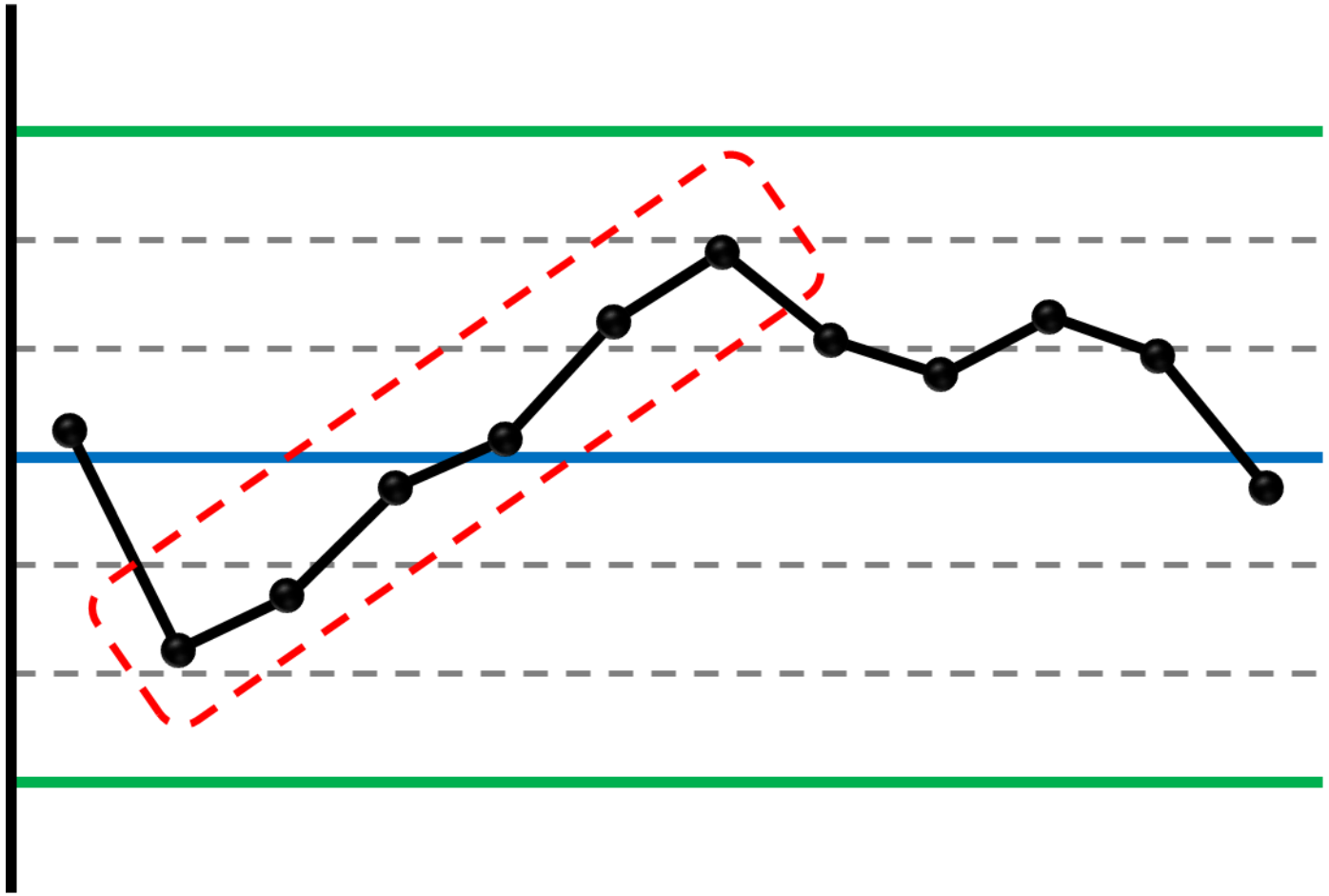
Western Electric Rules : Test 3

Test 3:

6 points in a row steadily increasing or steadily decreasing

- Identifies significant trends in performance



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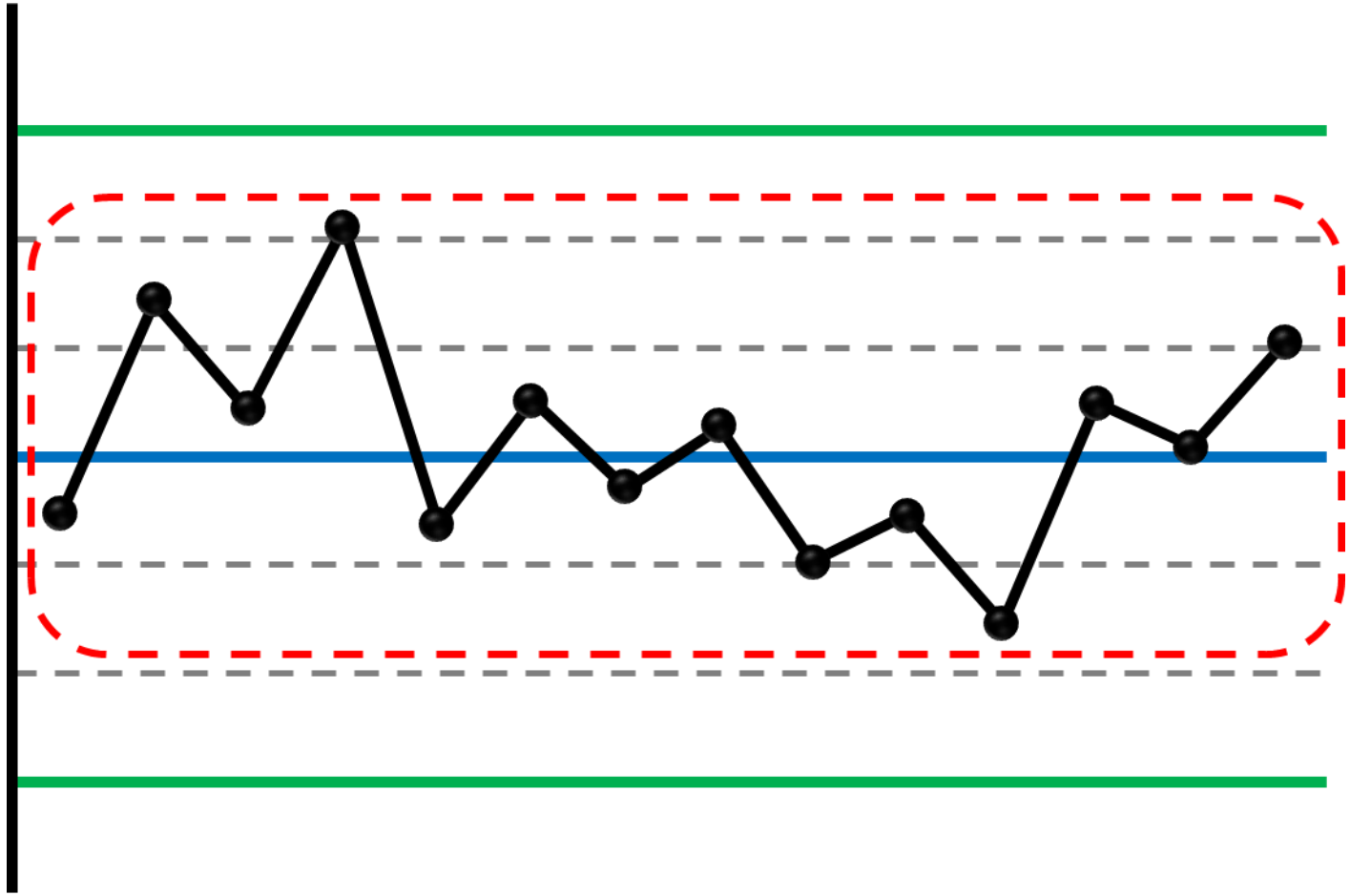
Western Electric Rules : Test 4

Test 4:

14 points in a row alternating up and down

- Indicates a cycle



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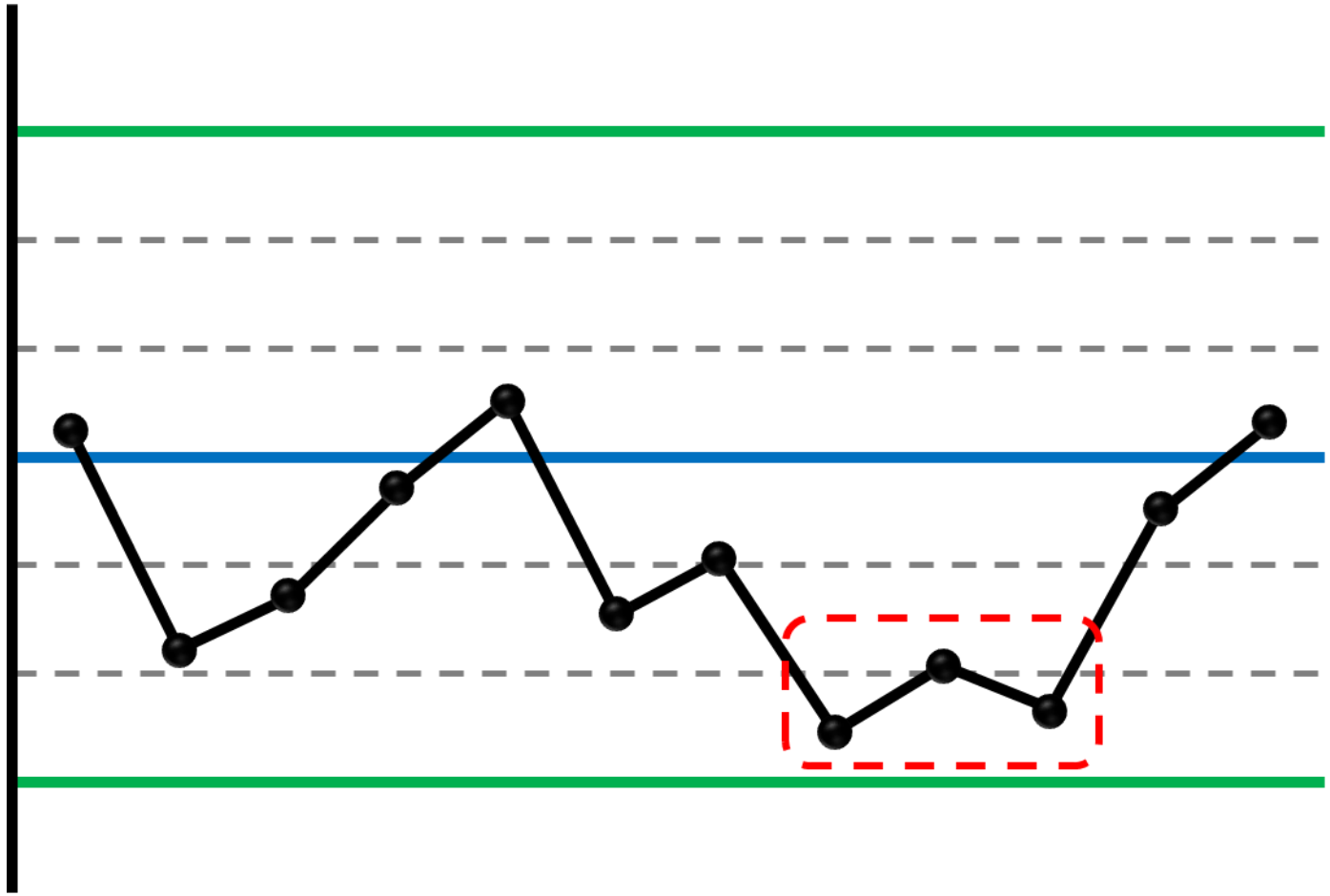
Western Electric Rules : Test 5

Test 5:

2 out of 3 points in a row at least 2 standard deviations from the center line



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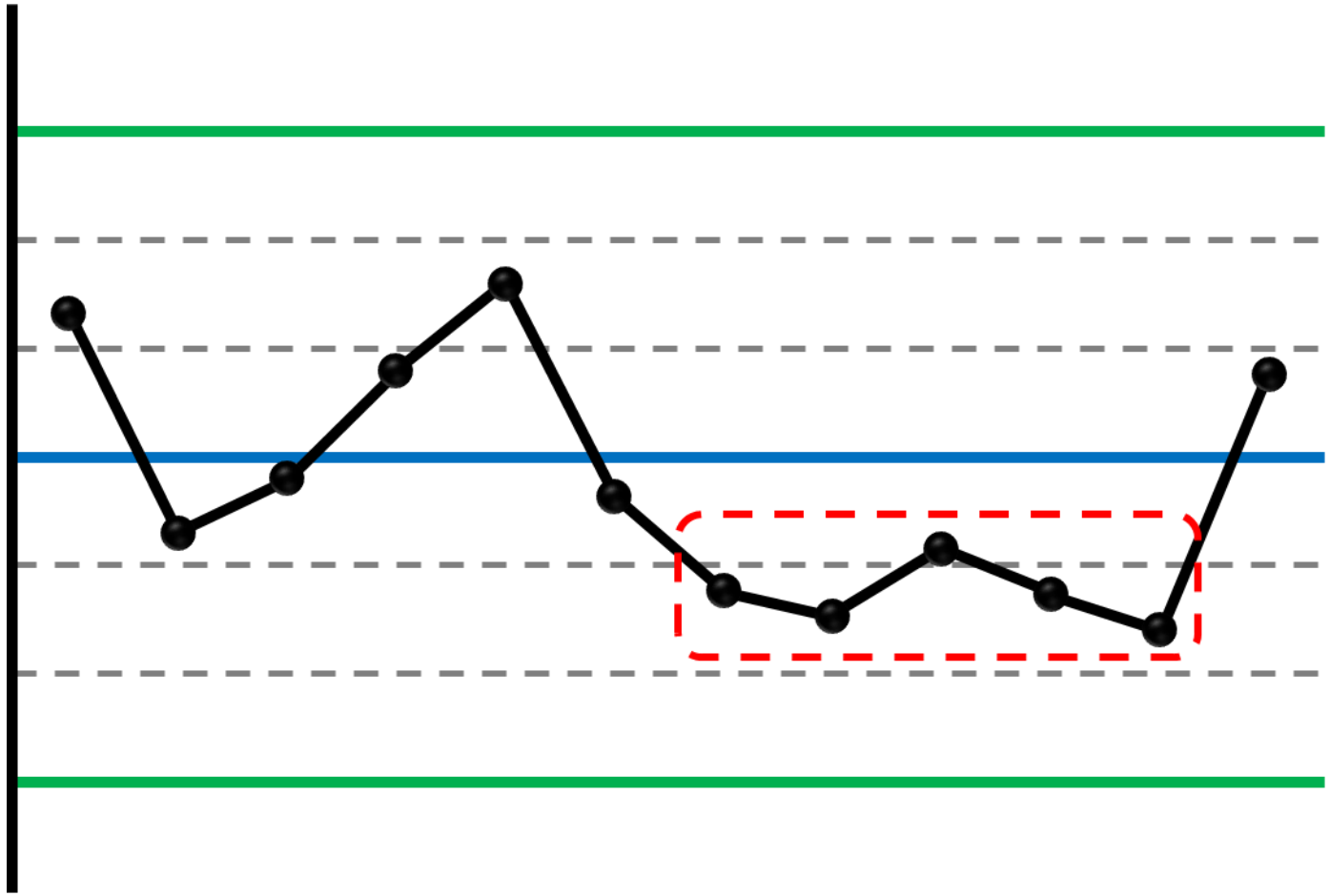
Western Electric Rules : Test 6

Test 6:

4 out of 5 points in a row at least 1 standard deviation from the center line on the same side of the center line



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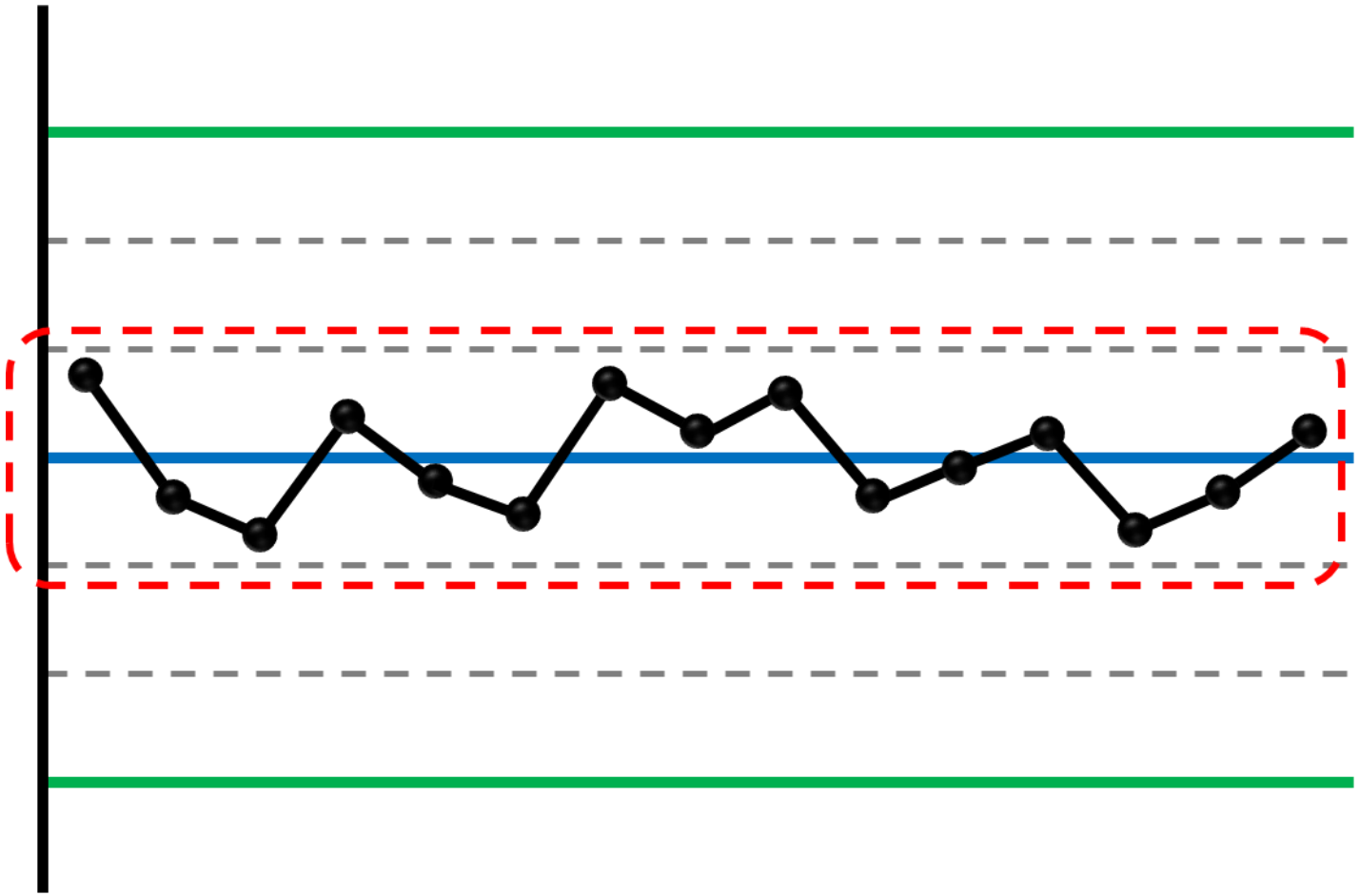
Western Electric Rules : Test 7

Test 7:

15 points in a row within 1 standard deviation from the center line on either side of the center line



- Test 7 indicates a change in the level of variation around the process mean

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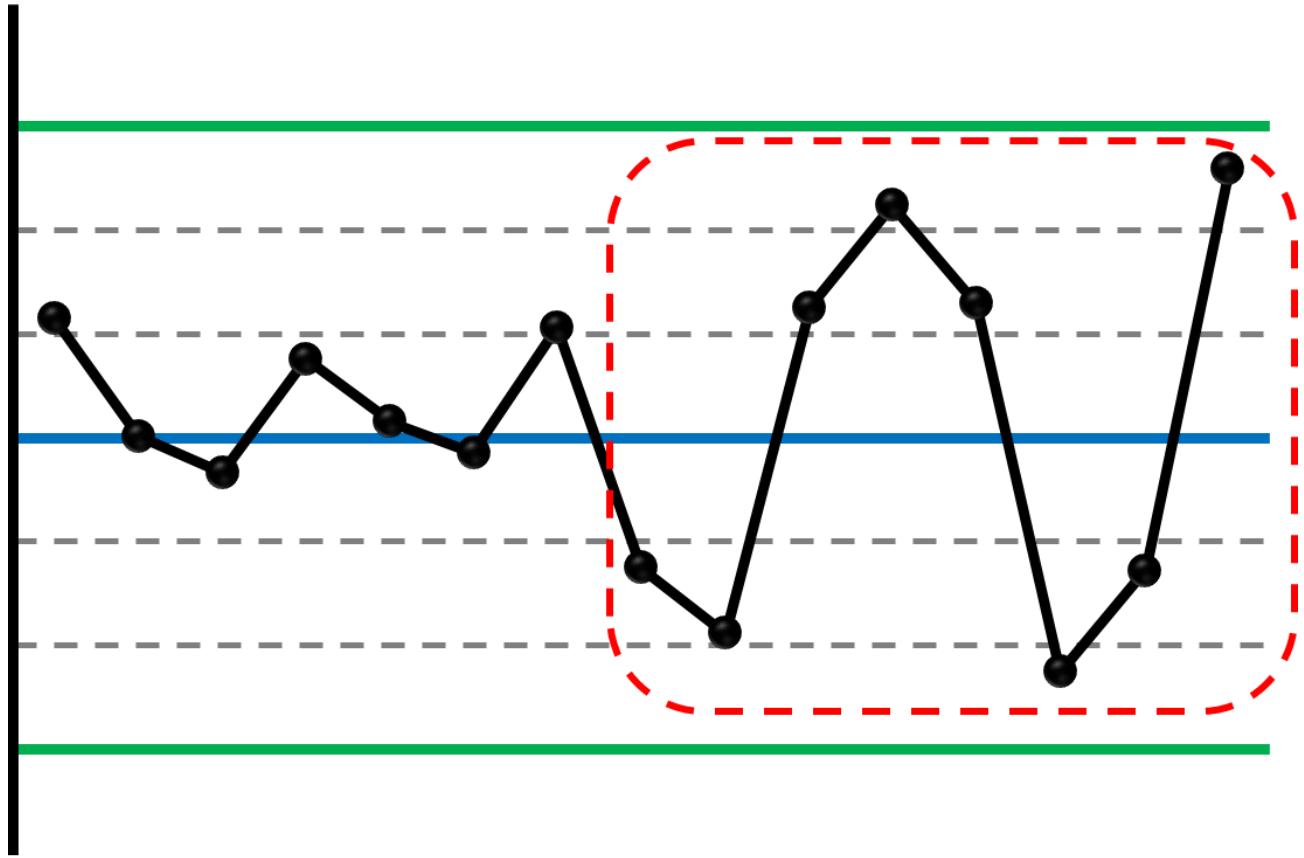
Western Electric Rules : Test 8

Test 8:

8 points in a row beyond 1 standard deviation from the center line on either side of the center line

- While test 7 indicates a narrowing of variation, test 8 **indicates an increase in variation**



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Control & Stability

- If no data points fail any tests for special causes, the process is in statistical control

- If any data point fails any tests for special causes, the process is unstable
  - ➢Process stability is the prerequisite of process capability analysis

- Set a periodic time frame for checking Control Limits and Center Lines

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Control Charts: Manual Construction

Manual Construction of Control Charts

1. Select characteristic (critical “X” or CTQ) to be charted
2. Determine the purpose of the chart
3. Select data-collection points
4. Establish the basis for sub-grouping (only for X’s)
5. Select the type of Control Chart
6. Determine the measurement method/criteria
7. Establish the sampling interval/frequency
8. Determine the sample size
9. Establish the basis of calculating the Control Limits
10. Set up the forms or software for charting data
11. Prepare written instructions for all phases
12. Conduct the necessary training

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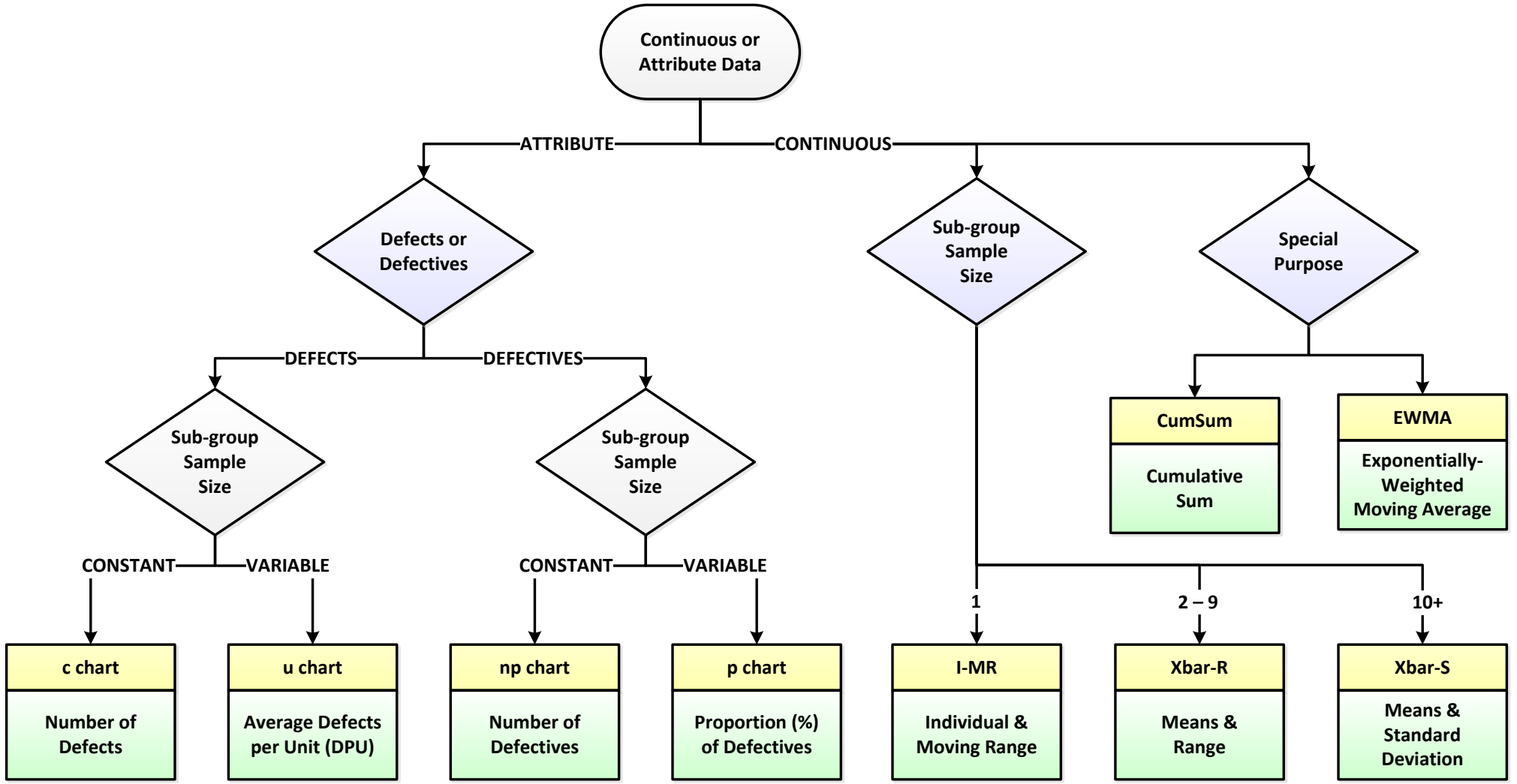
SPC Control Chart Selection



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5.1.3 I-MR Chart

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I-MR Chart

The I-MR chart (**I**ndividual-**M**oving **R**ange; IR chart) is for continuous data

- "**I**" chart plots an individual observation as a data point
- "**MR**" chart plots the absolute value of the range between two consecutive observations in individual charts as a data point

- I chart is valid only if the MR chart is in control
- It is the **most sensitive** of all charts to changes
- Underlying distribution is **normal**

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Reference: I-MR Chart Equations

I Chart (Individuals Chart)

- **Data Point:** x_i
- **Center Line:** $\frac{\sum_{i=1}^{n} x_i}{n}$
- **Control Limits:** $\frac{\sum_{i=1}^{n} x_i}{n} \pm 2.66 \times \overline{MR}$

MR Chart (Moving Range Chart)

- **Data Point:** $\left| x_{i+1} - x_i \right|$
- **Center Line:** $\frac{\left| x_{i+1} - x_i \right|}{n-1}$
- **Upper Control Limit:** $3.267 \times \frac{\left| x_{i+1} - x_i \right|}{n-1}$
- **Lower Control Limit: 0**

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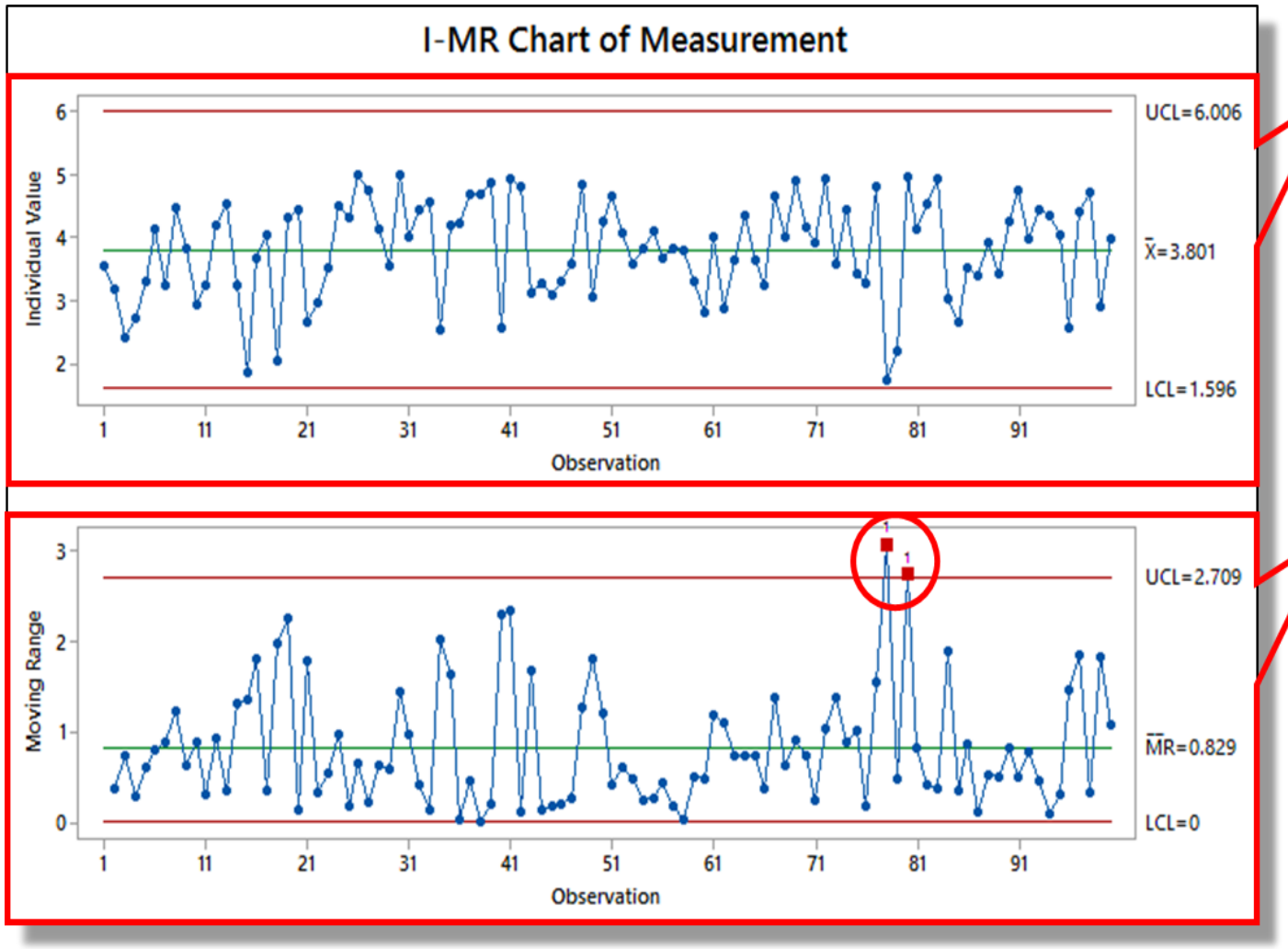
I-MR Charts Diagnosis



I Chart (Individuals' Chart)

- Since the MR chart is out of control, the I chart is invalid

MR Chart (Moving Range Chart)

- Two data points fall beyond the upper control limit. The MR chart is out of control

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5.1.4 Xbar-R Chart

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Xbar-R Chart

The Xbar-R chart is a control chart for continuous data with a sample size (subgroup) between 2 – 9

- The "**Xbar**" chart plots the mean of a sample as a data point
- The "**R**" chart plots the difference between the highest and lowest values within a subgroup as a data point

- The Xbar is valid only if the R chart is in control
- Underlying distribution is **normal**

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Xbar-R Chart Equations

Xbar chart

- **Data Point:** $\overline{X_i} = \frac{\sum_{j=1}^{m} x_{ij}}{m}$
- **Center Line:** $\overline{\overline{X}} = \frac{\sum_{i=1}^{k} \overline{X}_i}{k}$
- **Control Limits:** $\overline{\overline{X}} \pm A_2 \overline{R}$

Where m is the subgroup size and k is the number of subgroups. A2 is a constant depending on the subgroup size

R chart (Range Chart)

- **Data Point:** $R_i = \underset{j\in[1,m]}{Max}(x_{ij}) - \underset{j\in[1,m]}{Min}(x_{ij})$
- **Center Line:** $\overline{R} = \frac{\sum_{i=1}^{k} R_i}{k}$
- **Upper Control Limit:** $D_4 \times \overline{R}$
- **Lower Control Limit:** $D_3 \times \overline{R}$

Where m is the subgroup size and k is the number of subgroups. D3 and D4 are constants depending on the subgroup size

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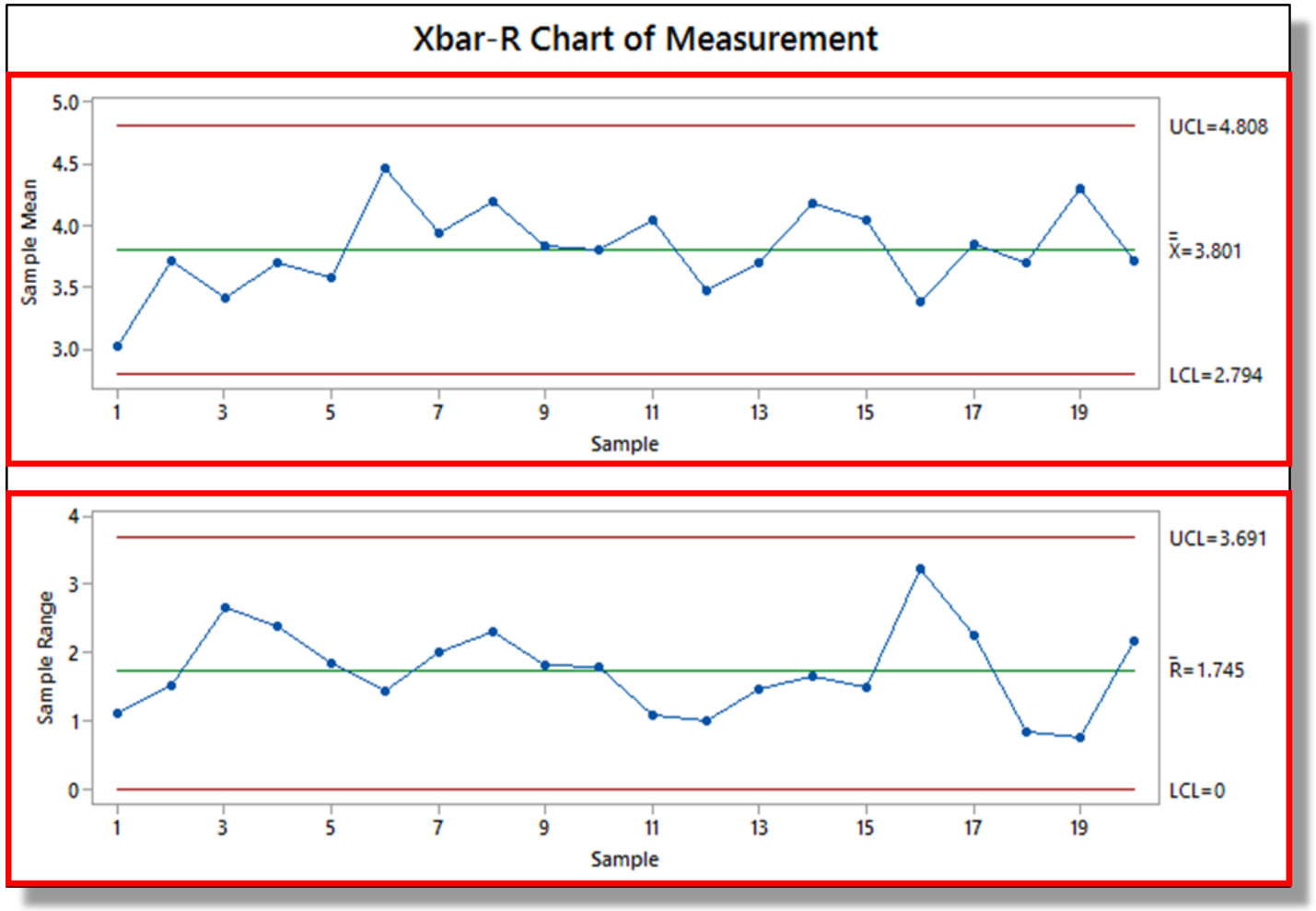
Xbar-R Charts Diagnosis



Xbar-R Charts

- Since the R chart is in control, the Xbar chart is valid
- In both charts, there are not any data points failing any tests for special causes
- We conclude that the process is in control

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5.1.5 Xbar-S Chart

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Xbar-S Chart

The Xbar-S chart (also called X-S chart) is a control chart for continuous data with a **constant** sample size greater than 9

- The "**Xbar**" chart plots the sample mean as a data point
- The "**S**" chart plots the standard deviation within a subgroup as a data point

- The Xbar chart is valid only if the S chart is in control
- Underlying distribution is **normal**

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Xbar-S Chart Equations

Xbar Chart

- **Data Point:** $\overline{X}_i = \frac{\sum_{j=1}^{m} x_{ij}}{m}$
- **Center Line:** $\overline{\overline{X}} = \frac{\sum_{i=1}^{k} \overline{X}_i}{k}$
- **Control Limits:** $\overline{\overline{X}} \pm A_3 \bar{s}$

Where m is the subgroup size and k is the number of subgroups. B3 and B4 are constants depending on the subgroup size

S Chart

- **Data Point:** $s_i = \sqrt{\frac{\sum_{j=1}^{m} (x_{ij} - \bar{x}_i)^2}{m-1}}$
- **Center Line:** $\bar{s} = \frac{\sum_{i=1}^{k} s_i}{k}$
- **Upper Control Limit:** $B_4 \times \bar{s}$
- **Lower Control Limit:** $B_3 \times \bar{s}$

Where m is the subgroup size and k is the number of subgroups. B3 and B4 are constants depending on the subgroup size

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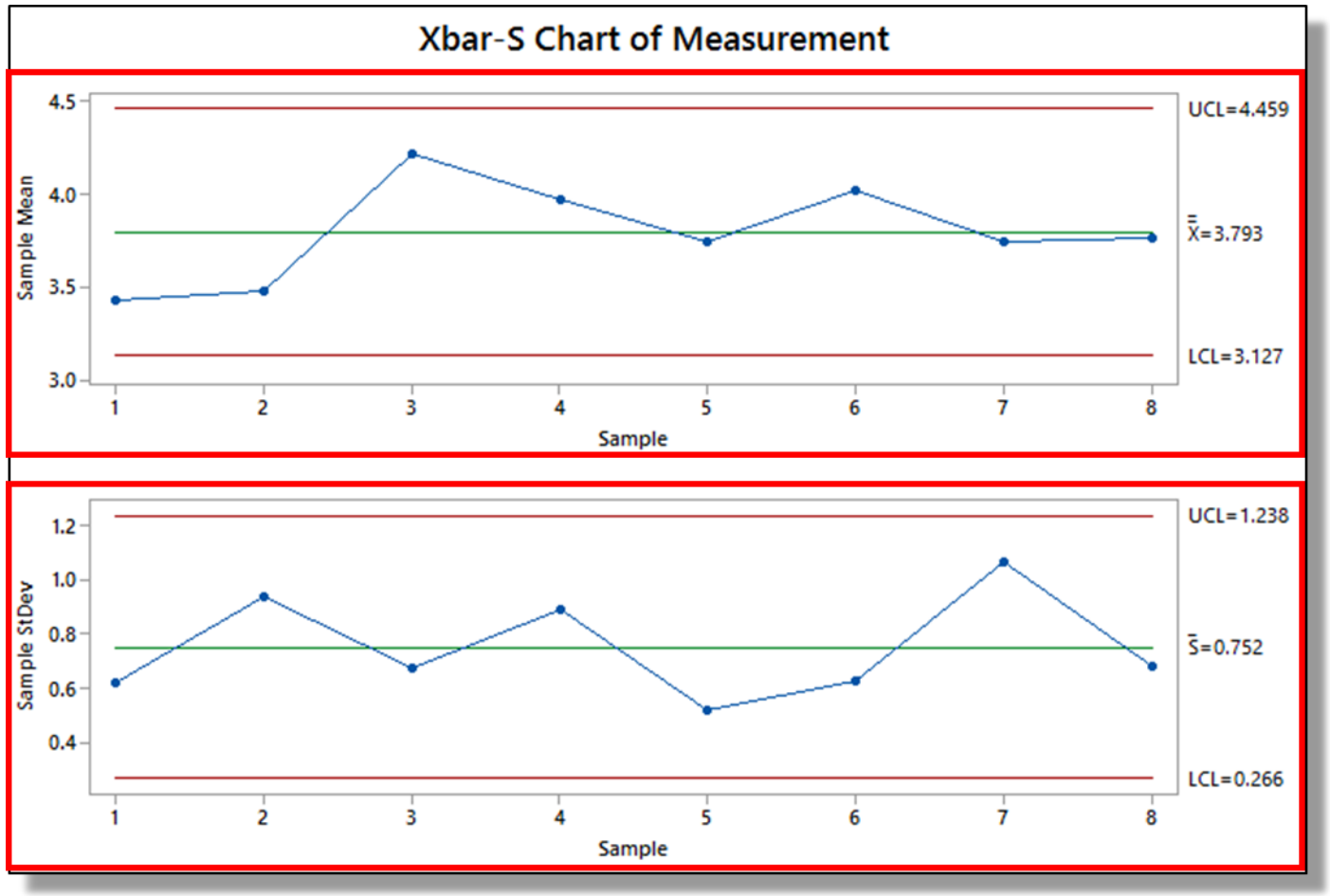
Xbar-S Charts Diagnosis



Xbar-S Charts

- Since the S chart is in control, the Xbar chart is valid
- In both charts, there are not any data points failing any tests
- We conclude that the process is in control

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5.1.6 NP-Chart

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NP Chart

The NP chart is a control chart monitoring the count of defectives

- The "**NP**" chart plots the **number** of defectives in one subgroup as a data point
- Sample size is **constant**
- Underlying distribution is **binomial** distribution

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NP Chart Equations

NP Chart

- **Data Point:** x_i
- **Center Line:** $\overline{np} = \frac{\sum_{i=1}^{k} x_i}{k}$
- **Control Limits:** $\overline{np} \pm 3 \times \sqrt{\overline{np}(1-\overline{p})}$

- Where n is the constant subgroup size
- k is the number of subgroups
- x_i is the number of defectives in the i^{th} subgroup

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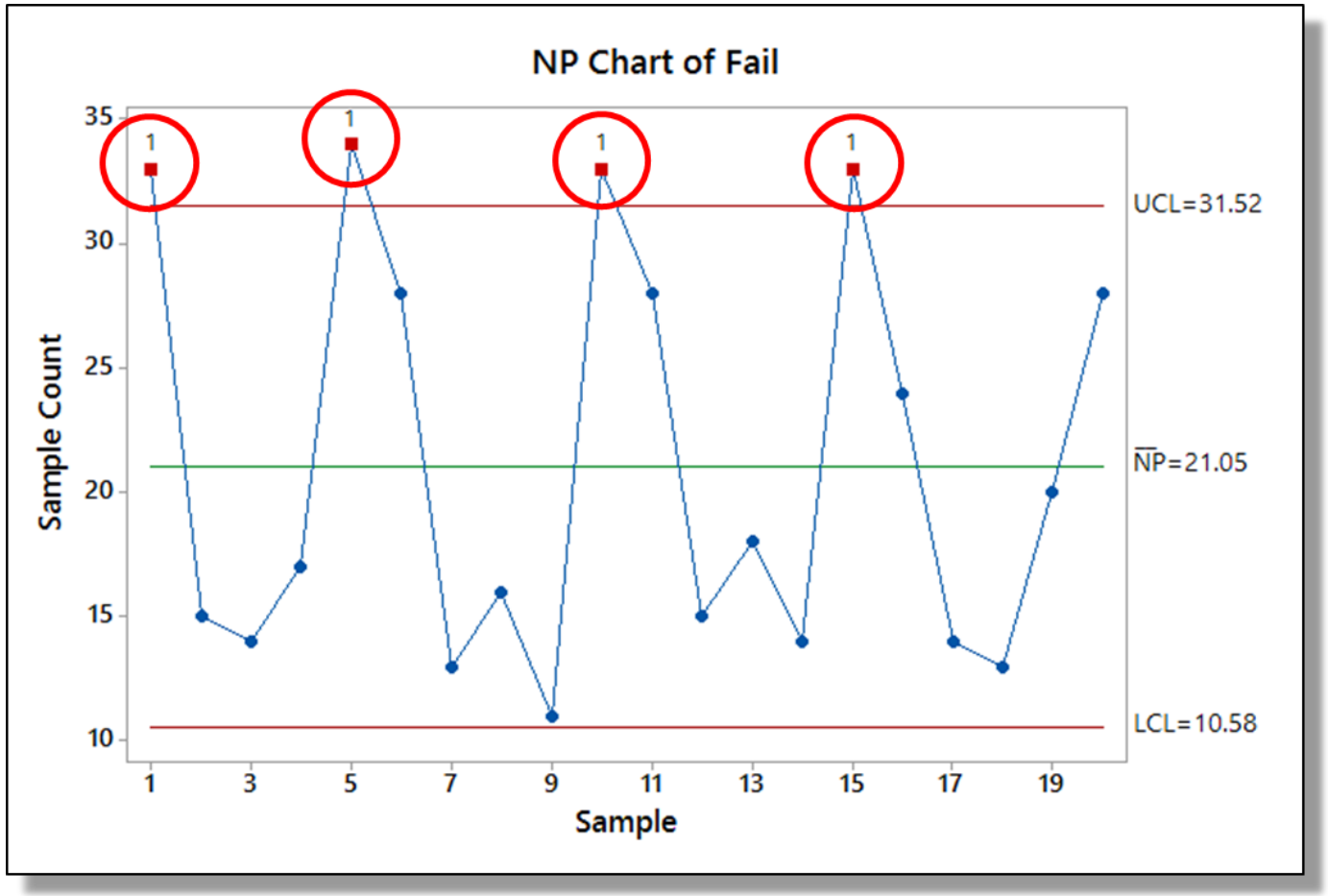
NP Chart Diagnosis



NP Chart

- Four data points, circled in red, fall beyond the upper control limit
- We conclude that the NP chart is out of control

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5.1.7 P-Chart

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P Chart

The P chart is a control chart monitoring the percentages of defectives

- Plots the **percentage** of **defectives** in one subgroup as a data point
- Sample size is **not constant**
- Underlying distribution is **binomial** distribution

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P Chart Equations

P Chart

- **Data Point:** $$p_i = \frac{x_i}{n_i}$$
- **Center Line:** $$\bar{p} = \frac{\sum_{i=1}^{k} x_i}{\sum_{i=1}^{k} n_i}$$
- **Control Limits:** $$\bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

- where n_i is the subgroup size for the i^{th} subgroup
- k is the number of subgroups
- x_i is the number of defectives in the i^{th} subgroup

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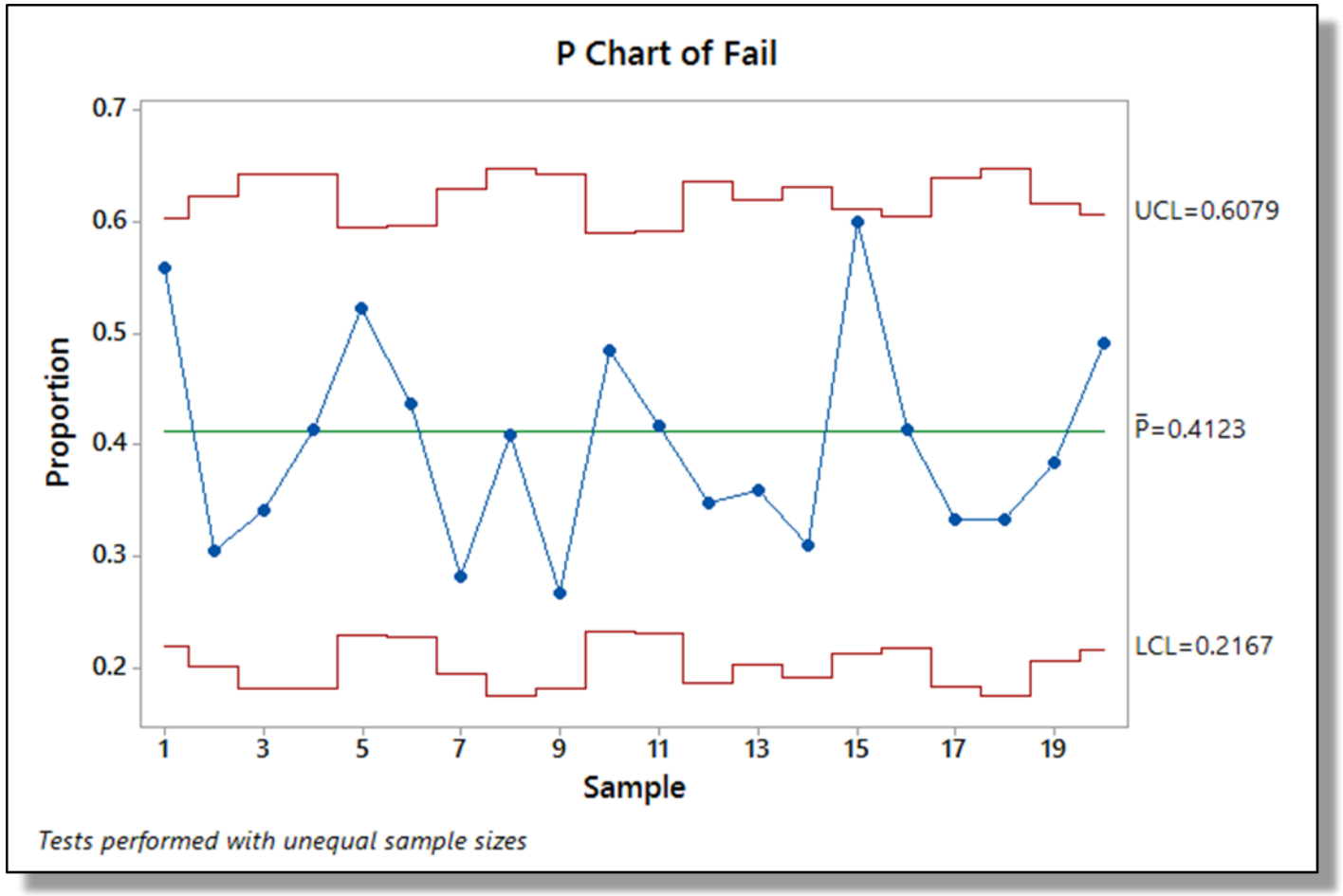
P Chart Diagnosis



P Chart

- Sample sizes are **not constant** over time, the control limits are **adjusted** to different values accordingly
- We conclude that the process is in control

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5.1.8 C-Chart

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C Chart

The C chart is a control chart monitoring the number of defects in a sample

- The "**C**" chart plots the **count** of defects per unit of a subgroup as a data point
- Sample size is **constant**
- Underlying distribution is **Poisson** distribution

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C Chart Equations

C Chart

- **Data Point:** c_i
- **Center Line:** $\bar{c} = \frac{\sum_{i=1}^{k} c_i}{k}$
- **Control Limits:** $\bar{c} \pm 3 \times \sqrt{\bar{c}}$

- Where n_i is the subgroup size for the i^{th} subgroup
- k is the number of subgroups
- x_i is the number of defects in the i^{th} subgroup

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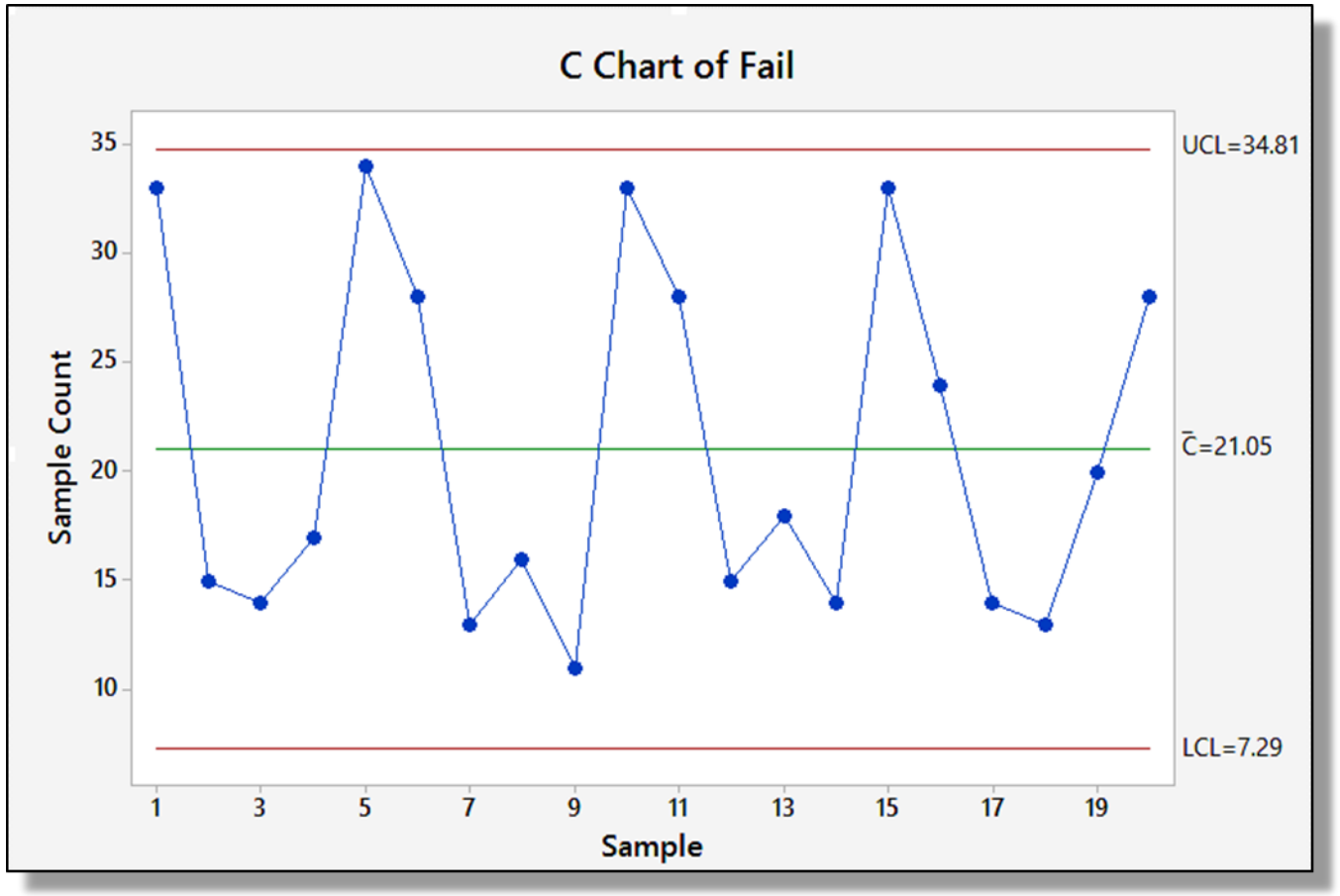
C Chart Diagnosis



C Chart

- Same function and look as the NP chart
- Has a constant sample size

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5.1.9 U-Chart

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U Chart

The U chart is a control chart monitoring the average defects per unit

- The "**U**" chart plots the **average number** of defects per unit of a subgroup as a data point
- Sample size is **not constant**
- Underlying distribution is **Poisson** distribution

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U Chart Equations

U Chart

- **Data Point:** $u_i = \frac{x_i}{n_i}$
- **Center Line:** $\bar{u} = \frac{\sum_{i=1}^{k} u_i}{k}$
- **Control Limits:** $\bar{u} \pm 3 \times \sqrt{\frac{\bar{u}}{n_i}}$

- Where n_i is the subgroup size for the i^{th} subgroup
- k is the number of subgroups
- x_i is the number of defects in the i^{th} subgroup

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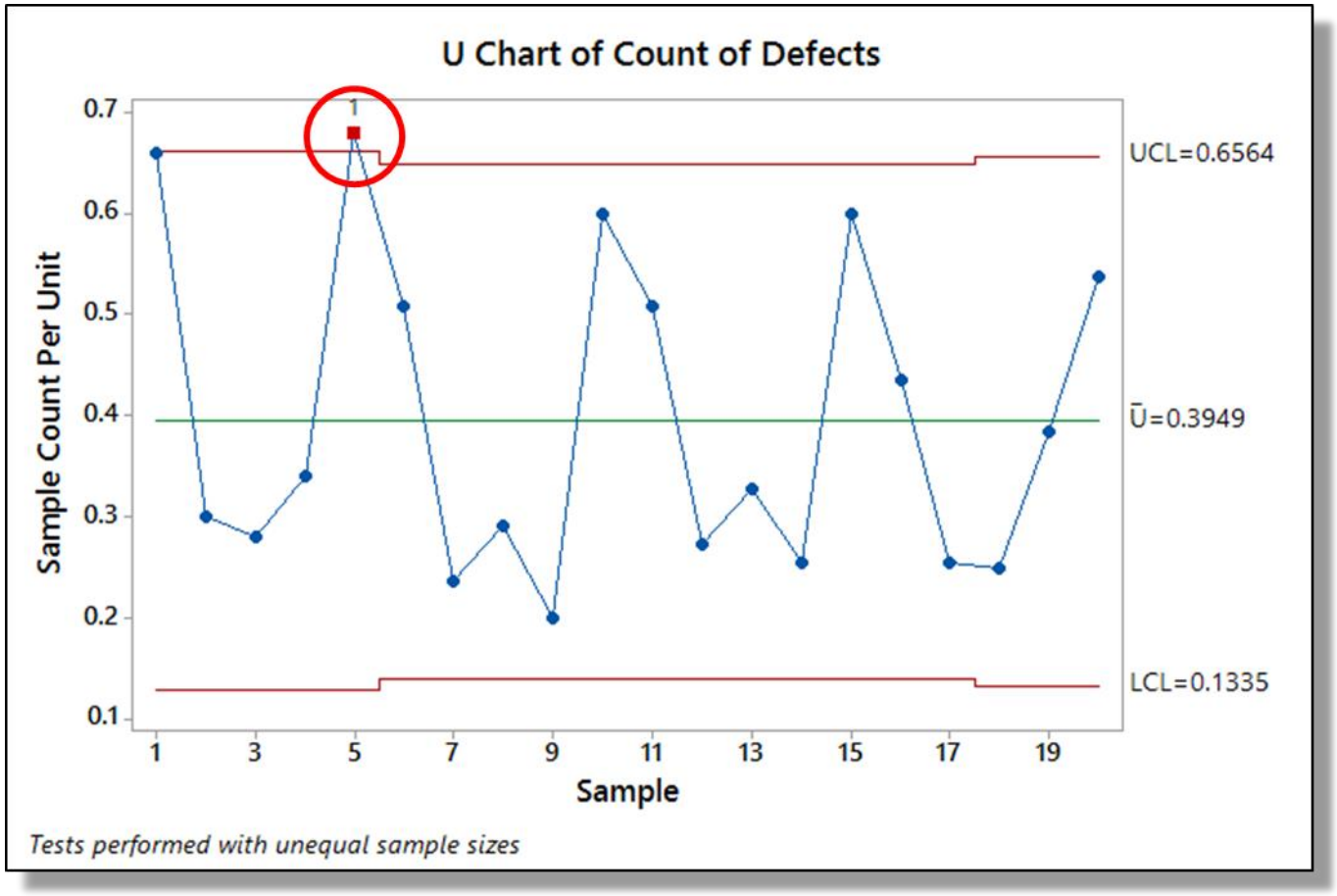
U Chart Diagnosis



U Chart

- Sample sizes are **not constant**
- Control limits are dynamically adjusted
- The data point circled in red falls beyond the upper control limit. We conclude that the process is out of control

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5.1.10 CumSum Chart

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CumSum Chart

The CumSum chart (Cumulative Sum control chart; CUSUM) is a control chart of monitoring the cumulative sum of the sample mean deviations from the process target

- Detects the **shift** of the process **mean** from the process target over **time**
- Underlying distribution is **normal** distribution

- Two types of CumSum charts:
  - ➢ One two-sided
  - ➢ Two one-sided
    - See following slides

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Two-Sided CumSum: V-Mask

In two-sided CumSum charts, use a V-mask to identify out-of-control data points

- V-mask is a transparent overlay shape of a horizontal “V”
- Superimposed the CumSum chart
- Origin point is placed on the last data point of the CumSum chart
- Center line is horizontal
- If all of the data points stay inside the V-mask, the process is in statistical control

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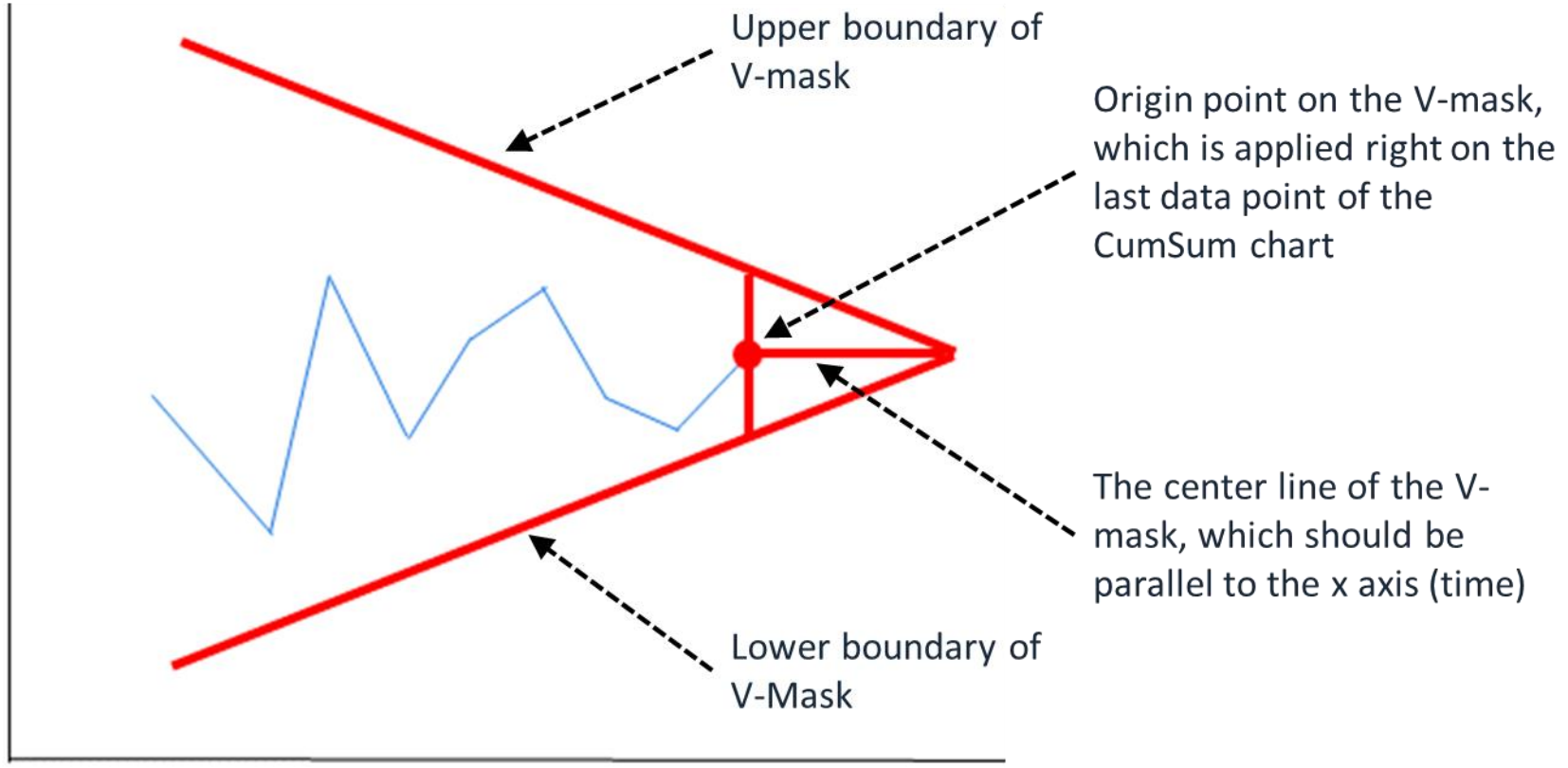
V-Mask

If any data points in the CumSum chart to the left of the origin point are outside the V-mask, the process is considered out of statistical control



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Two-Sided CumSum Equations

Two-Sided CumSum

- **Data Point:**

$$\begin{cases} c_i = c_{i-1} + (\bar{x}_i - T) & i > 0 \\ c_i = 0 & i = 0 \end{cases}$$

- **V-Mask Slope:**

$$k\frac{\sigma}{\sqrt{m}}$$

- **V-Mask Width at the Origin Point:**

$$2h\frac{\sigma}{\sqrt{m}}$$

- Where $\bar{x}_i$ is the mean of the i^{th} subgroup
- T is the process target
- σ is the estimation of process standard deviation
- m is the subgroup size

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5.1.11 EWMA Chart

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EWMA Chart

The EWMA chart (Exponentially-Weighted Moving Average Chart) is a control chart monitoring the exponentially-weighted average of previous and present subgroup means

- More recent data get more weight than more historical data
- Detects the shift of the process mean from the process target over time
- Underlying distribution is **normal** distribution

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EWMA Chart Equations

EWMA Chart

- **Data Point:** $z_i = \lambda \bar{x}_i + (1-\lambda) z_{i-1}$ where $0 < \lambda < 1$
- **Center Line:** $\overline{\overline{X}}$
- **Control Limits:** $\overline{\overline{X}} \pm k \cdot \frac{s}{\sqrt{n}} \sqrt{\left(\frac{\lambda}{2-\lambda}\right)\left[1-(1-\lambda)^{2i}\right]}$

- Where $\bar{x}_i$ is the mean of the ith subgroup
- λ and k are user-defined parameters to calculate the EWMA data points and the control limits

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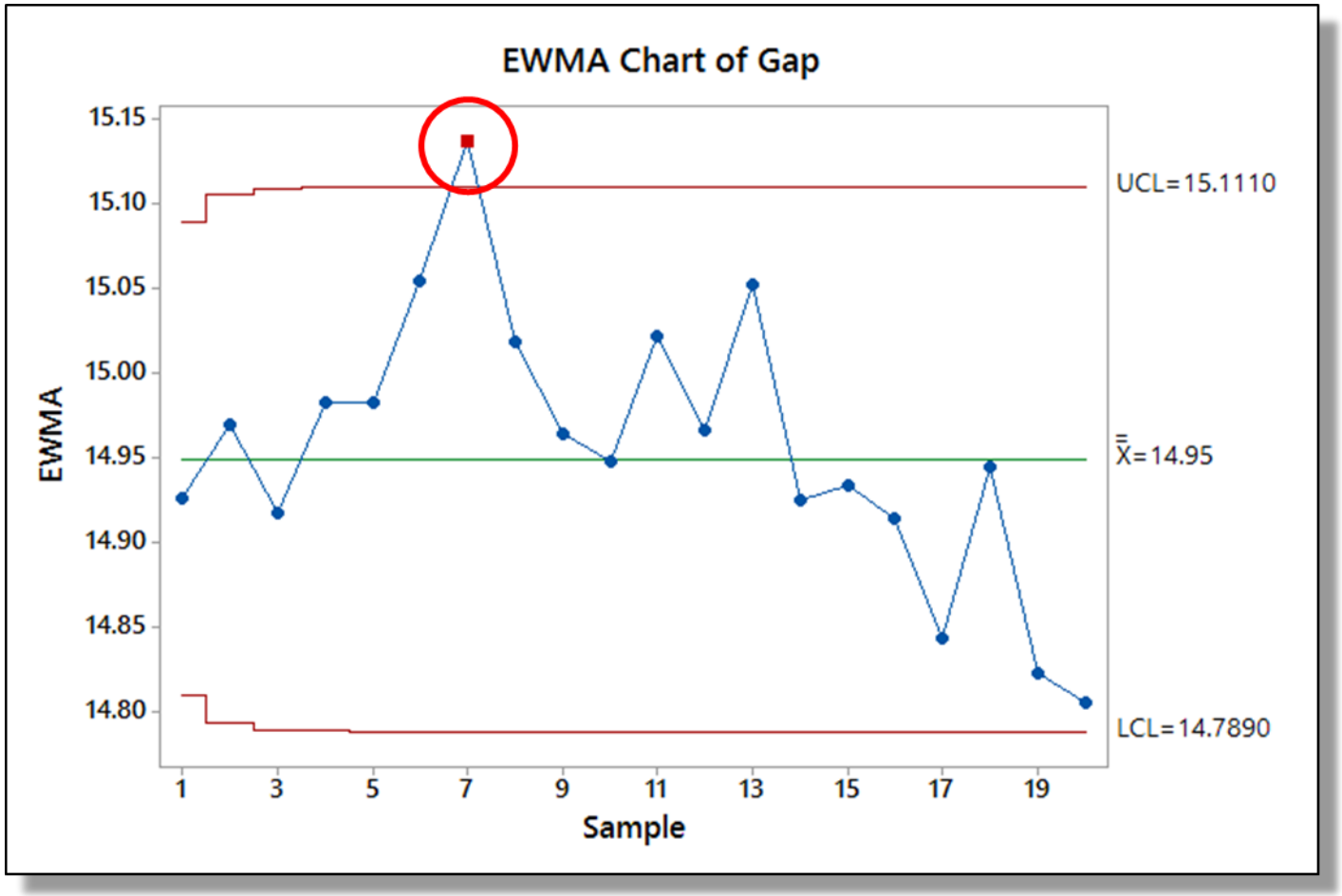
EWMA Chart Diagnosis



EWMA Chart

- As with other charts, the control limits are calculated
- One data point falls beyond the upper control limit, and we conclude that the process is out of control

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5.2 Six Sigma Control Plans
5.2.1 Control Plans

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Control Plans

Control plan objective is to sustain Six Sigma project results

- Benefits of the Control phase and plans include:

- Methodical roll-out of changes
- Standardize and communicate processes and work procedures
- Ensure compliance with changes through methods like auditing and corrective actions
- Transfer solutions and learning across the enterprise

- Coordinate ongoing team and individual involvement
- Standardize data collection and procedures
- Measure process performance, stability, and capability
- Plan actions that mitigate possible out-of-control conditions

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Control Plan Elements

Control Plan

- Clear and concise summary document that details key process steps, CTQs metrics, measurements, and corrective actions

Standard Operating Procedures (SOPs)

- Supporting documentation showing the “who does what, when, and how” in completing the tasks

Communication Plan

- Outlining messages to be delivered and to who

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Control Plan Elements

Training Plan

- Document outlining the necessary training for employees to successfully perform new processes and procedures

Audit Checklists

- Document that provides auditors with the audit questions they need to ask

Response Plan

- Activities that need to be conducted when an audit fails

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What is a Control Plan?

- A control plan is a **management planning tool** used to:
  - Identify critical process steps that have significant impact on the products or services
  - Establish appropriate controls mechanisms
  - Identify, describe, and monitor the process performance
  - **Meet the customer specifications consistently**
- The **most overlooked** element of most projects

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Control Plan

Key processes and process steps are identified

Critical information regarding key measurements are documented and clarified

Process	Step ID	Process Step	CTQ Characteristic	CTQ Metric	CTQ Target	Specification/ Requirement		Measurement Method	Sample Size	Measure Frequency	Responsible for Metric	Where Recorded	Decision Rule/ Corrective Action	Responsible for Action
						LSL	USL							

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Control Plan, cont.

Measurements are clearly defined with the target value

Customer upper and lower specifications are marked

Other key measurement information is documented: sample size, measurement frequency, people responsible for the measurement, etc

Process	Step ID	Process Step	CTQ Characteristic	CTQ Metric	CTQ Target	Specification/ Requirement		Measurement Method	Sample Size	Measure Frequency	Responsible for Metric	Where Recorded	Decision Rule/ Corrective Action	Responsible for Action
						LSL	USL							

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Control Plan, cont.

Where will this measurement or report be found? Good control plans provide linking information or other report reference information

Control plans identify the mitigating action or corrective actions required in the event the measurement falls out of spec or control. Responsible parties are also declared

Process	Step ID	Process Step	CTQ Characteristic	CTQ Metric	CTQ Target	Specification/ Requirement		Measurement Method	Sample Size	Measure Frequency	Responsible for Metric	Where Recorded	Decision Rule/ Corrective Action	Responsible for Action
						LSL	USL							

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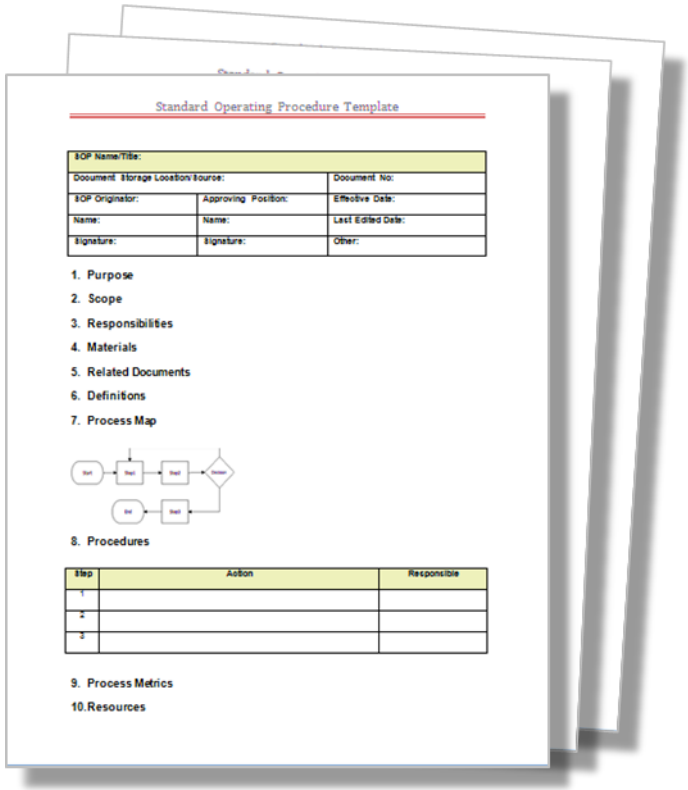
Standard Operating Procedures (SOPs)

Standard Operating Procedures (SOPs) are documents focused on process steps, activities, and specific tasks required to complete an operation:

- Not be much more than two to four pages
- Written to the user's level of required detail and information
- Auditable, easy to follow, and not difficult to find
- Auditable characteristics are: observable actions and countable frequencies



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Communication Plan

Communication plans focus on planning and preparing for the dissemination of information

- Ensure that the proper audiences receive timely and accurate messages
- A good communication plan identifies:

➢ Audience
➢ Key points/message
➢ Medium
➢ Delivery schedule
➢ Messenger
➢ Dependencies and escalation points
➢ Follow-up messages and delivery mediums

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Training Plan

- Training plans are used to manage the delivery of training for new processes and procedures

- Lack of proper training is one of the primary reasons that successful project results eventually **fail**

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Training Plan, cont.

Training plans address the following items:

- Who/What organizations require training?
- Who will complete the training?
- When will training be conducted?
- What is the timeline to train everyone on the new process(es)?

- What will trigger ongoing training?
  - New employee orientation?
  - Refresher training?
  - Part of the response plan when monitoring shows performance degrading?
- Training logistics

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Audits

ISO 9000 defines an Audit as “a systematic and independent examination to determine whether quality activities and related results comply with planned arrangements and whether these arrangements are implemented effectively and are suitable to achieve objectives.”

- Ensure actions, processes, procedures, and other tasks are performed as expected
- Must be well-defined and executed by independent, unbiased personnel
- Confirm compliance or declare non-compliance

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Audit Checklist

Auditors should review the SOPs before preparing checklists or ensure that existing checklists properly reference SOPs

- Audit checklists may include:
  - Review of training records
  - Review of maintenance records
  - Focus on expected behaviors
  - Open-ended where possible
  - Definitive observations yes/no, true/false, present/absent, etc.

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Response Plan

Response plans are a management planning tool that include:

- Identified risk / failure mode to monitor
- Triggers for a response
- Recommended response
- Who is responsible for the response
- Timeframe for contingency (corrective action) completion

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Control Phase: Roadblocks

Potential Roadblocks to Success:

- Lack of project sign-off
- Lack of involvement in Control Plan design
- Management does not have knowledge regarding the what's and how's to sustain
- Financial benefits not integrated into business and tracked
- Lack of buy-in from staff

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5.3 Lean Controls

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Lean: Control Tools

- **5S** is the **starting point**
  - Cleaning and Organizing
- **Visual Factory**
  - Notice of status
  - Alert of variation
  - Examples: Sign boards, reader boards
- **Standardized work** provides **consistency**
  - Measurable
  - Required to sustain system benefits

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Lean 5S: Summary Table

STEP	ENGLISH TERM	QUICK DEFINITION	COMMENTS
Step 1	**Sort**	Clearing up	"When in doubt, move it out"
Step 2	**Straighten / Set in Order**	Organize	Organize all necessary tools, parts, and components of production. Use visual ordering techniques wherever possible
Step 3	**Shine**	Clean	Clean machines and/or work areas. Set regular cleaning schedules and responsibilities
Step 4	**Standardize**	Standardize	Solidify previous three steps, make 5S a regular part of the work environment and everyday life
Step 5	**Sustain**	Training & Discipline	Audit, manage, and comply with established five-s guidelines for your business or facility

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Lean: Control Tools, cont.

- **Kaizen**
  - Philosophy of Continuous Improvement
  - Kaizen Blitz / Event is the application of Kaizen designs
- **Kanban**
  - “Signboard” communication of material movement
  - Allows for [customer] demand driven production
- **Poka-Yoke**
  - Mistake proofing
  - Defect prevention

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Poka-Yoke: Two Types

- **Prevention**
  - ➢Preventing defect occurrence
  - ➢Removing the possibility of errors
  - ➢Make occurrence of errors impossible
- **Detection**
  - ➢Detecting defects as they occur
  - ➢Highlighting defects, drawing attention immediately
  - ➢Correct defects before they reach the next stage

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Poka-Yoke: Three Methods

- **Contact Method**
  - Use of shape, color, size, or any other physical attributes
- **Constant Number Method**
  - Use of a fixed number to make ensure a certain number of motions are completed
- **Sequence Method**
  - Use of a checklist to make ensure prescribed process steps are followed in order

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Preventative Maintenance & Jidohka

- **Preventative Maintenance**
  - Focus on reducing and eliminating equipment malfunctions
  - Proactive approach
  - Inspecting, monitoring, and auditing
  - Checklists
- **Jidohka**
  - Stopping of production immediately once a process failure occurs
  - Process doesn't begin again until resolved

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Thank You

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